A One-Equation Transition Model: Autogenous Inception from the Spalart–Allmaras Working Variable

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We present a one-equation Reynolds-averaged Navier–Stokes (RANS) transition model in which the Spalart–Allmaras (SA) working variable $\tilde{\nu}$ serves as its own transition indicator. We repurpose its sub-$O(1)$ range—inert in the baseline model—as a Tollmien–Schlichting amplification factor: a blended production term—with the destruction tied to it and the molecular diffusion coefficient reduced in the $\tilde{\nu}$ equation only—drives the SA transport equation, unchanged in structure, through laminar instability growth and turbulent eddy-viscosity production, handing over near $\chi = \tilde{\nu}/\nu \sim 1$. No auxiliary field is introduced. The amplification rate depends only on the local mean flow—a vorticity Reynolds number and two profile-shape indicators, a vorticity fraction and an accumulated-inflection (Rayleigh) coordinate—with no pressure-gradient parameter anywhere, and is anchored to canonical stability limits: the free-shear growth eigenvalue, the linear-stability onset of the Falkner–Skan family, and the Blasius e^N envelope; the freestream seed is set by Mack's receptivity correlation. The model is tested on three cases spanning zero, favorable, and adverse pressure gradients—a flat plate, the NLF(1)-0416 airfoil at $Re = 4{\times}10^6$, and the Eppler 387 at $Re = 6{\times}10^4$–$4.6{\times}10^5$—across wide incidence ranges, on structured and unstructured meshes at three refinement levels, with the same constants throughout. The Eppler Reynolds sweep locates the model's bubble-bursting boundary one Reynolds step above the measured one and traces the displacement to the deliberately unmodeled transition zone. The model is also deployed in three dimensions on the wing of the Daedalus human-powered aircraft.

埏埴以為器，當其無，有器之用。

Clay is shaped into a vessel:

it is the emptiness within that makes it useful.

—Daodejing, ch. 11

Nomenclature

$\tilde{\nu}$	SA working variable

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$\tilde{S}$	SA modified vorticity
$\nu,\ \nu_t$	kinematic (laminar) and eddy viscosity ($\nu_t = \tilde{\nu}[(1-b)f_{v1} + b]$, Eq. (18))
$c_{v1},\ f_{v1}$	SA near-wall constant and function
$P,\ D$	production and destruction of $\tilde{\nu}$
$P_{\mathrm{SA}},\ P_{\mathrm{AI}}$	turbulent (SA) and laminar-amplification production terms
$c_{\nu,\mathrm{ai}}$	laminar-viscosity scale in the SA diffusion term only ($1/6$)
$\hat{\nu}$	disturbance field of the parabolized march, Eq. (14)
$\chi,\ \chi_\infty$	$\tilde{\nu}/\nu$ and its freestream value
d	distance to the nearest wall
ω	vorticity magnitude
Re_Ω	vorticity Reynolds number, $d^2\omega/\nu$
$X,\ Y,\ Z$	velocity, vorticity (shear), and curvature indicators, Eq. (4)
$\hat{\Omega},\ \lambda$	vorticity fraction $Y/\sqrt{X^2+Y^2} = \sin\lambda$ and longitude, Eq. (5)
$I,\ \hat{I}$	accumulated inflection $I = d\,u' - u - \frac{1}{2}d^2u''$ and its normalized (sphere) coordinate $\hat{I} = Y - X - Z$, Eqs. (6) and (7
$\hat{\Omega}\hat{I}$	amplifying (rate) coordinate: the even product of the vorticity fraction and the accumulated-inflection coordinate, Ec
$\hat{\mathbf{n}}$	unit wall-normal direction (gradient of d)
$a,\ a_{\max}$	laminar amplification rate and its free-shear ceiling (0.19)
$Re_\Omega^{\mathrm{c}}(\hat{\Omega}\hat{I})$	vorticity-Reynolds onset threshold, Eq. (13)
$C,\ A,\ B,\ k$	onset-threshold shape constants and scale, Eq. (13)
$Re_{\theta 0}(H)$	Drela–Giles critical (first-instability) momentum-thickness Reynolds number
w	onset ramp width scale, Eq. (11)
$\sigma_P, \sigma_D;\ \tau$	production and destruction handover weights and the production width, Eq. (16)
$N,\ N_{\mathrm{crit}}$	amplification factor and its critical value (e^N method)
Tu	freestream turbulence intensity
$p,\ \rho$	static pressure and density
$\lvert u\rvert$	velocity magnitude relative to the nearest-wall point
$\mathbf{d},\ \boldsymbol{\omega}$	wall-distance vector $d\,\nabla d$ and vorticity vector
$\boldsymbol{\ell}$	curvature vector $\frac{1}{2}\langle \mathbf{d}, \mathbf{d}\rangle\, \nabla^2\mathbf{u}$, Eq. (E4)
$\langle\cdot,\cdot\rangle$	Euclidean inner product of vectors
$b;\ q,\ s,\ G$	f_{v1}-bypass blend weight and its gate inputs, Eqs. (17)–(18)
H	boundary-layer shape factor
α	angle of attack

$c,\ x/c$	airfoil chord and chordwise coordinate
$x_{\mathrm{tr}},\ x_R$	transition-onset and turbulent-reattachment locations
$Re,\ M$	chord Reynolds number and Mach number
$Re_x,\ Re_\theta$	streamwise and momentum-thickness Reynolds numbers
$C_f,\ C_d,\ C_l$	skin-friction, drag, and lift coefficients

I. Introduction

Laminar flow over a lifting surface offers one of the largest available viscous-drag reductions, and predicting where it ends is a first-order requirement for natural-laminar-flow design. This is not only a transport-aircraft concern: at the low-to-moderate chord Reynolds numbers of general-aviation and sport aircraft and of small—increasingly electric—rotors and drones, the laminar run is a large fraction of the chord, and its extent sets efficiency, range and endurance, and acoustic signature. In a low-disturbance environment the laminar boundary layer first becomes unstable to gentle two-dimensional Tollmien–Schlichting (TS) waves [1, 2]: as the layer thickens, the envelope of the strongest TS mode grows quasi-exponentially over a long streamwise stretch until, at an amplitude of a few percent of the mean, three-dimensional secondary instabilities trigger localized turbulent spots that spread and merge into a fully turbulent layer [2, 3]. The slow TS-envelope stage can occupy most of the streamwise distance and set where transition is observed, so it is the stage a transition model must reproduce. The classical tool is the e^N method [4, 5]: linear stability theory is integrated along the boundary layer to accumulate an amplification factor N, and transition is declared where N reaches an empirical threshold N_{crit} set by the disturbance environment [1]. Drela and Giles reduced the stability integration to an approximate envelope correlation, making e^N practical inside an integral-boundary-layer method [6, 7].

The e^N method is non-local: it presumes a boundary layer that can be marched and integrated, which is foreign to a general unstructured RANS solver. The community has therefore recast transition as locally computable transport equations, along three lines. Amplification factor transport carries N itself: Coder and Maughmer's AFT model transports an amplification factor whose source reproduces the e^N envelope and couples it to SA through the f_{t2} term, the variant catalogued as SA-AFT2017b on the NASA Turbulence Modeling Resource adding a second (intermittency) equation [8, 9]; Halila et al. smoothed this model (AFT-S) for implicit Newton–Krylov solution in gradient-based optimization, retaining the separate amplification-factor equation [10]. Local-correlation intermittency transport instead carries an intermittency: the Langtry–Menter γ–Re_θ model adds two transport equations [11, 12], reduced to one in a later γ-only formulation [13] and coupled to SA by Medida and Baeder [14]; Walters and Cokljat transport a laminar-fluctuation energy [15]. Finally, algebraic models add no transport equation: the Bas–Cakmakcioglu model and its revision (BC/BCM) multiply the SA production by an intermittency computed from purely local correlations

[16, 17], an approach extended in several recent SA modifications [18, 19].

These nearest neighbors share a premise we depart from: the SA working variable $\tilde{\nu}$ is treated as a purely turbulent quantity, switched on either by an extra transported field (AFT, γ–Re_θ) or by an external algebraic intermittency (BC/BCM). Equation count alone therefore does not separate the present approach from the algebraic family, which is also one-equation. The distinction is in what the field is used for. A scaling argument motivates—it is a motivation, not a representation claim—placing a Tollmien–Schlichting envelope variable in the sub-$O(1)$ range of $\tilde{\nu}$. The SA working variable has the dimensions of an eddy viscosity—a product $u'\ell$ of a fluctuation-velocity scale u' and a coherence length ℓ. In a Blasius-like layer just before secondary instability, the TS-envelope amplitude u' is a few percent of the mean and its wall-normal coherence length ℓ is a fraction of the boundary-layer thickness, at $Re_\theta \sim O(10^2)$–$O(10^3)$ [2, 20]. Their product is therefore at most a few times the molecular viscosity,

$$\frac{u'\ell}{\nu} \lesssim 1, \tag{1}$$

so a disturbance-viscosity scale fits within $\chi = \tilde{\nu}/\nu$ of order one or below through the pre-transitional layer.

This is the part of the SA state space the baseline model barely uses: for $\chi \lesssim 1$ the field is dynamically inert [21]—a useful emptiness (無之以為用, “usefulness comes from what is not there”; Daodejing 11) we exploit by letting that range be the amplification factor. In the laminar regime the production of $\tilde{\nu}$ is replaced by a local, envelope-method-inspired instability rate, so $\ln \tilde{\nu}$ accumulates monotonically in the manner of an amplification factor; as $\tilde{\nu}$ approaches its turbulent level the production reverts smoothly to the standard SA form. One transport equation—the SA equation, unchanged in structure—spans laminar instability and turbulence with no auxiliary field. The same working variable whose magnitude measures how turbulent the flow already is thus also carries the inception of that turbulence: the field gives rise to its own transition from within.

We refer to the model as SA-AI* (Spalart–Allmaras with Autogenous Inception), distinguishing it from Coder’s SA-AFT. Table 1 had one empty cell—amplification-factor (e^N-like) physics carried by the turbulence working variable itself, through a local, continuous source—which the last row fills, and this paper is an exploration of it: what has to be added for the working variable to carry its own inception, and what comes out when it does.

SA-AI targets natural, Tollmien–Schlichting transition in two-dimensional, low-speed flow, and it aims at the transition location and the macroscopic aerodynamics that follow from it—not at the transitional zone’s internal structure (Sec. II). In the same spirit, we do not constrain $\tilde{\nu}$ to represent how large the disturbance is; the model is built to reproduce the effect of the disturbances, and the scaling of Eq. (1) says only that the range in which it does so is dimensionally natural. Bypass transition (high freestream turbulence, $Tu \gtrsim 1\%$, where the mechanism is streak-induced

*Where the prevailing pursuit of artificial intelligence is all Yang—ever more parameters, more compute—this model tries the Yin: transition arising autogenously—of itself, ziran (自然, “self-so”)—from a variable the baseline equation already carries, with no equation added. Here “AI” abbreviates Autogenous Inception, not the other kind.

Table 1 RANS-compatible transition-model families and where the present model (SA-AI) sits. "Extra equations" counts transport equations beyond the baseline turbulence model.

Family	Extra eqns	Transition mechanism
e^N envelope [4, 6]	— (non-local)	stability-integral N
AFT [8, 10]	1–2	transported amplification factor
γ–Re_θ [12, 14]	2	transported intermittency, $Re_{\theta t}$
one-equation γ [13]	1	transported intermittency
algebraic SA [16, 18]	0	local intermittency gates production
SA-AI (present)	**0**	working variable is the factor

and demands a closure driven by freestream turbulence energy; Sec. III), crossflow transition on swept geometries, and compressible/transonic regimes lie outside the present formulation and are left to future work; the validation here is incompressible ($M = 0.1$). Section II formulates the model and fixes all its constants but the freestream seed in one pass, each constant anchored, as its form is introduced, to a canonical stability limit or established correlation rather than to the transition test cases. Section III fixes the remaining input—the freestream seed, via Mack's receptivity map—and verifies the resulting flat-plate onsets against the Abu-Ghannam & Shaw correlation—the zero-pressure-gradient regime. The same solver, constants (collected in Appendix E), and seeding then carry the two airfoil regimes, each on both mesh families at three refinement levels and all incidences: the favorable rooftop on the NLF(1)-0416 (Sec. IV) and the adverse-gradient, laminar-separation-bubble regime on the Eppler 387 (Sec. V), with the Eppler then swept across the tunnel's Reynolds range (Sec. VI), locating the model's bubble-bursting boundary and the mechanism that misplaces it. Section VII takes the same machinery to the circular cylinder and traverses the drag crisis in Reynolds number over ten decades on a single continuation ladder. Section VIII isolates a bistability the model inherits from standard SA itself—an attachment-anchored turbulent branch at the leading edge—and states the convergence protocol every result in this paper uses. Section IX presents a three-dimensional deployment on the Daedalus human-powered-aircraft wing, and Sec. X closes with the inclined 6:1 prolate spheroid, probing the model's stated scope boundary where transition turns crossflow-dominated.

II. The amplification model

The model is a single algebraic source term added to the Spalart–Allmaras transport equation, built entirely from the local mean-velocity profile. This section constructs it from what it must reproduce, on a geometric picture of the local profile that we build up in turn.

The Spalart–Allmaras model [22] solves a transport equation for $\tilde{\nu}$,

$$\frac{D\tilde{\nu}}{Dt} = P_{\mathrm{SA}} - D_{\mathrm{SA}} + \frac{1}{\sigma}\nabla\cdot\left[(\nu+\tilde{\nu})\nabla\tilde{\nu}\right] + \frac{c_{b2}}{\sigma}|\nabla\tilde{\nu}|^2, \qquad P_{\mathrm{SA}} = c_{b1}\tilde{S}\,\tilde{\nu}, \quad D_{\mathrm{SA}} = c_{w1} f_w\left(\frac{\tilde{\nu}}{d}\right)^2, \tag{2}$$

with the modified vorticity $\tilde{S}$, the wall function f_w, the constants $c_{b1}, c_{b2}, c_{w1}, \sigma, \kappa$, and the eddy viscosity $\mu_t = \rho\,\tilde{\nu} f_{v1}(\chi)$, $\chi = \tilde{\nu}/\nu$, all as defined there; we use the negative-$\tilde{\nu}$ safeguards of the standard implementation [23]. The eddy-viscosity function $f_{v1}(\chi) = \chi^3/(\chi^3 + c_{v1}^3)$ vanishes cubically for $\chi \ll c_{v1}$ (with $c_{v1} = 7.1$, $f_{v1} \approx 2.8\times10^{-3}\,\chi^3$): even at $\chi = 1$ the eddy viscosity $\mu_t = \rho\,\tilde{\nu} f_{v1}$ is $\sim 0.3\%$ of $\rho\tilde{\nu}$. The field $\tilde{\nu}$ therefore contributes negligibly to the mean flow while $\chi \lesssim 1$: this sub-$O(1)$ range is dynamically inert in the baseline model and is what we repurpose.

In the laminar range we replace the turbulent production P_{SA} by a laminar amplification production of the same algebraic form,

$$P_{\mathrm{AI}} \;=\; a\,\omega\,\tilde{\nu}, \tag{3}$$

with ω the vorticity magnitude supplying the inverse time scale in place of $c_{b1}\tilde{S}$ and a a nondimensional rate; the remainder of this section constructs a from the local mean-velocity profile.

A. The profiles the kernel must match

The canonical laminar instabilities are each a self-similar mean-velocity profile, and for each, linear stability theory tells the kernel two things: the growth rate, and where, across the layer, the disturbance extracts its energy. We take the profiles in order of increasing instability. The attached Falkner–Skan family [24] runs from favorable gradients, whose full, convex profiles are stable, through the Blasius layer (Tollmien–Schlichting), to adverse gradients, where the instability waves amplify faster than the Tollmien–Schlichting waves of a flat plate. Past separation the reversed-flow lower branch of Falkner–Skan (Stewartson [25]) is the detached shear layer of a laminar separation bubble, the fastest-growing wall-bounded case. Its unbounded limit is the free mixing layer, the most unstable of all; its Kelvin–Helmholtz eigenvalue sets the rate ceiling (§II.C).

The second requirement—resolving where the disturbance grows—is easy to overlook but also important, because the rate a of (3) is a temporal one. With the diffusion and destruction of (2) negligible against production in the amplifying range, (3) drives the working variable along particle paths, $D\tilde{\nu}/Dt = a\,\omega\,\tilde{\nu}$; in a steady, thin shear layer the material derivative collapses to streamwise convection ($\partial_t = 0$, $v\,\partial_y \ll u\,\partial_x$), giving $u\,\partial_x\tilde{\nu} = a\,\omega\,\tilde{\nu}$. Transition, however, is set by the spatial amplification—the N-factor accumulated along the wall—and the two differ by the local advection speed: the spatial growth the model deposits at height y is $(a\,\omega)/u(y)$. Placing the rate at the wrong wall-normal station therefore does not merely slightly mis-scale the growth; through the $1/u$ weighting it can over-amplify unboundedly wherever u is small. The laminar separation bubble is the sharp case: the physical wavepacket rides the detached shear layer at a phase speed near half the edge velocity ($c \approx 0.5\,U_e$ for the detached shear layer [26]), but a kernel that lets growth slide toward the dividing streamline ($u \approx 0$) deposits a diverging spatial rate. The kernel must resolve not only how fast a disturbance grows but where, and that calls for local sensors of the profile shape rich enough to place it.

B. The indicator sphere

The local sensors are few. In a thin, quasi-two-dimensional boundary layer only one velocity component is significant and only its wall-normal derivatives are large, so to second order about any wall-normal point the profile is captured by exactly three velocity scales formed from the wall distance d—the velocity u, the shear (in-plane vorticity) $d\,u'$, and the curvature $\frac{1}{2}d^2u''$ (the Taylor coefficients weighted by d^k, the $\frac{1}{2}$ the quadratic term's actual velocity contribution). There is a fourth independent local number, the vorticity Reynolds number $Re_\Omega = d^2\omega/\nu$; but the amplification rate is Reynolds-independent beyond its viscous cutoff [6], so Re_Ω does not enter the rate and is reserved for the onset (§II.D). The three velocity scales, nondimensionalized by their common magnitude R, place the local state on the unit sphere of directions

$$(X,Y,Z) = \big(u,\ d\,u',\ \tfrac{1}{2}d^2u''\big)/R, \qquad R = \sqrt{u^2 + (du')^2 + (\tfrac{1}{2}d^2u'')^2}. \tag{4}$$

Because the growth is invariant under $u \to -u$ (a layer and its mirror are equally unstable), the state is a direction defined only up to sign—a point of the real projective plane $\mathbb{RP}^2$, the space of lines through the origin in $\mathbb{R}^3$.†

The kernel built on this sphere must be smooth: a profile evolves continuously along the wall and up through the layer, tracing a continuous curve on $\mathbb{RP}^2$, so a rate with jumps would inject non-physical discontinuities into the source. That requirement is strict over the bulk of the sphere, but it relaxes at the curvature poles, for a concrete reason. The curvature carries a d^2 scale where velocity and shear carry d^0 and d^1, so the curvature pole ($Z \to \pm 1$, $X, Y \to 0$) is typically reached in the shears and vortices standing away from the wall, where the growing d^2 inflates Z until it dominates the velocity and shear: a real wall-bounded profile approaches it but never sits on it. The pole is thus a measure-zero limit the kernel is never evaluated at; only the approach matters, and along it the longitude $\lambda = \operatorname{atan2}(Y, X)$ stays finite and distinguishes a detached shear layer standing off the wall from a genuine no-shear extremum. The kernel may therefore be left continuous but non-differentiable at the poles while smooth everywhere a profile actually visits. The vorticity fraction

$$\hat{\Omega} \;=\; \frac{Y}{\sqrt{X^2+Y^2}} \;=\; \sin\lambda, \tag{5}$$

is an example: continuous everywhere on the sphere but non-differentiable at the curvature pole—where $X, Y \to 0$ and the ratio depends on the direction of approach—yet legitimate on $\mathbb{RP}^2$. A function of longitude alone, it does not vanish merely because a layer is far from the wall, unlike the components normalized by R in (4).

The projective identification also fixes an algebra the model must obey. A single indicator, or any degree-one combination of them, reverses sign under the antipodal identification $(X,Y,Z) \sim (-X,-Y,-Z)$ and so is not a single-valued function on $\mathbb{RP}^2$; only even products—of two such odd quantities—are well defined. Every modeling term must therefore be built as such a product. This is not cosmetic: it is what lets the kernel carry to three dimensions, where there is no global orientation (no "$u > 0$") and only rotation-invariant products of the indicator vectors survive.

†$\mathbb{RP}^2 = S^2/\{\pm\}$; the sphere double-covers it. We plot one hemisphere, with the antipode of any point understood to be identified with it.

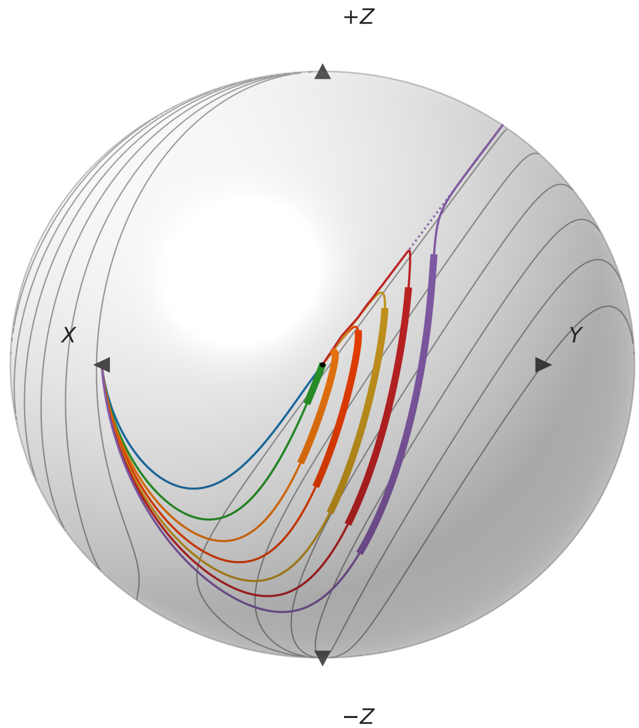

Fig. 1 The profiles the kernel must match, on the $\mathbb{RP}^2$ indicator sphere (4) (orthographic x-ray view: curvature vertical, vorticity−velocity horizontal; near hemisphere solid, far side dotted; triangles mark the four visible poles—velocity (X, left), vorticity (Y, right), ±curvature ($\pm Z$, top and bottom)). Seven Falkner–Skan profiles run as thin curves from the wall (center) toward the freestream: favorable ($\beta = +0.30$, blue), Blasius ($\beta = 0$, green), moderate adverse ($\beta = -0.10$, orange), strongly adverse ($\beta = -0.16$, red-orange), near-separation attached ($\beta = -0.19$, gold), incipient separation ($\beta = -0.1988$, red), and a separated reversed-flow shear layer—the Stewartson lower-branch solution at $\beta = -0.19$, $H \approx 4.9$ (purple). The thick segment on each marks the wall-normal band where its linear-stability (Orr–Sommerfeld Reynolds-stress production) density exceeds half its peak—where the disturbance actually grows; the attached profiles are evaluated at $Re_\theta = 500$ (the favorable layer is still subcritical there and carries no band), the reversed-flow profile at $Re_\theta = 200$. Gray lines are contours of the rate coordinate $\hat{\Omega}\hat{I}$ at $\{0.025, 0.2, 0.4, 0.6, 0.8, 1.0\}$.

The profiles of §II.A become curves on this sphere (Fig. 1), and following one fixes the coordinate the rate will need. Every profile begins at the same wall point: the linear sublayer $u \approx u'_w d$ makes velocity and shear equal, and although the curvature u'' is finite there, its d^2 weight sends the indicator $Z \to 0$, so the state sits at $X = Y$, $Z = 0$ (longitude $\lambda = 45°$, on the equator). What every profile shares as it leaves is not a common direction but a constraint. To second order the near-wall profile is a parabola, and a parabola $u = By + Cy^2$ satisfies, through (4), the single linear relation

$$\hat{I} \;\equiv\; Y - X - Z \;=\; \text{vorticity} - \text{velocity} - \text{curvature} \;=\; 0 \tag{6}$$

identically, so the parabola great circle $\hat{I} = 0$ is the neutral locus. A real profile matches its osculating parabola through second order; $\hat{I}$ and its first two wall derivatives therefore vanish, and $\hat{I}$ departs zero only at third order (a Taylor identity). The profiles do not all leave the wall point along the same sphere-tangent—only those with wall curvature ($u''_w \neq 0$, a pressure-gradient effect) leave tangent to the circle, while a zero-gradient layer, having $u''_w = 0$, peels off at a small angle—but every one departs with $\hat{I}$ third-order small. What distinguishes them is how they leave that circle, and a profile can leave it only through its third derivative: integrating (6) by parts against $u(0) = 0$ gives, for the unnormalized combination,

$$I(d) \;\equiv\; R\,\hat{I} \;=\; d\,u' - u - \tfrac{1}{2}d^2 u'' \;=\; -\int_0^d \tfrac{1}{2}\,s^2\,u'''(s)\,\mathrm{d}s, \tag{7}$$

so I is exactly the residual of the near-wall parabolic fit: the accumulated inflection—the integrand $-\frac{1}{2}s^2u''' = \frac{1}{2}s^2\omega''$ is the third-derivative contribution to the wall-to-point change of the vorticity, and I accumulates it from the wall to the probe point. Since $u''' = -\omega''$ (with the two-dimensional sign convention $\omega = -u'$ used in this section; elsewhere ω denotes the vorticity magnitude), it is the accumulated curvature of the vorticity profile, the inflectional content Rayleigh's theorem ties to inviscid instability [2]; $\hat{I} > 0$ marks the inflectional side.

Read against Fig. 1, the family orders itself by how its wall curvature evolves, since (7) integrates the slope of that curvature. The favorable layer carries a negative wall curvature ($u''_w < 0$) that only relaxes toward zero at the edge; its slope u''' stays positive, so $\hat{I}$ leaves the circle tangentially and dives at once to the stable $\hat{I} < 0$ side, never turning inflectional. Blasius starts flat ($u''_w = 0$): its curvature first builds up—dipping negative a short way off the wall—before decaying back, so $u''' < 0$ near the wall lifts $\hat{I}$ into a small positive bump (the weak instability) before $u''' > 0$ aloft draws it down again. An adverse gradient deepens and widens that curvature deficit, and the s^2 weight and thicker layer amplify its contribution, so the bump grows into a large positive hump: the adverse and incipient-separation profiles sit well onto the inflectional ($+\hat{I}$) side. The separated profile leaves the wall the other way (dotted, reversed flow) along the same parabola, and along that arc it too is stable; but as it crosses the velocity-zero longitude—the vorticity pole on the equator—$\hat{I}$ turns sharply positive, past even the separation threshold, the most unstable member of the family. Every profile finally settles at the velocity pole ($X \to 1$, $\hat{I} \to -1$) as the shear dies at the edge; what distinguishes them is only the size of the positive excursion in between.

C. Where and how much the instabilities amplify

On each of these curves we thicken the wall-normal band where the disturbance actually grows—the contiguous interval over which the most-unstable Orr–Sommerfeld mode's production density $\frac{\alpha_x}{2}\,|\mathrm{Im}(\phi'\phi^*)|\,|U'|$ (α_x the streamwise wavenumber) stays above half its peak (the thick segments in Fig. 1). Two facts then read straight off the plot. First, every amplifying band lies east of the velocity-pole longitude, toward the vorticity pole $+Y$ and away from the pure-curvature poles. Second, the amplifying region is bounded below by exactly the parabola great circle $\hat{I} = 0$ of (6): a parabola being inviscidly neutral by Rayleigh's theorem, $\hat{I} = 0$ is the neutral boundary and every band sits on the inflectional

$\hat{I} > 0$ side identified in §II.B. Where amplification can begin follows from the same integral (7): the s^2 weight and the steady-wall compatibility $u_w''' = 0$ [27] start every profile at $\hat{I} = 0$, tangent to the neutral circle. This has two consequences. A layer reaches the inflectional side only once accumulated curvature reversal carries it across $\hat{I} = 0$—the lower bound the bands of Fig. 1 sit above. And because the rate coordinate $\hat{\Omega}\hat{I}$ inherits $\hat{I} = 0$ at the wall, the kernel produces nothing there: amplification is held off the wall by construction, never seeded at the no-slip line where the eddy viscosity vanishes. The growing region sits at a viscous standoff fixed by the Re_Ω onset gate (§II.D), which is what makes the inception Reynolds-number dependent even though the rate itself is not.

The rate joins the two odd sphere functions—the vorticity fraction $\hat{\Omega}$ and the accumulated-inflection coordinate $\hat{I}$—into their even product $\hat{\Omega}\,\hat{I}$, the single amplifying coordinate the projective algebra allows: positive on the inflectional (shear) side, negative on the parabola's stable side, zero on the neutral circle, and—being weighted by the longitude-only $\hat{\Omega}$—finite at the curvature pole, so a detached shear layer standing off the wall still registers. The rate is

$$a \;=\; a_{\max}\ \mathrm{clip}\big\langle\, \hat{\Omega}\,\hat{I}\,\big\rangle_0^1, \qquad \mathrm{clip}\langle x\rangle_0^1 \;=\; \min\big(\max(x,0),\ 1\big), \tag{8}$$

linear in the even product, with $a_{\max}$ the free-shear ceiling (fixed just below), reached where $\hat{\Omega}\hat{I} \geq 1$ at the vorticity pole ($\hat{\Omega} \to 1$, the free-shear-layer limit). No near-neutral shaping is needed: measured against the Drela–Giles envelope, the linear rate tracks it across the attached shape-factor family, softening only at the separation edge (below). Because a favorable gradient keeps the profile's curvature single-signed (the accumulated curvature reversal that carries a layer across the neutral circle never builds), $\hat{\Omega}\hat{I} \to 0$ on the favorable side and the rate vanishes with it; no separate suppression term is needed.

The prefactor $a_{\max}$ is a natural physical limit, set by the structure of the amplification itself. To the accuracy of the e^N method, the envelope amplification a profile earns per unit Re_θ is a function of its shape alone—Reynolds-independent [6]. The canonical layers therefore form a single shape-driven continuum: from the near-parabolic Blasius profile, through the adverse layers as an inflection point emerges and the growth turns inviscid (Rayleigh), to the detached shear layer of a separation bubble and its unbounded limit, the free mixing layer. The coordinate $\hat{\Omega}\hat{I}$ runs along exactly this continuum—zero on the neutral parabola, unity at the fully-inflectional vorticity pole—so one ceiling closes it.

At that pole the profile is a free shear layer, and its amplification is the inviscid Kelvin–Helmholtz instability: a temporal eigenvalue with no free constant. For the hyperbolic-tangent layer the most-amplified mode grows at $\omega_{i,\max} = 0.1897\,U_0/\delta$ against a peak vorticity $\omega_{\text{peak}} = U_0/\delta$ [26], a ratio independent of the velocity and thickness normalization, so

$$a_{\max} \;=\; \omega_{i,\max}/\omega_{\text{peak}} \;=\; 0.19,$$

an eigenvalue—the head of the continuum. The model grows $\tilde{\nu}$ along particle paths at $a(\hat{\Omega}\hat{I})\,\omega$, a temporal rate that

Table 2 The inviscid frozen-profile slope of the rate (8), $\max_y a\,\omega/u$, as a ratio to the Drela–Giles spatial envelope rate s_{DG}/θ, on five Falkner–Skan profiles (starred: the reversed-flow, Stewartson lower-branch solution of the same β). $\max_y \hat{\Omega}\hat{I}$ is each profile's peak rate coordinate—the scale the airfoil visualizations of Secs. IV–V are read against. The ratio is independent of Reynolds number and of any diffusion: it is the most growth per unit length the rate can sustain on the profile. The diffusion retained by the working variable, taken up next, lowers the realized slope toward and below these values at transitional Reynolds numbers (Table 3).

profile	β	H	$\max_y \hat{\Omega}\hat{I}$	$\theta\ \max_y(a\,\omega/u)\ /\ s_{\mathrm{DG}}$
moderate favorable	+0.10	2.48	0.037	1.30
Blasius	0	2.59	0.078	1.42
moderate adverse	−0.10	2.80	0.162	1.30
separation limit	−0.1988	3.98	0.510	0.97
separated, shallow*	−0.19	4.92	0.657	0.97

convection lays down as the spatial N-factor (Eq. (3)): the same temporal-to-spatial passage that connects a growth eigenvalue to an accumulated amplification.

How well does the closed rate hold up, quantitatively, against the correlation it must reproduce? The comparison can be made analytically, before any transport equation is solved. Hold a Falkner–Skan profile frozen and parallel: a disturbance riding it grows at $a(\hat{\Omega}\hat{I})\,\omega$ while it advects at the local u, so the growth the rate can hand the transport per unit length—before any viscous loss—is the pointwise supremum $\max_y a\,\omega/u$. Drela's envelope correlation states the same quantity as an explicit spatial rate, s_{DG}/θ with $s_{\mathrm{DG}} = \frac{m(H)+1}{2}\,\ell(H)\,\mathrm{d}N/\mathrm{d}Re_\theta$ (ℓ, m the Falkner–Skan similarity factors of Drela's spatial conversion [6]), so the two compare with no wave-speed convention on either side. Table 2 lists the ratio across the family—which stops at $H \approx 5$, where Drela's Orr–Sommerfeld data end and no similarity profile remains validated against measured bubble profiles—and two facts read off it. Across the attached branch the inviscid slope rides a nearly uniform 1.30–1.42× above the correlation—headroom, and what consumes it is the subject of the next paragraphs. At the separation limit and onto the shallow reversed-flow branch the ratio is 0.97.

The headroom is spent on viscosity. The working variable that carries the disturbance lives in the SA equation (2), and it diffuses with the molecular coefficient all the while it grows. On the frozen profile the contest is an eigenvalue problem: growth, diffusion, and advection compose the generalized problem

$$\left[a(\hat{\Omega}\hat{I})\,\omega \;+\; \frac{c_{\nu,\mathrm{ai}}\,\nu}{\sigma}\,\partial_y^2\right] v \;=\; s\,u\,v, \tag{9}$$

whose leading eigenvalue s is the growth per unit length the transport can actually realize on that profile at that Reynolds number—the inviscid supremum of Table 2 recovered as $Re_\theta \to \infty$, approached like $Re_\theta^{-1/2}$ as the eigenfunction narrows onto its peak. The problem is well posed and cheap: $v = 0$ at the wall and the far field, the reversed-flow advection floored at $0.02\,U_e$ (a choice the result is insensitive to: floors 0.01–0.05 U_e move no printed digit), and the substitution $v \to u^{-1/2} v$ then makes the operator symmetric, so each entry below is one symmetric tridiagonal eigenvalue. Here $c_{\nu,\mathrm{ai}}$

is a scale on the molecular coefficient, equal to one until decided otherwise. Table 3 carries the result, at the stations where each profile is unstable by the Drela–Giles criterion ($Re_\theta > Re_{\theta 0}(H)$, third column): below its critical Reynolds number a profile earns no envelope growth, and there is nothing to compare against. With the coefficient untouched ($c_{\nu,\mathrm{ai}} = 1$, left half), the equation declines the correlation everywhere the family grows: half of Drela's slope just past the Blasius critical station (0.49 at $Re_\theta = 400$), 0.63 at the favorable wedge's lone unstable station, and 0.64–0.85 even on the separation branch, where diffusion should matter least. At full molecular diffusion the drain does not perturb the growth; it overwhelms it.

So we let the disturbance keep less of it. The laminar viscosity in the SA diffusion term—and only there—is scaled by a constant factor $c_{\nu,\mathrm{ai}} < 1$,

$$(\nu + \tilde{\nu}) \;\longrightarrow\; (c_{\nu,\mathrm{ai}}\,\nu + \tilde{\nu}) \quad \text{in the diffusion term of Eq. (2) only,} \tag{10}$$

leaving the mean-flow viscosity, the SA production and destruction, and the definition $\chi = \tilde{\nu}/\nu$ (formed with the true ν) untouched. Doing so barely disturbs the original SA calibration—the scaling multiplies a term that is negligible or exactly zero wherever that calibration acts, quantified in Sec. II.G. How much less, the eigenvalue decides, read against a constraint that pulls the other way. Halving the coefficient down the ladder $c_{\nu,\mathrm{ai}} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \frac{1}{12}$, the station just past the Blasius critical station ($Re_\theta = 400$) climbs 0.49, 0.75, 0.87, 1.02, 1.13: the growth arrives on the correlation at $\frac{1}{6}$ and overshoots past it. The family arrives together—the favorable wedge's lone unstable station reads 1.02 at the same $\frac{1}{6}$, and the moderate adverse crosses unity between $\frac{1}{3}$ and $\frac{1}{6}$ (1.04 at $Re_\theta = 400$). Pressing further toward the inviscid ceiling would demand ever more grid resolution: the molecular floor sets the smallest wall-normal scale the $\tilde{\nu}$ field can develop, the equilibrium band thickness scaling as $\sqrt{c_{\nu,\mathrm{ai}}}$, so every halving thins the disturbance band the mesh must resolve by $\sqrt{2}$. We stop at

$$c_{\nu,\mathrm{ai}} \;=\; \tfrac{1}{6},$$

where the just-past-critical stations sit on the correlation and the separation branch holds 0.82–0.91 (right half of Table 3); what remains is a mild drift above it later in each profile's unstable decade (Blasius 1.21 by $Re_\theta = 1600$, where transition on that layer is already complete), climbing toward the finite ceiling of Table 2 as the diffusion keeps vanishing. The negative-$\tilde{\nu}$ safeguards [23] carry over with one consistency requirement: the negative-branch factor f_n is formed with the scaled coefficient $c_{\nu,\mathrm{ai}}\,\nu$, which keeps the total diffusivity positive, $c_{\nu,\mathrm{ai}}\nu + f_n\tilde{\nu} > 0$.

What the rate and its retained diffusion do not set is where the growth may begin. A laminar layer does not amplify from birth: linear theory gives each shape a floor, the critical Reynolds number $Re_{\theta 0}(H)$ below which every disturbance decays (the third column of Table 3), and growth starts only past it. The frozen profile of Eq. (9) knows no such floor: growth beats the drain from Re_θ of a few tens (≈ 20 on Blasius at the adopted coefficient)—far below the physical

Table 3 The frozen-profile eigenvalue of Eq. (9)—the growth per unit length the transport realizes with diffusion retained—as a ratio to the Drela–Giles rate s_{DG}/θ across the transitional decade Re_θ = 200–1600, at the untouched molecular coefficient ($c_{\nu,\mathrm{ai}} = 1$) and at the adopted $c_{\nu,\mathrm{ai}} = \frac{1}{6}$. Profiles are identified by their peak rate coordinate $\max_y \hat{\Omega}\hat{I}$ (the β map is Table 2). Entries appear only at stations above the profile's Drela–Giles critical Reynolds number $Re_{\theta 0}(H)$ (third column): where the profile is stable, the correlation has no growth to compare against. Each entry is one symmetric tridiagonal eigenvalue ($u^{1/2}$ similarity; $v=0$ at the wall and the far field; reversed-flow advection floored at $0.02\,U_e$, floors 0.01–0.05 moving no printed digit). The starred row is the reversed-flow (Stewartson lower-branch) solution at $\beta=-0.19$; the $Re_\theta \to \infty$ limit of every row is Table 2.

			$c_{\nu,\mathrm{ai}} = 1$				$c_{\nu,\mathrm{ai}} = 1/6$			
profile	$\max_y \hat{\Omega}\hat{I}$	$Re_{\theta 0}$	200	400	800	1600	200	400	800	1600
moderate favorable	0.037	827	–	–	–	0.63	–	–	–	1.02
Blasius	0.078	242	–	0.49	0.75	0.94	–	1.02	1.13	1.21
moderate adverse	0.162	97	0.50	0.72	0.87	0.99	0.94	1.04	1.11	1.16
separation limit	0.510	36	0.64	0.72	0.79	0.83	0.82	0.86	0.89	0.91
separated, shallow*	0.657	26	0.68	0.75	0.80	0.85	0.83	0.87	0.89	0.91

floor—so the laminar regime cannot be held by the drain alone: the inception is a threshold of its own, and it is the subject of the next subsection.

D. The onset threshold

In the e^N envelope description a laminar layer amplifies in two regimes: nothing grows until the layer reaches a shape-dependent critical Reynolds number $Re_{\theta 0}(H)$, and past it the envelope earns amplification at a rate that is, to the accuracy of the method, Reynolds-independent—a function of the shape alone [6]. The algebraic rate (8) carries the second regime and no explicit Reynolds number; the first—the viscous inception—is a separate, multiplicative gate on the vorticity Reynolds number, rising smoothly from 0 to 1,

$$S\big(Re_\Omega/Re_\Omega^{\mathrm{c}}(\hat{\Omega}\hat{I})\big), \qquad S(\zeta) = \tfrac{1}{2}\big[1 + \tanh\big((\zeta-1)/w\big)\big], \qquad Re_\Omega = d^2\omega/\nu, \tag{11}$$

with w the onset ramp width scale and its threshold a function of the same product $\hat{\Omega}\hat{I}$ (the form (13), fixed below). The width is not a fitted quantity. We set $w=0.35$—the gate runs from 12% to 88% open as Re_Ω crosses from 0.65 to 1.35 of its threshold—and the model barely notices: Re_Ω sweeps through the ramp quickly along a growing layer (it grows like Re_θ^2 at fixed shape), and the marched $N=1$ stations of the family (Sec. II.D) move by under 1% when w is halved and under 2.5% when it is doubled—shifts the anchor absorbs. The finite width itself is right: stability onset is not sharp (the graze scatter below is 6–14%), and a near-discontinuous source is hostile to the coupled solver's meshes and Newton convergence. Rate and onset reinforce: near the parabola a low rate and a high threshold give the late, slow transition of a zero-pressure-gradient layer; inflectional profiles get a high rate and a low threshold, hence early, fast

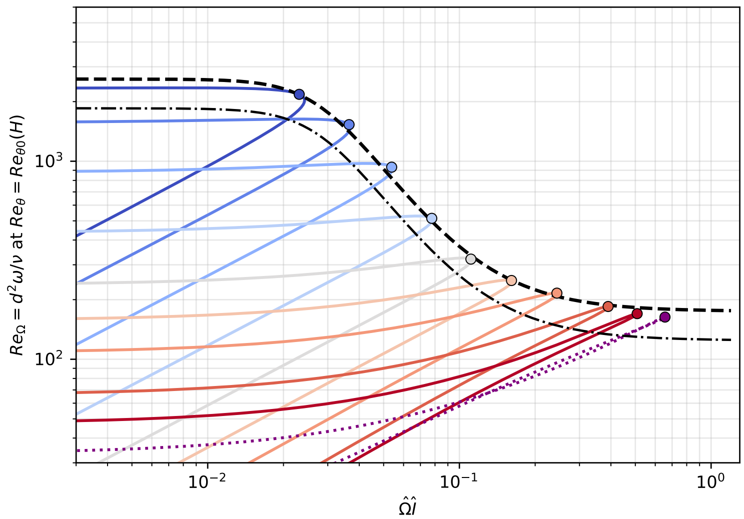

Fig. 2 The onset threshold's shape from the linear-stability graze. Each Falkner–Skan profile, at its Drela–Giles critical Reynolds number $Re_{\theta 0}(H)$, traces a curve in the $(\hat{\Omega}\hat{I},\ Re_\Omega)$ plane: solid, colored blue to red with decreasing β, the attached family β = +0.15, +0.10, +0.05, 0 (Blasius), −0.05, −0.10, −0.15, −0.19, −0.1988 (incipient separation); dotted, a separated reversed-flow profile, the Stewartson lower-branch solution at $\beta = -0.19$ ($H \approx 4.9$). The attached family shares the dashed outer envelope, which each member grazes at its marked closest-approach point: graze ratios 0.93–1.14 (root-mean-square ∼6%) from incipient separation through β = +0.15. The soft-min shape of Eq. (13), at $k = 1$, is that envelope: a ceiling on the favorable (small $\hat{\Omega}\hat{I}$) side, an additive floor toward separation; dash-dot, the model threshold—the same shape scaled by the calibrated $k = 0.712$ of the march anchor below.

transition. The complete source is

$$P_{\mathrm{AI}} \;=\; a_{\max}\ \mathrm{clip}\big\langle \hat{\Omega}\hat{I}\,\big\rangle_0^1\ S\big(Re_\Omega / Re_\Omega^{\mathrm{c}}(\hat{\Omega}\hat{I})\big)\ \omega\,\tilde{\nu}. \tag{12}$$

Two functions of the one even coordinate $\hat{\Omega}\hat{I}$—the rate and the onset threshold—carry the whole model. A favorable gradient drives $\hat{\Omega}\hat{I} \to 0$, so both shut off geometrically, with no pressure-gradient parameter anywhere.

The one free function is the onset threshold $Re_\Omega^{\mathrm{c}}(\hat{\Omega}\hat{I})$ of the gate (11), and its shape follows from linear theory. Figure 2 makes the construction concrete. Each attached Falkner–Skan profile, evaluated at its Drela–Giles critical Reynolds number $Re_{\theta 0}(H)$—the station where the layer first turns linearly unstable—traces a curve in the $(\hat{\Omega}\hat{I},\ Re_\Omega)$ plane, and the family shares a common outer envelope: a threshold curve that each profile grazes, touching without crossing, at its own most-critical point. The graze collapses the family onto one monotone shape. On the favorable side the threshold saturates toward a ceiling—an accelerated layer must grow thicker before any point of it turns critical, but not indefinitely so—while near separation it flattens onto an additive floor, inception becoming essentially immediate. The collapse does not stop at the separation limit: the reversed-flow profile grazes the same envelope at its own critical Reynolds number, and because that critical value is tiny ($Re_{\theta 0} \approx 26$), at any Reynolds number a bubble actually reaches the profile rides far above the floor and ignites at once, as it should. We represent the shape by a soft minimum of the two limits,

$$Re_\Omega^{\rm c}(\hat\Omega\hat I) \;=\; k\ \mathrm{softmin}_2\Big(C,\ A + B\,\langle\hat\Omega\hat I\rangle_+^{-2}\Big), \qquad (C,\ A,\ B) \;=\; (2600,\ 175,\ 2), \tag{13}$$

with $\mathrm{softmin}_2(x_1, x_2) = x_1 x_2/\sqrt{x_1^2 + x_2^2}$; the graze fixes the three shape constants to a root-mean-square graze error of $\sim 6\%$ from incipient separation through $\beta = +0.15$ (the soft-min exponent 2 is the best of the sharpnesses tried), and they are never refit; strongly favorable members ($\beta \geq +0.25$, beyond the plotted family) graze 1.27–1.34 high, consistent with the favorable-branch departure the calibration will meet below.

That leaves the single scale k, and its role is physical. Linear theory knows only the profile; the working variable that must carry the amplification also diffuses, with the retained molecular coefficient $c_{\nu,\rm ai}\nu$ of Sec. II.C, all the while it grows. The drain is felt most where inception happens at low Reynolds number—in a thin layer the young disturbance loses to viscosity for a while after the model first lets its growth rate turn positive, so net growth starts late. The compensation is to permit inception earlier than the inviscid graze: one factor $k < 1$ scales the whole threshold, so that the accumulated growth reaches the correlation's first station on time. Its size has to be measured, with the model's own equation marched down the layer.

The simplest such instrument is a parabolized march of that equation: in non-dimensional boundary-layer variables (length in ν/U_∞, velocity in U_∞),

$$u\,\frac{\partial\hat\nu}{\partial x} + v\,\frac{\partial\hat\nu}{\partial y} \;=\; \frac{c_{\nu,\rm ai}}{\sigma}\,\frac{\partial^2\hat\nu}{\partial y^2} + a(\hat\Omega\hat I)\,S\big(Re_\Omega/Re_\Omega^{\rm c}(\hat\Omega\hat I)\big)\,\omega\,\hat\nu, \tag{14}$$

keeping the amplification and the molecular part of the SA cross-stream diffusion. The $\tilde\nu\,\nabla^2\tilde\nu$ self-diffusion and c_{b2} gradient terms are $O(\chi)$ smaller in this range and are dropped, in the manner of the classical parabolized boundary-layer equations [27]. No SA production or destruction enters: production is switched off outright where $\chi \leq 1$ ($\sigma_P = 0$, Sec. II.E), and the tied destruction floor is inert at the disturbance's scale ($< 10^{-3}$ of the amplification, same section). The march is seeded at the leading edge with one unit of an arbitrary infinitesimal: the equation is linear in $\hat\nu$, so the seed's magnitude cancels from every ratio and only the accumulated growth is meaningful. Measured in that unit, $N = \ln\hat\nu$ is the amplification factor by construction, and its envelope can be laid directly against Drela–Giles [6]. (The crossings quoted in this subsection are grid-converged values of the march; the $N = 1$ crossing in particular demands a fine wall-normal grid, the amplifying band being thin where inception occurs.)

Marching the equation untouched ($c_{\nu,\rm ai} = 1$, $k = 1$) on the wedges of Fig. 3 confirms the eigenvalue's verdict of Sec. II.C in accumulated form (not plotted): the envelope falls far short of the correlation, its early secant (the $N \in [1, 5]$ span of the Blasius layer) running 24% below the late one ($N \in [5, 9]$), where the envelope's own curvature—the $c_{\nu,\rm ai} \to 0$ limit of the same march—is under 1%. The march also adds what the asymptotic eigenvalue cannot see: at full molecular diffusion the inception fails outright, the Blasius $N = 1$ crossing sitting at $Re_\theta = 392$ even with the

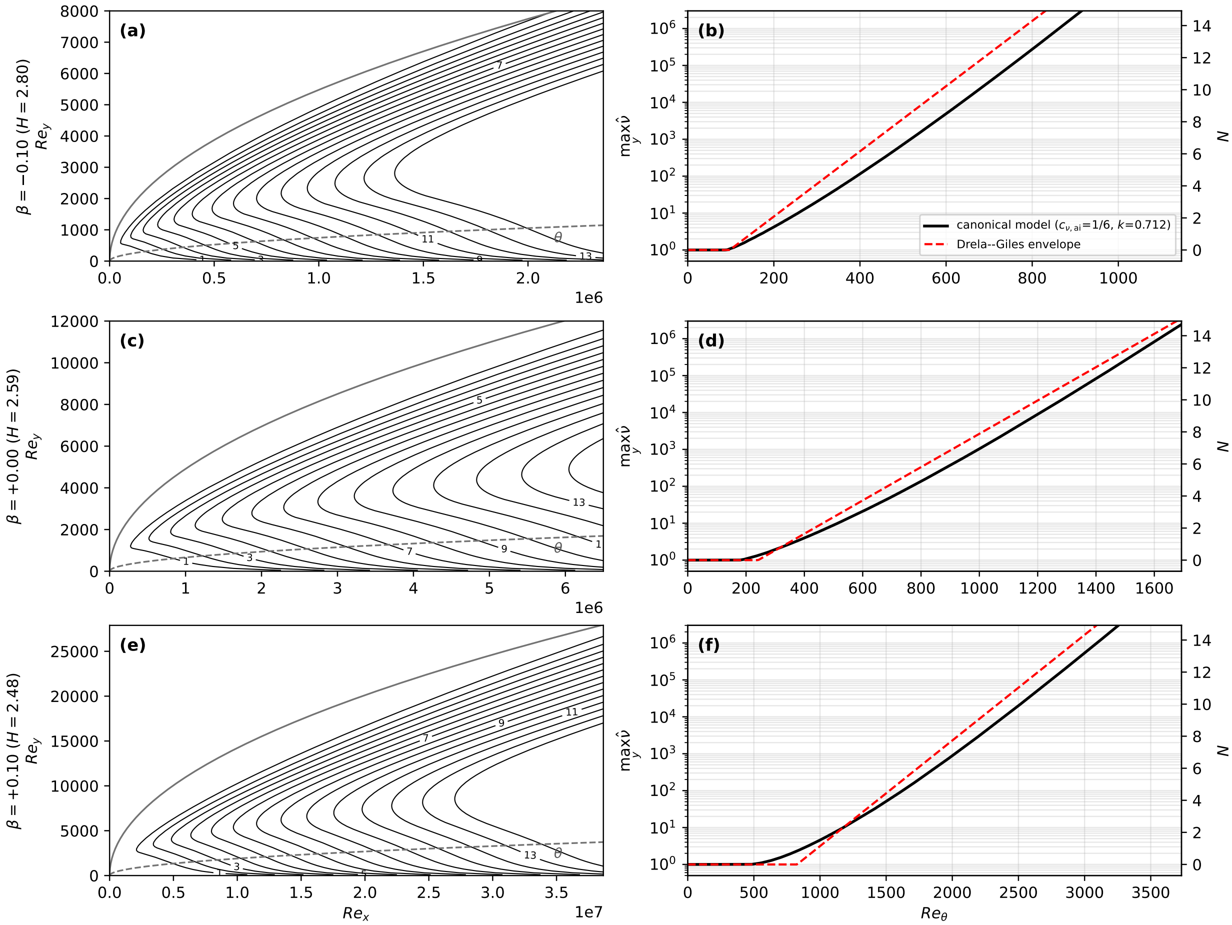

Fig. 3 The disturbance transport (14) on three wedge flows—adverse ($\beta = -0.10$, top), Blasius ($\beta = 0$, middle: the calibration anchor), and mildly favorable ($\beta = +0.10$, bottom; $\max_y \hat{\Omega}\hat{I}$ = 0.162, 0.078, and 0.037, respectively)—seeded $\hat{\nu}_\infty = 1$, $a_{\max}$ = 0.19. Left column (a,c,e): contours of $N = \ln \hat{\nu}$ at the canonical $(c_{\nu,\mathrm{ai}}, k) = (\frac{1}{6}, 0.712)$ in the local-Reynolds plane, concentrated between the overlaid momentum thickness θ (dashed) and edge thickness δ_{99} (solid) where the amplifying band sits. Right column (b,d,f): the marched envelope $\max_y \hat{\nu}$ vs Re_θ (log left axis; N linear right axis) at the canonical $(c_{\nu,\mathrm{ai}}, k) = (\frac{1}{6}, 0.712)$ (black), against each profile's Drela–Giles envelope [6] (red dashed). On the anchor (d) the canonical envelope meets the correlation at $N = 1$ by construction (Sec. II.D); its $N = 9$ crossing is an untuned prediction, 7% past Drela's. The adverse wedge (b) ignites earlier and grows faster. The mildly favorable wedge (f) stays close to the correlation—its $N = 1$ station lands ~11% early—with the suppressed rate stretching the growth over a run nearly twice Blasius's in Re_θ (over four times in Re_x), and no pressure-gradient parameter anywhere. A physical case seeded with χ_∞ hands over where its envelope crosses $N_{\mathrm{crit}} = \ln(c_{\nu 1}/\chi_\infty)$ ($\chi = c_{\nu 1}$; the seed convention of Sec. III).

threshold removed entirely ($k \to 0$)—the drain, not the threshold, sets the inception, and no compensation exists. At the committed $c_{\nu,\mathrm{ai}} = \frac{1}{6}$ the drain is light enough that the threshold is back in command, and k has a value to take.

The compensation takes its value from a single anchor, stated operationally: k is chosen so that the march (14), on the Blasius layer at the committed $c_{\nu,\mathrm{ai}} = \frac{1}{6}$, crosses $N = 1$ at the Drela–Giles station $Re_\theta = 338$; this gives $k = 0.712$—the threshold permitting inception 29% earlier than the inviscid graze, the drain consuming the difference.

Everything after that anchor is prediction. The marched Blasius envelope's $N = 9$ crossing lands at $Re_\theta = 1181$, 7%

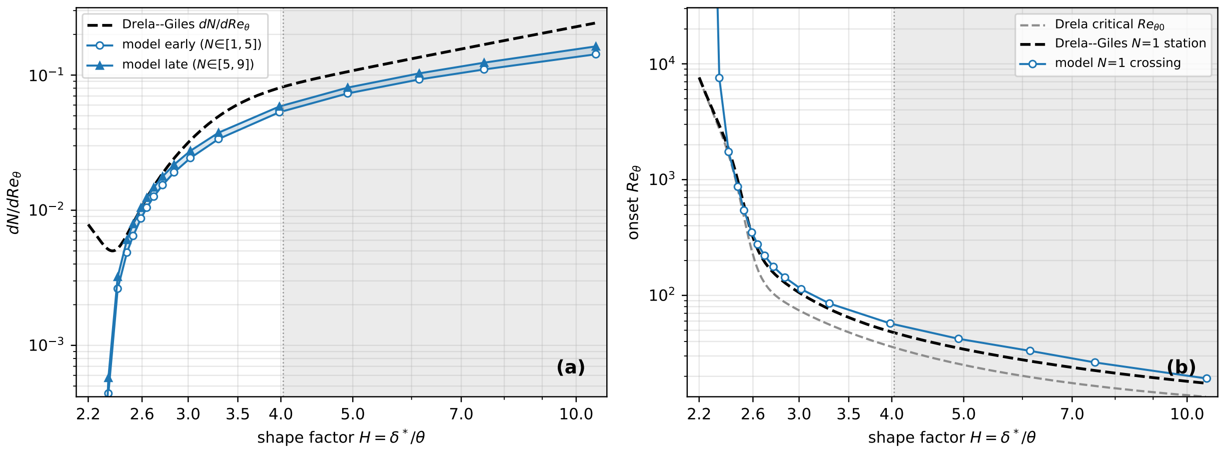

Fig. 4 The calibrated model compared with the Drela–Giles envelope across the full Falkner–Skan family, from the stagnation profile ($H\approx 2.2$) through Blasius ($H\approx 2.59$) and the separation limit ($H=3.98$, dotted) onto the reversed-flow (Stewartson lower) branch ($H\approx 10.6$; separated region shaded). (a) model rate $\mathrm{d}N/\mathrm{d}Re_\theta$ as an early ($N\in[1,5]$) and a late ($N\in[5,9]$) secant, band shaded—set entirely by the fixed eigenvalue $a_{\max}=0.19$ and $c_{\nu,\mathrm{ai}}=1/6$, with nothing tuned to this curve. (b) model onset Re_θ (the $N=1$ crossing) against the Drela–Giles $N=1$ station $Re_{\theta 0}+1/(dN/dRe_\theta)$: anchored at Blasius, tracking the stations within $\sim\pm 12\%$ across the attached family, from $H=2.44$ ($\beta=+0.15$) to $H=3.98$ (the separation limit). Both vertical axes extend slightly past the Drela range to show the favorable-regime departure—the rate falling below, the onset rising above—before the curves leave the panels.

past Drela's 1108—the residual footprint of the retained diffusion (the late-envelope slope itself matches Drela's to a fraction of a percent; the offset is accumulated in the early, drain-affected span). Across the attached family the $N=1$ stations land on the Drela–Giles stations to 6% root-mean-square: the adverse wedges within ±4%, the favorable pair 11–12% early (Fig. 4b)—the graze shape, anchored at one point, carrying the family's two-decade threshold variation.

With $a_{\max}$ and the threshold both fixed, the marched envelope tracks Drela–Giles across the attached family and softens through the separation limit onto the reversed-flow branch (Fig. 4a); the softening is structural—the amplifying band thins as the profile approaches full inflection, so the integrated envelope falls below the pointwise rate. The favorable branch is where the model degrades, and the rate coordinate grades the degradation (Fig. 4). At $\max_y \hat{\Omega}\hat{I}\approx 0.04$ ($\beta=+0.10$, the bottom row of Fig. 3) the model is still roughly right, the marched $N=1$ station $\sim$11% early; the family holds that level through $\max_y \hat{\Omega}\hat{I}\approx 0.02$ ($\beta=+0.15$). Beyond, the rate falls faster than Drela's and the onset climbs above the critical Reynolds number: by $\max_y \hat{\Omega}\hat{I}\approx 2\times 10^{-3}$ ($\beta=+0.35$, $H\approx 2.34$) the rate is $\approx 0.1\times$ Drela and the onset $\approx 3\times$ its critical Re_θ; by $\beta\approx 0.45$ ($\max_y \hat{\Omega}\hat{I}\approx 5\times 10^{-4}$) transition has retreated beyond any station of practical relevance (the marched $N=1$ station past $Re_\theta\sim 5\times 10^5$), and stronger favorable wedges are suppressed outright. The linear $\hat{\Omega}\hat{I}$ has no near-neutral floor to hold the small favorable growth Drela's envelope keeps; matching the strongly favorable branch is left to future work.

E. The blend and its handover

A handover weight ties the laminar and turbulent regimes to the single state variable through χ,

$$\sigma_t(\chi;\tau) = \max\left[\,1 - e^{-(\chi-1)/\tau},\ 0\right] \;=\; \begin{cases} 0, & \chi \le 1, \\ 1 - e^{-(\chi-1)/\tau}, & \chi > 1. \end{cases} \tag{15}$$

The production and destruction of Eq. (2) are blended, the production with a width of its own, the destruction tied to it,

$$P = \max\left[(1-\sigma_P)\,P_{\mathrm{AI}},\ \sigma_P\,P_{\mathrm{SA}}\right], \qquad D = \sigma_D\,D_{\mathrm{SA}}, \qquad \sigma_P = \sigma_t(\chi;\tau), \quad \sigma_D = 1 - \frac{c_{b1}}{\kappa^2 c_{w1}}\,(1-\sigma_P). \tag{16}$$

The max hands the production from amplification to SA at the crossover: for $\chi \le 1$ ($\sigma_P = 0$) it returns P_{AI}, leaving the e^N amplification unchanged, and once $\sigma_P P_{\mathrm{SA}}$ overtakes it just past $\chi = 1$ the source reverts to standard SA with no laminar residue. The production width τ is chosen for numerics rather than physics, and is bracketed by its failure modes: too wide throttles the SA production past $\chi = c_{v1}$, where the eddy viscosity is already dynamically active; too narrow snaps to full SA at $\chi = 1$, compressing the transition zone into a front steeper than the streamwise mesh can resolve. $\tau = 4$ sits between them—the handover is 78% complete where f_{v1} is half active ($\chi = c_{v1}$) and 97% complete where f_{v1} reaches 90% ($\chi \approx 14.8$). The destruction is not gated off but tied, and needs no width of its own: σ_D tracks σ_P with slope $c_{b1}/(\kappa^2 c_{w1}) \approx 0.249$ from the floor $(1 + c_{b2})/(\sigma c_{w1}) \approx 0.751$, the two summing to one by the definition of c_{w1}. The tie is the weighting under which Spalart's linear wall solution $\tilde{\nu} = \kappa u_\tau y$ survives the gates exactly, at every height and for any width τ—derived in Sec. II.G. In the laminar range ($\sigma_P = 0$) the destruction now runs at its floor instead of vanishing, but the floor is inert below the crossing: D_{SA} is quadratic in $\tilde{\nu}$, and its ratio to the amplification production stays below 10^{-3} up to $\chi = 1$ (quantified below), so the e^N growth to the crossing is untouched. Past the crossing the retained SA terms are no longer passive—they pace the growth of χ through the handover, and at low Reynolds numbers, where the handover occupies a macroscopic streamwise extent, that pacing becomes dynamically significant (Sec. VI). The limit $\sigma_t \to 1$ recovers the unmodified SA equation and its fully turbulent calibration.

The floor's inertness deserves its numbers, since it is the tie's one back-reaction on the laminar branch. Measured at the $\tilde{\nu}$ peak of the frozen-profile marches of this section, across wedges from $\beta = +0.10$ to the separation limit and seeds $N_{\mathrm{crit}} = 7$–11 (`repro/analytic/floor_backreaction_table.py`), its ratio to the amplification production—quadratic in χ along the envelope—stays below 10^{-3} for every wedge and seed up to the $\chi = 1$ crossing (largest at the lowest seeds), and at $\chi = c_{v1}$ is still at most 1.4% of the σ_P-weighted SA production it is tied against (largest at the separation limit, where the tied SA production is weakest). The calibration instrument of Sec. II.D omits the term; marching with it included moves the anchor and prediction stations by less than 1% (within the solve residuals; on an earlier anchor variant the favorable-family stations' mean moved −2.6%, most of it accumulated past the $\chi = 1$ crossing).

Between its $\sigma_t = 0$ and $\sigma_t = 1$ limits the blend is an interpolation, and we claim no physics for it. The model does not attempt to represent the transitional zone's internal dynamics—turbulent-spot formation, growth, and merging, or the intermittency distribution they produce—and nothing above constrains what $\hat{\Omega}\hat{I}$ and D should be mid-handover. The target is deliberately narrower: the transition location, and the macroscopic aerodynamics that follow from it. Because the actively transitional region is often short on the scale of the geometry, the integrated quantities are often insensitive to how the interpolation crosses it, and σ_t is asked only to cross smoothly and monotonically. One consequence of handing everything above $\chi \sim 1$ to standard SA deserves flagging here: at a leading edge, standard SA itself admits a spurious self-sustained turbulent state at high Reynolds number, so the coupled model is bistable there and the solution an implementation finds depends on how it is initialized; Sec. VIII quantifies the branch and states the convergence protocol used throughout this paper. The exception is a flow whose transitional zone itself occupies a macroscopic extent—a long laminar separation bubble may close or burst depending on how fast the handover completes; Sec. VI meets exactly such a flow, on the Eppler 387 at $Re \leq 10^5$, where the handover zone grows to a large fraction of the chord and sets the model's bubble-bursting boundary.

F. The eddy-viscosity assembly in lifted transitional layers

One further term acts during the handover: the near-wall damping f_{v1} in the eddy-viscosity assembly $\mu_t = \rho\tilde{\nu} f_{v1}(\chi)$. Its derivation is the attached-wall-layer identity $\chi = \kappa y^+$, under which f_{v1} enforces the $\mu_t \propto y^3$ sublayer scaling; its half-saturation at $\chi = c_{v1}$ is $y^+ \approx 17$, the middle of the buffer layer, by construction. In a lifted transitional shear layer—the free shear layer over a laminar separation bubble in mid-handover—that identity is void at every χ: there is no wall scale for f_{v1} to enforce, and its damping is an unearned sink on exactly the young turbulence that must close the bubble. Whether a point is in an attached wall layer is decidable from the state itself, with no new calibrated constants: writing $y^+ := \chi/\kappa$ (exact in any SA-consistent attached layer, where $\tilde{\nu} = \kappa y u_\tau$ through buffer and log regions), the reconstruction

$$q = \frac{|\mathbf{u}|\, d/\nu}{y^+\, U^+(y^+)}, \tag{17}$$

with U^+ the Reichardt law of the wall, equals unity identically in an attached equilibrium layer and is many times larger in a lifted layer, whose $|\mathbf{u}|$ and d are set by the bubble geometry rather than the wall unit. The assembly is therefore blended,

$$\mu_t = \left[(1-b)\, f_{v1}(\chi) + b\right] \rho\tilde{\nu}, \qquad b = s(\chi)\, G(q), \tag{18}$$

where G ramps on over $q \in (2, 4)$—the margin covers how far attached layers stray from Reichardt under pressure gradient: a composite log–wake profile (`repro/analytic/fv1_qmargin_composite.py`) keeps $q < 2$ in the gated band for wake parameters up to $\Pi \approx 3.5$, while near-separation wakes can open G partially, with the batteries bounding the consequence—and $s = \mathrm{clip}(\chi - 1, 0, 1)$ is a smoothing ramp across the amplification threshold. The s ramp is deliberately faster than the σ_t handover: σ_t encodes turbulence maturity and is already applied to the production, while f_{v1} encodes wall proximity—gating the bypass on maturity would charge the young turbulence twice for the same immaturity. The construction is inert everywhere it should be: for $\chi < 1$, μ_t is negligible under either assembly; for $\chi \gtrsim 30$, $f_{v1} \to 1$ and the blend is the identity; and in the inner attached wall layer $q = 1$ by construction and G stays shut. At an attached layer's outer edge, where χ descends back through the handover band, the gate does open—there f_{v1} is already close to one, and the flat-plate and NLF batteries bound the net effect below 1.5 drag counts (`data/fv1bypass_battery_results.json`). Its effect is confined to lifted transitional layers. Status of the results that follow: the bypass entered the model after the original campaigns were computed; the controlled on/off batteries (flat plate and NLF within 1.5 counts at their cold-budget protocol, Eppler moved toward the measurement at every condition) motivated the promotion, and the full recomputation is now adopted: every two-dimensional figure, table, and quoted number in Secs. III–VI is from the bypass-active model (the converged recomputation moved the finest-grid results by more than the cold-budget batteries bounded: up to 8 counts on the NLF at high incidence—27 on its coarsest grids—and -10 counts at the Eppler benchmark's $\alpha = 5°$, all toward the measurement); the three-dimensional deployment of Sec. IX is likewise recomputed in full: its tabulated totals and every figure and number are the final kernel.

G. Preservation of the Spalart–Allmaras calibration

The reduced diffusion (Eq. (10)) and the two handover gates (Eq. (16)) must leave SA's calibrated behavior alone. Neither enters the momentum equation, which keeps the true molecular viscosity: $c_{\nu,\mathrm{ai}}$ rescales the molecular coefficient in the $\tilde{\nu}$-transport equation only, and the gates scale that equation's source terms, so the velocity field can change only through the eddy viscosity $\nu_t = \rho\tilde{\nu} f_{v1}$. This subsection introduces nothing new; it walks Spalart and Allmaras's own calibration stages (§II.2–II.5 of Ref. [21]) and shows each is preserved—exactly, not approximately.

The high-Reynolds-number stages are untouched by construction. §II.2, Free shear flows, fixes c_{b1}, σ, c_{b2} on mixing-layer and wake spreading rates with the molecular viscosity excluded from the equation outright; §II.3, Near-wall region, high Reynolds number, fixes c_{w1} and the f_w calibration on the log layer and flat-plate skin friction at $\chi = \kappa y^+ \gg 1$. Both are formulated where χ is large or ν absent: there the gates are fully open ($\sigma_P = \sigma_D = 1$) and the molecular diffusion is at most a sub-percent addend to $\tilde{\nu}$, so neither modification acts. §II.5, Laminar region and trip term, is replaced outright: the trip devices are dropped and the laminar region becomes the model's active domain. What must be managed is §II.4, Near-wall region, finite Reynolds number—the calibration that introduces χ itself and sets the viscous functions f_{v1}, f_{v2} through the buffer layer, where χ passes through $O(1)$ and the gates run partially open.

That calibration rests on a single flow: the constant-stress inner layer (sublayer, buffer layer, log layer), on which SA's solution is the linear profile $\tilde{\nu} = \kappa u_\tau y$. Through the handover band ($\chi = 1$–20, $y^+ \approx 2$–50) the amplification branch P_{AI} is negligible—the near-wall profile is nearly parabolic, so $\hat{\Omega}\hat{I} \le 0$ over most of the band (where the clip returns exactly zero) and the onset threshold $Re_\Omega^{\rm c}$ sits far above the local Re_Ω, which stays below $\sim 10^2$ across the band—so what runs there is standard SA with both sources reduced, $\sigma_P P_{\rm SA}$ and $\sigma_D D_{\rm SA}$. On the linear solution Spalart's $\tilde{S}$ construction holds $\tilde{S} = \tilde{\nu}/(\kappa^2 d^2)$—hence $r \equiv 1$ and $f_w \equiv 1$—at every height, so all three terms of the balance are height-independent: $P_{\rm SA} = c_{b1} u_\tau^2$, $D_{\rm SA} = c_{w1} \kappa^2 u_\tau^2$, and the self-diffusion of the linear profile contributes $(1 + c_{b2})\, \kappa^2 u_\tau^2/\sigma$. Requiring the gated balance $\sigma_P P_{\rm SA} - \sigma_D D_{\rm SA} + \text{diffusion} = 0$ to hold pointwise gives

$$\sigma_D \;=\; \frac{(1 + c_{b2})/\sigma + \sigma_P\, c_{b1}/\kappa^2}{c_{w1}} \;=\; 1 - \frac{c_{b1}}{\kappa^2 c_{w1}}\,(1 - \sigma_P), \tag{19}$$

the tie of Eq. (16). (Equal-fraction weighting, $\sigma_D = \sigma_P$, would instead remove about four times as much destruction as production—on the linear solution $P/D = c_{b1}/(c_{w1}\kappa^2) \approx 1/4$—and leave a spurious buffer surplus of $\tilde{\nu}$.) With the tie, $\tilde{\nu} = \kappa u_\tau y$ is an exact solution of the gated equation at every height and for any τ; and because χ is formed with the true ν, the damping function $f_{v1} = \chi^3/(\chi^3 + c_{v1}^3)$ still switches on the eddy viscosity at exactly the calibrated χ. The inner-layer ν_t—and with it the velocity solution—is standard SA's.

The preserved solution is linear, and that settles the diffusion reduction: the molecular diffusion multiplies a curvature that is identically zero, $(c_{\nu,\rm ai}\nu/\sigma)\, \partial^2\tilde{\nu}/\partial y^2 = 0$ at any coefficient, so $c_{\nu,\rm ai}$ does not act on the inner layer at all. Nowhere in SA's calibrated domain does the reduced coefficient multiply anything nonzero. The preservation is discrete as well as continuous: a linear field has exact gradients and fluxes under any consistent discretization and the source terms are pointwise, so the linear profile is an exact solution of the discretized gated equation too, and SA-AI's inner-layer solution coincides with standard SA's to round-off independent of the numerical scheme and resolution (verified to below 10^{-10} in profile and log-law intercept with two unrelated discretizations of the constant-stress layer).

The assembled equation, its closure functions, and the complete constant set are collected for reference in Appendix E.

III. Zero pressure gradient: the Blasius flat plate

The coupled-RANS results in this and the following sections were produced with Flow360, an unstructured, node-centered, second-order finite-volume solver with a Roe flux; the freestream modified-viscosity ratio χ_∞ sets both the SA freestream value and the disturbance seed (Sec. II).

A higher freestream turbulence intensity is a larger seed χ_∞ and thus fewer e-folds to the handover, moving transition upstream: a case seeded with χ_∞ reaches the handover where its envelope crosses $N_{\rm crit} = \ln(c_{v1}/\chi_\infty)$—the half-saturation threshold $\chi = c_{v1}$ at which the eddy viscosity engages. We map the turbulence intensity to the seed by

$$N_{\rm crit}(Tu) = -8.43 - 2.4\,\ln(Tu_{\rm frac}), \qquad \chi_\infty = c_{v1}\, e^{-N_{\rm crit}}, \tag{20}$$

with $Tu_{\rm frac}$ the rms turbulence intensity as a fraction. This is Mack's [1, 9] e^N critical-N-factor correlation. The

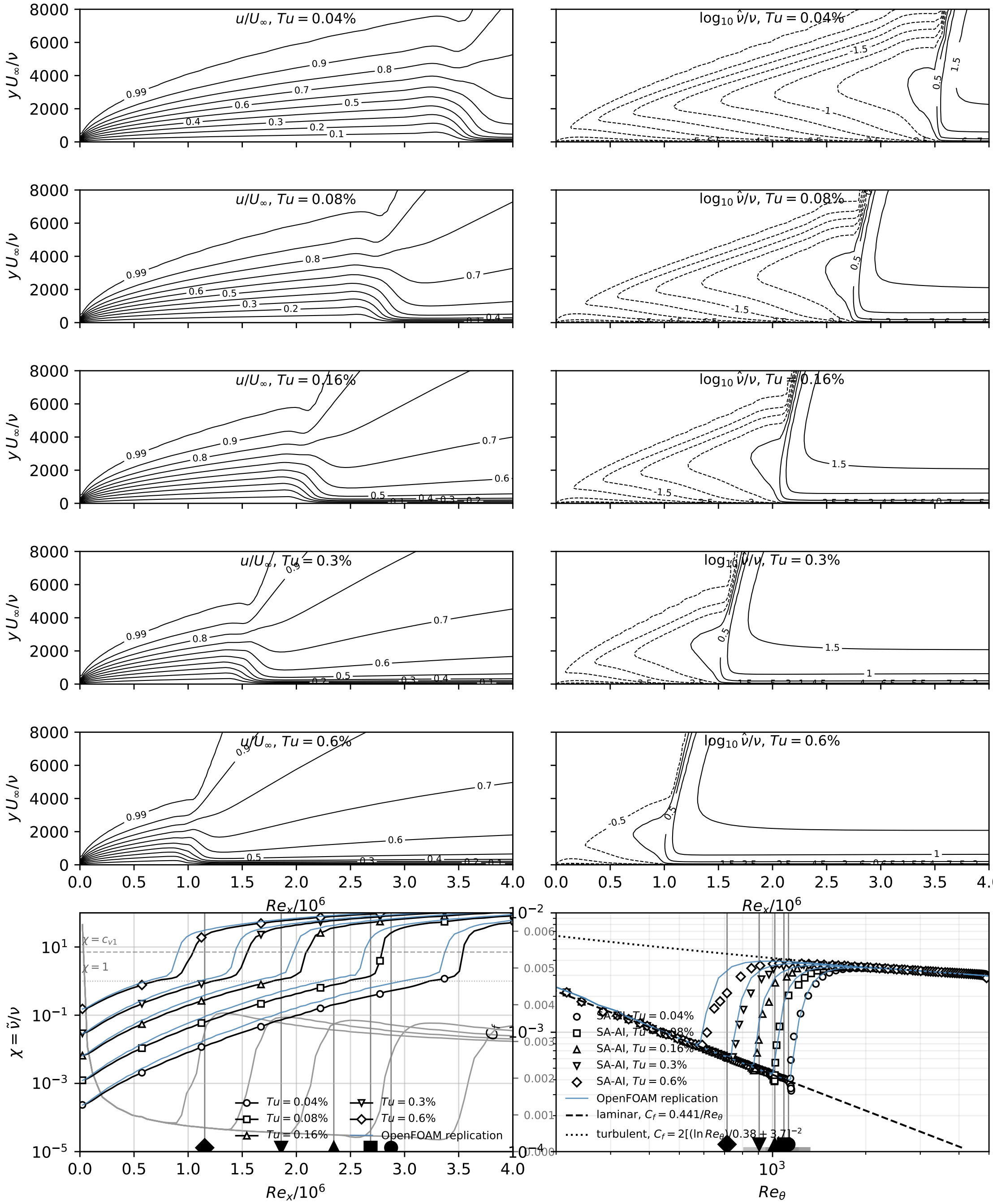

Fig. 5 Freestream-turbulence sweep on the flat plate at five levels in the natural-transition range $Tu = 0.04$–0.60% (rows). Left columns: contours of u/U_∞ and $\log_{10}\chi$ in the (Re_x, Re_y) plane ($Re_x = x \cdot 10^6$). Bottom-left: in-BL maximum $\chi = \tilde{\nu}/\nu$ vs Re_x, with horizontal references at $\chi = 1$ (onset of the σ_t blend) and $\chi = c_{v1} = 7.1$ (eddy-viscosity half-saturation). Bottom-right: C_f vs Re_θ with the laminar ($C_f = 0.441/Re_\theta$, dashed) and turbulent ($C_f = 2[(\ln Re_\theta)/0.38 + 3.7]^{-2}$, dotted) correlations. Both bottom panels use black curves with one symbol per Tu (legend); the bottom-left panel overlays C_f (gray, right axis) on the χ trajectories, locating each case's C_f rise between its $\chi = 1$ and $\chi = c_{v1}$ crossings; solid-gray verticals and filled bottom-edge symbols mark the Abu-Ghannam & Shaw [20] transition-onset reference, and the bottom-right adds the digitized Schubauer–Skramstad Fig. 7 band (faint bars) for context. Thin steel-blue curves in both bottom panels: the independent cell-centered incompressible (OpenFOAM) implementation of the model on the same grid specification (full-detail counterpart in Appendix G; both solvers' crossings tabulated in Table F1). With the receptivity taken from Mack's e^N correlation, the model's half-saturation ($\chi = c_{v1}$) onset brackets AGS within 10% across the range—late at the lowest Tu, early at the higher levels—while the $\chi = 1$ blend-onset reading drifts early as the seed approaches unity (markers placed via the laminar Re_θ–Re_x relation; the quoted residuals use the integrated Re_θ of the text); the C_f follows the laminar branch then recovers the turbulent correlation (baseline SA, with no laminar branch, does not produce this).

correspondence is not exact: e^N methods declare transition at a sharp N_{crit}, while the model's threshold is smeared over the $\chi=1\to O(c_{v1})$ handover—several e-folds of the envelope. Sensitivity to the seed is that of any e^N method: one e-fold in χ_∞ is one unit of N_{crit}, which on the Blasius envelope ($dN/dRe_\theta\approx 10^{-2}$) shifts onset by $\Delta Re_\theta\approx 100$.

In the solver the indicator triple is assembled from four local vector-calculus objects and their invariant pairings alone: the velocity $\mathbf{u}$ (taken relative to the nearest wall point, which makes the triple Galilean invariant; for the static walls of this paper, the absolute velocity), the vorticity $\boldsymbol{\omega}$, the wall-distance vector $\mathbf{d} = d\,\nabla d$, and the curvature vector $\boldsymbol{\ell} = \frac{1}{2}(\mathbf{d}\cdot\mathbf{d})\,\nabla^2\mathbf{u}$, where the divergence-free identity $\nabla^2\mathbf{u} = -\nabla\times\boldsymbol{\omega}$ supplies the second derivative; the onset gate's $Re_\Omega=d^2|\omega|/\nu$ completes the set. The three-dimensional realization—its division-free Gram form, Eq. (E4), how the magnitude-based components realize the signed algebra of Sec. II exactly on all but a thin mixed-sign layer of reversed profiles, and the two further pairings the same four vectors admit—is stated in Appendix E. Because the rate depends on the frozen mean flow and never on $\tilde\nu$, the production's $\tilde\nu$-derivative is analytic—the handover derivative is its only stiff piece—and it is treated implicitly with the same clipping as the baseline SA production.

Figure 5 computes the model on the flat plate at five freestream-turbulence levels spanning the natural-transition range, $Tu=0.04\%$, 0.08%, 0.16%, 0.30%, 0.60%. The plate is resolved by a structured 320×80 grid (streamwise $\times$ wall-normal). In units of the reference length (unit Reynolds number 10^6), the domain spans $0\le x\le 6$ ($Re_x\le 6\times10^6$) streamwise and 0.5 in the wall-normal direction; the streamwise spacing is clustered at the leading edge (from $\Delta x\approx 8\times10^{-4}$, coarsening toward the outflow) and the wall-normal spacing grows geometrically at ratio 1.12 from a first-cell height of $\approx 7\times10^{-6}$ ($y^+\lesssim 0.3$), resolving the laminar and transitional boundary layer. The boundary conditions are no-slip on the plate, a characteristic freestream on the far boundary, and slip walls on the two span planes (quasi-2D); the freestream Mach number is 0.1. An independent cell-centered, pressure-based incompressible implementation of the model, run on this same grid specification, reproduces the onset crossings quoted below within 7% in Re_θ at the $\chi=1$ crossing, consistently slightly early across the whole sweep (Table F1; the full-detail counterpart figure is in Appendix G), so neither the grid nor the discretization carries the result. We compare the resulting transition onset to the Abu-Ghannam & Shaw zero-pressure-gradient correlation [20], $Re_\theta^{\mathrm{tr}}=163+\exp(6.91-Tu[\%])$ — the community-standard low-disturbance reference, itself a smooth fit through flat-plate data including Schubauer & Skramstad [28]. We use that correlation rather than Schubauer & Skramstad's raw Fig. 7 because that figure's low-Tu plateau (transition fixed near $Re_x\approx 2.8\times10^6$ below $Tu\approx 0.08\%$) was later shown by Wells [29] to be largely an artifact of wind-tunnel acoustic noise rather than a true insensitivity to turbulence — a caveat Schubauer & Skramstad themselves raise.

The Tu $\to\chi_\infty$ map of Eq. (20) sends each case to a single freestream seed ($\chi_\infty\approx 2.3\times10^{-4}$, 1.2×10^{-3}, 6.3×10^{-3}, 2.9×10^{-2}, 1.5×10^{-1} at the five Tu; the delivered ambient matches these set values to four digits at the boundary-layer edge at every station). Starting from this small laminar seed, $\tilde\nu$ remains inert while the skin friction follows the laminar Blasius law (dashed); the amplification (12) grows the disturbance through the unstable region until χ reaches unity, after which C_f rises toward the turbulent log-law correlation (dotted). As Tu rises the transition moves forward.

The two bottom panels collect the in-BL maximum χ (vs Re_x) and the resulting $C_f(Re_\theta)$. The operational transition — the C_f rise — runs between the $\chi = 1$ crossing (the start of the σ_t blend) and the $\chi = c_{v1} = 7.1$ crossing (the eddy-viscosity half-saturation point): the bottom-left panel overlays C_f (gray, right axis) on the χ trajectories, and each case's rise starts at its $\chi = 1$ crossing and levels off at its $\chi = c_{v1}$ crossing. Because the modeled transition has finite extent, quoting a single onset number against AGS requires choosing a crossing, and the comparison depends on the choice; we state the convention and give both readings, with Re_θ integrated from the computed profiles at the crossing station (edge-normalized momentum thickness, integral cut at the boundary-layer edge). At the $\chi = 1$ crossing—the start of the blend—the model reads +2% late at the lowest disturbance level ($Tu = 0.04\%$) and drifts steadily early as the seed grows, −8% at 0.08% to −27% at 0.60%: the seed itself spans 15× and approaches unity, so the first crossing measures receptivity as much as growth and is seed-level-sensitive by construction. At the quoted convention, the $\chi = c_{v1}$ crossing—the half-saturation point, mid-way up the C_f rise, and the threshold the receptivity map itself is anchored to ($\chi_\infty = c_{v1} e^{-N}$)—the model brackets the correlation within 10% across the whole sweep: +7% late at $Tu = 0.04\%$, −2% at 0.08%, and 4–10% early over 0.16–0.60%. The onset residuals have different roots: the late bias at low Tu is the retained-drain footprint of the calibration (the marched Blasius envelope runs 7% past Drela's station by $N = 9$, Sec. II.D, and the low-Tu seeds hand over near that depth), while the early bias at the higher levels traces to the receptivity map (20), taken from Mack's e^N correlation without adjustment here; tightening either is left to future work.

Strongly bypass-dominated conditions ($Tu \gtrsim 1\%$, e.g. the ERCOFTAC T3A and T3B test cases at $Tu = 3\%$ and 6%) lie outside this scope: at $Tu = 1\%$ the map (20) leaves only $N_{\text{crit}} \approx 2.6$—a seed already within a few e-folds of handover—and by $Tu \approx 3\%$ the seed χ_∞ exceeds c_{v1} outright, no longer a meaningful kinematic parametrization of bypass transition. We therefore do not present T3A/T3B comparisons.

IV. The NLF(1)-0416 airfoil at $Re = 4\times10^6$

The natural-laminar-flow airfoil NLF(1)-0416 demands an e^N-style instability tracker. Figure 6 tells most of the story: the computed transition fronts across the incidence matrix, on every grid of both mesh families, against the microphone measurements and the e^N references. The model is evaluated at $Re = 4\times10^6$, $M = 0.1$, $\chi_\infty = c_{v1}\, e^{-9} \approx 8.76\times10^{-4}$ (the $N_{\text{crit}} = 9$ anchor, set through the map of Sec. III; via Mack's map, Eq. (20), it corresponds to $Tu \approx 0.07\%$, consistent with the low-turbulence tunnel), across $\alpha \in \{0°, 4°, 9°, 15°\}$, on a three-level isotropic refinement ladder of both mesh families (unstructured cavity-mesher and structured O-grid, L0–L2, twenty-four cases total). The same six grids extend the matrix to the negative-incidence pair $\alpha = -8°$ and $-4°$ (twelve further cases; Fig. 8 shows the finest pair, and the wall-anchored contour sheets of Appendix A carry the full ladder). The unstructured mesher uses a shared NLF contour (a natural cubic spline in arc length through the published coordinates, resampled with piecewise-cosine clustering at the LE and TE) with blunt-trailing-edge–aware spacing (h_{TE} scales with first-cell height h_0 at every level; the TE ramp

Table 4 NLF(1)-0416 mesh ladder: size, wall-tangential and wall-normal resolution, and resulting y^+. Tangential spacing $\Delta s/c$ is measured on the upper surface; the wall-normal first-cell height h_0 and the cell size $0.5c$ off the wall are read from the center-span slice; the y^+ percentiles aggregate the four incidences ($\alpha = 0°, 4°, 9°, 15°$). The cavity meshes are all-triangular, the structured O-grids all-quadrilateral.

	Unstructured cavity			Structured O-grid		
	L0	L1	L2	L0	L1	L2
Nodes	15,268	53,182	193,476	15,920	63,840	255,680
Elements	29,855	105,481	385,665	15,721	63,441	254,881
$\Delta s/c$ @ LE ($\times 10^{-3}$)	0.55	0.28	0.14	5.31	2.66	1.33
@ $0.05c$	5.73	2.85	1.42	10.5	5.23	2.61
@ $0.25c$	11.3	5.59	2.78	17.6	8.81	4.40
@ $0.98c$	6.57	6.22	4.10	2.74	1.38	0.69
@ TE	2.70	1.75	0.98	0.72	0.38	0.20
h_0/c ($\times 10^{-6}$)	12.6	5.6	2.6	7.0	3.5	1.8
cell @ $0.001c$ ($\times 10^{-6}c$)	227	97	49	159	82	43
cell @ $0.5c$ ($\times 10^{-2}c$)	8.45	4.41	2.21	2.19	1.07	0.52
y^+ (95th pct)	2.99	2.10	1.22	1.90	0.92	0.46
y^+ (99th pct)	4.26	3.12	2.05	2.53	1.39	0.70

ratio r_{TE} decays from 2.0 at L0 to 1.25 at L2 so that the local cell size at fixed chord-distance from the TE also halves); the structured O-grid is generated with the same contour by Construct2D [30]. Every length scale—surface point count, h_0, growth-ratio offset (the wall-normal growth ratio minus one), TE step, and TE ramp—refines by a factor of two per level. The two families also separate the solver classes: the cavity meshes' anisotropic boundary-layer triangles are tractable for the node-centered least-squares operators used here but defeat a cell-centered compact-stencil discretization outright (the independent cell-centered implementation of Sec. III diverges on both the primal triangulation and its median dual, for standard SA as much as for SA-AI), so the cross-solver replication carries the structured family only. Figures A1 and A2 (appendix) show the two mesh families at the leading and trailing edges across the three refinement levels, and Table 4 lists their resolution quantitatively.

The unstructured family is deliberately adversarial. Unlike the usual "unstructured" airfoil mesh—which grows a structured layer of wall-normal quadrilaterals or prisms through the boundary layer and Delaunay-triangulates only the isotropic exterior—the cavity mesher applies the same constrained-Delaunay triangulation everywhere, the boundary layer included. There the metric is stretched by $O(10^2–10^3) : 1$, and the cavity point-insertion uses an anisotropic distortion metric (with a skewness/normal-deviation objective) to promote near-right-angled stretched triangles ($\sim 90°$-$90°$-$0°$) over flat slivers ($\sim 180°$-$0°$-$0°$); but this is a preference, not a guarantee, and the resolved boundary layer carries many near-degenerate triangles (visible in Fig. A2, and reflected in the elevated peak y^+ of the cavity grids in Table 4). This directly stresses the model, whose indicators (Re_Ω and the sphere coordinate $\hat{\Omega}\hat{I}$, the latter carrying a second velocity derivative in its curvature term $\frac{1}{2}d^2\nabla^2 u$) are all local and gradient-based, so sliver-induced

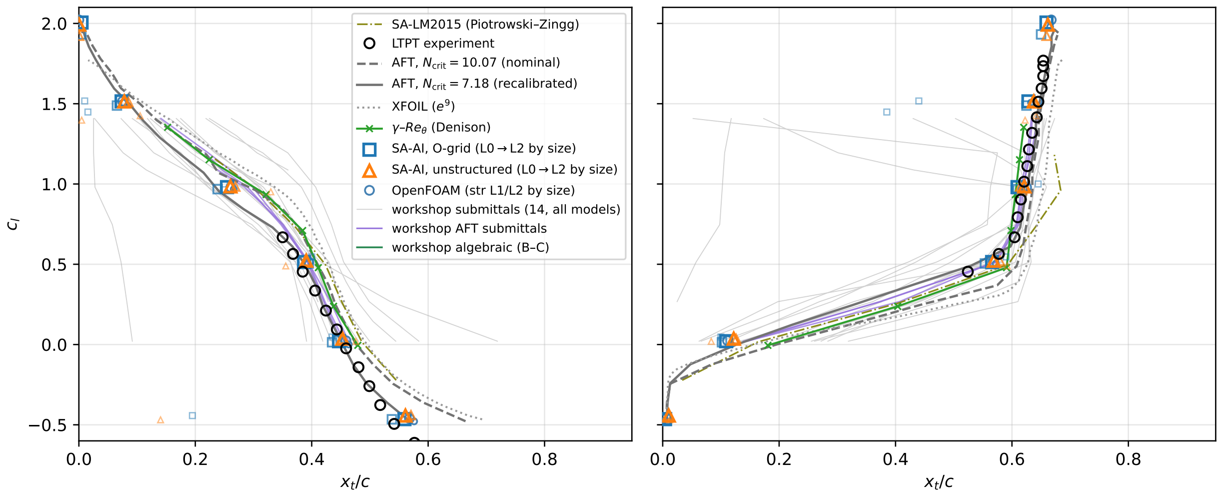

Fig. 6 NLF(1)-0416 at $Re = 4\times10^6$: transition location against lift coefficient, upper (left) and lower (right) surfaces. Circles: LTPT microphone measurements. Gray curves: the AFT model in OVERFLOW from Coder's dissertation [31] at its nominal $N_{\text{crit}} = 10.07$ (dashed) and after the source's recalibration to $N_{\text{crit}} = 7.18$ (solid), with its XFOIL reference (dotted). Thin underlaid curves: all fourteen submittals to the First AIAA CFD Transition Modeling and Prediction Workshop's NLF(1)-0416 incidence sweep ($\alpha = -4°$–$8°$) [32]; the AFT submittals are purple, the algebraic Bas–Cakmakcioglu submittal thin sea-green, all others light gray. Olive dash-dot: the SA-LM2015 fine-grid sweep of Piotrowski and Zingg [33]. Dark-green crosses: the Langtry–Menter γ–Re_θ model in OVERFLOW (fine grid) from the transition-workshop exercise of Denison [34]. Extraction provenance and per-source caveats for every digitized series: Appendix F. No published SA-BCM incidence sweep exists at this condition; the one open SA-BCM result is a single-incidence ($\alpha = 0°$) mesh-adaptation study [35]. The isolated coarse-grid points at incidence are path-sensitive front breakaways that the coarse grids admit but do not force: restarting the structured-L0 9° case from its converged flow with $\tilde{\nu}$ reset to the seed recovers the fine-grid fronts on the same mesh (0.66/0.075 c lower/upper against L1–L2's 0.63–0.64/0.07–0.08 c), machine-converged; the cavity-L1 15° lower front is the one unsettled case of the twenty-four, still drifting over 0.65–0.67 c (the final value is plotted). Open squares/triangles: SA-AI on all six grids (L0–L2 by marker size, both families) at $\alpha = 0°, 4°, 9°, 15°$, plus the negative-incidence pair $\alpha = -8°$ (well into the negative branch of the measured polar) and $-4°$ (zero lift), all at the paper-wide untuned seed. The computed fronts here and in every airfoil comparison of this paper are the near-wall $\chi = 1$ crossing—stated because the choice matters when comparing implementations: the $\chi = c_{v1}$ crossing sits up to 0.05 c aft where the front grows slowly (the $-8°$ pressure side is the extreme case). Small steel-blue circles: the independent cell-centered incompressible (OpenFOAM) implementation of Fig. 5 on the same structured L1 and L2 grids (by marker size) at all six incidences, fronts read from its fields at the same near-wall $\chi = 1$ convention (the port omits the bypass of Eq. 18); its L0 solutions are excluded—they include the same coarse-grid breakaway states discussed above. On the finest grid every front lies within 0.016 c and the lift within 0.9% (absolute 0.003 at zero lift) of the SA-AI values (Table F3); the cross-solver replication spans thirty cases in all—both of the paper's airfoil benchmarks, every incidence, all three grid levels (the Eppler counterpart: Fig. 11 and Table F6). At $-8°$ the drag sits 8–11 counts below the measured polar at matched lift, the suction (lower) surface trips at 0.004–0.011 c, on the references' leading-edge trip (e^9: 0.015–0.018 c), and the pressure-side (upper) front converges at 0.559–0.561 c (families agreeing to 0.002 c)—on the measured ≈0.54 c, with the recalibrated AFT at 0.57 and e^9 at 0.66 c.

errors in the computed gradients feed straight into the amplification rate. Figure 6 shows the result: at L1 and L2 the cavity markers land on the O-grid markers on every front.

This case also permits the paper's one direct comparison with the AFT model, whose dissertation-stage validation ran the same airfoil at the same Reynolds number in OVERFLOW [31]. Figure 6 overlays the two (the AFT, XFOIL, and measurement curves are vector-exact extractions from the source's figures). At its nominal tunnel calibration ($N_{\text{crit}} = 10.07$) the AFT fronts sit systematically downstream of the microphone measurements, and the source closes the gap by recalibrating to $N_{\text{crit}} = 7.18$ ($Tu = 0.15\%$), chosen from the transition-location discrepancy itself. SA-AI's fronts, computed with the untuned $N = 9$-class seed ($\chi_\infty = c_{v1} e^{-9}$), land on the measurements where the recalibrated AFT lands, on both surfaces and both mesh families. We do not read the untuned agreement as a ranking, because it may be partly compensatory: the model's laminar-to-turbulent handover is least decisive at low Re_θ (Sec. VI), the c_{v1}-based N_{crit} convention was anchored on flat-plate transition at lower Re_θ than these fronts, and at the NLF's higher-Re_θ fronts the same convention is effectively sharper—while AFT's recalibrated freestream may equally be read as absorbing the tunnel's true disturbance level rather than as a per-case tuning.

Two entries in the polar (Fig. 7) need qualification. The $\alpha = 15°$ point lies beyond the measured stall ($C_{l,\max} \approx 1.68$ in the tunnel data), where the quasi-2D computation does not capture stall and returns a higher C_l at elevated drag; and the coarse L0 mesh over-predicts drag at $\alpha \geq 9°$ (partly the front breakaway of Fig. 6's caption, though most of the drag excess survives the recovered front—0.0208 against the breakaway state's 0.0225 and L1's 0.0144 at 9°—and is resolution proper). The absence of an mfoil reference above $\alpha = 7°$ is a solver failure, not a method-class limit: FlexFoil converges at the identical conditions ($C_l = 1.51$ at $\alpha = 9°$, consistent with the present RANS), as does XFOIL (its digitized e^9 curve in Fig. 6 spans these incidences). This is the paper-wide substitution rule for the e^9 reference: mfoil [37] wherever its coupled solve converges, XFOIL [7] (repaneled) or FlexFoil—Flexcompute's separately written Rust implementation of the XFOIL closure set—where it fails. All three implementations were run at every condition where mfoil serves as a reference in this paper and agree to better than 1% in c_l and c_d and $0.005\,c$ in x_{tr}, with two catalogued exceptions: the bistable $Re = 6 \times 10^4$ (the footnote of Table C1) and the NLF $\alpha = 0°$ lower-surface front, whose near-flat envelope amplifies implementation differences (mfoil and FlexFoil read $0.62\,c$; an XFOIL run of the original study read $0.52\,c$). The fully turbulent SA polar on the same structured L2 grid closes on SA-AI only as the laminar run shrinks (+144% at 0° narrowing to +67–74% at 9°–15°).

The airfoil surface distributions are presented as five-row figures (Figs. 9–10 here and Figs. 12–13 for the Eppler 387) with a common layout: row 1 the wall-normal-probe maximum of Re_Ω (log, the onset gate); row 2 the wall-normal-probe maximum of the sphere rate coordinate $\hat{\Omega}\hat{I}$ (Eq. (8))—the amplifying inflectional coordinate the kernel acts on, on a log axis spanning the amplifying decades (suppressed and favorable stations, $\hat{\Omega}\hat{I} \leq 0$, fall below the plotted range), with the rate $a_{\max}\,\text{clip}\langle\hat{\Omega}\hat{I}\rangle$; the field is neighbor-smoothed (≈0.5% chord streamwise, 5% of the probe wall-normal) before the maximum is taken, because the transported $\tilde{\nu}$ integrates production along streamlines, so isolated single-point

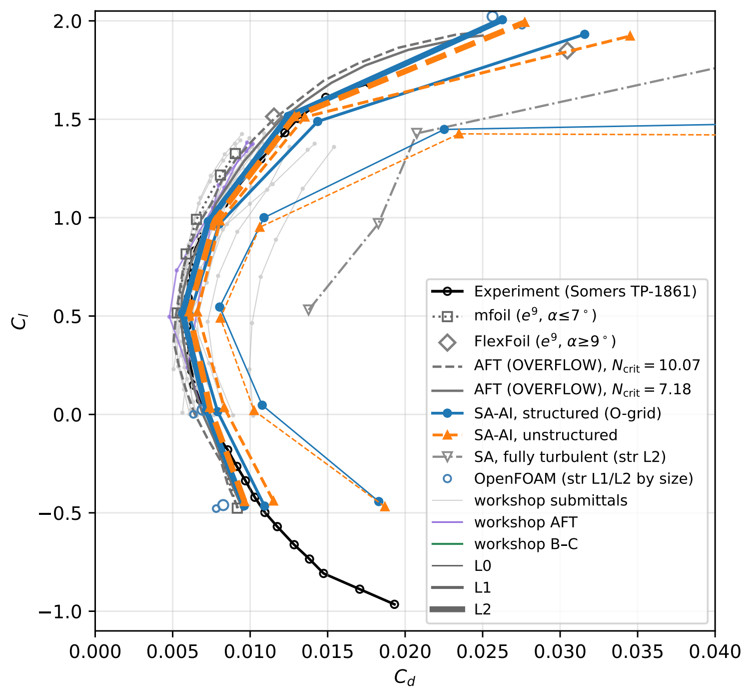

Fig. 7 Drag polar of the NLF(1)-0416 at $Re = 4\times10^6$: the present SA-AI prediction (structured O-grid and unstructured meshes, all three refinement levels) at $\alpha \in \{0^\circ, 4^\circ, 9^\circ, 15^\circ\}$ (all six grids), extended on the finest grids through $\alpha = -4^\circ$ and -8° to $c_l \approx -0.46$, overlaid on the digitized experimental section characteristics (Somers, NASA TP-1861, Fig. 11d [36]) and the mfoil e^9 reference. Mesh family sets the color (blue = O-grid, orange = unstructured); refinement sets the line weight (L0 thin → L2 thick). The mfoil reference (gray squares) is shown where its coupled solve converges to an attached solution ($\alpha \leq 7^\circ$); at $\alpha = 9^\circ$ and 15°, where mfoil fails, the e^9 reference is FlexFoil's (gray diamonds; the substitution rule and three-way cross-check are in the text). The two gray AFT (OVERFLOW) polars are digitized from the same dissertation source as Fig. 6 (dashed: $N_{\mathrm{crit}} = 10.07$; solid: the recalibrated 7.18); they end mid-plot because the source figure's axis stops at $c_d = 0.025$. Thin underlaid curves: the workshop submittals' polars, as in Fig. 6 (AFT purple, Bas–Cakmakcioglu sea-green, others light gray; segments break where the source's marker identity is ambiguous). The gray dash-dot curve is fully turbulent SA ($\chi_\infty = 3$, same solver, structured L2 grid): without the laminar run the drag sits 1.6–2.5× above the SA-AI polar (+144/+150% in the bucket at $\alpha = 0^\circ/4^\circ$, +67/+74% at $9^\circ/15^\circ$). Small steel-blue circles: the independent OpenFOAM implementation on the structured L1/L2 grids (by size), as in Fig. 6 (Table F3).

rate values—mesh noise on the unstructured family—have no effect on the solution and a raw pointwise maximum would over-report them; row 3 the in-BL maximum $\chi = \max_y \tilde{\nu}/\nu$ on a log axis with the amplification factor N from mfoil [37], a coupled viscous–inviscid e^N panel solver, on a linear right axis, the two linked by $N = \ln(\chi/\chi_\infty)$ so one unit on the right equals one e-fold on the left (vertical dotted lines: mfoil x_{tr}); row 4 $-C_p$; row 5 C_f. Line conventions, here and in every five-row figure: upper surface blue, lower red; O-grid solid, cavity dashed; L0–L2 by line weight; the e^9 reference dotted; where measured onsets are plotted, triangles on the C_f row mark them, read at the computed lift (open where extrapolated beyond the recorded data). Wherever the reference C_f comes from an XFOIL or FlexFoil surface dump (mfoil's is already freestream-normalized), it is converted from the dump's edge normalization to the freestream convention of the computed rows ($\times u_e^2$).

Because a favorable gradient keeps the profile's curvature single-signed ($\hat{\Omega}\hat{I} \to 0$), the model holds the favorable

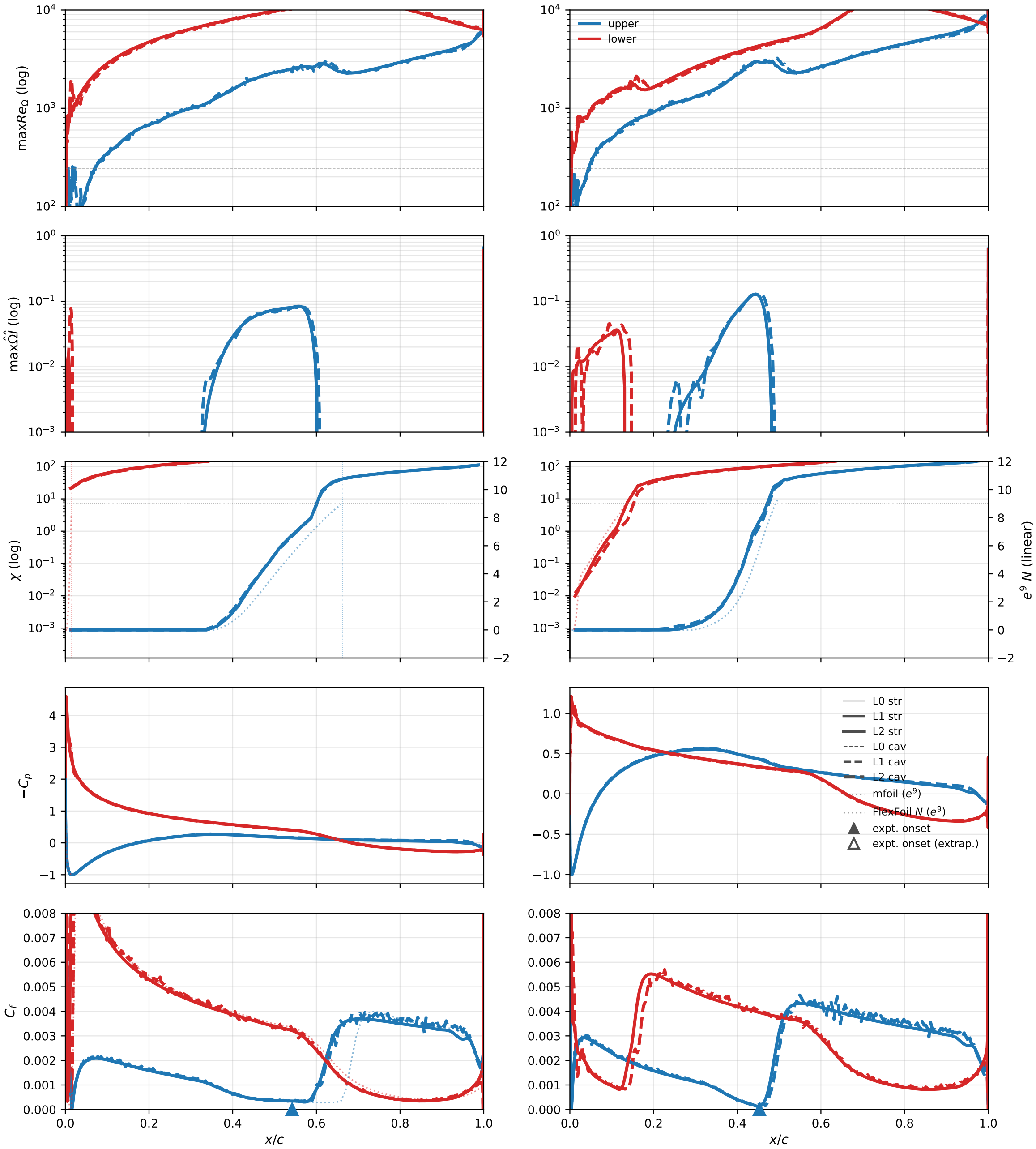

Fig. 8 NLF(1)-0416 at the negative-incidence pair $\alpha = -8°$ (left, well into the negative branch of the measured polar) and $-4°$ (right, zero lift), finest (L2) grids of both mesh families, in the five-row layout and line conventions of the text. The e^9 overlay is mfoil at $-8°$ and FlexFoil at $-4°$, where mfoil's coupled solve does not converge. At $-4°$ both fronts land on the references; the $-8°$ front and drag comparison is in Fig. 6's caption. The wall-anchored contour sheets appear in Appendix A (Figs. A3–A6).

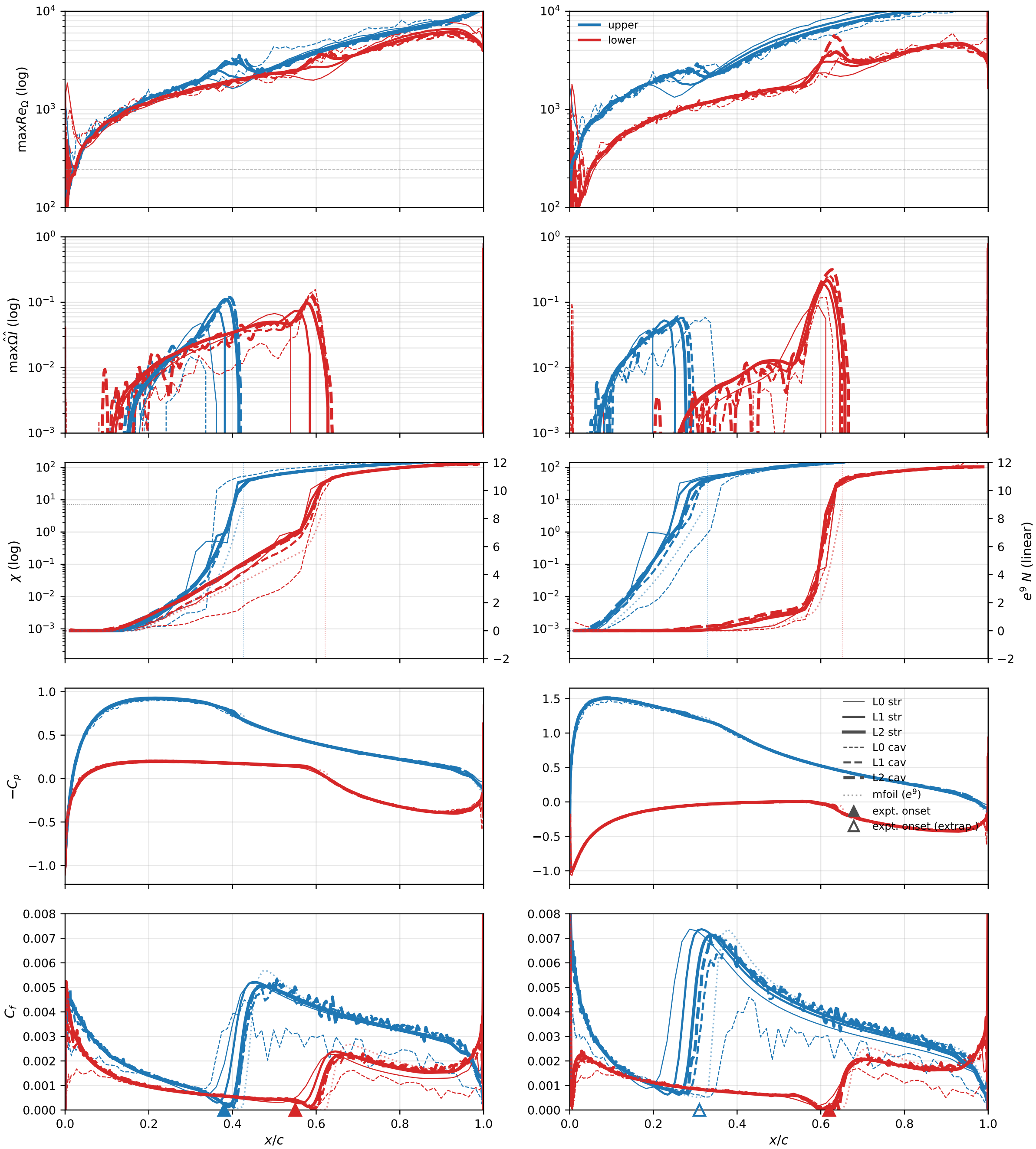

Fig. 9 NLF(1)-0416 surface distributions at $\alpha \in \{0°, 4°\}$, $Re = 4\times10^6$, $\chi_\infty = 8.76\times10^{-4}$. Five-row layout and line conventions as in the text; the measured onsets are Somers's transition orifices (Fig. 9d of Ref. [36]). At $\alpha = 0°$ both surfaces run laminar past a third of the chord (upper $x/c \approx 0.38$–0.39, lower 0.56–0.57), recovering the low-drag bucket; at $\alpha = 4°$ the upper-surface transition has marched forward to $x/c \approx 0.25$ while the lower stays near design. The stall-and-dip of the L0 χ curves over the last two cells before transition is a purely numerical artifact of the under-resolved laminar–turbulent jump (σ_t is not involved: $\chi < 1$ there). Compressed into less than one cell, the jump's discretization error reaches two to three cells upstream and distorts the velocity profile; the rate coordinate $\hat{\Omega}\hat{I}$ at the amplifying point collapses toward zero, stalling the amplification. On L1 and L2, $\hat{\Omega}\hat{I}$ holds steady through the same station, χ rises monotonically, and the small L0 onset shift converges out.

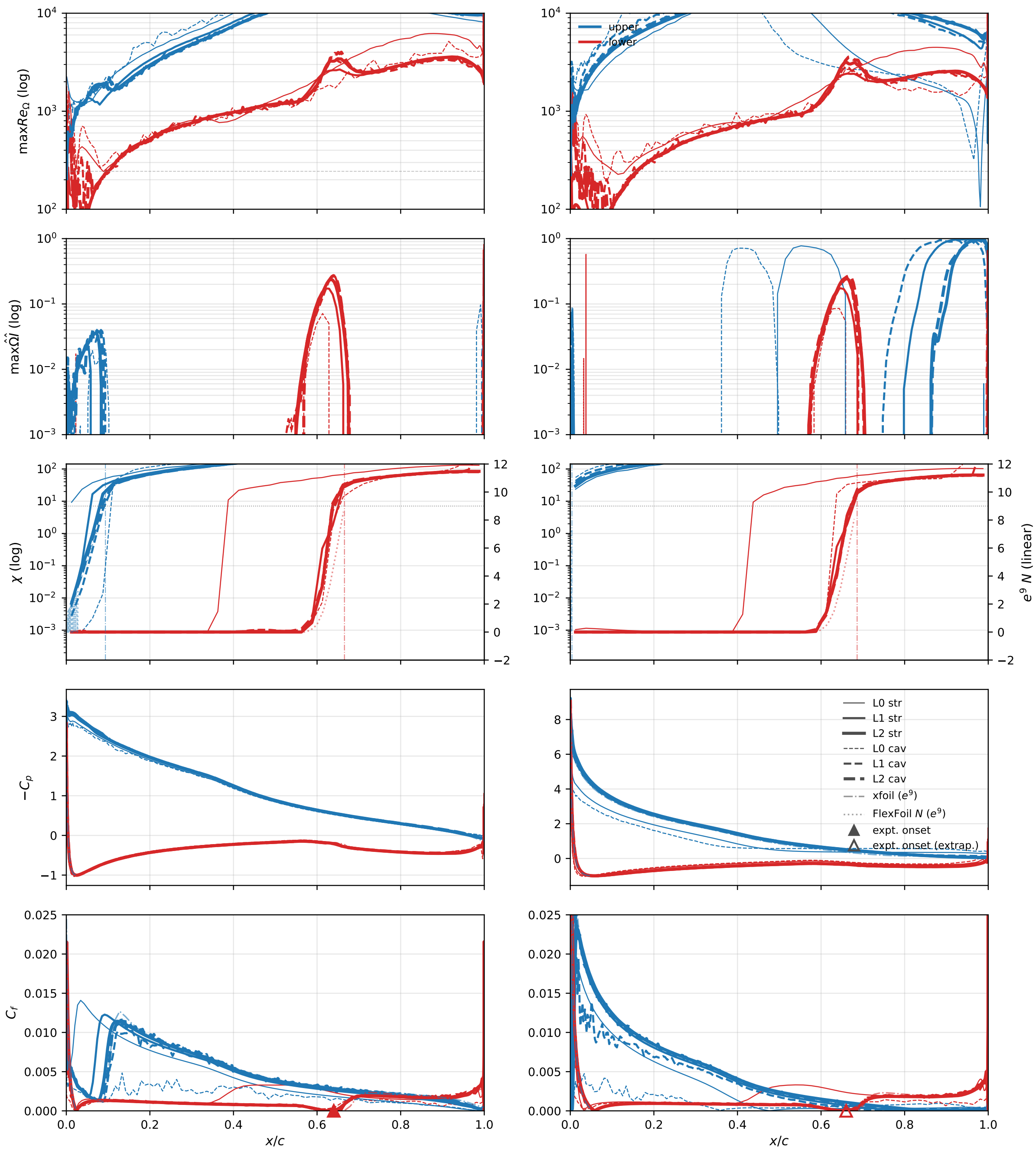

Fig. 10 As Fig. 9, at $\alpha \in \{9°, 15°\}$. The upper-surface transition has marched to $x/c \lesssim 0.05$ at $\alpha = 9°$ and is at the LE by $\alpha = 15°$, while the lower transition remains aft of mid-chord. mfoil does not converge at $\alpha = 9°$ and stalls at 15°, so the e^9 overlay (dash-dot: C_p, C_f, transition location) is XFOIL's, and the continuous amplification envelope $N(x)$ (row 3, dotted) is FlexFoil's (the substitution rule is in the text). The L0 unstructured under-resolves the LE suction peak by ~ 1 in $-C_p$ at $\alpha = 15°$ (thin dashed blue); L1/L2 collapse on both grids. The high-incidence cases test the amplification model under a strong adverse gradient: χ rises through c_{v1} within the first few percent of chord on the upper surface, but the model still resolves a finite laminar dip and the laminar/turbulent handover stays at one chordwise location across the refinement series.

rooftop laminar with no pressure-gradient parameter anywhere: amplification is confined to the recovery stretch, where the profile turns inflectional ($\hat{\Omega}\hat{I} > 0$) and the fronts are set.

V. The Eppler 387 low-Reynolds-number airfoil

The Eppler 387 tests the low-Reynolds-number, bubble-dominated regime. The McGhee, Walker, and Millard experiment in the NASA Langley Low-Turbulence Pressure Tunnel [38] tested this airfoil from $Re = 6\times10^4$ to 4.6×10^5 at tunnel turbulence levels between 0.06% and 0.18%; the data are the canonical reference for low-Re transition modeling and supply the comparison condition used by Coder and Maughmer's AFT-2014, Coder's AFT-2017, the Langtry–Menter γ–Re_θ benchmark, and the Bas–Çakmakçıoğlu algebraic SA-BCM variant [8, 10, 12, 16]. At these conditions the upper surface carries a long laminar-separation bubble (LSB) that controls the section drag: laminar separation at the shoulder of the suction peak, an inflectional shear layer in which the rate kernel must accelerate amplification, turbulent reattachment, and a short turbulent wedge to the trailing edge.

The model is evaluated at the de-facto benchmark condition of the transition-modeling literature: $Re = 2\times10^5$, $M = 0.1$, across $\alpha \in \{0°, 2°, 5°, 7°\}$, with the freestream seed at the same $N_{\text{crit}} = 9$, LTPT-class anchor as the NLF (Sec. IV). Three refinement levels on two independently-generated mesh families (unstructured cavity, structured O-grid) give twenty-four cases. The finest (L2) pair extends the matrix with six further incidences, $\alpha = -2°$, 1°, 3°, 4°, 6°, and 8.5° (twelve additional solutions), and the two incidences that bound the sweep, $\alpha = -2°$ and 8.5°, are additionally computed on the L0 and L1 pairs (eight further solutions), so that every grid spans the measured polar end to end and the mesh-refinement comparison is available at the endpoints as well as in the interior. The forty-four solutions were all converged with a single solver build.

Mesh generation reuses the NLF recipe verbatim. The 56-point Eppler 387 contour is read from the NASA TM 4062 Table I (design coordinates with the trailing edge thickened from 0 to 0.01 in blended at $x/c = 0.95$; $z_{\text{TE}} = \pm0.000833\,c$, half the NLF blunt-thickness), splined and resampled as the NLF contour at $N \in \{200, 400, 800\}$ points, and fed to both the in-house unstructured cavity mesher and Construct2D; the far-field is fixed at $100\,c$. The resulting grids range from 1.5×10^4 nodes at L0 to 2.6×10^5 at L2. The mesh close-ups mirror the NLF's (Figs. A1 and A2, the blunt-TE handling aside); Table 5 lists the Eppler resolution quantitatively.

On the finest grids the SA-AI polar (Fig. 11) tracks the measured low-drag bucket over $\alpha = 0°$–7° ($c_l \approx 0.38$–1.15) at matched lift—every percentage reads the experimental polar's drag at the computed lift: 3.7–6.4% below the data at the bucket edges (−5.8%/−4.3% structured/unstructured at $\alpha = 0°$, −4.4%/−3.8% at 2°, −3.7%/−6.4% at 7°) and +3.7%/+1.3% at the $\alpha = 5°$ bubble apex, where the pressure-drag footprint of the slightly over-long laminar-separation bubble (§II.C) remains the largest positive residual. At the two extension endpoints the comparison changes character: at −2° the computed drag sits ≈14% below the measured curve—the model's bucket extends below $c_l \approx 0.25$, where the LTPT's low-drag corner has already ended—and at 8.5° a matched-lift read is ill-posed (the computed lift reaches the

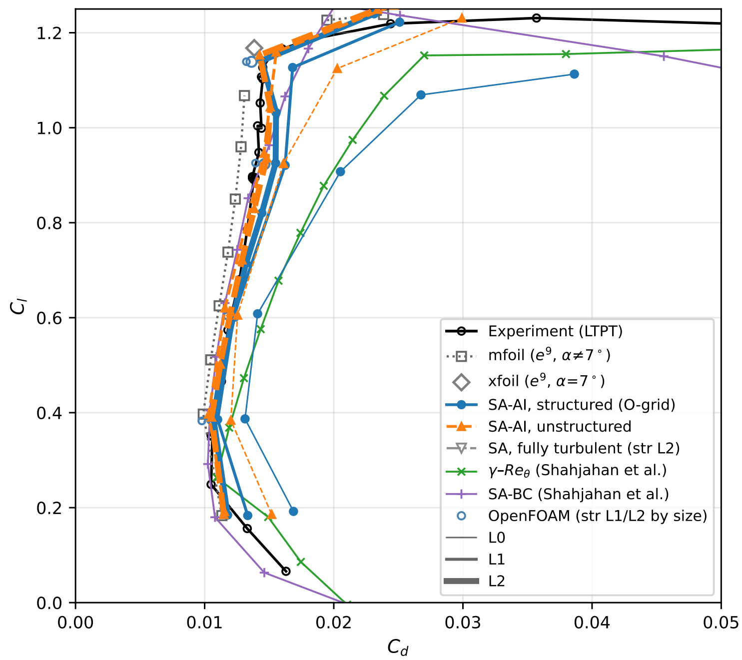

Fig. 11 Drag polar of the Eppler 387 at $Re = 2{\times}10^5$: the present SA-AI prediction (all six grids over $\alpha = -2°–8.5°$; every grid carries $\alpha \in \{0°, 2°, 5°, 7°\}$ plus the two bounding incidences $-2°$ and $8.5°$, and the finest L2 pair carries four further interior incidences) overlaid on the digitized experimental section characteristics from the Langley Low-Turbulence Pressure Tunnel (McGhee et al., NASA TM-4062, App. B, $R = 2{\times}10^5$) and the mfoil e^9 reference, spanning the same extended incidence range (XFOIL fills $\alpha = 7°$, where mfoil's coupled solve fails). The computed points track the low-drag laminar bucket; at L2 the two mesh families agree to within $\Delta C_d \le 6.9 \times 10^{-4}$ across $\alpha = -2°–7°$ (largest at $\alpha = 5°$) and to $2.7{\times}10^{-4}$ at the unsteady 8.5° point. The gray dash-dot curve is fully turbulent SA ($\chi_\infty = 3$, same solver, structured L2 grid), 66–87% above the SA-AI drag in the bucket and +149% at $\alpha = 7°$. The green and purple curves are the γ–Re_θ (Langtry–Menter, OpenFOAM) and SA-BC (SU2) transition models at the same condition and $Tu = 0.1\%$, digitized from Ref. [39] (Appendix F); the SA-BC intermittency correlation was calibrated in part on this airfoil [40]. Small steel-blue circles: the independent cell-centered incompressible (OpenFOAM) implementation on the structured L1/L2 grids (by size), as in Fig. 7; the port omits the bypass of Eq. 18, worth −6 to −10 counts at $\alpha = 0–5°$ here, and drag agreement is not claimed—the visible offsets are consistent with that omission in sign and magnitude (Table F6).

measured maximum and the solutions are unsteady; the plotted points are medians over an oscillating window). The one published SA-based transported-intermittency (γ-equation) result on this benchmark—the γ–SA coupling of Ref. [41] (foam-extend, $Tu = 0.07\%$)—reports $c_l = 0.91 \pm 0.02$ with $c_d = 0.0133–0.0140$ at matched lift, the natural family comparison for SA-AI's ≈0.0142 (structured L2) interpolated at the same lift. The coarsest grid (L0) over-predicts drag markedly; refinement is essential in this bubble-dominated regime.

The skin friction (Figs. 12–13, bottom rows) shows the LSB directly: the upper-surface C_f drops below zero over a band that spans up to $\sim 0.25\,c$ at low incidence—a shallow laminar separation feeding a compact turbulent recirculation core—and collapses toward $\alpha = 7°$. Both Flow360 and mfoil show the same recirculation structure at $\alpha \le 5°$ on both mesh families.

The rate coordinate $\max_y \hat{\Omega}\hat{I}$ (row 2 of Figs. 12–13) shows the adverse-gradient response the Eppler 387 probes: on

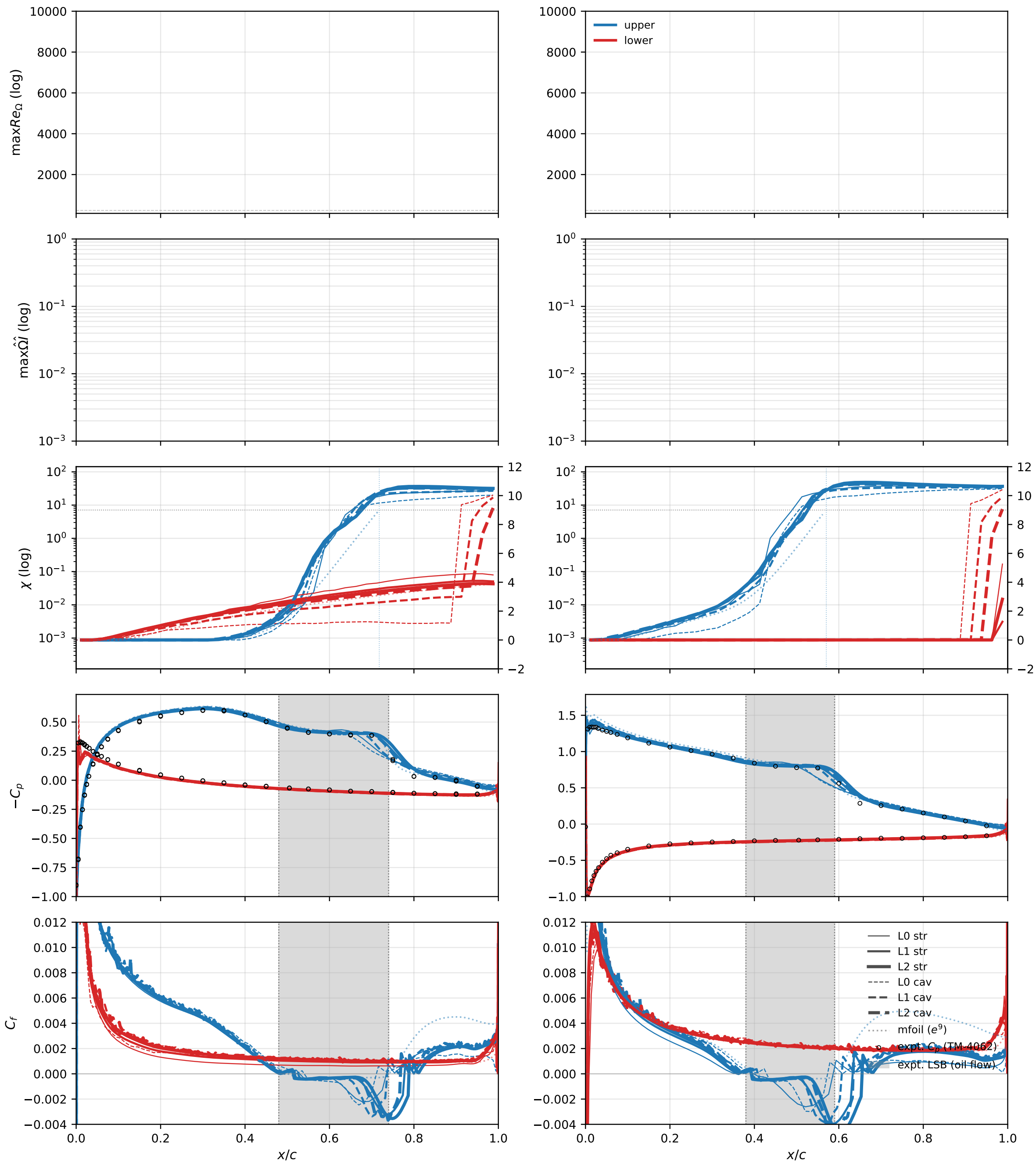

Fig. 12 Eppler 387 surface distributions at $\alpha \in \{0°, 5°\}$, $Re = 2\times10^5$, $M = 0.1$, $\chi_\infty = 8.76\times10^{-4}$ (the paper-wide seed, Sec. IV). Five-row layout and line conventions as in the text; open black circles on the $-C_p$ row: the tabulated experimental C_p (NASA TM-4062 App. D, column with the nearest angle of attack). The signed C_f (row 5) shows the laminar-separation bubble directly as the dip below zero on the upper surface. The bubble sits at $x/c \approx 0.52$–0.76 at $\alpha = 0°$ and marches forward to $x/c \approx 0.40$–0.63 at $\alpha = 5°$; both Flow360 and mfoil show the same recirculation structure.

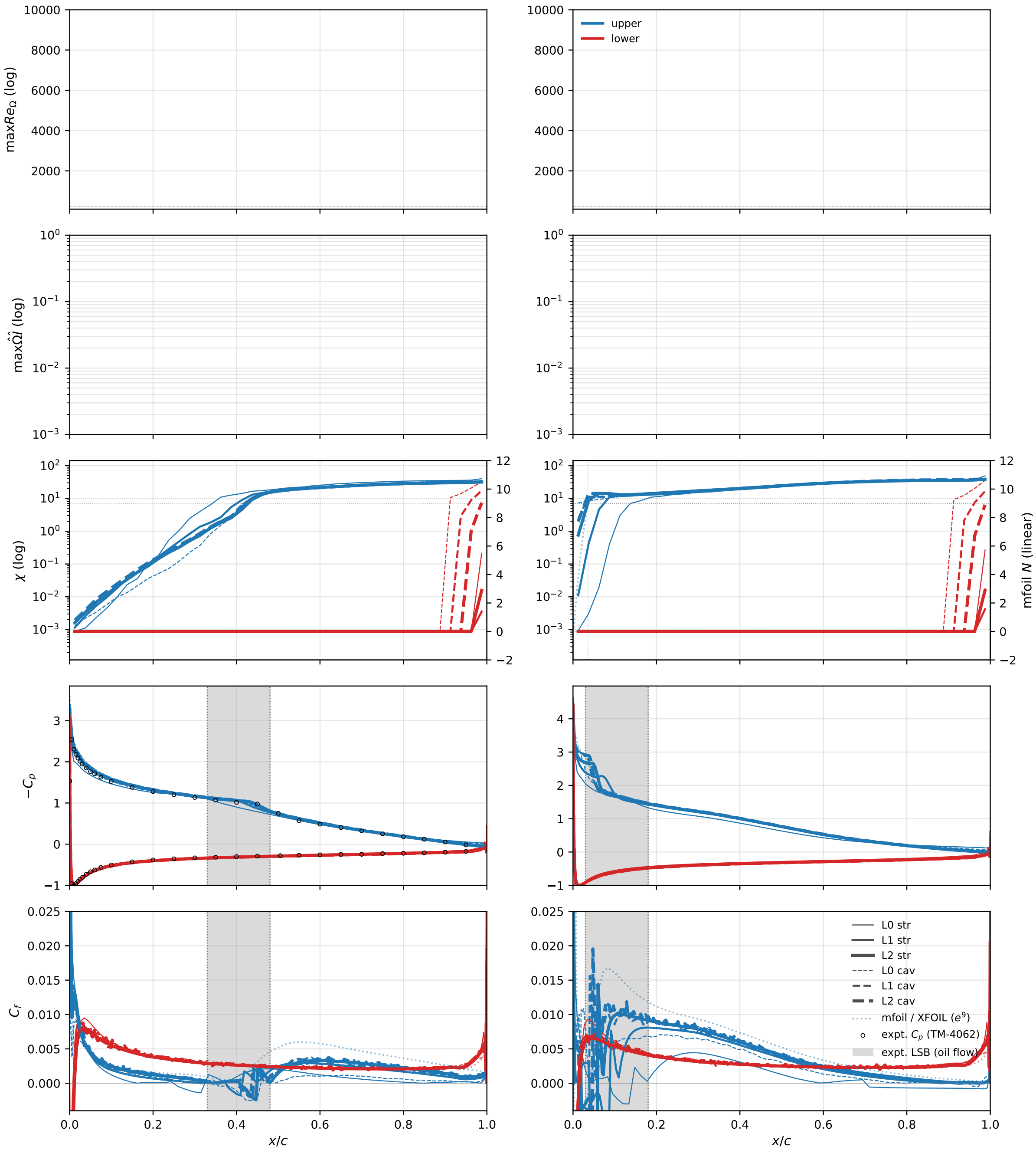

Fig. 13 As Fig. 12, at $\alpha \in \{7°, 8.5°\}$ (8.5° on the finest L2 pair only). At $\alpha = 7°$ the suction peak is near $-C_p \approx 3$ and the model sits at the edge of bubble collapse: the upper-surface C_f crosses zero only on three of the six grids (structured L0/L2 and cavity L0)—on the structured L2 a $x/c \approx 0.39$–0.41 sliver, reattaching ~0.07 c ahead of the measured oil-flow reattachment (0.48 c)—while on the other three, transition preempts separation entirely, one incidence ahead of the experiment's $\alpha = 8°$ natural-transition point. At the post-stall 8.5° both families trip in the leading-edge bubble (gray band: the oil flow's 0.03–0.18 c)—χ crosses c_{v1} within the first few percent of chord and the flow is turbulent and attached over the rest of the surface. The lower-surface boundary layer remains attached and laminar to $x/c \approx 0.99$ across the entire α range, matching the e^9 prediction that the lower transition lies aft of the airfoil. At $\alpha = 7°$ the dotted reference is XFOIL rather than mfoil, which stalls numerically there (Fig. 14); the surface distributions carry no N curve, so row 3 shows no envelope at 7° (at 8.5° mfoil converges and the envelope returns).

Table 5 Eppler 387 mesh ladder: size, wall-tangential and wall-normal resolution, and resulting y^+, measured as in Table 4; the y^+ percentiles aggregate the four incidences ($\alpha = 0°, 2°, 5°, 7°$). The lower Reynolds number (2×10^5) keeps y^+ well below unity on every grid. The cavity meshes are all-triangular, the structured O-grids all-quadrilateral.

	Unstructured cavity			Structured O-grid		
	L0	L1	L2	L0	L1	L2
Nodes	15,219	53,109	192,714	15,920	63,840	255,680
Elements	29,757	105,335	384,141	15,721	63,441	254,881
$\Delta s/c$ @ LE ($\times 10^{-3}$)	1.13	0.57	0.28	4.94	2.48	1.24
@ $0.05c$	5.28	2.63	1.31	9.82	4.88	2.43
@ $0.25c$	11.0	5.46	2.72	17.4	8.63	4.30
@ $0.98c$	15.1	5.04	3.32	2.68	1.32	0.69
@ TE	2.27	2.97	0.86	0.58	0.32	0.19
h_0/c ($\times 10^{-6}$)	14.3	5.7	2.8	6.6	3.3	1.7
cell @ $0.001c$ ($\times 10^{-6}c$)	226	98	51	168	86	44
cell @ $0.5c$ ($\times 10^{-2}c$)	8.20	4.33	2.23	1.91	0.94	0.45
y^+ (95th pct)	0.35	0.17	0.08	0.11	0.05	0.03
y^+ (99th pct)	0.43	0.27	0.13	0.15	0.08	0.04

the upper surface $\hat{\Omega}\hat{I}$ rises from near zero (the attached, near-neutral layer) into the strongly amplifying range across the laminar-separation bubble (the inflectional profile), then relaxes after reattachment, while the lower surface stays near zero. (The conspicuous double peak the upper-surface trace shows inside the bubble—and the sharp notch between the two—is anatomized in Appendix B against the contour sheet of Fig. B4.)

The bubble stations are collected in Fig. 14, against the oil flow at its eight bubble incidences, the independent Selig et al. measurement, the dense-α mfoil sweep, and the two digitized transition-model references. The separation location is reproduced by every method on the plot, SA-AI included, to a few percent of chord at every incidence. The reattachment—the bubble's aft end, set by the post-transition turbulent recovery—is the quantity this test hinges on, and it is where the model shows its cost: at $\alpha \leq 5°$, at grid convergence, both families reattach late against the oil flow—a bubble 0.02–0.04 c too long—while at the near-stall 7° the model sits at the edge of bubble collapse: three of six grids (the unstructured finest among them) transition ahead of separation and plot no stations—one incidence ahead of the experiment's $\alpha = 8°$ natural-transition point—and the surviving structured-L2 sliver (0.014 c of recirculation) closes 0.07 c early. That low-α margin sits inside the facility-to-facility spread on the figure: the Cole–Mueller pressure-station measurements at the same Reynolds number [42] read reattachment 0.03–0.08 c aft of the LTPT oil flow at and between its tabulated incidences ($\alpha = -2°$–$3°$; the oil flow interpolated where the incidences do not coincide).

Past stall the trend continues: at the 8.5° extension both families reproduce the experiment's post-stall leading-edge-bubble topology (oil flow: separation at 0.03 c, reattachment at 0.18 c) but reattach at 0.032–0.049 c—roughly four times short of the measured recovery—carrying the $\alpha \geq 7°$ early-reattachment direction through the stall. The structured

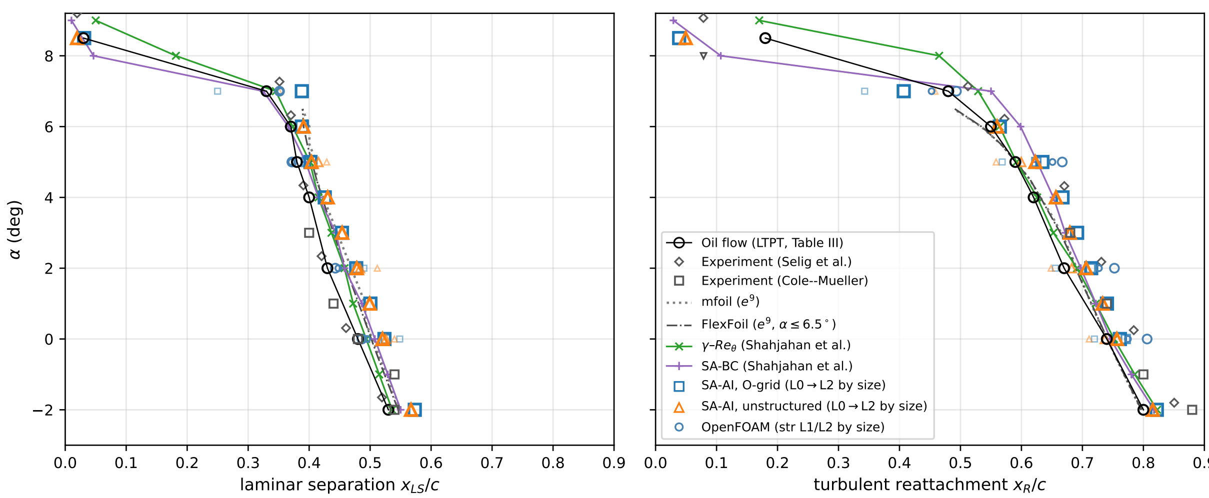

Fig. 14 Upper-surface laminar-separation (left) and turbulent-reattachment (right) locations on the Eppler 387 at $Re = 2\times10^5$ against incidence. For a laminar-separation bubble the physically observable stations are the separation and reattachment points, not the amplification onset; in the computation they are the signed-C_f zero crossings (separation: first downward; reattachment: most-downstream upward—the physical return of attached flow aft of the bubble), and the same measure is taken from the e^9 references: mfoil (dotted; drawn to $\alpha = 6.5°$, beyond which it is at the edge of its convergence regime—at $\alpha = 7°$ it returns $C_l = 0.928$, below its 5° value, while XFOIL converges there with $C_l = 1.17$) and FlexFoil (dash-dot). Past $\alpha = 6.5°$ the e^9 closure transitions ahead of laminar separation—its signed C_f never crosses zero (FlexFoil and XFOIL agree, minimum $+4.7\times10^{-4}$ in the freestream convention at 7°)—and so predicts no bubble where the oil flow still shows one; the isolated FlexFoil marker at 8° is a marginal three-point sliver. Black circles: the oil-flow visualization (McGhee et al., NASA TM 4062, Table III; its nine tabulated incidences less $\alpha = 8°$, where the report notes natural transition at $0.32\,c$ and no bubble exists to plot—the 8.5° point is the post-stall leading-edge bubble), i.e. the edges of the gray band on the C_f rows of Figs. 12–13. Gray diamonds: the independent Selig et al. measurement, digitized with the γ–Re_θ and SA-BC curves from Ref. [39] (Appendix F). Larger gray squares: the Cole–Mueller pressure-station measurements [42] at the same Reynolds number (their $\alpha \geq 7°$ entries are attached-transition values and plot no bubble; Appendix F). Open squares/triangles: SA-AI on all six grids (L0–L2 by marker size, both families); as on the polar, the $-2°$–$8.5°$ extension incidences exist on the finest (L2) pair only, L0/L1 keeping $\alpha \in \{0°, 2°, 5°, 7°\}$. Small steel-blue circles: the independent cell-centered incompressible (OpenFOAM) implementation on the structured L1/L2 grids (by size), stations from the same signed-C_f crossings; at L2 its separation sits 0.030–$0.037\,c$ ahead of and its reattachment 0.033–$0.045\,c$ behind the SA-AI stations ($0.086\,c$ at 7°)—the bypass-less port (Eq. 18) reattaching later at every incidence (Table F6). The lower surface stays attached and laminar to the trailing edge at every incidence.

family's bubble is a marginal few-sample sliver, and the solutions there are unsteady; Fig. 14 plots the stations of the time-median surface solution.

This low-α over-length is not an amplification deficit: the $\chi = 1$ front sits mid-bubble, slightly ahead of the e^9 crossing (Figs. 12–13, row 3), and the inflectional profile saturates the rate coordinate ($\hat{\Omega}\hat{I} \approx 1$) across the mid-bubble. What is slow is the transition itself: the model carries no closure for the transition process, and the turbulent stress that closes the bubble arrives aft of where spot-driven breakdown completes in the tunnel—reattachment, the quantity the oil flow measures, lands 0.02–$0.04\,c$ late even though the front is placed correctly. The mechanism (the transported-χ handover's fixed e-fold count against its growing physical length), and where it leads as the Reynolds number falls—to a bursting boundary, and to the loss of the two-state structure the experiment shows—is quantified in the next section

(Sec. VI).

VI. Reynolds-number study of the Eppler 387 at $\alpha = 5°$

To assess whether the same constants follow the bubble across the Reynolds-number range of the McGhee experiment, we hold everything of Sec. V's study fixed—the incidence at $\alpha = 5°$ and the paper-wide freestream seed (Sec. IV)—and sweep the four other tabulated Reynolds numbers, $Re \in \{6\times10^4,\ 1\times10^5,\ 3\times10^5,\ 4.6\times10^5\}$, reusing the grids of both mesh families at all three refinement levels from the $Re = 2\times10^5$ study unchanged (only the reference viscosity changes)—24 solutions in all. Figures 16 and 17 show the resulting $-C_p$ and signed C_f, overlaid with the exact experimental surface pressures (open markers, from the NASA TM-4062 Appendix D tables at the column with the nearest angle of attack—every repeat dataset the report holds at these conditions) and the mfoil/XFOIL e^9 reference where it converges; wall-anchored contour sheets for every solution of the sweep appear in Appendix C. Every solution away from 10^5 is steady—force coefficients flat to $\Delta C_l \leq 3\times10^{-4}$ over the final 2000 pseudo-steps, most below 10^{-4} (the loosest are the structured L1 benchmark at 2.6×10^{-4} and the structured L0 at 3×10^5 at 1.8×10^{-4}). At 10^5, the near-bursting condition discussed below, the cavity family carries the residual unsteadiness ($\Delta C_l \approx 3\times10^{-4}$ at L1, 7×10^{-4} at L2) while the structured grids sit at $\leq 3\times10^{-5}$; the structured-L1 bubble limit cycle of the pre-bypass model does not recur under the recomputation.

The section lift, drag, and quarter-chord pitching-moment coefficients are collected in Fig. 15, and tabulated at L2 against the e^9 reference and the tunnel data in Table C1 (Appendix C). The three-way e^9 cross-check of Sec. IV has its one force-level exception here: at $Re = 6\times10^4$ mfoil visibly departs from XFOIL and FlexFoil, and XFOIL—whose forces also sit closest to the measurement—is presented as the reference there, in the table (mfoil in the footnote) and in Fig. 16.

The one systematic failure is at $Re = 10^5$, where the computation sits past the model's bursting boundary (the opening fan of Fig. 15): on every grid the upper-surface recovery is incomplete—the signed C_f regains zero only within 0.05 c of the trailing edge, against the measured oil-flow reattachment near 0.67 c—and the L2 drag sits +30% (unstructured) to +73% (structured) above the measurement, refinement pushing both families toward the burst state rather than the measured short bubble (the coarser grids only mask the failure: the unstructured L1 lands +4% high). The over-length has two contributions—the under-amplified detached shear layer of §II.C, dominant at 2×10^5, and the stretching handover zone of Sec. VI, which takes over as the Reynolds number falls: at this Reynolds number the model's extended laminar separation fails to close at all. The remainder of this section takes this failure apart: the computed state is unique—every initialization admitted by the convergence protocol of Sec. VIII converges to it, on relaxation times an order of magnitude longer than anywhere else in the sweep—and what makes that unique state the burst one is the length of the model's transition zone. The measured pressures confirm the computed bubble structure where the bubble does close: the suction-side plateau of the separated shear layer appears in the open-circle data at 3×10^5 exactly

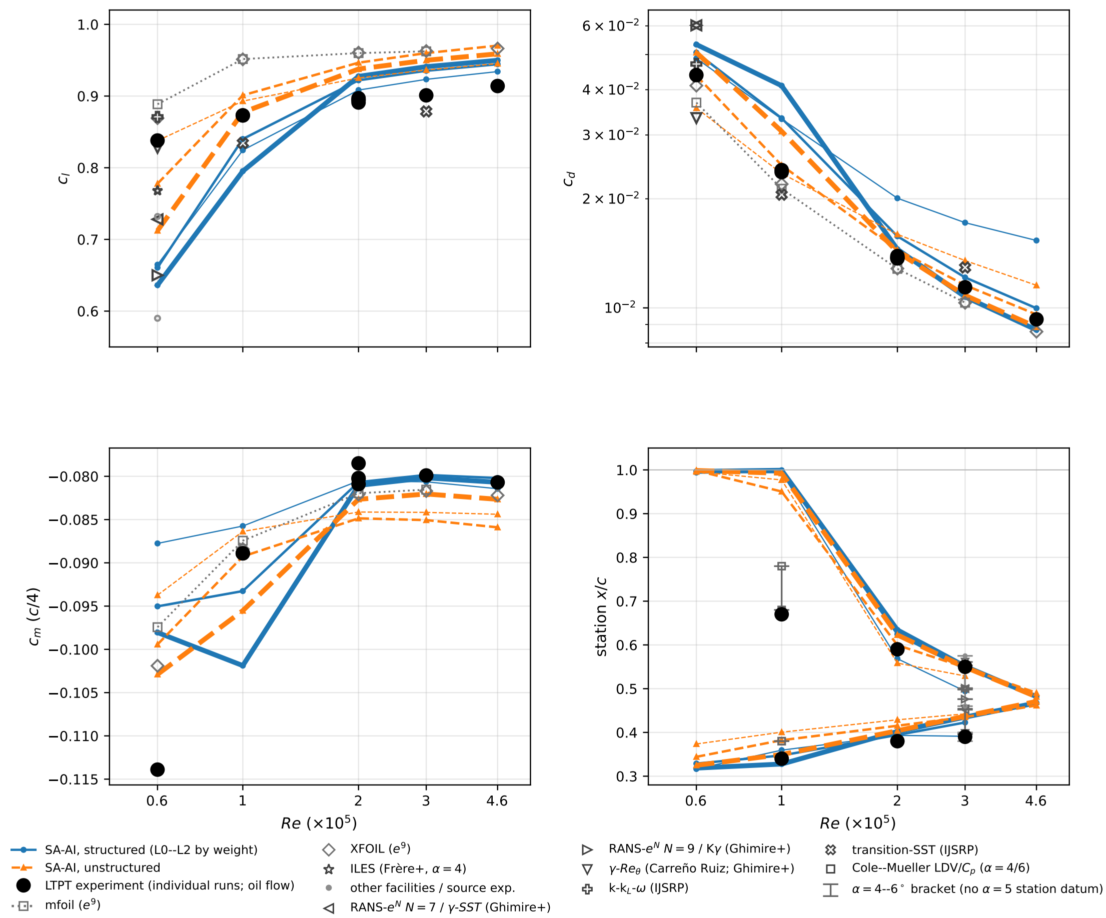

Fig. 15 The $\alpha = 5^\circ$ Reynolds sweep. Top: section lift and drag (log). Bottom left: quarter-chord moment. Bottom right: upper-surface laminar-separation and turbulent-reattachment stations on a shared x/c axis (lower band separation, upper reattachment; SA-AI values pinned near 1.0 mean no closure ahead of the trailing edge). All panels show the 24 SA-AI solutions (structured O-grid solid blue, unstructured cavity dashed orange, L0–L2 by line thickness), the e^9 panel reference (mfoil squares at 0.6–3×10^5 where its solve returns forces, XFOIL diamonds at all five Reynolds numbers, as in Table C1; c_m from both), and the LTPT measurement as large filled black circles—the dominant experiment: each individual TM-4062 Table B1 run reading within 0.1° of $\alpha = 5^\circ$ is its own circle (1/2/4/1/1 runs at $0.6/1/2/3/4.6\times10^5$), so repeat runs show as genuine measurement scatter rather than a mean with an assumed error bar (coincident where the readings agree), with the oil-flow separation and reattachment stations (Table III, tabulated at $\alpha = 5^\circ$ for $Re = 1$–3×10^5 only). Small open gray symbols are published calculations read at $\alpha = 5^\circ$: at 6×10^4 the ILES ($\alpha = 4^\circ$, ahead of its reattachment jump) and coupled RANS-e^N ($N = 7/9$) of Ref. [43] with that source's compiled Delft and Stuttgart lift (small dots), the γ–Re_θ of Ref. [44], and the k-k$_L$-ω of Ref. [45] whose transition-SST appears at 1 and 3×10^5 (its 6×10^4 series excluded; Appendix F). The capped range bars in the station panel are $\alpha = 4$–6° brackets straddling the computed $\alpha = 5^\circ$, drawn because no source tabulates a station at exactly $\alpha = 5^\circ$: Cole–Mueller pressure/LDV at 10^5 [42] and the γ-variant tables of Ref. [46] at 3×10^5. The refinement fan closes onto the measurement at 2–4.6×10^5 and opens at 10^5—the model's bursting boundary (Sec. VI)—where refinement drives both families away from the measured short-bubble state.

where the computed C_p flattens, and at 10^5 over the bubble's front half, before the computed and measured recoveries part company.

The solution is markedly more mesh-sensitive at 6×10^4 than at the higher Reynolds numbers: the open separation gives $C_l = 0.71$ unstructured vs. 0.64 structured at L2, against $\Delta C_d \leq 2\times10^{-4}$ (8×10^{-4} at L1) at 3–4.6×10^5. The experiment itself is bistable there: in the TM-4062 listing at $Re = 6\times10^4$, three consecutive repeat points at $\alpha = 4.00°$ give $c_l = 0.643$, 0.697, and 0.721 ($c_d = 0.0431$, 0.0386, 0.0400), and the measured lift collapses from 0.838 at $\alpha = 4.99°$ to 0.639 at $\alpha = 5.51°$—run-to-run scatter of the same order as the spread between the two computed solutions. The condition sits near the bubble-bursting limit, where the measurement is bistable and the computation most mesh-sensitive (the branch question is taken up below); the mesh scatter and the offset from the e^9 reference there reflect that regime together with the model's over-long bubble (§II.C) and the post-separation turbulent recovery, and we claim only the macroscopic trend.

The sweep traces the classic low-Re bubble progression at fixed incidence: as the Reynolds number rises the transition front moves forward and the laminar-separation bubble shortens. The upper-surface C_f shows an open, non-reattaching separation at $Re = 6\times10^4$—the flow separates near $x/c \approx 0.32$ and does not recover ahead of the trailing edge, the low-Re bubble-bursting limit—then an incompletely closing separation at 10^5, the C_f regaining zero only within $0.05\,c$ of the trailing edge on every grid (the model carries the bursting regime one Reynolds step past the measurement), and then short, shallow bubbles—the computed C_f only grazes zero—recovering, on the two finer levels, at $x/c \approx 0.55$ (3×10^5) and $x/c \approx 0.48$ (4.6×10^5).

The deep reverse-C_f surge inside the long bubbles at $Re = 6\times10^4$ and 10^5 ($C_f \approx -0.005$, several times the turbulent peak that follows) is qualitatively physical—bubble DNS shows a sharp negative surge near the rear of strongly separated bubbles [47, 48]—but its magnitude is outside the model's declared scope: the mid-bubble C_f profile is an uncalibrated byproduct of the σ_t handover and of standard SA's separated-flow behavior, and we claim no accuracy for it. What this section tests is the macroscopic content: separation, transition, reattachment, and the forces that follow.

The measured flow is itself two-valued at 6×10^4: the LTPT observed laminar separation with and without turbulent reattachment at the same angle of attack over $\alpha \approx 3°$–$7°$ (McGhee et al.'s Fig. 11(a)), the selection was environmental (at the slightly noisier tunnel condition the flow always failed to reattach), repeat polars scatter an order of magnitude more than at any higher Reynolds number (Table C1), and the decreasing-α hysteresis runs read a measurably weaker, less-recovered upper surface than the up-sweep at the same angle (ΔC_p rms 0.06, four times the repeat scatter; squares vs. circles in Fig. 16)—while above 6×10^4 "turbulent reattachment always resulted" and the 10^5 repeat conditions coincide. The model reproduces neither the second branch nor the hysteresis: direct forks of each family's converged 2×10^5 solution, slow one-step Reynolds descents and climbs, and 40 000-iteration extensions of every 10^5 endpoint all converge to one family-specific state at 10^5 (structured: lift agreeing to ≤ 0.003, drag to $\approx 5\times10^{-4}$; cavity: ≤ 0.005 among cold, fork, and descent, with the slow climb still 0.012 away and drifting toward the pack at the end of its 40 000-

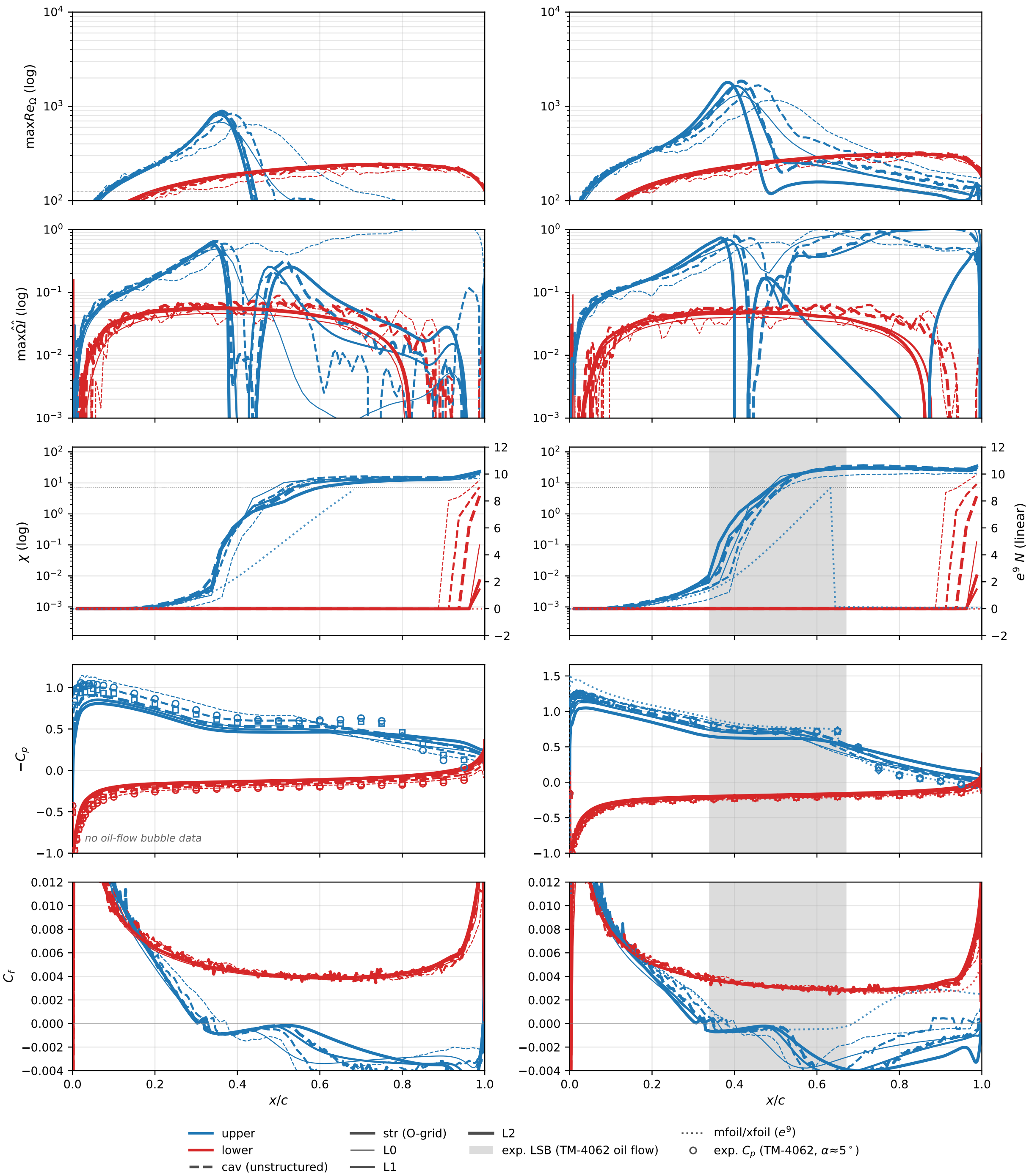

Fig. 16 Eppler 387 at $\alpha = 5°$, $Re = 6\times10^4$ (left) and 1×10^5 (right): the five-row suite and line conventions of the benchmark figures (Sec. IV), both mesh families. Dotted: the e^9 reference—XFOIL at 6×10^4 (Table C1), mfoil at 10^5; the N envelope at 6×10^4 is FlexFoil's. Open markers: every TM-4062 repeat pressure dataset at these conditions. At 6×10^4: circles the up-sweep, squares the decreasing-α hysteresis branch, whose weaker plateau and less complete recovery measure the experimental branch scatter at the bistable condition (see text). At 10^5: circles/squares/diamonds are three tunnel conditions that coincide to repeat accuracy—the measured flow there is single-branch. Gray band: oil-flow bubble extent (TM-4062 Table III; exists only at 10^5 among these two). At 6×10^4 the computed separation does not reattach ahead of the trailing edge; at 10^5 the recovery is incomplete on every grid (see text).

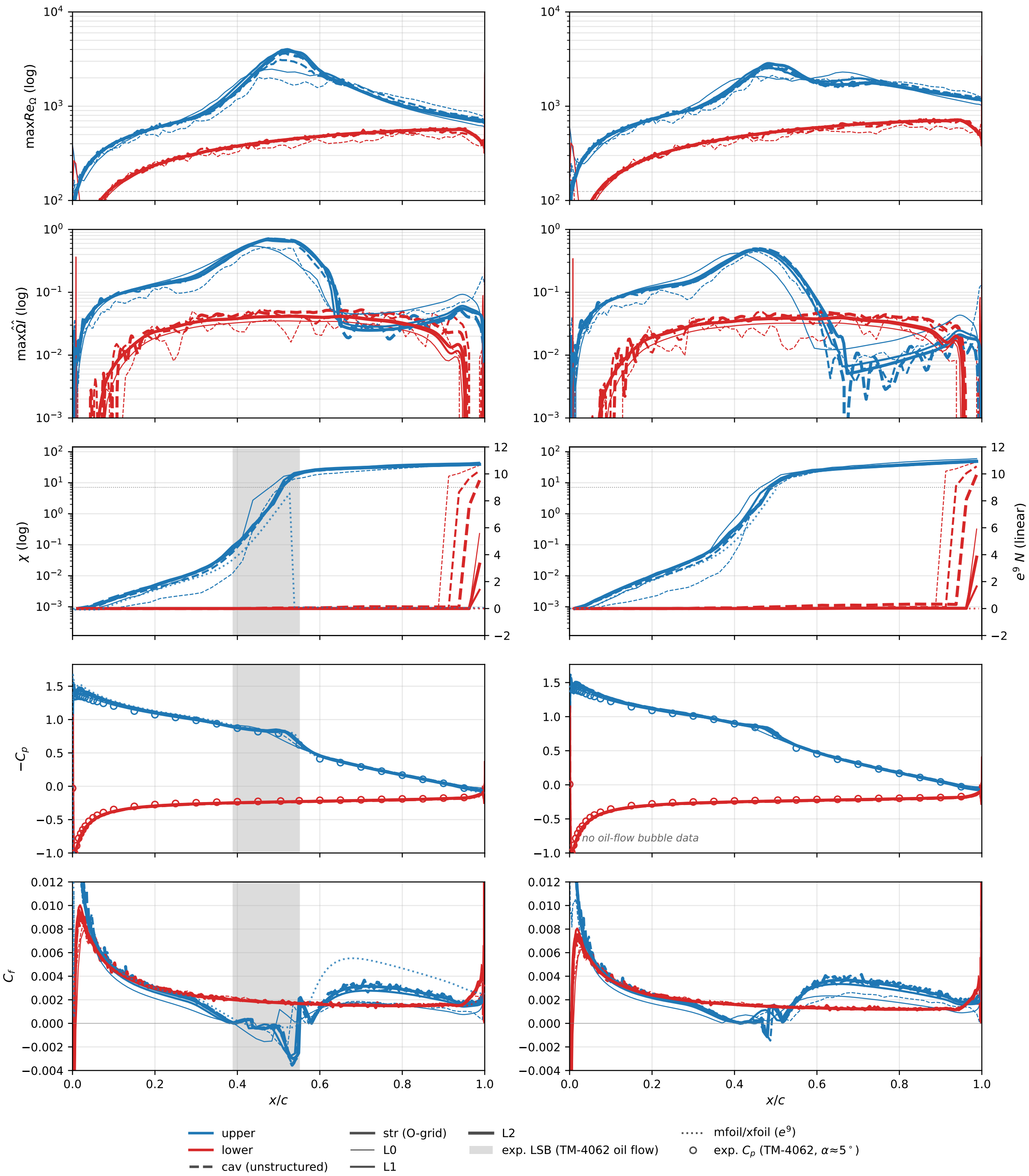

Fig. 17 As Fig. 16, at $Re = 3\times10^5$ (left) and 4.6×10^5 (right). The bubble is short and moves forward with Reynolds number, recovering (on the two finer levels) near $x/c \approx 0.56$–0.57 at 3×10^5 and ≈ 0.48 at 4.6×10^5 (at these two Reynolds numbers the computed C_f only grazes zero). The mfoil e^9 panel solver converges at 3×10^5 but diverges on the thin bubble at 4.6×10^5, where XFOIL supplies the e^9 C_p/C_f reference (dotted) and FlexFoil the N envelope (χ row) instead. The oil-flow bubble band is available at 3×10^5 only.

iteration extension) and collapse onto the burst branch at $6{\times}10^4$ (repro: `repro/cfd/run_bistability_forks.py`, `run_continuation_ladders.py`, `run_fork_extensions.py`, `run_dn_extensions.py`; the probes were repeated in full at the recomputed canon and the conclusion carried over). The model is monostable throughout the sweep; the only trace of the nearby fold is the relaxation time at 10^5—the warm-started paths need their full $4{\times}10^4$-iteration extensions to close on the cold state (the cavity climb is still drifting at the end of its), an order of magnitude longer than anywhere else in the sweep—and the experiment's two-state structure, hysteresis included, lives at $6{\times}10^4$, one Reynolds step below the model's bistable-free burst.

Why does the model burst where the measurement reattaches? The account begun at the 2×10^5 benchmark (Sec. V)—separation on time, the front on time to early, reattachment late—completes here as a length scale. The handover from amplification to SA production must climb a fixed number of e-folds of χ: $\ln c_{v1} \approx 2.0$ from the $\chi = 1$ crossing to eddy-viscosity half-saturation, another ≈ 1.4 to effectively full production. The length one e-fold costs, roughly $u_{\text{conv}}/(a\,\omega)$, is not fixed: it stretches like $\delta/c \propto Re^{-1/2}$ as the layers thicken, and faster once the bubble grows. Measured on the computed structured-L2 solutions (repro: `repro/cfd/analyze_handover_length.py`), the half-saturation length is 0.025–0.05 c at 2–4.6$\times10^5$, where the bubble closes 0.04–0.15 c past the crossing; at 10^5 half-saturation alone costs 0.05–0.10 c, full production arrives only 0.18–0.30 c downstream, and the bubble never closes; at 6×10^4 full production is never reached on the airfoil. The e^9 reference—whose closure switches to fully turbulent at the transition point—reattaches within 0.05 c of its transition station at 10^5, on the measured reattachment, and spot-driven breakdown in bubble DNS is similarly abrupt [47, 48]. It is therefore not the onset placement but the length of the modeled transition zone—the deliberately unphysical σ_t interpolation—that starves the bubble of eddy viscosity, makes the model's unique state the burst one, and pushes the bursting boundary one Reynolds step above the measurement while erasing the two-state structure the experiment shows at 6×10^4. Narrowing the interpolation is not the remedy: at $\tau = 1$ (a four-times-narrower ramp) the handover becomes a relaxation oscillator—production closes the bubble, closure destroys the amplification fetch that fired the gate, and the bubble reopens—with $\Delta C_l \approx 0.3$ at 10^5 and ≈ 0.06 even at the otherwise steady 2×10^5 benchmark (cases `tautest_strL2_Re{100,200}k_a5_tau1`, repeated at the recomputed canon, where the oscillation persists). The gate is memoryless where breakdown is irreversible. The identified remedy, left to future work, is a transition-zone closure with physical content: an outer-scaled breakdown trigger (a Reynolds-stress proxy such as $\tilde{\nu}\omega/U^2$ reaching a mixing-layer saturation level, in place of the molecular-units χ ladder) combined with the transported memory of breakdown that spot-growth and intermittency models carry.

The starved handover is one of two low-Reynolds defects, and the second runs in the opposite direction: below the bursting boundary the amplification the model books inside the bubble outpaces the e^N envelope. The mechanism is not the rate—the kernel's production in the reversed "dead air" is onset-gated to effective zero—but the topology of the transport: in the elliptic solution the bubble is a closed cell, and $\tilde{\nu}$ produced in the shear layer diffuses across it into the reversed stream, convects back upstream, and re-enters the amplifying band already elevated. Measured on the structured-L2 sweep (`repro/cfd/measure_lsb_reseeding.py`), the median χ in the reversed region over the bubble's first third sits 21–27× above the incoming-envelope level just upstream of separation at $Re \leq 10^5$, but 27× below it already at 2×10^5 and 70–170× below at 3–4.6×10^5; the flip is sharp at the bursting boundary because the conducting overlap—an amplifying layer running over un-closed reversed flow for half the chord—is itself created by the starved handover, so the two defects are one causal chain. The signature is readable directly in the χ sheets of Appendix C (Figs. C1 and C4): at low Re the χ front is bottom-led—a cold front, the reversed flow carrying χ upstream along the wall—while at high Re it is top-led, a warm front riding the outer layer. No profile-based calibration can see this defect: the frozen-profile instrument of Sec. II.C marches one-way by construction, and prices the band's slow convection but not the spiral ledger path of a closed cell.

The two defects meet, at the lowest Reynolds numbers, in a fortunate cancellation, and the honest reading of this section is worth stating plainly. At 2–4.6×10^5 both defects are small and the model is genuine to the physics: the booked amplification tracks the e^N envelope through bubbles and attached recoveries alike, and the handover is short on the bubble's scale. At $Re \lesssim 10^5$ the model still places separation and the transition front, and lands within 11% (family-dependently) on lift—but by compensation: the re-seeded climb fires the $\chi = 1$ front $\sim 0.16\,c$ ahead of the physical envelope's equivalent station (measured against the mfoil envelope at $N = \ln(1/\chi_\infty)$: 0.41 vs 0.57 c at 10^5, 0.44 vs 0.60 c at 6×10^4), a head start that the over-long handover then spends—and, below the boundary, overspends—compounded by standard SA's own low-Re anemia: in a constant-stress layer the equilibrium $\tilde{\nu}$ supports only $\chi \approx 0.09\,Re_\tau$ (the friction-velocity Reynolds number; `repro/analytic/sa_sustain.py`), so at the 6×10^4 bubble shear layer's $Re_\theta \approx 220$–330 the reattached turbulence would equilibrate near $\chi \sim c_{v1}$ with f_{v1} only half active. Low-Re solutions in this range should be read as the sum of two canceling errors, not as resolved transition physics.

VII. The cylinder drag crisis

The oldest widely taught manifestation of boundary-layer transition—the drag crisis of smooth cylinders and spheres [49, 50]—remains outside the validation canon of RANS transition modeling. Its computational replications are recent and almost exclusively scale-resolving [51, 52], reproducing the crisis with the asymmetric single-bubble states and force jumps observed through the critical range [53, 54]; the founding detached-eddy study selected the laminar- or turbulent-separation branch by hand [55], and the few correlation-based RANS attempts report strong sensitivity exactly there [56], where separation-induced transition enters those models through a reattachment correction calibrated on turbine cascades and the regime itself is bistable [54]. The present model brings no separated-flow correction to calibrate: amplification is generated geometrically by the inflectional coordinate—the mechanism the Eppler campaign exercised at these same Reynolds numbers—so the cylinder is a parameter-free traverse of the crisis, run here entirely on the steady branch and extended to the full span of the classical composite curve: $Re_D = 1$–2×10^7, 7.3 decades, one closure and one set of constants. The measured composite is itself a stitch of many facilities [49, 54, 57–59], no one of which spans more than about two and a half decades; on the computational side, wall-resolved LES reaches 8.5×10^5 [51, 52], wall-modeled LES lost accuracy by 2×10^6 [60], and correlation-based RANS studies report isolated points near the crisis [56, 61]. To our knowledge no single model of any kind has produced this curve across these regimes before, and the concessions that price the present one are stated below.

The traverse is a single freestream seed, $\chi_\infty = 1.1\times10^{-2}$ (Mack's map, Sec. III, at $Tu = 0.2\%$), swept as one upward continuation ladder warm-started from the neighboring lower Reynolds number, on a matched grid of twenty-five diameter Reynolds numbers spanning $Re_D = 1$–10^{10} (ten decades, roughly three points per decade), with the solver settings, constants, and branch-selection protocol used everywhere in this paper (Sec. VIII). Four grid families of the same analytic O-grid construction cover the span: a large-domain laminar grid (600×121, first spacing $10^{-3}\,D$, outer boundary $1000\,D$) for $Re_D \le 10^3$, the working grid (1200×148, first spacing $8\times10^{-6}\,D$, outer boundary $100\,D$; $y^+ \le 0.64$ at 2×10^6) for 3×10^3–1.78×10^6, a fine supercritical grid (1600×170, first spacing $10^{-6}\,D$) for 3×10^6–10^7, and an ultra grid (1600×170, first spacing $10^{-9}\,D$; measured $y^+ \le 0.47$ at 10^{10}) for 3×10^7–10^{10}—twenty-five steady cases on one closure. A case is accepted when C_d drifts by less than 10^{-3} over 1000 pseudo-steps; otherwise it runs to a fixed cap and is flagged, and five of the twenty-five—concentrated in and just above the crisis band, with two grid caps—carry that flag. Below $Re_D \approx 7\times10^5$ every case lands on the symmetric branch, $|C_L| < 2\times10^{-6}$; the branch multiplicity that appears above it is the sixth concession below.

Figure 18 shows the single-seed up-ladder traverse. The crisis falls between $Re_D = 3\times10^5$ ($C_d = 0.773$) and 1.78×10^6 (0.228), the steepest drop spanning 5.62×10^5–10^6; the supercritical floor then settles near $C_d = 0.21$–0.23 from 1.78×10^6 to 10^7. The post-crisis wall sequence is a genuine laminar-separation bubble (Fig. 20): at $Re_D = 1.78\times10^6$ the first wall separation sits at 98° with turbulent separation following near 117°, and by the supercritical grid the turbulent separation settles at 118–121°. Through the crisis the shoulder suction deepens from $C_p = -0.90$ to -2.47 and the base pressure recovers from -0.62 to -0.25. Figure 19 collects the transition and separation angles across the whole traverse, with the radial-ray χ profiles that define the transition angle. Figure 20 shows the fields at three stations of the traverse: the separated shear layer transitioning at separation over the subcritical dead-air wake, then transition reaching the separation point, the wall turning turbulent through a shrinking bubble, and the wake narrowing.

The two ends of the span read differently from the middle. Below the shedding onset ($Re_D \approx 46$ [64]) the steady solution is the physical flow and the traverse is a laminar validation with the model provably passive: $C_d = 10.67$ at $Re_D = 1$ and 2.797 at 10, −1.7% against the steady benchmark of Dennis & Chang [68] (+6.4% at 100, where the lengthening steady eddy outruns the grid's radial resolution), with near-wall $\chi_{\max} \le 10^{-4}$ through $Re_D = 30$ and $\le 2.5\times10^{-2}$ through 10^3—no switch tells the model to stand down there. Doubling the far-field radius from 300 to $1000\,D$ moves C_d by 0.54% at $Re_D = 1$. Between the onset and roughly 10^3 the solver converges to the steady symmetric branch that nature abandons for shedding—and not to the exact decaying branch of that family either, whose recirculation length outgrows any practical domain [69]—so those points are drawn for continuity, not validation. The

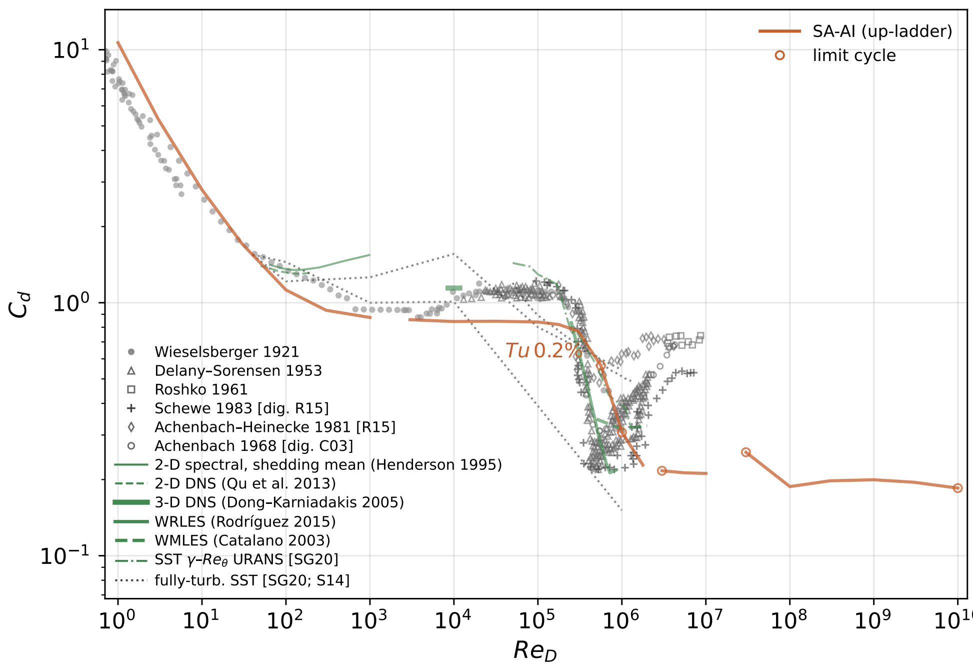

Fig. 18 Steady-branch drag coefficient of the circular cylinder, $Re_D = 1$–10^{10}: the single freestream seed ($Tu = 0.2\%$, χ_∞ from Mack's map), swept as an upward continuation ladder in Re_D over four contiguous grid families, overlaid on the experimental and computational literature. Open rings mark the cases the steady monitor did not accept (peak-to-peak in Table F8). Below $Re_D \approx 46$ the steady solution is the physical flow; between there and $\sim 10^3$ the steady-branch points are drawn for continuity, not validation. Symbols are experiments; lines are computations. Gray and black symbols, digitized independently from the sources: Wieselsberger's nine-diameter composite [57] (faired low-Re_D curve rendered as sparse symbols along its polyline), Delany & Sorensen [58], Roshko's pressurized-tunnel points (splitter-plate runs excluded) [59], Schewe [54] and Achenbach & Heinecke [62] as replotted by Rodríguez et al. [51], the Achenbach drag curve [49] as reproduced by Catalano et al. [60] (likewise sparse symbols), and the low-Re_D laminar drag collected by Veysey & Goldenfeld [63]. Green lines are transition-resolving computations. In the shedding band the unsteady two-dimensional shedding-mean drag—Henderson's spectral-element curve [64] (solid) and the DNS sweep of Qu et al. [65] (dashed)—recovers the measured level exactly where the steady branch reported here sits below it by construction. At the crisis: wall-resolved LES [51] (solid), wall-modeled LES [60] (dashed), and the SST γ–Re_θ transition URANS sweep of Stabnikov & Garbaruk [61] (dash-dotted); the short green stub at $Re_D = 10^4$ marks the finest-grid value of the three-dimensional spectral DNS of Dong & Karniadakis [66] (its $N_z \geq 64$ resolution study brackets $C_d = 1.110$–1.143). Charcoal dotted lines are fully-turbulent SST URANS [61, 67]. The steady solutions lie below the shedding-mean experimental band at subcritical Re_D by construction; the comparison targets the crisis location and the supercritical level. Every SA-AI point is tabulated in Table F8; digitization provenance in `repro/cfd/litdata/dragcrisis/`.

ray transition angle first registers in the far wake near $Re_D = 10^2$; the near-wall $\chi = 1$ front reaches the shoulder only near 10^5, at the approach to the crisis, so the subcritical plateau carries essentially no wall amplification.

Six things are conceded, all visible in the figures. First, through the crisis band the steady branch is weakly unsteady: five of the twenty-five up-ladder cases fail the steady acceptance monitor (tail peak-to-peak median 0.044, maximum 0.171 at 3×10^7 in C_d—the open rings of Fig. 18), concentrated in and just above the crisis band and at two grid caps; no hysteresis claim is made, and the measured folds and lift jumps [54] belong to unsteady continuation, not to this

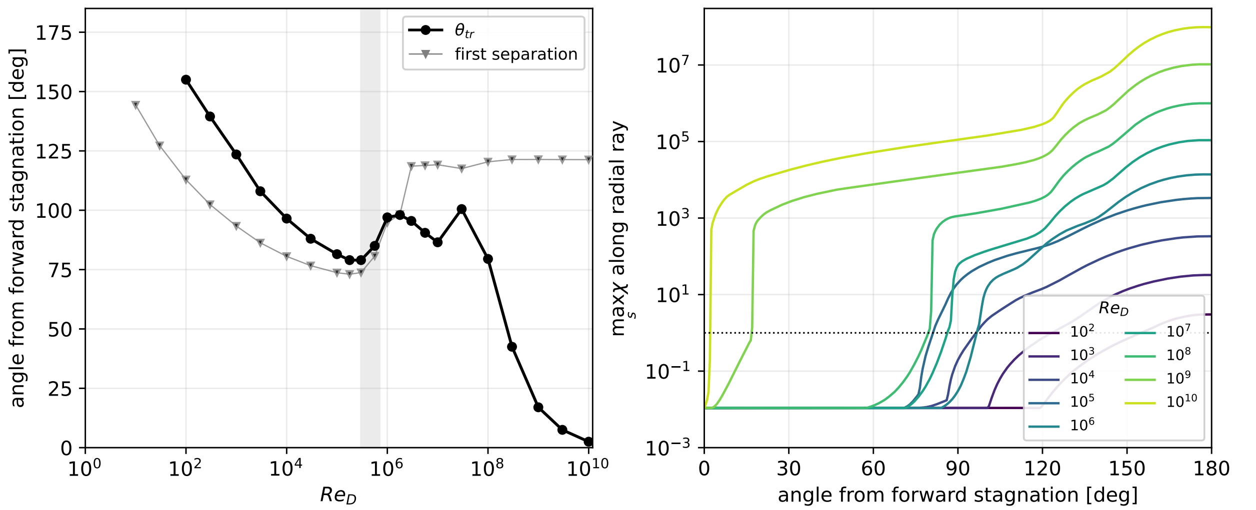

Fig. 19 Transition and separation angles across the up-ladder traverse ($Tu = 0.2\%$). Left: transition angle θ_{tr}—the smallest angle at which the maximum of χ along an outward radial ray from the wall reaches 1—and the first wall separation angle (tangential-C_f sign change) against Re_D, authoritative grid family at each Re_D. Below the crisis θ_{tr} saturates a few degrees aft of the laminar separation angle: the separated shear layer transitions essentially at separation. Through the crisis θ_{tr} retreats to its maximum as transition locks onto the separation bubble, then marches forward monotonically through the transcritical range, reaching the nose (2.5°) by 10^{10}. The radial-ray convention agrees with the near-wall $\chi = 1$ front to ±1.5° above 1.5×10^5 and, below, catches the lifted-shear-layer transition that the near-wall convention reports late or not at all. Right: $\max_s \chi$ along the ray versus angle at one Re_D per decade (10^2–10^{10}; $\chi = 1$ dotted).

steady solver. Second, the symmetric steady protocol cannot see the one-bubble asymmetric states of the critical range [53, 54]: $|C_L| < 2\times10^{-6}$ across all twenty-five cases (below 6×10^{-8} on the converged plateau) is a statement about the protocol, not the physics. Third, the subcritical plateau, $C_d \approx 0.84$, sits well below measurement, as any steady symmetric (shedding-free) wake solution must; the boundary-layer-side quantities are the product there. Fourth, the supercritical turbulent separation angles (117–121°) remain early against measurement—the familiar class of turbulent-C_f-on-curved-wall error, compounded by the steady dead-air base. Fifth, there is no transcritical recovery: measurement rebuilds toward $C_d \approx 0.7$ [54, 59] while the computed floor sits flat between 0.19 and 0.23 all the way to 10^{10}. That this is not the transition content is now settled two ways. The $\chi = 1$ front marches monotonically to the nose—98° at 1.78×10^6, 86.5° at 10^7, 2.5° at 10^{10}—and yet the drag does not move; and a fully-turbulent control (the same solver with the amplification kernel disabled, freestream $\chi_\infty = 3$) separates earlier (105–112°) and drags higher (0.25–0.37) than the transitional model on matched branches—so SA-AI is not merely reproducing the turbulent baseline—yet even that control sits about a factor of two below the measured transcritical level and separates at 105–123° against the observed ≈147°. The missing rise is therefore the baseline turbulent-C_f-on-curved-wall separation of concession four plus the steady symmetric dead-air wake, not the transition closure, whose front is already e^N-consistent and, marched to the nose at 10^{10}, leaves C_d unchanged. Sixth, the supercritical regime carries a second, lower-drag steady state that the up-ladder selects against: a downward continuation ladder sits 33–52% below the up-ladder from

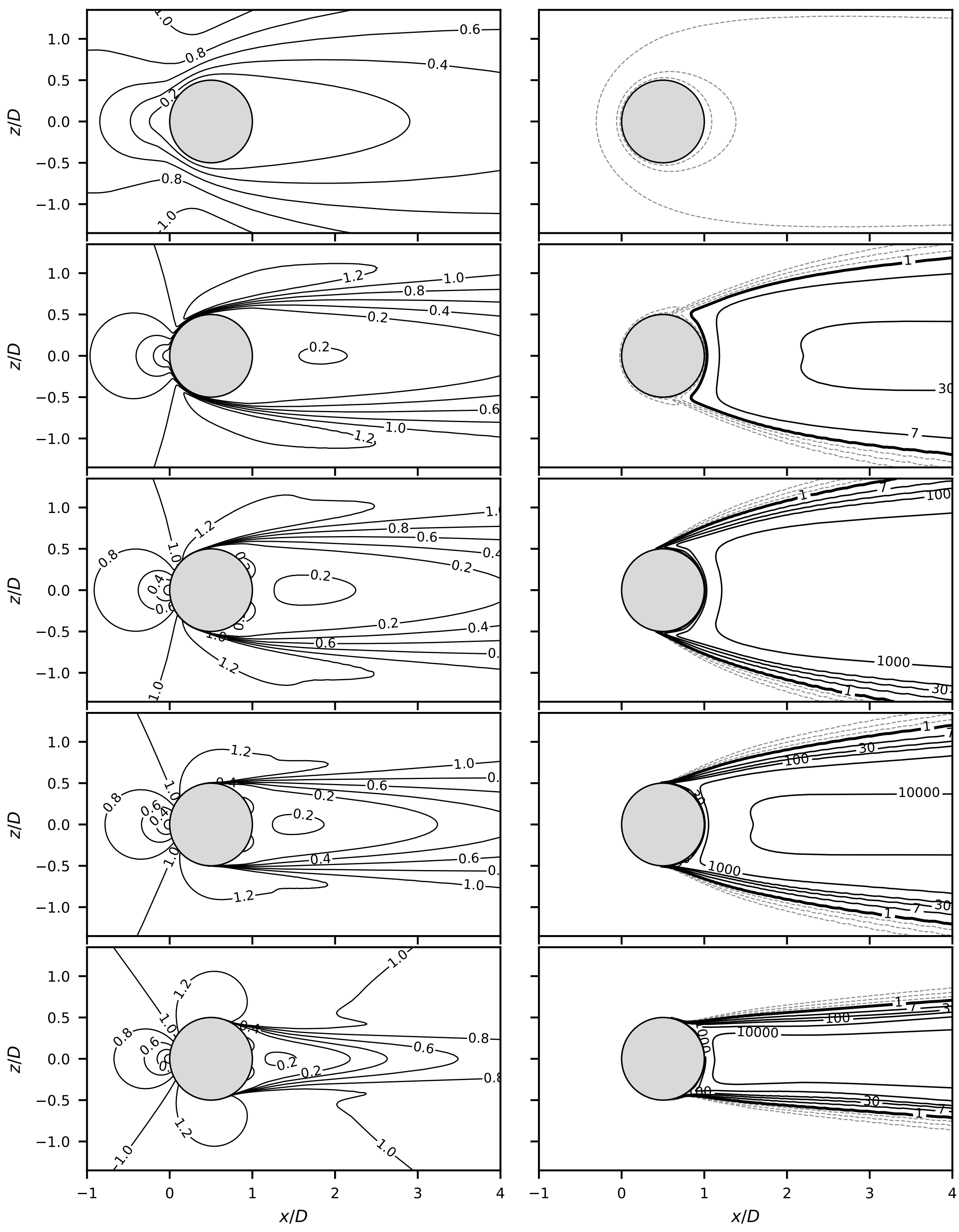

Fig. 20 Velocity magnitude (left) and χ (right; dashed $\chi < 1$, solid $\chi = 1$, c_{v1}, 30, 10^2–10^4) at center span, $Tu = 0.2\%$ up-ladder; rows $Re_D = 10$, 10^3, 10^5, $5{\times}10^5$, $2{\times}10^6$. Top, $Re_D = 10$: steady attached flow below the shedding onset, χ at the freestream seed throughout, $C_d = 2.80$. $Re_D = 10^3$: a broad laminar wake with the detached shear layers amplifying downstream, $C_d = 0.87$ (near the subcritical plateau). Subcritical 10^5: the laminar wall separates at 74° and the lifted shear layer amplifies immediately—the free-shear pole of the kernel—over a dead-air wake, $C_d = 0.84$. Mid-crisis $5{\times}10^5$: transition has reached the separation point and the wake narrows ($C_d = 0.62$; final snapshot of a limit-cycling case, band in Fig. 18). Bottom, supercritical $2{\times}10^6$: a 1.5° laminar-separation bubble at 98–100° hands the wall to turbulence, final separation 118°, $C_d = 0.22$.

10^8 onward ($C_d \approx 0.10$ vs 0.19), and at 10^{10} relaxes to $C_d = 0.108$ against the up-ladder's 0.185 —a spread wider than the crisis band, so the ultra-Re asymptote is a protocol-dependent band (0.11–0.19), not a level, and its adjudication belongs to unsteady continuation. The traverse plotted here is the up-ladder throughout.

The fifth concession invites the question of whether the transition front itself should be further forward, and two frozen-profile audits answer it. On Falkner–Skan wedges the model's envelope slope tracks the Drela–Giles fit through $\beta \leq 0.1$, halves at $\beta = 0.2$, and collapses three orders of magnitude by $\beta = 0.5$—in strongly favorable gradients the inflectional coordinate has no amplification to book while the e^N envelope retains a floor near $5{\times}10^{-3}$ per unit Re_θ; the onset gate is not the binding element (it opens ahead of the envelope's critical Re_θ through $\beta = 0.2$ and saturates at $Re_\Omega = 1851$ beyond). On the cylinder nose, however, this deficit does not drive the drag: integrating the Drela envelope along a validated laminar march of the measured nose flow puts the $N = \ln(1/\chi_\infty)$ crossing behind the computed front at all three Reynolds numbers audited ($2{\times}10^6$, $7{\times}10^6$, $2{\times}10^7$)—at $2{\times}10^7$ the solver front sits 10° ahead of the e^N station on its own mean flow—and where the ultra-Re arm does march the front onto the nose (2.5° at 10^{10}), C_d is unmoved (concession five). Within envelope-method physics at this seed the front has nowhere useful forward to go; the far-forward transition of the transcritical measurements at accessible Re_D carries disturbance content beyond the envelope class.

VIII. The attachment-anchored turbulent branch, and the convergence protocol it dictates

Every laminar solution in this paper may coexist with a second, spurious fixed point of the discretized equations: a turbulent wedge anchored at the leading-edge attachment point, sustained by standard SA production with no help from the transient or the amplification kernel—at high enough nose Reynolds number with no help from the freestream either (the model problem below), and at this paper's wing-scale conditions by regrowth from the seed, whose e-fold budget the nose over-supplies. This section demonstrates the branch on a model problem, explains why it exists and why it is unphysical, and states the convergence protocol used throughout this paper to exclude it—which a reader implementing the model in any solver will need in some form, because the printed transport equation alone does not determine which fixed point an initialization finds.

The two halves of this bistability have separate histories. The laminar-side fixed point is documented: Rumsey's apparent-transition study [70] showed SA converging to a laminar-behavior state when the freestream level falls below a threshold, with grid-dependent extent and non-unique onset, and the follow-up with Spalart [71] concluded the model is not intended for transitional use—while Crivellini and D'Alessandro [72] turned the same low-seed behavior into a working laminar-separation-bubble predictor, the nearest published ancestor of the present model's premise. The propagation of model-turbulence fronts into quiescent fluid was analyzed by Cazalbou, Spalart, and Bradshaw [73] for two-equation models. What appears to be undocumented—and is supplied here—is the turbulent-side fixed point: that at a stagnation point the standard model sustains an anchored turbulent state at exactly zero freestream seed, above a

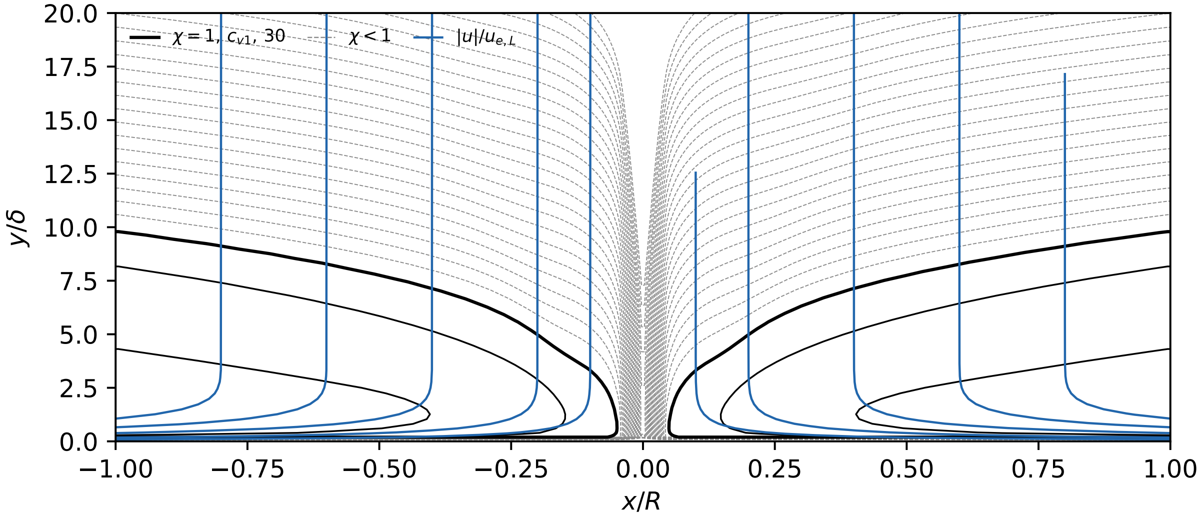

Fig. 21 Standard SA on the frozen Hiemenz field with $\chi_\infty = 0$ exactly. Left: the steady state reached from a saturated initialization—collapse to the laminar solution for $L \lesssim 685$, a self-sustained turbulent wedge above (gray band: the bisected critical interval, $Re_r = 4.66$–4.74×10^5), with the sustained level approaching max $\chi \approx 0.025\,L$ over the upper decade. Right: the sustained wedge at $L = 3000$ ($Re_r = 9 \times 10^6$), max $\chi = 76$; line contours of $\log_{10}\chi$, dashed below $\chi = 1$, solid at 1, c_{v1}, and 30. Nothing enters from the freestream; the wedge root re-seeds itself.

computable critical Reynolds number, in a region whose conditions the same literature studies as relaminarizing [71, 74].

The model problem. Plane stagnation flow $U_e = ks$ is the prescribed pressure distribution of a leading edge of radius r at strain rate $k \approx U_\infty/r$. The velocity field is the exact Hiemenz similarity solution, frozen; only the SA transport equation is solved on it—standard SA (no f_{t2}, the solver's 0.3Ω floor on $\tilde{S}$; none of this paper's amplification terms)—by a plain Cartesian finite-difference solver (`repro/analytic/stagnation_bistability.py`). Nondimensionalized on the stagnation-layer thickness $\delta = \sqrt{\nu/k}$ and time $1/k$, the steady equation for χ is parameter-free except for the domain half-extent: with the nose extent $x_{\max} \approx r$, $L \equiv x_{\max}/\delta = \sqrt{Re_r}$, $Re_r = U_\infty r/\nu$. The freestream boundary value is exactly zero: $\chi = 0$ everywhere is then an exact fixed point at every L (production is multiplicative), and the experiment initializes a saturated layer ($\chi = 50$) and asks whether it decays or survives.

Result. Figure 21: below a critical extent the wedge washes out; above it (Re_r between 4.66 and 4.74×10^5, each near-critical case classified by chained continuation until it either collapses or demonstrably locks) it persists, steady to the solver's tolerance, with max χ growing roughly linearly in L—$\chi \approx 24$ at $Re_r = 10^6$, 76 at 9×10^6—from an exactly zero seed. The mechanics are those of a pulled reaction–diffusion front. The $\tilde{\nu}$ equation is FKPP-like: linear reaction $r = c_{b1}\tilde{S}$ plus diffusion $D = (\nu + \tilde{\nu})/\sigma$, giving a front that propagates into the laminar state at $c^* = 2\sqrt{Dr}$; in an equilibrium turbulent layer ($\tilde{\nu} = \kappa u_\tau y$, $\tilde{S} = u_\tau/\kappa y$) the y-dependence cancels and $c^* = 2\sqrt{c_{b1}/\sigma}\,u_\tau \approx 0.90\,u_\tau$—finite and physically plausible. The trap is a geometric race: on the Hiemenz wall $u_\tau \propto \sqrt{s}$ while the opposing advection grows like s, so close enough to the attachment point any finite front speed wins and the front holds station; the wedge in Fig. 21 anchors at exactly such a standoff. The same race read at wing scale banks $N \sim c_{b1}\sqrt{Re_r}$ e-folds of $\tilde{\nu}$ growth along near-wall streamlines through the nose—far more than the $\ln(1/\chi_\infty) \approx 7$–12 that natural transition needs.

Why it is spurious. Physically a two-dimensional stagnation zone relaminarizes: the acceleration parameter $K = (\nu/U_e^2)\, dU_e/ds = \nu/(ks^2)$ diverges toward the attachment point and exceeds the relaminarization threshold ($K_c \sim 3 \times 10^{-6}$ [74]) throughout a quench zone one to a few leading-edge radii wide at this paper's conditions, and outside that zone a real turbulent front spreading at $O(u_\tau)$ loses to $O(U_\infty)$ advection. The comparison is damning in the model problem's own units: the quench boundary $K = K_c$ sits at $s/\delta = \sqrt{1/K_c} \approx 580$, while the computed wedge of Fig. 21 anchors its near-wall $\chi = 1$ root at $x/\delta \approx 160$—the model parks its wedge exactly where real turbulence is forbidden. (The genuinely bistable configuration is the swept attachment line, where flow along the line sustains contamination—a three-dimensional phenomenon of which the SA trap is a zero-sweep impersonation.) SA's production carries no acceleration information, so the model happily sustains what the physics quenches. Two corroborations at wing scale: restarting this paper's structured-L0 NLF $\alpha = 9°$ case from its converged flow with $\tilde{\nu}$ reset to the seed moves both fronts from the trapped values to the fine-grid locations on the same mesh (Fig. 6); and the independent OpenFOAM implementation of Appendix G, started impulsively, first landed on the trapped branch (transition at $0.05\,c$ with plausible-looking polars) before adopting the staged start below. The diagnostic is direct: under an established wedge the amplification rate of Sec. II is ≈ 0 at the front—a trapped state, not modeled transition.

The protocol. Because the branch belongs to standard SA at high χ, any transition model that leaves SA's fully turbulent behavior unmodified is bistable at high Reynolds number, and every result in this paper was therefore converged with the laminar dynamics deliberately slowed: the $\tilde{\nu}$-equation residual is scaled by $f^{\,1-g}$, where g rises smoothly from 0 (laminar) to 1 (turbulent) with χ and $f = 10^{-2}$ (the solver uses $g = 1 - e^{-(\chi - c_{v1})/4}$ above $\chi = c_{v1}$, zero below; the gate's exact shape is immaterial because the scaling preserves fixed points). The scaling multiplies the residual, so every converged solution is a fixed point of the unmodified equations; it lets the pressure field and the slow laminar amplification equilibrate before the fast turbulent branch can act on the startup transient—a branch selector, not an accelerator. Because the freestream boundary condition flows through the same scaling, the freestream viscosity ratio supplied to the solver is $\chi_\infty f$; the seeds quoted throughout this paper are the physical χ_∞, recovered at the boundary-layer edge (Sec. III). Equivalent protocols exist for solvers without residual scaling: advancing the turbulence equation at a reduced pseudo-time rate (CFL) relative to the mean flow during initial convergence, or—the route the OpenFOAM replication of Appendix G takes—converging the flow, resetting $\tilde{\nu}$ to the seed, and continuing to convergence, at the cost of one extra partial solve. Under any of these the uniqueness statements of Sec. VI are to be read as uniqueness among the states these protocols admit; the trapped branch is excluded by construction, as Fig. 21 says it must be.

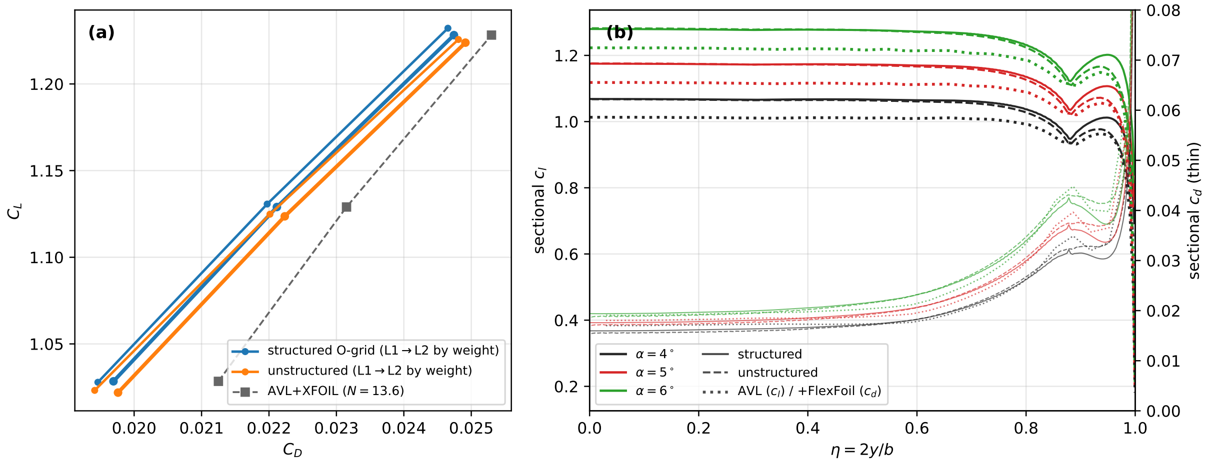

Fig. 22 The macroscopic comparison. (a) Wing polar: both mesh families at L1 and L2 (color = family, line weight = level) against the AVL+XFOIL (vortex-lattice + section polars) reference at matched N_{crit}, trimmed to the RANS lift at each incidence (matched-lift reference). (b) Sectional lift on the finest grid per family and incidence against the AVL distribution (dotted), with sectional drag on the right axis (thin lines; the dotted reference is AVL induced drag, Trefftz-rescaled, plus the FlexFoil section profile drag): the loading agrees inboard to $\sim 1\%$; the dip–recovery structure at the taper break ($\eta = 0.88$) is mesh-family independent.

IX. Three-dimensional deployment: the Daedalus wing

(This section's campaign—both mesh families, all levels—is recomputed at the final kernel of §II.F.)

The model's first three-dimensional application is the wing of the Daedalus human-powered aircraft: half-span 17.07 m, root chord 0.917 m, aspect ratio 37.8, Drela's DAE-11/21/31 low-Reynolds sections [75] blended along the span, constant chord to $\eta = 0.88$ and a straight taper to the tip. The operating point is the aircraft's: $Re_{\mathrm{root}} = 5\times10^5$, $\alpha \in \{4°, 5°, 6°\}$ spanning $C_L \approx 1.02$–1.23 about the design lift, with the flight-quiet freestream seed $\chi_\infty = 8.76\times10^{-6}$ ($N_{\mathrm{crit}} \approx 13.6$ by the map of Sec. III, the flight-environment amplification level of the Daedalus design work [75]). The solver and kernel are exactly those of the airfoil studies: an earlier predecessor-kernel campaign exposed the compressed-bubble deficiency discussed below (which kernel element it lacked, and what that cost, is quantified there), and this section reports the recomputation with the final model. Two mesh families are used: a structured O-grid stacked spanwise and pinched to a line at the tip, and a classical three-dimensional unstructured mesh—anisotropic boundary-layer prisms at the wall, hexahedra in the far field, and tetrahedra joining the two—at the L0–L2 refinement levels of Table 6 (construction in Appendix D); the reference is the same e^N framework used throughout—a vortex-lattice model (AVL [76]) with strip-wise XFOIL section polars at matched local c_l and N_{crit}, and FlexFoil amplification envelopes at section conditions (Sec. VI). Eighteen final-kernel RANS solutions underlie the comparison: both families at L0, L1, and L2 at all three incidences.

The macroscopic ledger comes first (Fig. 22; the per-level coefficients are tabulated in Appendix D, Table D1). The two families agree at L1 to 0.46–0.53% in C_L and 0.4–1.6 counts (0.18–0.65%) in C_D—tighter than the predecessor

Table 6 Daedalus wing mesh ladder: size, spanwise and around-the-section resolution, and resulting y^+. Both families carry discrete spanwise stations—the structured family its stacked O-grid planes, the unstructured family the surface rings of its triangulated skin—uniform over the constant-chord panel and clustered through the taper and tip closeout; the Δy rows are the exact station spacings. The around-the-section rows are measured at the $\eta = 0.30$ section as in Tables 4 and 5; the y^+ percentiles are from the predecessor-kernel campaign's $\alpha = 5°$ solutions on these same meshes (a mesh-resolution property, insensitive to the kernel revision).

	Unstructured (prism/tet/hex)			Structured O-grid		
	L0	L1	L2	L0	L1	L2
Nodes	1.70M	8.27M	54.9M	0.72M	5.05M	36.7M
Elements	3.44M	17.0M	111M	0.70M	4.96M	36.4M
Spanwise stations	81	161	321	42	82	162
Δy @ root [mm]	335	167	84	654	331	167
@ $\eta = 0.98$	62	32	17	112	61	33
@ tip	3.3	0.8	0.2	12.5	3.2	0.81
$\Delta s/c$ @ LE ($\times 10^{-3}$)	7.67	3.85	1.96	4.49	2.28	1.13
@ 0.05c	11.5	5.85	2.92	8.26	4.13	2.06
@ 0.25c	21.3	10.7	5.38	14.3	7.16	3.58
@ 0.98c	6.66	3.18	1.55	3.51	1.76	0.85
@ TE	5.47	2.59	1.29	2.71	1.22	0.61
h_0/c ($\times 10^{-6}$)	39.9	20.0	10.0	39.9	20.0	10.0
cell @ 0.001c ($\times 10^{-6} c$)	208	112	56	212	116	58
cell @ 0.5c ($\times 10^{-2} c$)	3.62	1.74	0.87	2.85	1.52	0.74
y^+ (95th pct)	1.68	0.89	0.44	1.68	0.88	0.44
y^+ (99th pct)	1.88	1.03	0.54	2.00	1.07	0.55

campaign's 0.6–0.8%/0.9–1.8%—and at L2 to 0.35–0.63% in C_L and 0.7–1.7 counts (0.34–0.69%) in C_D (the predecessor's finest-grid pair agreed to 0.6–0.7% in C_L). At L0 the families sit 0.74–0.97% apart in C_L and 18–21 counts (8–9%) apart in C_D (Table D1), the unstructured grid the high-drag member at every incidence; those rows complete the ladder but carry the coarse-grid caveat of Sec. IV—grids this coarse admit path-sensitive front breakaways (Fig. 6)—so no convergence claim rests on them. Against the AVL+XFOIL reference trimmed to the same total lift (the lift-consistent comparison; the vortex lattice needs $\sim 0.5°$ more incidence to carry the RANS C_L), the RANS drag runs 15.6 counts (7.3%) low at $\alpha = 4°$, 10.3 (4.4%) at 5°, and 5.6 (2.2%) at 6° on the finest grid—the deficit shrinking with incidence, and entirely in the induced term's territory, since the reference's profile drag barely moves with the trim. (At matched incidence the same reference sits 10 counts lower still—the lattice carries 5% less lift and hence $\sim 10\%$ less induced drag—which would flatter the RANS into coincidence; the matched-lift numbers are the honest ones.) The drag slope disagrees more than the levels: mid-span sectional drag grows at ≈ 0.018 per unit c_l in the RANS against the reference's 0.011, all of whose growth is induced—the strips' XFOIL profile drag is flat in lift, while the RANS profile drag climbs ≈ 6 counts per 0.1 of c_l. The sectional diagnostic sheets of Appendix D (Figs. D3–D5) locate the mechanism: at fixed separation and reattachment—both match the strips at every incidence—the model's handover-stretched pressure recovery inside the bubble deepens with incidence, the $-C_p$ plateau persisting further and recovering more gradually than the strip solution's abrupt-closure recovery, and the area between the two curves growing with α: bubble pressure drag that an instant-transition closure structurally cannot produce, the same mechanism as Sec. VI at the sections' $Re_c = 5 \times 10^5$. (The $\chi = 1$ lead over $N = N_{\mathrm{crit}}$ holds its $\sim 0.09c$ convention offset at every incidence and contributes no slope.) An accounting note for reading sectional comparisons (`repro/cfd/daedalus_strip_accounting.py`): the RANS strip curves reproduce their own force totals to a count, but the reference's plotted sectional curve integrates a nearly incidence-independent 18–21 counts below its own ledger total—its induced shape is Trefftz-exact only in the integral, and its profile term lives on nine FlexFoil stations that do not tile the span—so sectional slopes are trustworthy where sectional level differences are not. At matched incidence the RANS lift runs $\sim 5\%$ above the lattice's—the excess a vortex lattice built on camber lines alone would be expected to miss. A single-station decomposition makes it quantitative: at $\eta = 0.31$, $\alpha = 4°$ ($\alpha_{\mathrm{eff}} \approx 3.5°$), the camber-only AVL strip carries $c_l = 1.011$, a thick-section inviscid panel solution of the same DAE-11 gives 1.119 (+11%: thickness), viscous FlexFoil at $N_{\mathrm{crit}} = 13.6$ gives 1.053 (−6%: decambering), and the RANS strip reads 1.066—the wing's 5% excess over the reference is the thin-airfoil truncation of the vortex lattice, not a RANS surplus. Domain confinement does not explain the lift excess: the two families' far fields differ by a factor of ~ 20—the O-grid's boundary is a cylinder of radius 100 root chords, 2.7 spans for this wing, against the unstructured octree's ±2 km (60 spans)—yet their lifts agree to 0.7%. A classical wall-interference estimate confirms the inference with room to spare: treating the O-grid's outer boundary as a closed test section of cross-section $C \approx 4 \times 10^4$ m^2, the upwash correction $\Delta\alpha = \delta\,(S/C)\,C_L$ with $\delta \approx 0.12$ gives $\Delta C_L / C_L \sim 0.06\%$ at these conditions—an order of magnitude below the families' 0.7% gap, which is therefore ordinary discretization difference, not confinement. (Against the matched-lift reference the predecessor-kernel campaign sat 15–26 counts below; like-for-like on each case, the final kernel's longer bubble—see below—returned 2–11 counts of bubble drag, leaving the 5.6–15.6-count residual above.) Sectional lift on the finest grids with sectional output available (Fig. 22b) matches the vortex-lattice distribution inboard to $\sim 1\%$ and reproduces, identically on both families, the dip–recovery structure at the taper break that the vortex lattice smooths over.

The flow itself is read from the surface maps of Fig. 23 and its $\alpha = 5°, 6°$ companions (Figs. D1 and D2, Appendix D): the finest RANS solutions against the e^N strips, column by column. The $-C_p$ maps agree on the loading and carry the bubble's pressure plateau at mid-chord. The $C_{f,x}$ maps carry the bubble itself: a spanwise-uniform negative band whose extent now matches the strips' station by station—at $\eta = 0.31$, $\alpha = 4°$ the RANS bubble spans 0.49–0.65 c against the strips' 0.47–0.66 c, at 5° 0.47–0.64 c against 0.47–0.65 c, and at 6° 0.46–0.62 c against 0.46–0.63 c: separation agrees to 0.017 c and reattachment to 0.009 c at every incidence. The reattachment is also the quantity still moving under refinement—0.630 at L1 to 0.653 at L2, approaching but not yet reaching the strips' 0.660—while separation and the χ field are level-converged. The two families agree at L1 to 0.03 c on both edges (the recomputation shortened the unstructured bubbles: gaps 0.023–0.032 c, up from the predecessor's 0.02 c); at L2 they tighten onto each other,

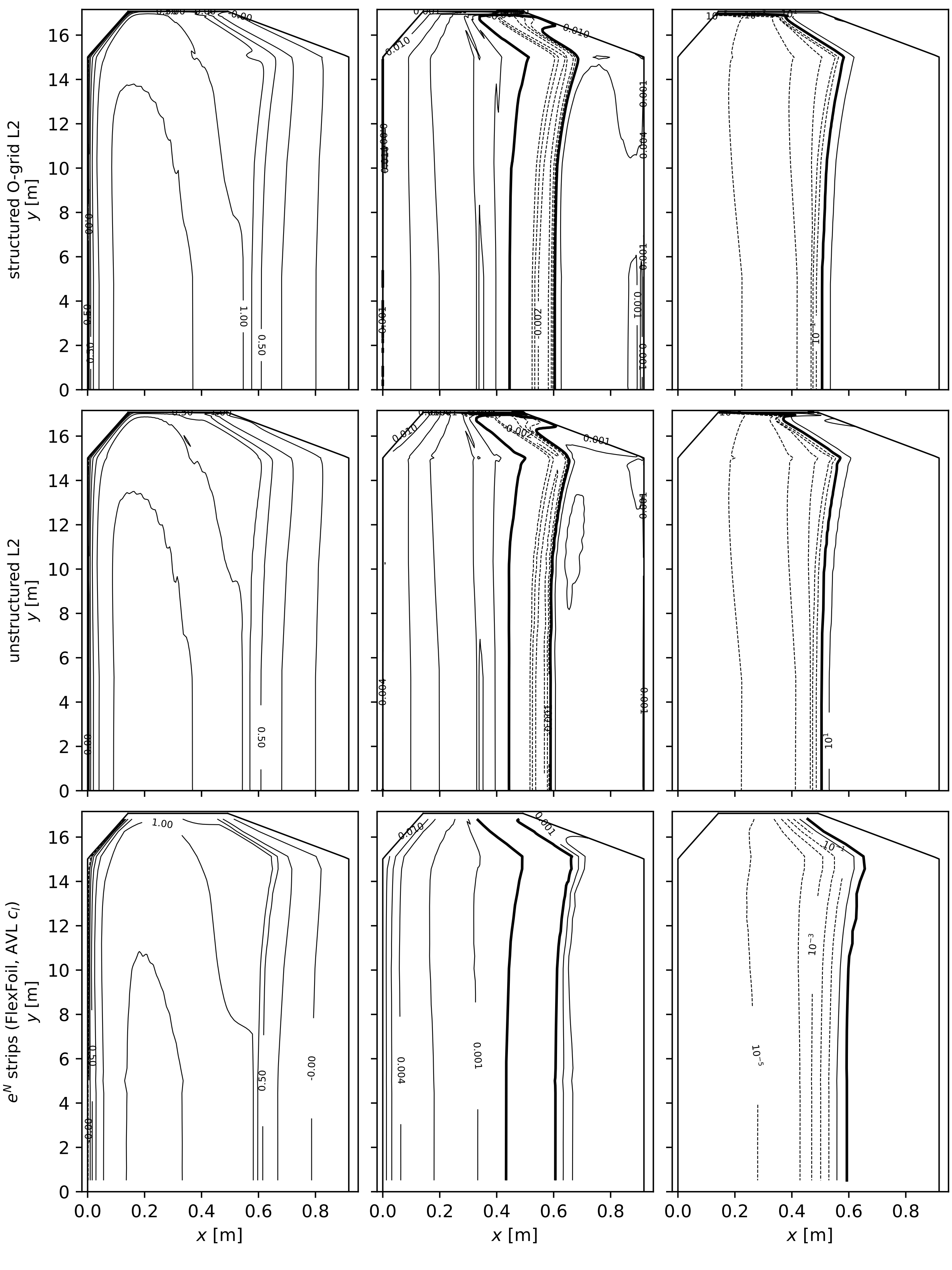

Fig. 23 Daedalus wing upper surface at $\alpha = 4°$, finest grids, chord horizontal and span vertical (contour lines): $-C_p$ (left), streamwise skin friction $C_{f,x}$ (middle; the laminar separation bubble is bounded by the bold $C_{f,x} = 0$ contour), and the near-wall amplification $\max_n \chi$ in decade contours (right; bold: the model's transition front $\chi = 1$). Rows: structured O-grid L2, unstructured L2, and the section-by-section e^N reference—FlexFoil at each station's AVL c_l, local chord Reynolds number, and $N_{\text{crit}} = 13.6$, its C_p from the Karman–Tsien-corrected edge velocity and its amplification drawn as $\chi_\infty e^N$ at the same decade levels (bold: $N = N_{\text{crit}}$).

separation agreeing to $0.008\,c$ and reattachment to $0.016\,c$ at every incidence. The χ maps show the interior anatomy: the bubble's extent matches the strips' station by station, but the model's transition front ($\chi=1$, bold in the figure) sits near mid-bubble, $\sim 0.09\,c$ ahead of the strips' N_{crit} contour, which sits just ahead of reattachment. At this chord Reynolds number the early front is the seed convention, not the low-Re defect pair of Sec. VI (whose re-seeding measurement is quiet at these Re): $\chi=1$ sits $\ln(1/\chi_\infty)=11.7$ e-folds up the same envelope against the strips' $N_{\text{crit}}=13.6$, and the in-bubble envelope climbs ~ 2 e-folds per $0.09\,c$. What the wing does share with Sec. VI is the commit half: the model spends the remainder of the bubble bringing the eddy viscosity to closing strength where the e^N reference commits at once—both halves short enough here that the sum, separation to reattachment, lands on the reference.

This bubble length is the final kernel's correction at work: the predecessor-kernel campaign on the same grids returned the same fronts—its $\chi=1$ contour sat the same $\sim 0.1\,c$ ahead of the strips' N_{crit} contour—but closed the bubble three to four times sooner (0.04–0.06 c). That kernel applied the free-shear ceiling from the separation point on, while a just-separated layer is still wall-damped—its instability approaches the free-shear rate only as it lifts several θ off the surface—and, as the frozen-profile ledger of Appendix D measures in temporal units, its transported $\tilde{\nu}$ peak rode below the physical critical layer at roughly half its convection speed, banking nearly twice the e-folds per unit length (an excess intrinsic to a temporal rate advected by the local velocity, §II.A). The in-bubble arm of the final onset threshold (the $B/\langle\hat{\Omega}\hat{I}\rangle_+^2$ branch of Eq. (13), which suppresses the rate while the separated layer is still wall-damped) was built to close exactly this blind spot, and the recomputation confirms it does: the wing bubble lengthened from 0.04–0.06 c to the strips' $\sim 0.2\,c$, and the 6–14 counts of bubble drag that the predecessor was missing returned to the polar.

Two spanwise phenomena, invisible to any sectional diagnostic, close the study; both are read directly off the surface maps (Fig. 23 and the Appendix D companions). Outboard of the taper break the falling chord Reynolds number stretches the bubble toward the trailing edge—at $\alpha=4^\circ$ its reattachment retreats from $0.66\,c$ at $\eta=0.31$ to $0.79\,c$ at $\eta=0.92$ and $0.95\,c$ at $\eta=0.97$, mirroring the strips' $0.66\to0.74\to0.84$, the $C_{f,x}$ bands bending aft in unison. And over the last half-percent of span the tip vortex trips the boundary layer, pulling transition to the leading edge in a compact patch at the tip (χ exceeds unity from the leading edge for $\eta\gtrsim0.992$; the bright sliver at the top of the RANS χ maps): a genuinely three-dimensional transition mechanism, captured by the transported amplification with no special treatment—and absent, necessarily, from the sectional e^N row.

X. Toward crossflow: the 6:1 prolate spheroid

The inclined 6:1 prolate spheroid is the canonical fully three-dimensional transition benchmark: the DFVLR campaign of Kreplin, Vollmers, and Meier [77] mapped the wall shear stress (magnitude and direction) over the whole body with surface hot films, and the transition-modeling literature validates against it in a standard presentation—the surface unrolled into the $(x/L,\phi)$ plane ($\phi=0$ the windward symmetry line), with skin-friction magnitude, wall-shear direction, and the transition front drawn as maps. The definitive stability analysis of the case is Stock's [78]: transition

Table 7 6:1 prolate spheroid totals (half-model, reference area = frontal): SA-AI at every DFVLR-measured condition on the L0–L2 O-grid ladder, every entry the median of the last fifth of a common fixed cold-start budget (20 000 iterations).

Re_L	α	L0		L1		L2	
		C_L	C_D	C_L	C_D	C_L	C_D
1.5×10^6	5°	0.0946	0.05412	0.0874	0.05381	0.0816	0.05736
1.5×10^6	10°	0.2054	0.08172	0.2081	0.08526	0.1861	0.08175
1.5×10^6	29.7°	1.1329	0.59527	0.9256	0.52169	1.2087	0.66027
7.2×10^6	0°	0.0062	0.01496	0.0000	0.01689	0.0000	0.02113
7.2×10^6	2.5°	0.0304	0.01892	0.0302	0.02386	0.0289	0.02714
6.5×10^6	5°	0.0629	0.03187	0.0533	0.03271	0.0463	0.03498
6.5×10^6	10°	0.0860	0.04136	0.1240	0.04941	0.1094	0.05145

is pure-TS at low incidence, mixed TS–crossflow at $\alpha \approx 5$–15° and $Re_L \gtrsim 6.5\times10^6$, pure-crossflow beyond—and at $Re_L = 1.5\times10^6$ the boundary layer stays laminar up to the open free-vortex-layer separation, with the measured fronts tracking the separation line itself. The model carries no crossflow closure—the amplification rate is calibrated on two-dimensional TS growth—so this body probes what that economy costs, regime by regime.

The geometry is the analytic spheroid ($x^2/A^2 + r^2/B^2 = 1$, $A = L/2$, $B = L/12$, half-model) on a structured O-grid ladder (L0–L2: 0.37/2.2/12.6 million nodes, first wall spacing halving per level from $5\times10^{-6}L$ at $Re_L = 1.5\times10^6$ and from $1.5\times10^{-6}L$ at the higher Reynolds numbers), at $M = 0.1$, with the airfoil-study solver settings, seed, and constants unchanged. The matrix covers the measured conditions spanning Stock's three regimes, and this section walks them by Reynolds number, low to high, and by incidence within each. First the low-Reynolds ladder at 1.5×10^6 ($\alpha = 5°$, 10° and 29.7°), where the layer runs laminar to the open free-vortex-layer separation at every measured incidence, so that the question is whether the model correctly withholds a front rather than where it puts one. Then the high-Reynolds cases, where transition is real, walked from the case the model was calibrated for to the case it was not: the pure-TS anchors at $\alpha = 0$ and 2.5° ($Re_L = 7.2\times10^6$, with a zero-incidence Reynolds-trend point at 6.5×10^6), then the mixed TS–crossflow pair at 6.5×10^6 and 5° and 10°. The DFVLR record's remaining incidences (15–29.5° at 3–8.5×10^6) stay outside the present matrix. Table 7 collects the totals (no force measurements exist at these conditions—the hot films and pressure taps are the data—so the totals serve the grid story only): the attached-flow incidences move by several percent per level—up to 11% in C_L at $\alpha = 10°$ on the finest step, not yet grid-converged—the massively separated 29.7° case remains strongly level-sensitive, and the axisymmetric case carries exactly zero lift on L1 and L2 (the L0 residual, $C_L \approx 0.006$, is the coarsest grid's asymmetry, vanishing under refinement) with its drag still rising with refinement (0.0150/0.0169/0.0211).

Table 8 6:1 prolate spheroid at the tunnel conditions: computed near-wall $\chi = 1$ front minus the measured front, at every azimuth carrying a measurement, on the L1 O-grid at the measured Mach number. Each condition is run at the facility-calibrated seed (the tunnel-calibrated limiting N_{TS}) and at the facility-measured hot-wire level. Δ lee / Δ wind are the residuals at the most leeward and most windward measured azimuth ($\phi = 0$ windward). The n column is the number of measured azimuths at which the computed front was found, over the number measured: where they differ the near-wall χ never reaches 1 within $x/L \leq 0.97$, i.e. the computed layer is still laminar at the tail and the residual there is a bound, not a value. The last two columns are the same residual for Stocks own two-N-factor e^N computation, digitized from the printed panels at the same azimuths; at $Re_L = 1.52\times10^6$ his computed transition line is the laminar separation line. Rows are grouped by Reynolds number: the low-Re band first, where the boundary layer runs laminar to the open free-vortex-layer separation at every measured incidence, then the high-Re band where transition is real; within each band the walk is by incidence.

α	Re_L [10^6]	tunnel	seed	Tu [%]	n	SA-AI − measured mean	rms	lee	wind	Stock − measured mean	rms
5°	1.52	Göttingen	measured	0.330	7/8	-0.020	0.068	+0.074	-0.037	+0.020	0.058
5°	1.52	Göttingen	calibrated	0.106	4/8	-0.006	0.028	+0.026	-0.040	-0.064	0.064
10°	1.52	Göttingen	measured	0.330	8/8	-0.013	0.027	+0.002	-0.041	-0.091	0.100
10°	1.52	Göttingen	calibrated	0.106	8/8	+0.019	0.082	+0.160	-0.053	-0.091	0.100
29.7°	1.53	Göttingen	measured	0.330	12/12	-0.048	0.098	+0.185	-0.045	–	–
29.7°	1.53	Göttingen	calibrated	0.106	12/12	+0.041	0.095	+0.264	-0.024	–	–
0°	7.2	Göttingen	measured	0.330	17/17	-0.097	0.097	-0.097	-0.098	-0.013	0.013
0°	7.2	Göttingen	calibrated	0.106	17/17	+0.077	0.077	+0.078	+0.076	-0.013	0.013
2.5°	7.2	Göttingen	measured	0.330	3/3	-0.093	0.102	-0.150	-0.049	+0.006	0.021
2.5°	7.2	Göttingen	calibrated	0.106	3/3	+0.082	0.096	+0.016	+0.132	+0.006	0.021
5°	6.49	Göttingen	measured	0.330	4/4	-0.098	0.177	-0.292	+0.007	-0.024	0.051
5°	6.49	Göttingen	calibrated	0.106	4/4	+0.107	0.201	-0.088	+0.246	-0.024	0.051
10°	6.56	Göttingen	measured	0.330	5/5	+0.365	0.440	-0.105	+0.375	+0.073	0.098
10°	6.56	Göttingen	calibrated	0.106	5/5	+0.421	0.463	+0.068	+0.387	+0.073	0.098
10°	6.56	ONERA F1	calibrated	0.161	5/5	+0.399	0.475	+0.002	+0.549	+0.088	0.134
10°	6.56	ONERA F1	measured	0.100	5/5	+0.420	0.486	+0.051	+0.549	+0.088	0.134

A. Low Reynolds number: laminar to the open separation

At $Re_L = 1.5\times10^6$, the lowest of the DFVLR matrix, the spheroid does not transition on the attached body at any measured incidence. Stock's stability analysis puts the layer laminar up to the open free-vortex-layer separation at $\alpha = 5°$—except on a single streamline, where the circumferential pressure gradient thickens the layer enough for TS to commit—at $\alpha = 10°$, and again at 29.7°, and at 10° he notes the measured points show the same tendency [78]. Crossflow is not what closes this branch off: at this Reynolds number neither amplification factor reaches its tunnel-calibrated limit ($N_{TS} = 8.0$, $N_{CF} = 5.5$) before the layer separates, which is why the branch stays mechanism-clean all the way to 29.7° and why it cannot be pushed to higher incidence for a crossflow-free transition front—there is no front to find. What it tests instead is whether the model correctly withholds one, and where it puts the laminar separation line. That is a weaker test than the high-Reynolds-number comparisons that follow, and one on which agreement is partly structural: a computed shear rise that lands on the measurement is in part reproducing separation.

At the lowest Reynolds number of the DFVLR matrix the boundary layer never transitions on the attached body at all: it stays laminar up to the open free-vortex-layer separation, and the measured points track the separation-line shear rise [78]. The comparison quantity is therefore the resultant-shear rise—the first station where c_f exceeds $k = 1.5\times$ its running minimum, the hot-film detection analogue, with the $k = 1.25$–2 sensitivity shown as a band—and not the model-native $\chi = c_{v1}$ front, which stays aft (0.92–0.99 x/L across the azimuth range).

Figure 24 makes that comparison at $\alpha = 10°$, against the DFVLR hot-film transition points [77] as reported in Stock's Fig. 15a [78] (measured at $Re = 1.52\times10^6$ against the computed 1.5×10^6; digitized to $\pm0.005\,x/L$), together with Stock's computed free-vortex separation line. The computed rise matches the measurement to within $\pm0.023\,x/L$ over the windward half ($\phi \leq 93°$) and lags by up to $0.21\,x/L$ at the leeward flank ($\phi = 132°$). The flank lag is not the slow transported-χ handover of Sec. VI: across the azimuth range the $\chi = 1$ and $\chi = c_{v1}$ crossings sit within $0.05\,L$ of each other (0.003–$0.016\,L$ at the measured azimuths), both between 0.90 and $0.99\,x/L$, so once the amplification commits, the handover completes quickly.

That lag, however, is largely a disturbance-level artifact rather than a kernel one. Figures 24–27 and Table F9 are computed at the airfoil campaign's flight-quiet seed ($\chi_\infty = 8.76\times10^{-6}$, $Tu = 0.01\%$), which is a factor twenty-two quieter in χ_∞ than the level Stock calibrates for this tunnel; at that seed $\chi = 1$ arrives 0.13–$0.56\,L$ aft of the hot-film line on the flank. Re-run at the tunnel conditions (Table 8) the lag closes: at the calibrated seed the front tracks the eight measured azimuths with mean +0.019 and rms 0.082, and at the measured hot-wire level with mean −0.013 and rms 0.027. Whatever the flank deficit is at this Reynolds number, most of it was the seed.

Two cautions keep that from being read as a transition-prediction success. First, at 1.5×10^6 the measured points lie on the laminar separation line [78], so a computed shear rise that lands on them is in part reproducing separation, not predicting instability-driven transition. Second, at $\alpha = 5°$ and the same Reynolds number the near-wall χ never reaches unity within $x/L \leq 0.97$ at the four most leeward measured stations, so the model carries no front there at all where the measurement places one at 0.74–0.94; the four stations it does resolve agree to rms 0.028. A front that is absent and a front that is late are not the same failure, and the table marks which is which.

Figure 25 shows the $\alpha = 10°$ case in detail, in the presentation of the measured campaign: the surface unrolled into $(x/L, \phi)$ with the skin-friction magnitude and friction lines (the oil-flow analogue), the wall-shear direction γ_w (the hot-film quantity), and the wall-normal maximum of χ with the model front—the measured transition points of Fig. 24 repeated on the c_f and χ panels. The friction lines converge onto the leeward separation band at $\phi \approx 100$–$135°$, the crossflow reverses under the vortex ($\gamma_w = 0$ line), and the amplification builds under the leeward vortex from mid-body without committing until near the tail—the deficit quantified above. At $\alpha = 29.7°$ (Fig. H3, Appendix H) the open separation covers most of the leeward half and the windward laminar run shortens; the level-to-level scatter of Table 7 at this incidence reflects the vortex-dominated flow.

Figures 26 and 27 present both solutions at the DFVLR hot-film measurement stations, in the waterfall presentation of the measured campaign [77, 78] (Stock's Figs. 4–5): the resultant skin friction and wall-shear direction (displaced

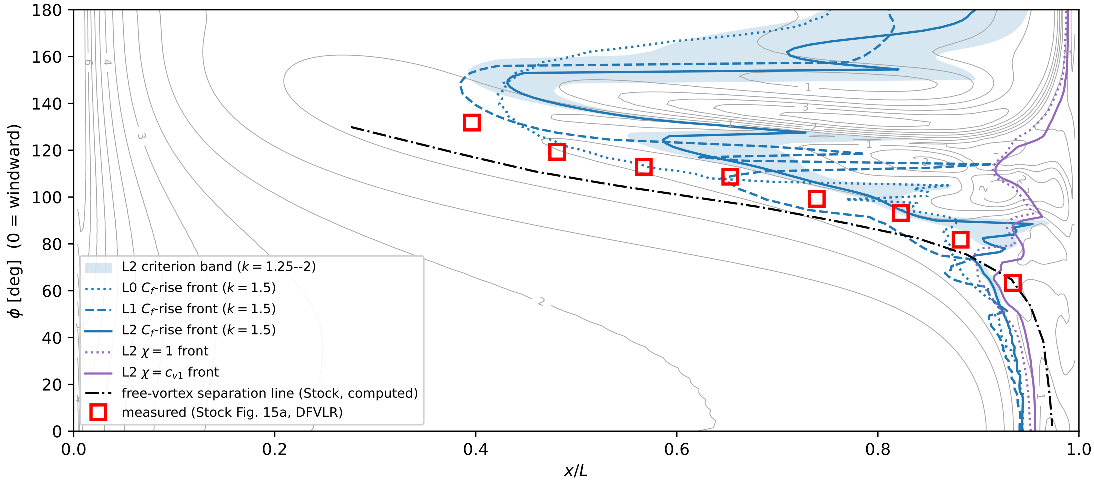

Fig. 24 6:1 prolate spheroid, $Re_L = 1.5{\times}10^6$, $\alpha = 10°$: the L2 c_f map with the computed C_f-rise fronts on the three grids ($k = 1.5$ criterion; the white band is the $k = 1.25$–2 sensitivity on L2), the model-native $\chi = c_{v1}$ front (solid cyan) with the $\chi = 1$ crossing (dotted cyan) hugging it—the fast handover—the measured DFVLR transition points (red squares; Stock Fig. 15a at $Re = 1.52{\times}10^6$ [78], Appendix F), and Stock's computed free-vortex separation line (gray dash-dot). The measured points at this Reynolds number lie on the laminar-separation line [78], which the computed c_f-minimum trench reproduces; the computed shear rise lags them by an amount growing from zero at the windward plane to $0.21\,x/L$ at the flank.

$-1.5{\times}10^{-3}$/$-15°$ per station at $\alpha = 10°$ and $-3{\times}10^{-3}$/$-20°$ at 29.7°, following the respective source figures; the hot-film array's fifth station moved between the two runs, $X/a = 0.304$ vs 0.130). Open red circles are the DFVLR measured chains, digitized from the printed source figures into the per-figure `stock2006_fign_digitized.json` files in `data/` (provenance and truncation policy in Appendix F). Filled squares, where present, mark the model's $\chi = c_{v1}$ front crossing a station: at $\alpha = 10°$ none appear—the front sits aft of every hot-film station, the amplification-deficit finding above in the campaign's own presentation—while at 29.7° the front crosses every station on the leeward side. (Computed views without a measured overlay belong to the appendix: the zero-incidence and $\alpha = 29.7°$ maps and the three-dimensional rendering are in Appendix H, together with the supporting pressure-station waterfalls.)

At $\alpha = 29.7°$ and $Re_L = 1.53{\times}10^6$ the branch closes. This is the largest incidence in the DFVLR record at this Reynolds number, the open free-vortex-layer separation covers most of the leeward half, and Stock predicts the layer laminar to that separation line here as at 5 and 10° [78]: the twelve hot-film stations are reading separation, not instability. The model resolves a front at all twelve, with mean residual +0.041 and rms 0.095 at the calibrated seed and −0.048 and 0.098 at the measured level (Table 8), so the two seeds bracket the measurement as they do at zero incidence. The residual is not uniform across the azimuth: at both seeds it is smallest on the leeward flank (−0.024 and −0.045 at the most leeward station) and largest on the windward one (+0.264 and +0.185), the same windward-late signature the high-Reynolds-number cases carry, here superposed on a separation line rather than a transition front. Stock digitized no computed curve for this panel, so the last two columns of the table are empty at this condition and the comparison is

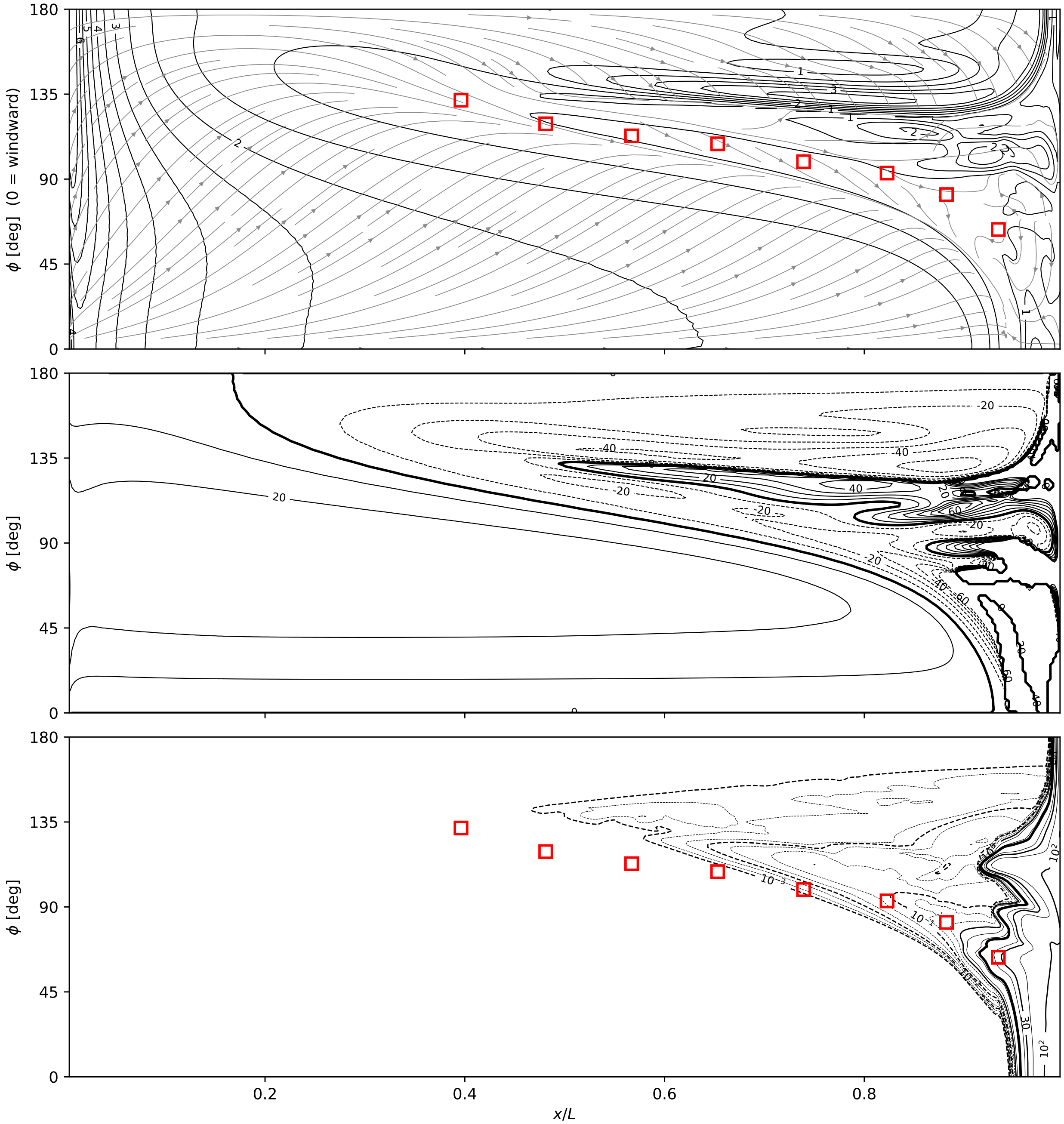

Fig. 25 6:1 prolate spheroid, $Re_L = 1.5{\times}10^6$, $\alpha = 10°$, SA-AI on the finest (L2) O-grid (12.6M nodes), surface unrolled into $(x/L, \phi)$, windward line at $\phi = 0$. Top: c_f magnitude (labeled contours) with skin-friction lines (gray; the oil-flow analogue); the friction lines converge onto the leeward separation-line band at $\phi \approx 100$–$135°$. Middle: wall-shear direction γ_w from the local meridian (positive toward leeward), the quantity the DFVLR hot films measured [77]; the bold $\gamma_w = 0$ line marks the crossflow reversal under the leeward vortex. Bottom: wall-normal maximum of χ with the $\chi = c_{v1}$ contour (bold) as the model-native transition front (sub-unity χ decades dashed, supercritical solid): the body runs laminar to $x/L \approx 0.9$ on the windward side, with amplification building (but not committing) under the leeward vortex from mid-body. Red squares on the c_f and χ panels: the measured DFVLR transition points of Fig. 24 (Stock Fig. 15a, Appendix F).

against the measurement alone.

B. The small-incidence limit: why every local model must transition first on the leeward side

Before the high-Reynolds-number cases, it is worth settling what incidence does to this boundary layer, because the answer is analytic and it predicts the sign of the error the rest of the section reports—and the sign of the error the rest of the literature reports as well.

Take the potential flow around the spheroid at incidence exactly, as a superposition of an axial and a transverse mode in prolate spheroidal coordinates. Let $\eta = 2x/L - 1$, so $\eta = -1$ at the nose, 0 at mid-body and +1 at the tail, and let φ be the azimuth measured from the windward generator, so $\varphi = 0$ is the windward symmetry line and $\varphi = 180°$ the leeward one. To first order in α the surface velocity has a meridional component u_s and an azimuthal one u_φ,

$$u_s = U_\infty [\, f_0(\eta) + \alpha f_1(\eta) \cos\varphi \,] , \qquad u_\varphi = U_\infty \, \alpha \, g \, \sin\varphi, \tag{21}$$

in which f_0 is the axisymmetric speed already used in Sec. X.C, f_1 is the incidence perturbation to it, and g sets the size of the circumferential flow. Because the perturbation carries $\cos\varphi$ and $\cos 0° = +1$ while $\cos 180° = -1$, the two symmetry meridians receive equal and opposite changes in u_s; because u_φ carries $\sin\varphi$, it vanishes identically on both of them.

Two properties of Eq. (21) are exact and both matter. First, $g = T/b$ is a constant along the body: the azimuthal edge velocity is $U_\infty \alpha (T/b) \sin\varphi$ at every station, with $T/b = 1.917$ for the $6:1$ body. That value is a check on the transverse mode rather than a fitted quantity—as $b/a \to 0$ a slender body in a crossflow αU_∞ becomes a two-dimensional cylinder, whose surface speed is $2U_\infty \sin\varphi$, and $T/b \to 2$ accordingly. Second, $f_1 \propto \eta$, so the meridional perturbation is antisymmetric about mid-body and passes through zero exactly at $x/L = 0.5$: over the front half the windward meridian is slower than the axisymmetric one and the leeward faster, and beyond mid-body they exchange roles. Physically the stagnation point has slid onto the windward side and the flow spills around toward the lee, so near the nose the leeward side has already been accelerated while the windward side is still near rest.

That gives two independent reasons why the leeward meridian should transition first, and they are worth separating because they belong to different parts of any transition model.

The first is the pressure gradient, and it is what an e^N envelope sees. Over the front half the windward meridian is on the slow side of f_0 and still accelerating, while the leeward meridian is on the fast side and already decelerating. Favourable gradient thins the layer and suppresses Tollmien–Schlichting growth; adverse gradient does the reverse. Marching a Mangler-weighted Thwaites integral on the exact u_e of Eq. (21) and integrating a Drela–Giles envelope [6] to $N = 8$—Stock's limiting value for this tunnel [78]—puts the leeward crossing $0.068\,L$ ahead of the windward one at $\alpha = 2.5°$, $Re_L = 7.2\times10^6$. Marching instead on our own computed c_p gives 0.067, so this channel is not an artifact of the analytic edge field: the two agree to 0.001, and the potential and computed windward-to-leeward u_e ratio at $x/L = 0.05$ agree to three digits at $\alpha = 5°$ (0.895 against 0.896).

The second is lateral straining, and it is the genuinely three-dimensional one. Since $u_\varphi \propto \sin\varphi$, its azimuthal derivative on the two symmetry planes is $\pm\alpha U_\infty g/r_0$—equal in magnitude, opposite in sign. On the windward line the surface flow leaves sideways in both directions, so a streamtube straddling that line spreads and the layer is thinner than the axisymmetric estimate; on the leeward line fluid arrives from both sides, the streamtube is squeezed, and the layer is thicker. Writing the streamtube width as $W = r_0 \exp\int k\, \mathrm{d}s$ with $k = (u_e r_0)^{-1}\partial u_\varphi/\partial\varphi$ and repeating the march, this channel alone moves the leeward crossing $0.219\,L$ ahead—three times the pressure-gradient channel, and in the same direction. Together they give $0.287\,L$.

The reason this is not a peculiarity of the envelope method is that Re_θ is the argument of both halves of any transition trigger: the critical condition that starts amplification, and the growth rate once it has started. A thicker layer at the same station has a larger Re_θ, so it reaches the critical value sooner and amplifies faster afterwards. Any model whose trigger is a local thickness or a local Reynolds number—an envelope, a transported amplification factor, an intermittency correlation—must therefore fire earlier on the squeezed leeward side and later on the spread windward side. The prediction is not a property of a calibration; it is a property of the model class.

The literature bears that out, and uniformly. In the first AIAA CFD transition modeling workshop's inclined-spheroid case at $\alpha = 5°$, $Re_L = 6.5\times10^6$, every submitted transport model that transitions on the interior of the body does so earlier at $\varphi = 180°$ than at $\varphi = 0$, by 0.24 to more than $0.7\,L$ [32]; the most extreme is an amplification-factor transport model, the published construction closest in spirit to the present one, which carries no windward front at all while placing the leeward front at $x/L = 0.19$. Langtry–Menter in OVERFLOW is windward-late by about $0.20\,L$ at the same condition and cannot be tuned out of it: raising the assumed roughness sixfold, to an unphysical $20\,\mu$m, pulls the leeward front upstream onto the measurement and leaves the windward front where it was [34]. A one-equation Spalart–Allmaras correlation model fires almost uniformly in azimuth at $x/L \approx 0.06$ [35]. Our own $\chi = 1$ front is windward-late by $0.147\,L$ at $\alpha = 2.5°$—inside that band, and less asymmetric than the textbook envelope estimate of 0.287 above.

The measurement, however, shows no such asymmetry: the two symmetry planes agree to $0.018\,L$ at $\alpha = 2.5°$ and $0.002\,L$ at 5° [77]. Neither does Stock's computation, which is leeward-late by 0.030 and 0.019; his own envelope N factors reach large values along the windward streamlines and the leeward ones alike, with the moderate values in between [78], and the N_{TS} values he recovers at every measured transition location across the campaign fall in a single band about the fitted limit rather than splitting into a windward and a leeward cloud. An asymmetry of $0.287\,L$ corresponds to about 4.8 in N at our own threshold sensitivity, which that band excludes.

So the sign and the mechanism of the error are settled, and the size of it is not attributable to the amplification rate: the same 0.15–0.29 L bias appears in an envelope method with no model constants at all. What distinguishes Stock is that he solves the three-dimensional laminar boundary layer by finite differences on a surface mesh with 121 wall-normal points and feeds the resulting profiles to local stability theory, rather than reducing each profile to a thickness and a shape factor. The lateral straining acts on the whole profile there, and stability responds to the profile's shape, which need not follow its width. That is the natural place to look next, and it is consistent with the observation of Sec. X.E that changing the amplification rate does not move the azimuthal spread. It should be said plainly that Stock nowhere writes the boundary-layer equations he solves or discusses what the lateral terms do to the profile, so the attribution is inference from what his method must contain, not a claim he makes.

C. Zero incidence: the pure-TS anchor and its disturbance environment

At zero incidence and $Re_L = 7.2\times10^6$ the spheroid is a pure Tollmien–Schlichting case [78], the cleanest test of the amplification rate stripped of crossflow, and the hot films place the measured front at $x/L = 0.438$ uniformly around the azimuth [77, 78]. The disturbance environment is therefore the whole question, and it is not a settled number: the tunnel's own hot-wire characterization gives 0.33–0.40% streamwise (with ~0.8% in the cross-stream components, an open-jet test section), while every transition study of this body instead uses a calibrated or prescribed value an order of magnitude quieter. This section therefore runs each condition twice, at the calibrated and at the measured level, and reports both.

Table 8 collects the comparison. At the facility-calibrated seed (Stock's limiting $N_{\mathrm{TS}} = 8.0$, $\chi_\infty = 2.38\times10^{-3}$, Mack $Tu = 0.106\%$) the computed $\chi = 1$ front lands at 0.516, 0.077 aft of the measurement; at the measured hot-wire level ($Tu = 0.33\%$, $\chi_\infty = 3.60\times10^{-2}$) it lands at 0.341, 0.097 ahead. The measurement is bracketed, and the bracket is tight enough to locate the effective level between the two—consistent with the $Tu \approx 0.17\%$ that an explicit seed sweep selects. Both fronts are uniform to ±0.001 across seventeen azimuths, as the axisymmetric flow requires.

Two properties of the comparison are worth recording because they bound what the rest of the section can claim. The seed is the dominant lever here: a factor 3.1 in Tu moves the front by $0.175\,L$. And the Mach number is not: repeating the calibrated case at $M = 0.1$ instead of the tunnel's 0.144 moves the front by 0.005.

D. $\alpha = 2.5^\circ$: where the azimuthal spread appears

Incidence separates the two effects immediately. At $\alpha = 2.5^\circ$ (still pure-TS by Stock's reading) the calibrated seed places the leeward front on the measurement (0.447 vs 0.432) but the windward front 0.132 aft of it (0.583 vs 0.450): the computed front spreads $0.136\,L$ around the azimuth where the measured one spreads $0.018\,L$. The sign is the one a purely two-dimensional kernel must give—the windward meridian of an inclined body carries the more favorable gradient, so the TS amplification it books is the more suppressed—and it is the measurement's near-uniformity that the model cannot reproduce. This is the crossflow economy's first visible cost, and the entry to the fully three-dimensional conditions that follow.

E. $\alpha = 5°$ at $Re_L = 6.49\times10^6$: the crossflow channel opens with Reynolds number

Raising the Reynolds number at fixed incidence introduces a mechanism that was absent, and Stock states the boundary directly: at 1.52×10^6 (Sec. X.A) the layer is laminar up to separation except on one streamline, while at 6.49×10^6 transition is triggered by TS near the windward and leeward symmetry planes and by TS and crossflow together over the remainder of the body [78]. The model carries only the mechanism that was already there.

The residuals follow that boundary (Table 8). Every measured station resolves here, and at the calibrated seed the residual opens from −0.088 at the most leeward to +0.246 at the most windward. At the measured hot-wire level the windward residual falls to +0.007 while the leeward goes to −0.292: the seed moves the whole front by 0.204–0.238 at every measured azimuth, so it sets the level but cannot correct the shape, and the shape is wrong. The computed front sweeps 0.36 L across the azimuth where the measurement sweeps 0.087 L. No single seed fits both ends.

F. $\alpha = 10°$ at $Re_L = 6.56\times10^6$: the deficit at its largest

This incidence at $Re_L = 6.56\times10^6$ produces the largest residual in the matrix. The most leeward station still agrees to +0.068, but from $\phi \approx 120°$ inward the computed front stands at 0.88–0.95 x/L—effectively at the tail—against a measurement at 0.31–0.56: a residual of +0.387 to +0.623, rms 0.463 (Table 8).

Stock's own mechanism coding, recovered from the dash styles of his printed computation, places the pure-TS band at $\phi = 172$–$145°$, mixed TS and crossflow from 144 to 95°, pure crossflow from 93 to 63°, and mixed again from 56° to the windward plane. Our single measured azimuth inside the pure-TS band, $\phi = 165°$, carries our best residual (+0.068); the four remaining stations lie in the mixed and crossflow region and carry +0.387 to +0.623. The correlation should not be pressed harder than that: the pure-crossflow band contains no hot-film station at all, and the worst station, $\phi = 59°$, falls in a segment whose style our digitization could not separate from mixed with confidence.

What distinguishes this condition is not the size of the residual but where the seed still has authority over the front, and the answer maps onto Stock's mechanism coding. Changing between the two facility seeds—a factor 3.1 in Tu—moves the front by 0.174 at every one of the seventeen azimuths of the axisymmetric case, and by 0.166–0.182 and 0.204–0.238 uniformly across azimuth at 2.5° and 5°. Here it does not act uniformly at all: −0.173 at $\phi = 165°$—full authority, indistinguishable from the axisymmetric value—then −0.049 at 59°, −0.027 at 40°, −0.016 at 29° and −0.012 at 20°. The azimuth that keeps its authority is the one inside Stock's pure-TS band; the azimuths that lose it are the ones he assigns to crossflow, alone or mixed with TS.

That is a sensitivity result rather than an accuracy one, and it is the stronger evidence: it does not ask whether the model is right, only what its front responds to. Where the measured transition is TS-driven the freestream seed retains its full leverage; where crossflow drives it the seed is inert, and a closure whose amplification is a function of the two-dimensional profile alone has nothing left to move. The kernel is not merely slow there: in a favorable streamwise gradient $\hat{\Omega}\hat{I} \leq 0$ and the amplification it books is identically zero, so the region the crossflow occupies is the region the rate is constructed to ignore.

Stock's computation is the reference that calibrates the cost, and the comparison cuts both ways (last two columns of Table 8). His two-N-factor e^N method, which carries an explicit crossflow channel, holds rms 0.098 at this condition against our 0.463—better by a factor of five. But it degrades in the same azimuth band: −0.032 at the pure-TS station against +0.082 to +0.150 across the mixed and crossflow stations. The condition is hard for the whole e^N class, not only for a closure that omits the mechanism. The ranking also inverts at the low Reynolds numbers, where transition is TS- or separation-set: at $\alpha = 10°$ and 1.52×10^6 our rms is 0.027 against his 0.100, and at $\alpha = 5°$ and the same Reynolds number 0.028 against 0.064. At zero and 2.5° he is the more accurate by four- and fivefold, which is the expected order given that his limiting N_{TS} was fitted to these same measurements.

The ONERA F1 measurement at this incidence and Reynolds number closes the section on a null result worth stating. Its condition differs from its Göttingen twin only in the seed—same geometry, incidence, Reynolds and Mach number—and the two measurements differ by 0.077 in front position at $\phi \approx 30°$, the quieter tunnel transitioning earlier. Across the two facility seeds the model returns 0.001. At that azimuth its front is pinned at the tail by the absent crossflow channel, so the seed has no leverage to express, and the cross-facility comparison cannot be settled here. It would need a condition on which the model is seed-sensitive, and no F1 measurement exists at one.

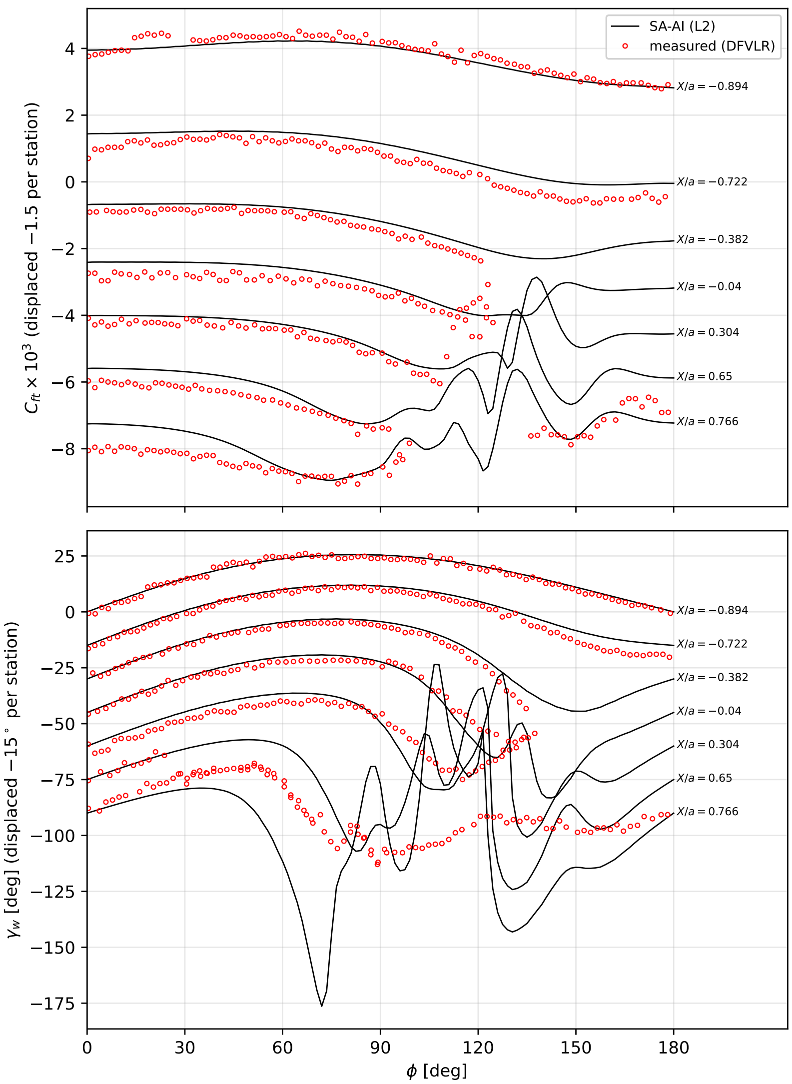

Fig. 26 6:1 spheroid, $Re_L = 1.5\times10^6$, $\alpha = 10°$ (L2): resultant skin friction (top) and wall-shear direction (bottom) at the DFVLR hot-film stations, presentation of Stock's Fig. 4. Lines: computed; open red circles: DFVLR measurement, digitized from the printed figure (Appendix F), truncated where the printed post-transition tangle makes chain identity unrecoverable.

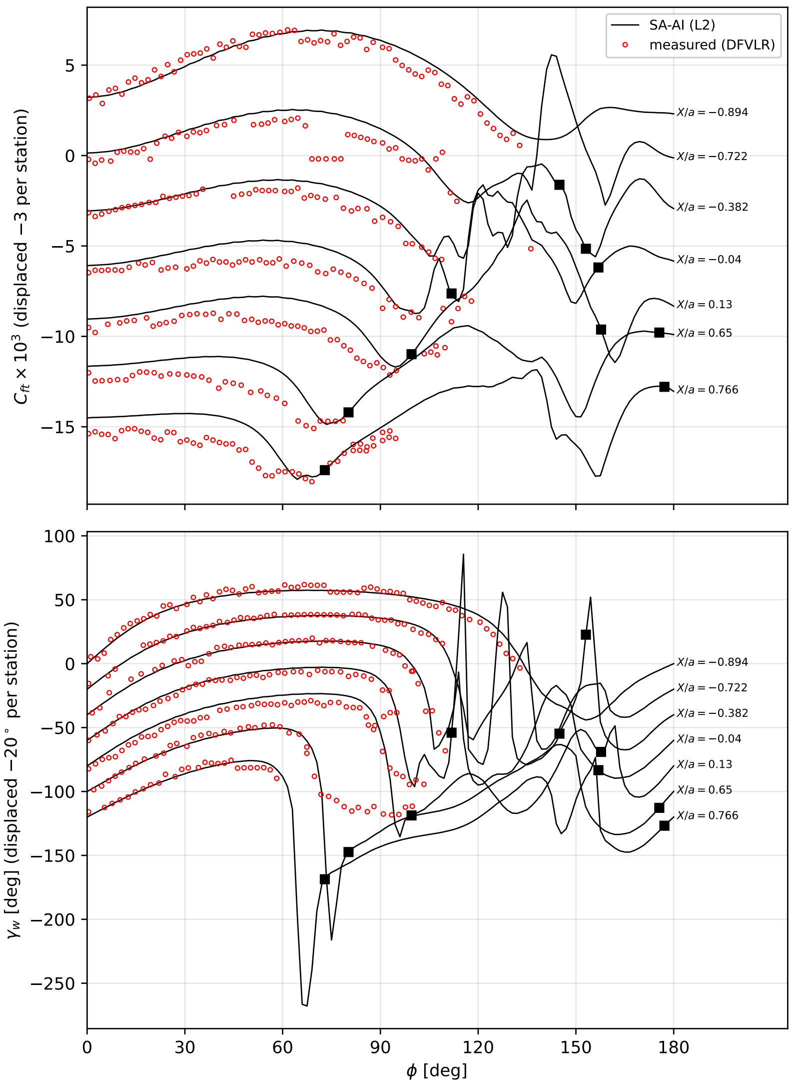

Fig. 27 As Fig. 26, at $\alpha = 29.7°$ (presentation of Stock's Fig. 5); measured chains truncated in the leeward tangle where printed chain identity is unrecoverable.

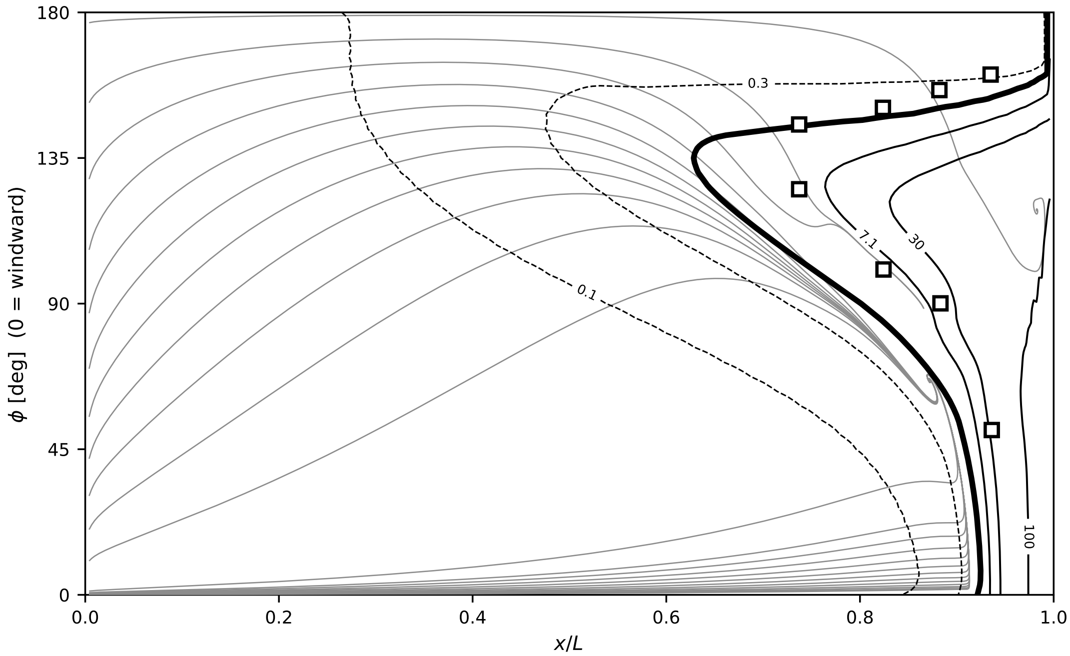

Fig. 28 $\alpha = 5°$, $Re_L = 1.52 \times 10^6$.

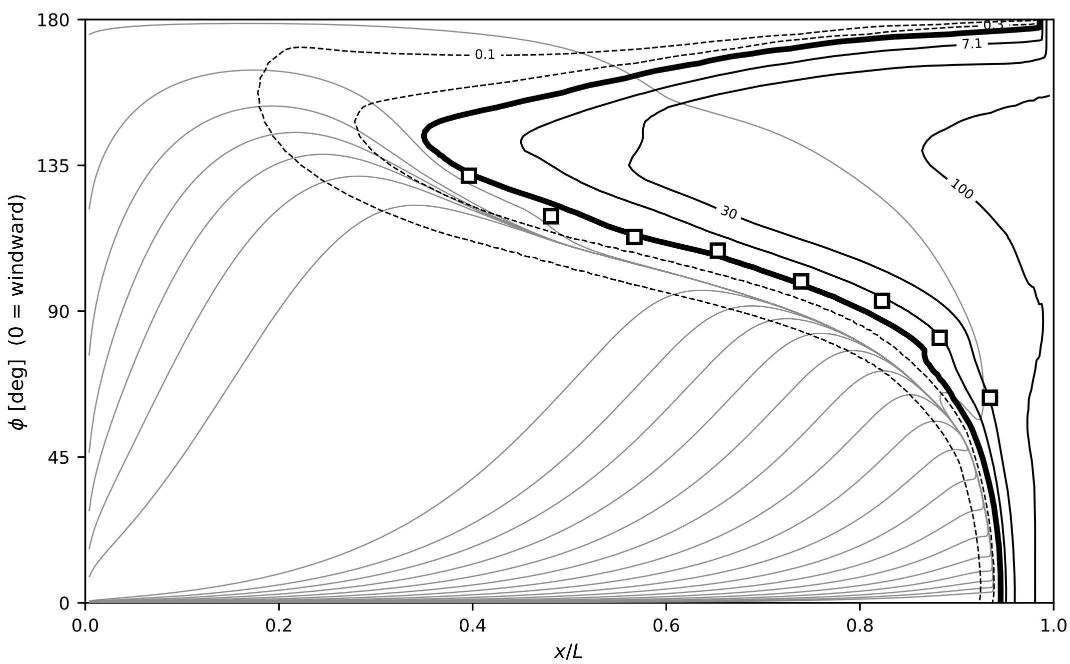

Fig. 29 $\alpha = 10°$, $Re_L = 1.52 \times 10^6$.

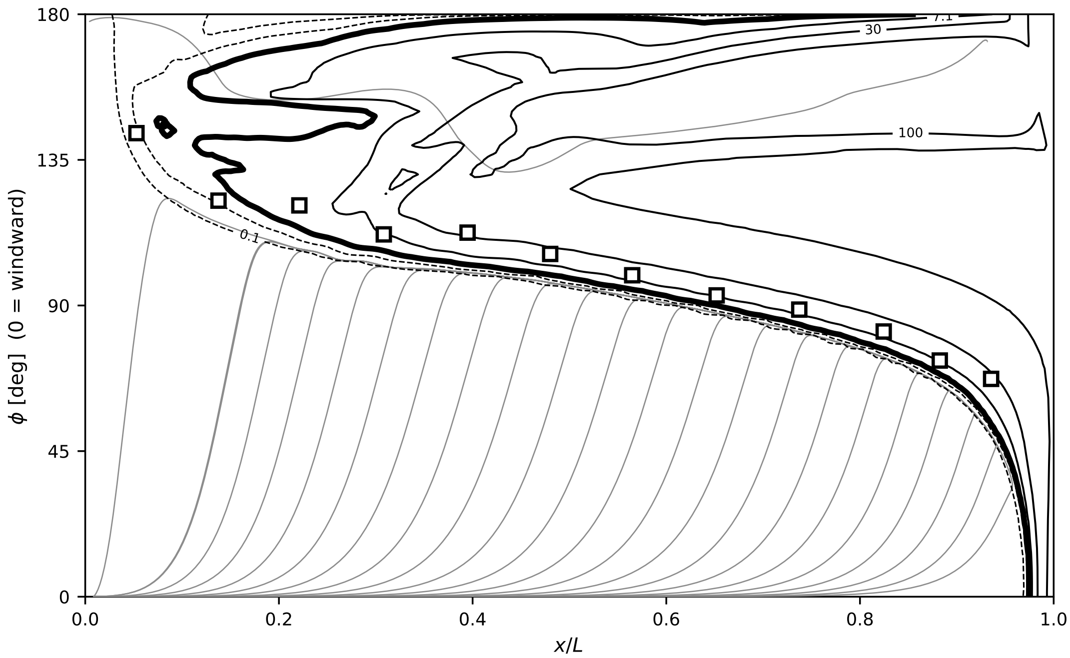

Fig. 30 $\alpha = 29.7°$, $Re_L = 1.53 \times 10^6$, the largest incidence measured at this Reynolds number. The skin-friction lines converge onto the open free-vortex-layer separation over most of the leeward half; the twelve measured stations lie on that line rather than on a transition front.

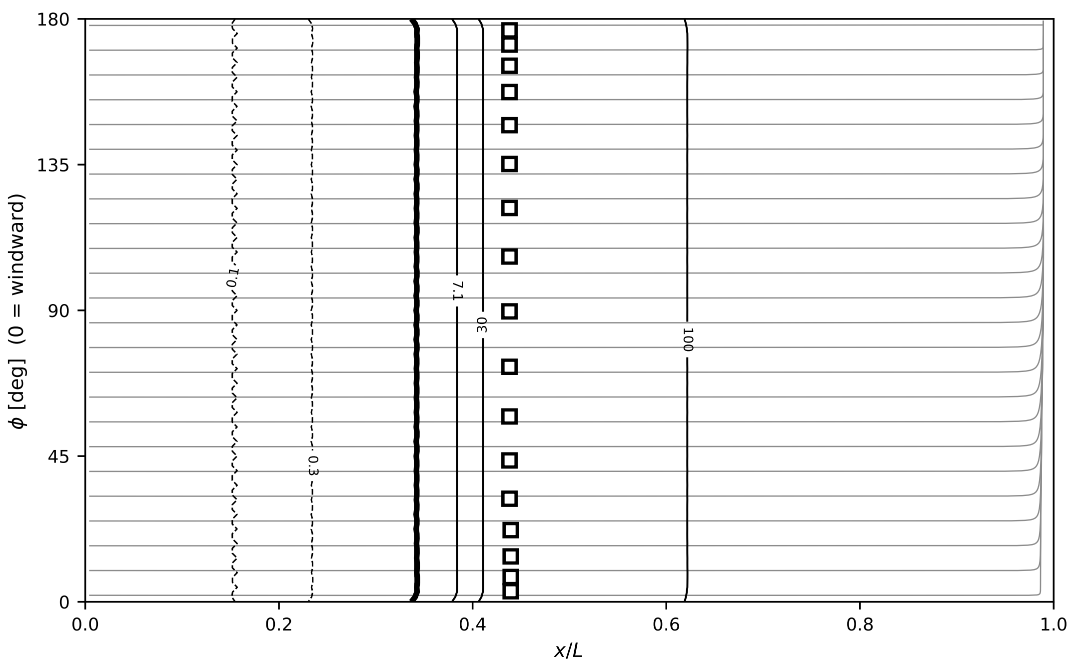

Fig. 31 Surface unrolled into $(x/L, \phi)$, $\phi = 0$ the windward symmetry line, at the measured-seed condition. Grey: skin-friction lines integrated from the wall shear vector itself—the oil-flow analogue, and not the inviscid streamlines of Stock Figs. 14–17—whose convergence marks separation. Black: the $\max_n \chi$ family on a wall-normal segment shot from each surface point, sub-unity decades dashed, $\chi = 1$ heavy, $\chi = c_{v1}$ and above solid. White-filled squares: the measured DFVLR transition points. Zero incidence, $Re_L = 7.2 \times 10^6$: both families are functions of x/L alone, as the axisymmetric flow requires.

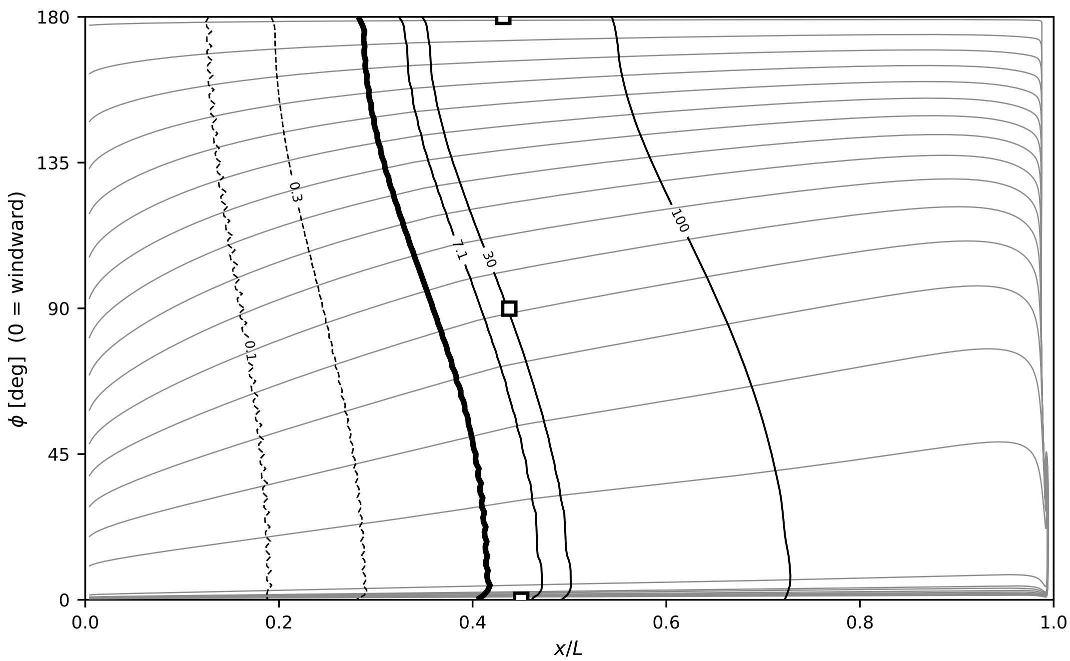

Fig. 32 $\alpha = 2.5°$, $Re_L = 7.2\times10^6$.

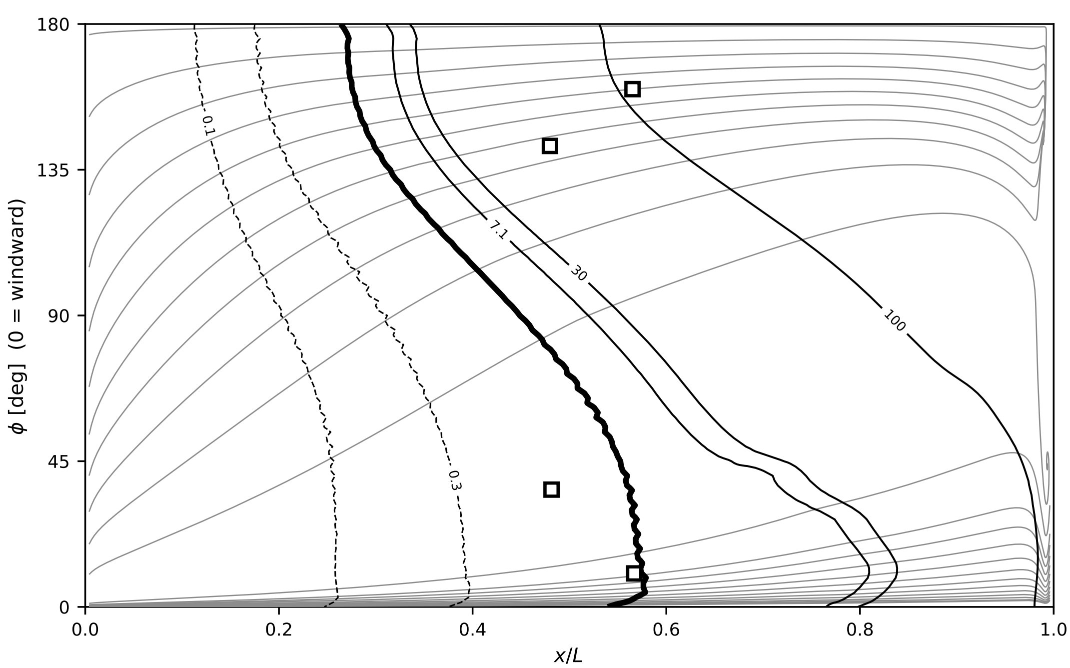

Fig. 33 $\alpha = 5°$, $Re_L = 6.49\times10^6$.

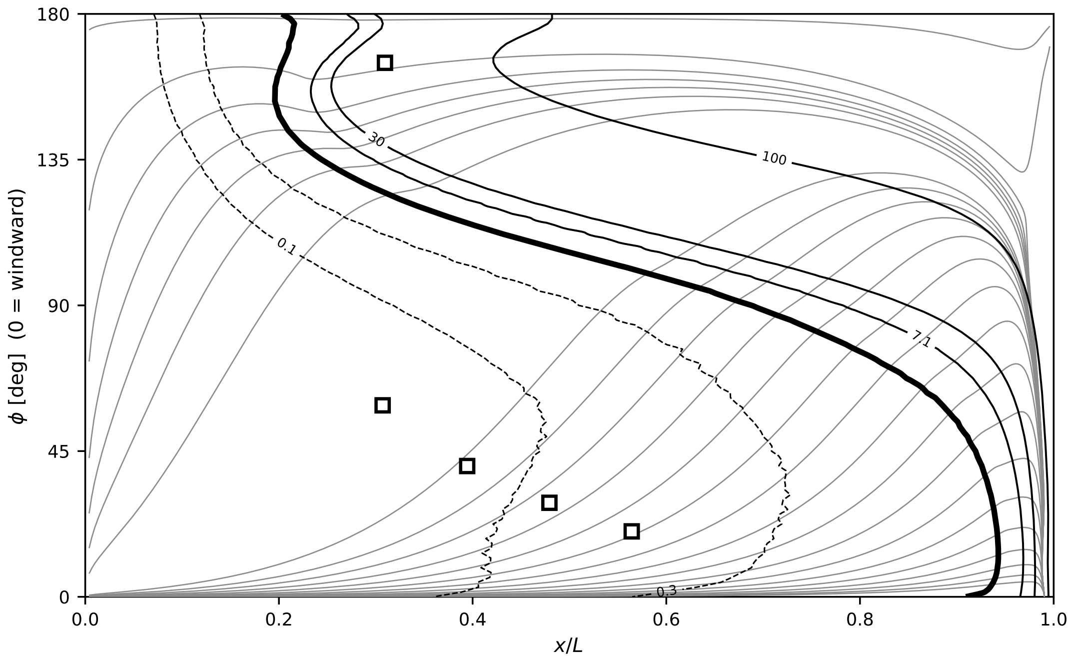

Fig. 34 $\alpha = 10°$, $Re_L = 6.56{\times}10^6$, Göttingen seed.

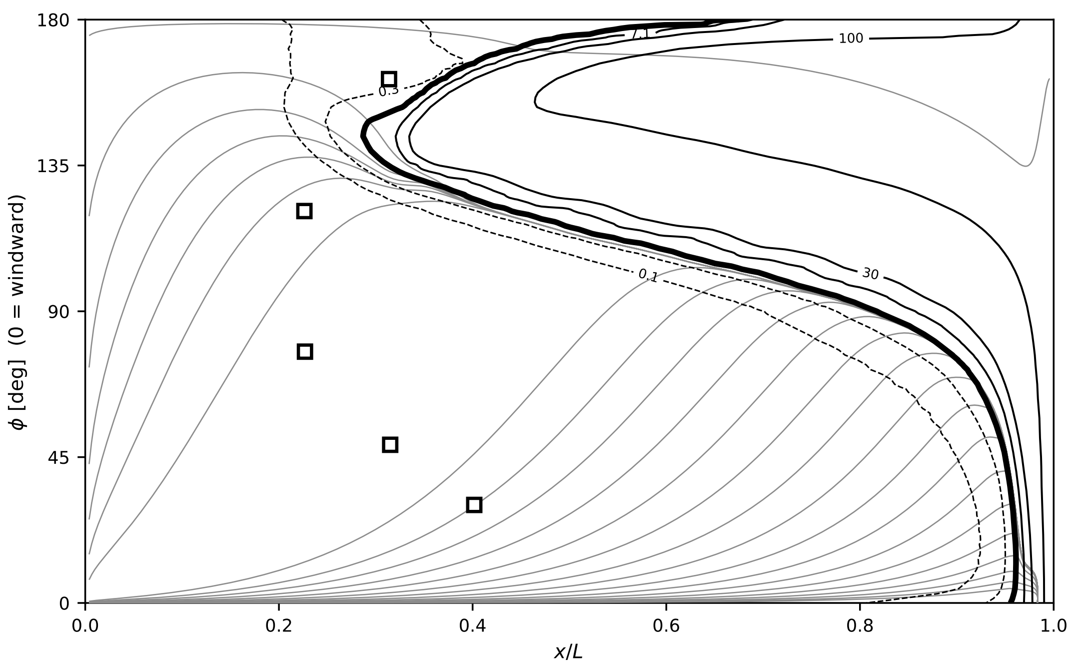

Fig. 35 $\alpha = 10°$, $Re_L = 6.56{\times}10^6$, ONERA F1 seed; the condition differs from its Göttingen twin only in the freestream seed.

XI. Conclusion

The Spalart–Allmaras working variable can, it turns out, serve as its own transition indicator. By replacing the production in the laminar range with a local, e^N-like amplification rate and reverting to standard SA near $\chi \sim 1$, a single transport equation predicts natural transition without any auxiliary field.

The amplification rate is anchored to canonical stability limits, not to the transition cases. Its ceiling $a_{\max} = 0.19$ is the tanh free-shear (Kelvin–Helmholtz) eigenvalue, reached undiluted at the vorticity pole; its geometry—the vorticity fraction $\hat{\Omega}$ and the accumulated-inflection coordinate $\hat{I}$, whose even product $\hat{\Omega}\hat{I}$ is the single amplifying coordinate the projective algebra allows—is fixed by the mean profile alone; and its viscous onset is a gate on the vorticity Reynolds number whose threshold $Re^{c}_{\Omega}(\hat{\Omega}\hat{I})$ takes its shape from linear stability theory and its single free scale from the $N = 1$ crossing of the Blasius e^N envelope alone—the $N = 9$ crossing is then an untuned prediction, landing 7% past Drela's—the residual footprint of the retained laminar diffusion. A favorable gradient keeps the curvature single-signed, so $\hat{\Omega}\hat{I} \to 0$ and the rate extinguishes geometrically, with no pressure-gradient parameter. The model was then exercised across three pressure-gradient regimes. The freestream-turbulence closure $\chi_\infty(Tu)$ is Mack's e^N correlation, and the destruction-gate weight is tied to the production weight by a wall-layer identity, so the linear $\tilde{\nu}$ wall solution—and with it the log law—is exactly standard SA's. On a zero-pressure-gradient flat plate (Sec. III) the Mack closure places transition onset (at the stated half-saturation $\chi = c_{v1}$ convention, Re_θ integrated from the computed profiles) within 10% of the Abu-Ghannam & Shaw correlation over the natural-transition range (7% late at the lowest disturbance level—the retained-drain footprint—within 2% at $Tu = 0.08\%$, and 4–10% early above), verifying the Blasius envelope. On the NLF(1)-0416 at $Re = 4\times10^6$ (Sec. IV) the model holds the designed laminar rooftop—the favorable run held by $\hat{\Omega}\hat{I} \to 0$ with no pressure-gradient term—and ignites at each recovery, tracking the Somers fronts (the lower front's aft-ward march with lift reproduced to within $0.02\,c$), with the handover held across the full incidence matrix, negative pair included. On the Eppler 387 at $Re = 2\times10^5$ (Sec. V) the model resolved the adverse-gradient, laminar-separation-bubble regime—the signed skin friction shows the recirculation directly—though it reattaches late at low incidence (0.02–0.04 c aft of the measured oil-flow reattachment at $\alpha \le 5^\circ$, inside the facility-to-facility spread of the two experiments on Fig. 14) and early at 7° (§II.C); an $\alpha = 5^\circ$ sweep across five Reynolds numbers, $Re = 6\times10^4$ to 4.6×10^5, follows the bubble's march and shortening. The same constants (Appendix E) were used throughout the airfoil campaign's ninety-six solutions (both mesh families, three refinement levels, every incidence and Reynolds number, with the finest-pair extensions). Across both airfoils the L1→L2 transition-location drift is at or below $0.01\,c$ (reaching $0.008\,c$ at the NLF's near-LE $\alpha = 9^\circ$ front and $0.011\,c$ on the Eppler reattachment), and the two independently-generated mesh families agree to within $0.02\,c$ of chord on every transition front and to 0.004–0.011 c on the Eppler reattachment at L2 (the edge-of-collapse 7° case excepted, where three grids retain no bubble). The Reynolds sweep also locates the model's low-Re boundary: the Eppler bubble fails to close at 10^5 on every grid, and an exhaustive initialization study (Sec. VI) shows the computed state is unique at every swept Reynolds number (among protocol-admitted states, Sec. VIII)—the model places its bursting boundary one Reynolds step above the measurement and carries no counterpart of the two-state structure the experiment shows at 6×10^4—traced quantitatively to the length of the deliberately unmodeled transition zone, which at these Reynolds numbers grows to a macroscopic fraction of the chord and starves the bubble of eddy viscosity—and to a second, opposite-signed defect it partially cancels against, the closed-cell re-seeding of the amplification by $\tilde{\nu}$ recirculating under the bubble's shear layer; low-Re bubbles the model gets right are the sum of these canceling errors (Sec. VI). In three dimensions, on the Daedalus wing at flight conditions, the completed runs of the final-kernel campaign put the mid-span bubble's separation within $0.02\,c$ and its reattachment within $0.01\,c$ of the strip-theory e^N reference, and capture two spanwise mechanisms no sectional method sees—the outboard bubble stretch past the taper break and the tip-vortex trip (Sec. IX).

Several extensions follow. The clearest limitation is the linear rate's under-amplification of strongly favorable and strongly separated profiles—the favorable envelope of Fig. 4 and the over-long Eppler bubble both trace to it—so a shaped rate that restores the near-neutral floor without a pressure-gradient parameter is the natural refinement. The second is the transition zone itself: an outer-scaled, memory-carrying closure for σ_t—a spot-growth or intermittency treatment replacing the memoryless fixed-χ-width interpolation, whose narrowing alone only trades the bias for a relaxation oscillation—would move the bursting boundary toward the measured one (Sec. VI). A third is the bistability of Sec. VIII: it belongs to standard SA, not to the amplification terms, so removing it—rather than excluding it by protocol—means modifying SA's behavior even at high χ; an acceleration-quench (relaminarization) gate on the production, driven by the parameter $K = (\nu/U_e^2)\, dU_e/ds$ that diverges toward an attachment point, is the candidate, since equilibrium turbulent layers sit one to two orders of magnitude below the quench threshold and would be expected to be untouched. A calibration against linear-stability and e^N databases beyond the anchors used here—and a direct comparison with the algebraic SA-BCM model on community-standard benchmarks—would establish accuracy beyond the present demonstration. The out-of-scope regimes noted in the introduction—bypass, crossflow, and compressible transition—are the natural next steps. For crossflow specifically, the Gram algebra of Eq. (E4) already contains the natural sensor: the odd pairing $\langle \nabla d,\, \mathbf{u} \times (\omega \times \mathbf{d})\rangle$, equal, for wall-parallel vorticity, to the helicity density times the layer depth, vanishing identically in two-dimensional flow, and measuring exactly the crossed shear a swept boundary layer adds—a second amplification channel in the same local language.

The cylinder traverse of Sec. VII—to our knowledge the first $C_d(Re)$ drag-crisis curve from a local single-equation transition closure—is deliberately a steady-branch result; its unsteady continuation, where the critical range's measured bistability and single-bubble lift states live [53, 54], the quantitative comparison against the measured pressure and skin-friction traverses [49], and the sphere are the natural next campaign.

天下萬物生於有，有生於無。

The myriad things are born of being;

being is born of non-being.

—Daodejing, ch. 40

Acknowledgments

The initial idea for this approach was sparked by conversations with Mark Drela and Philippe R. Spalart, whose insights on the amplification-factor envelope and on the structure of the SA equation were formative.

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Appendix A: the NLF(1)-0416—meshes and wall-anchored contour sheets

Appendices A–D collect, for one main-text section each, the mesh views and the wall-anchored contour sheets of its cases: Appendix A for the NLF(1)-0416 study (Sec. IV), Appendix B for the Eppler 387 benchmark (Sec. V), Appendix C for the Eppler Reynolds sweep (Sec. VI), and Appendix D for the Daedalus wing (Sec. IX). Appendix E assembles the model equations and constants, Appendix F tabulates the raw data behind the one-dimensional comparison figures, Appendix G carries the cross-solver replication detail, and Appendix H the spheroid pressure-station waterfalls and supplementary surface maps.

Figures A1 and A2 show the leading- and trailing-edge close-ups of the structured O-grid and unstructured cavity families at the three refinement levels. The cavity mesher is driven by the anisotropic metric

$$\mathsf{M}(\mathbf{x}) = R\,\mathrm{diag}\Big(h_n^{-2},\ h_t^{-2}\Big)\,R^{\mathsf{T}}, \qquad h_n = \min\big(h_0 + (\gamma-1)\,d,\ h_\infty\big), \quad h_t = \min\big(h_w + (\gamma-1)\,d,\ h_\infty\big),$$

where d is the distance to the nearest wall point, R rotates into the (wall-normal, wall-tangential) frame of that nearest point, h_0 is the first-cell height of Tables 4/5, γ the growth ratio, h_w the wall-tangential metric input, and h_∞ the isotropic far-field size; per level L0/L1/L2 the inputs are $\gamma = 1.2/1.1/1.05$, $h_w = 0.008/0.004/0.002$, and $h_\infty = 3.0/2.0/1.5$ (chords). The metric governs the interior only—the boundary spacing comes from the resampled contour—and h_∞, unlike every near-wall scale, does not halve per level (it is reached only far outside the region the sheets and tables measure).

The contour sheets present the computed fields in the manner of the flat-plate figure (Fig. 5): line contours of the streamwise velocity u_x/U_∞ (left column) and of $\log_{10}\chi$ (right column; dashed contours are the laminar levels $\chi < 1$, solid are $\chi = 1$, c_{v1}, 30, and 10^2—the last two tracking the handover's completion and the turbulent interior), on wall-anchored coordinates—the abscissa is the x/c of the wall anchor, the ordinate the wall-normal distance from that anchor, so each wall-normal probe scan is one vertical line of the panel. Rows pair the two mesh families at each refinement level (cavity, then O-grid; L0, L1, L2). The wall-normal range is chosen to hold the laminar band in frame ($0.0015c$ upper and $0.0025c$ lower for the NLF(1)-0416, scaled with $1/\sqrt{Re}$ for the other cases), letting the turbulent layer overshoot the frame top—except the Eppler upper surface, whose frame is raised a further 5× ($0.0335c$ at 2×10^5) so the handover above the lifted bubble shear layer stays in frame.

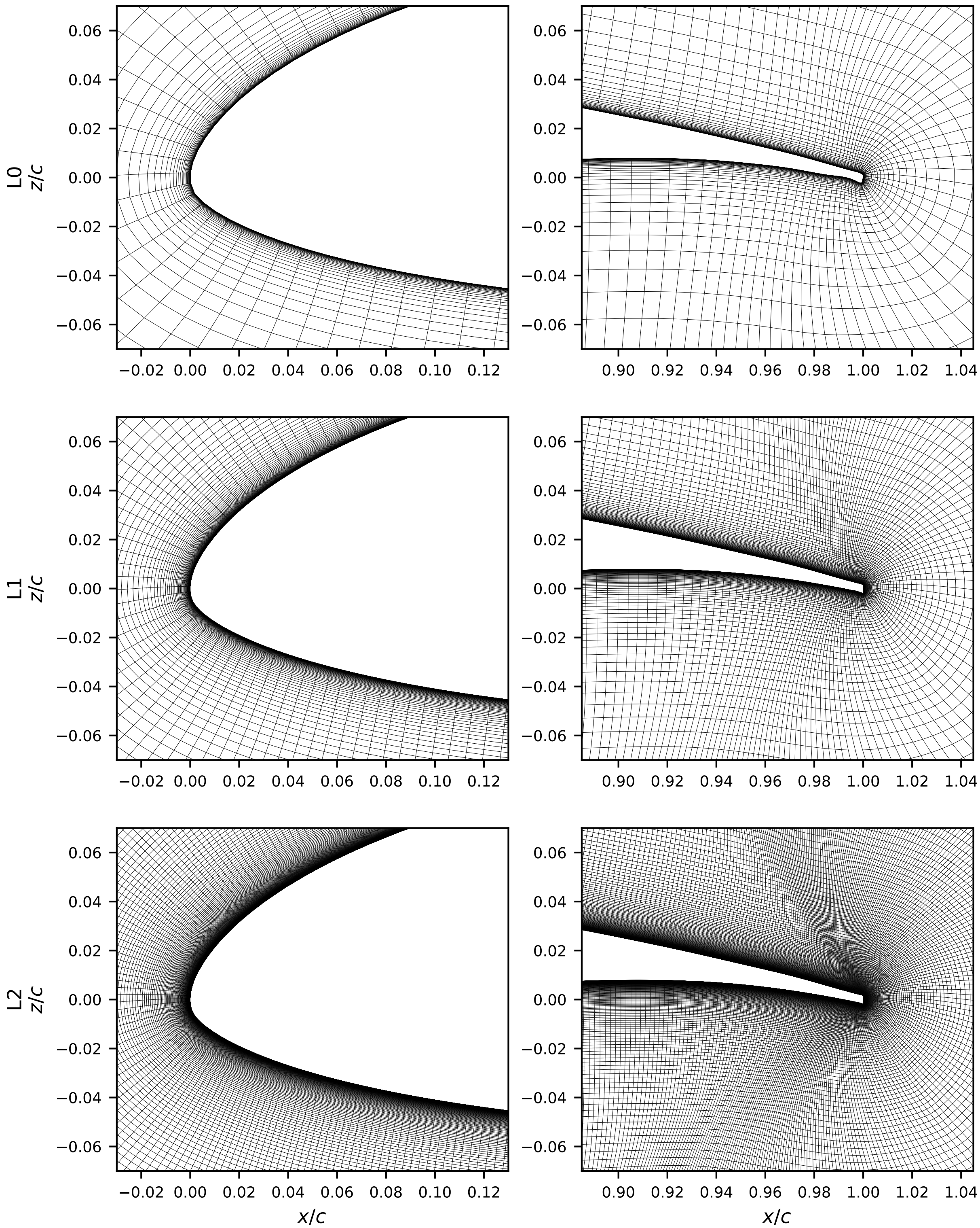

Fig. A1 NLF(1)-0416 structured O-grid mesh at the leading edge (left column) and blunt trailing edge (right column) for the three refinement levels L0–L2 (rows). The O-grid wraps the section with wall-normal clustering for the boundary layer; every length scale halves per level. Chord normalized to unity (z vertical).

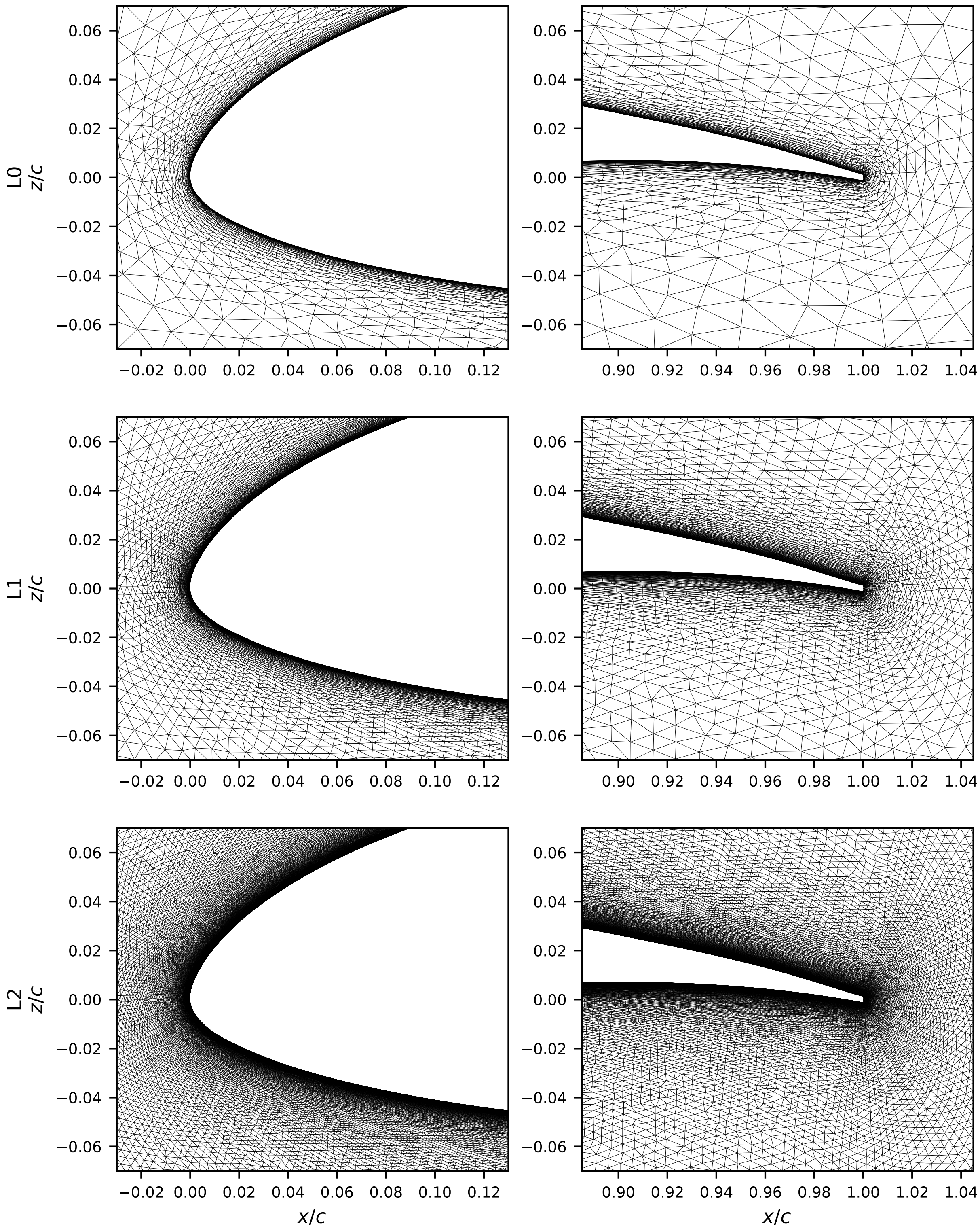

Fig. A2 NLF(1)-0416 unstructured cavity mesh (leading and trailing edges, levels L0–L2), built from the same arc-length cubic-spline contour as the structured grid of Fig. A1 with blunt-trailing-edge–aware spacing. Every length scale halves per level.

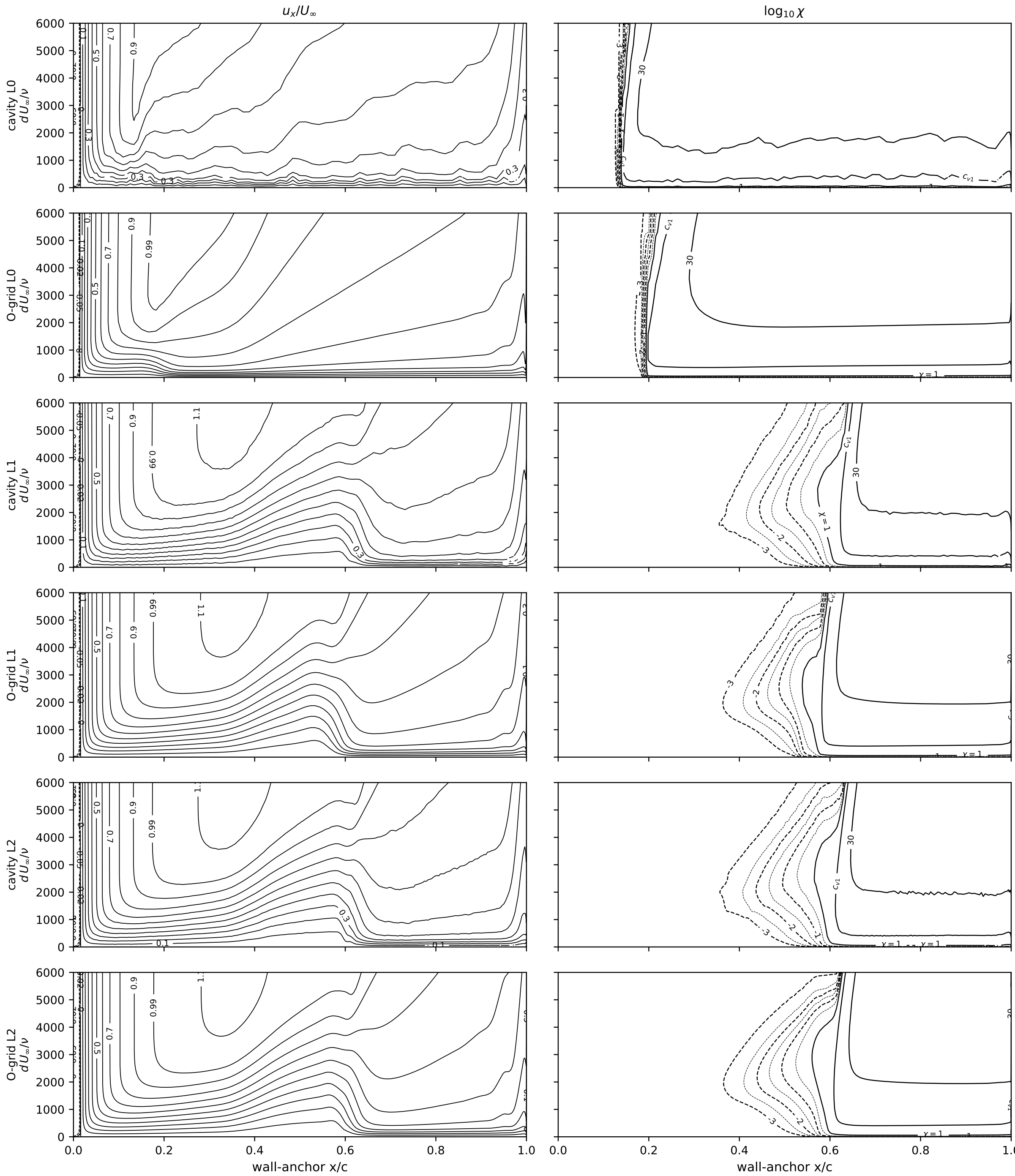

Fig. A3 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = -8°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids. This (pressure) surface holds a long laminar run before its front settles at 0.54–0.57 c on L1/L2 (Fig. 8); the L0 grids trip at 0.14–0.20 c, the coarse-grid front breakaway of Fig. 6's caption.

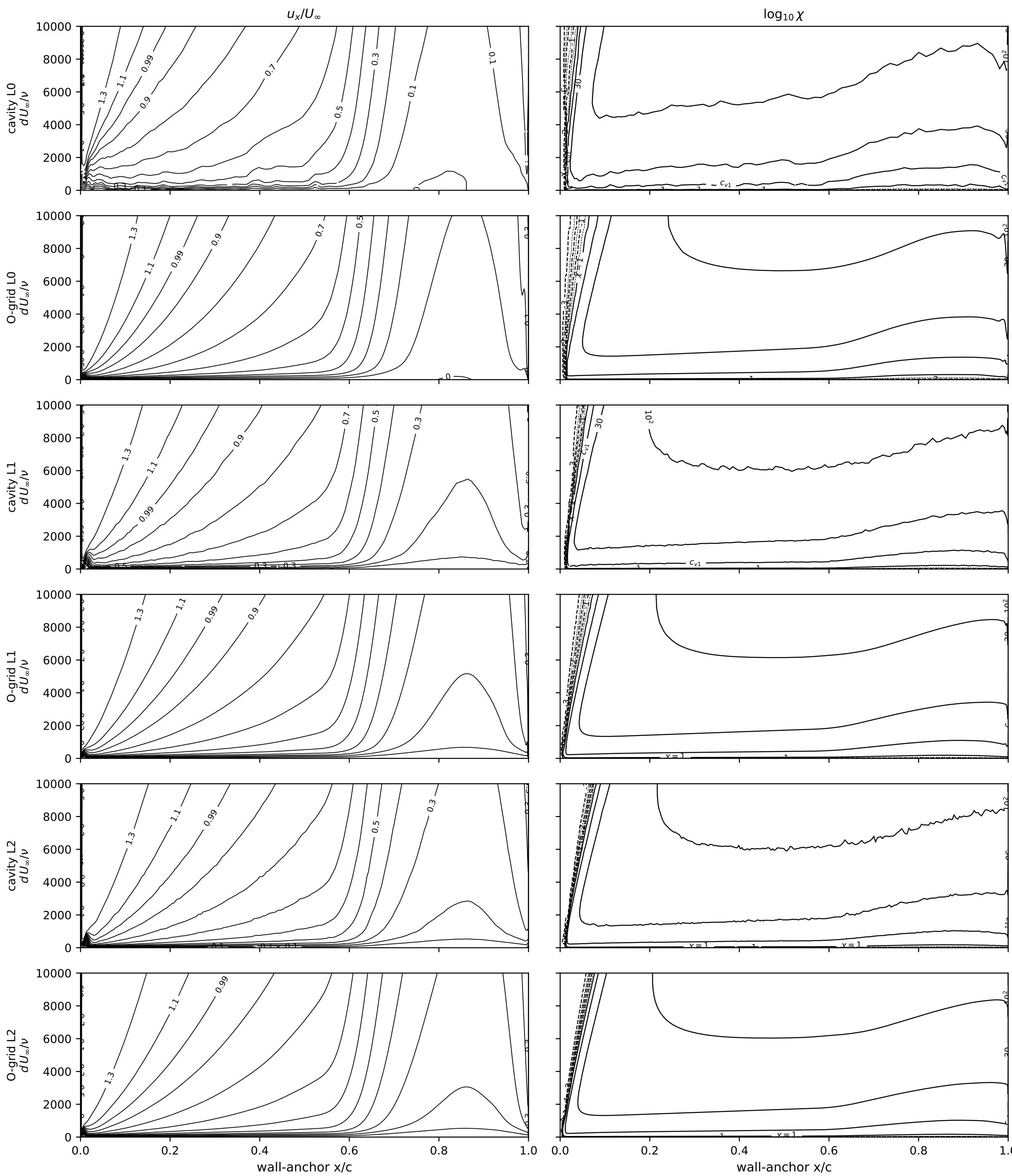

Fig. A4 As Fig. A3, lower surface (the suction side at −8°): tripped at the leading edge on every grid.

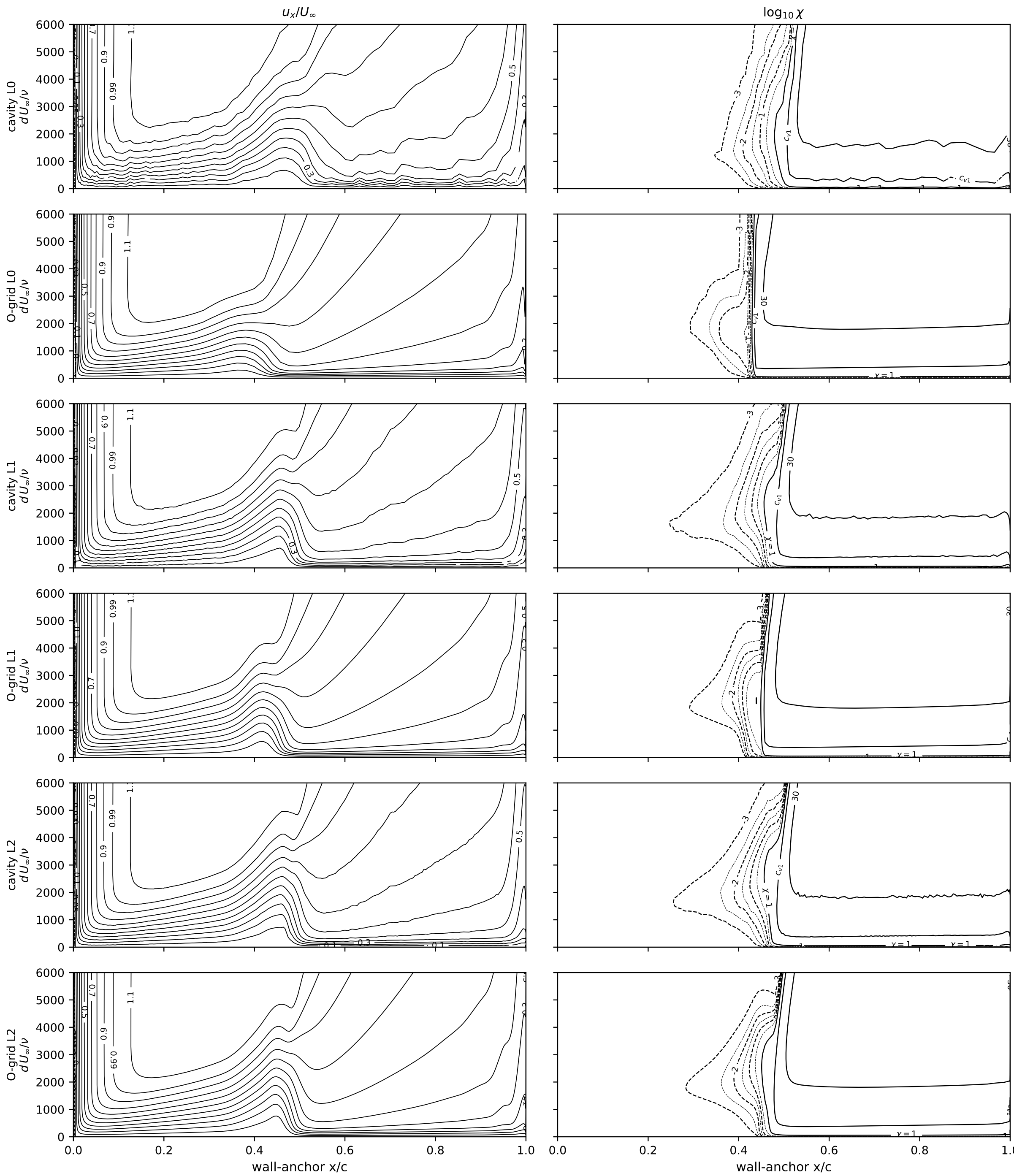

Fig. A5 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = -4°$ (zero lift), upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids; fronts at 0.44–0.48 c across the ladder.

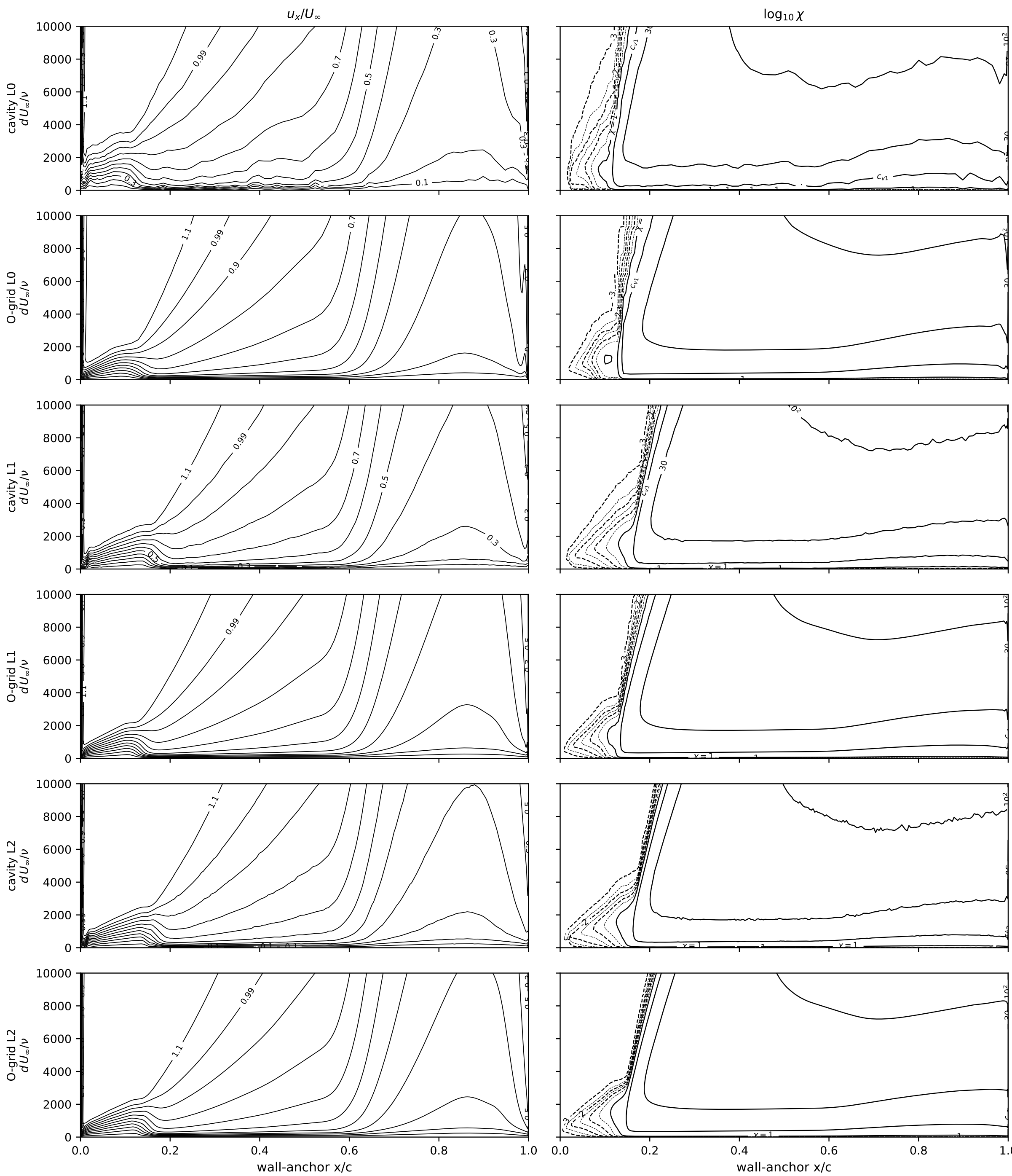

Fig. A6 As Fig. A5, lower surface: laminar to ≈0.08–0.13 c on every grid.

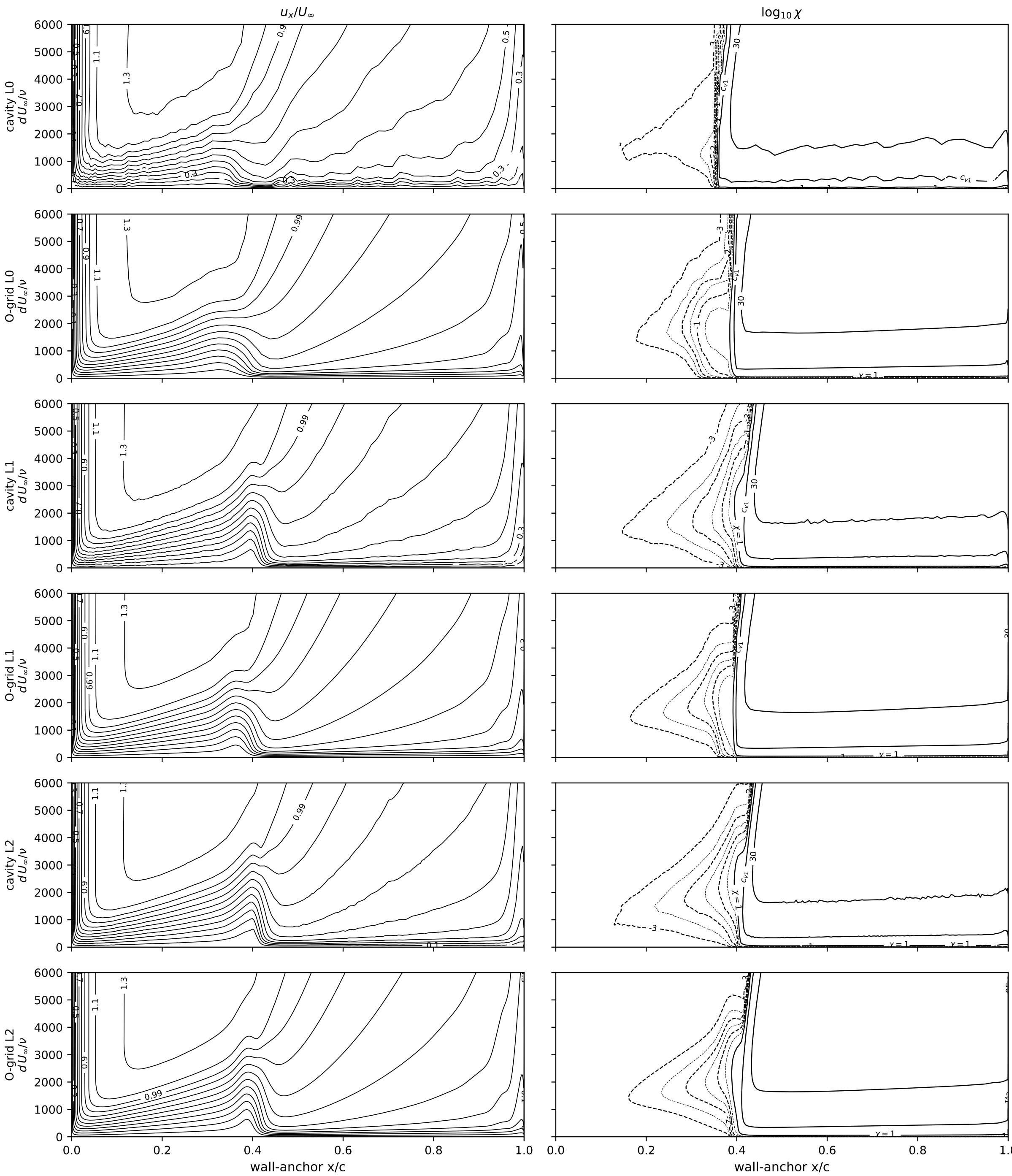

Fig. A7 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = 0°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

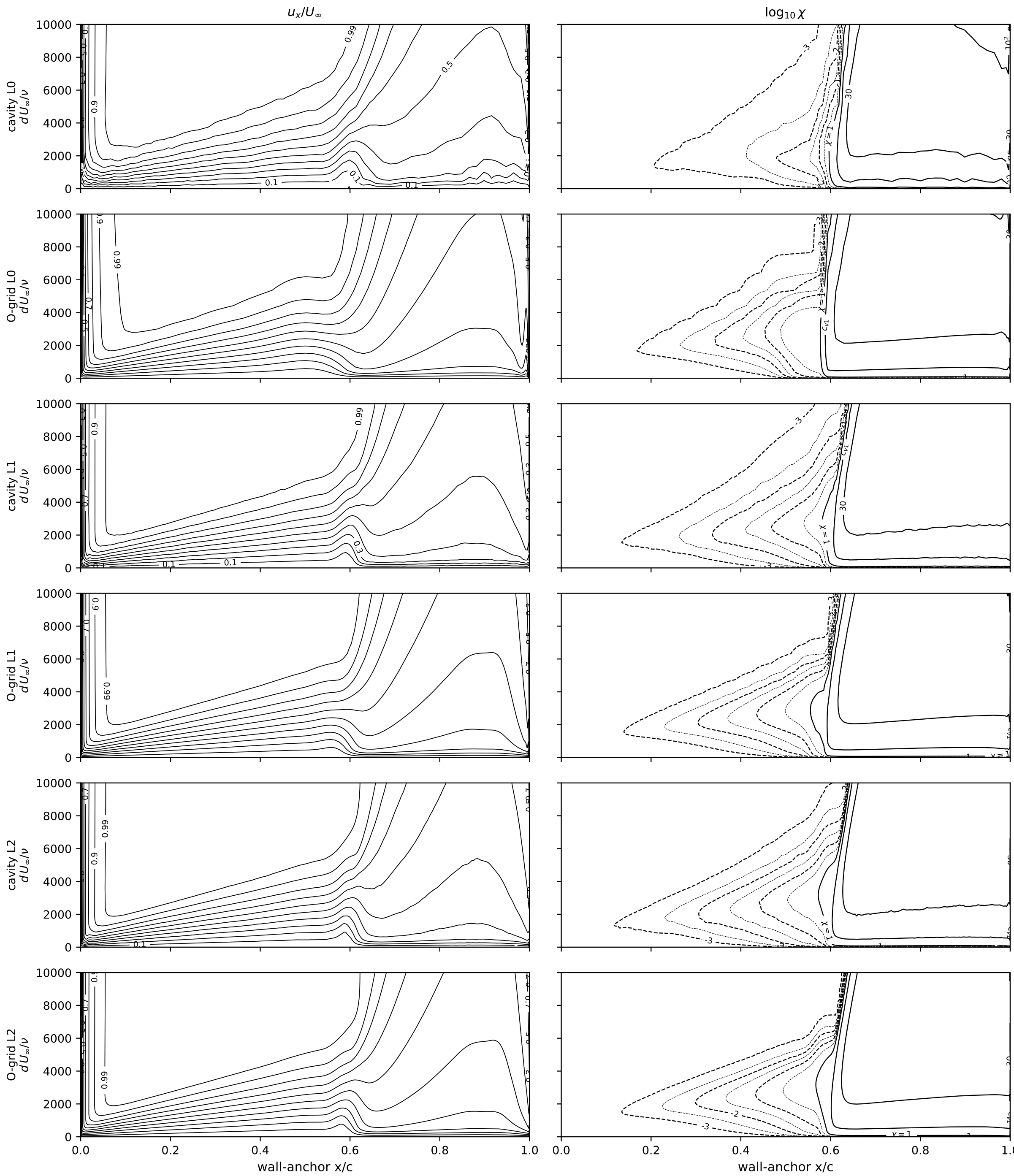

Fig. A8 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = 0°$, lower surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

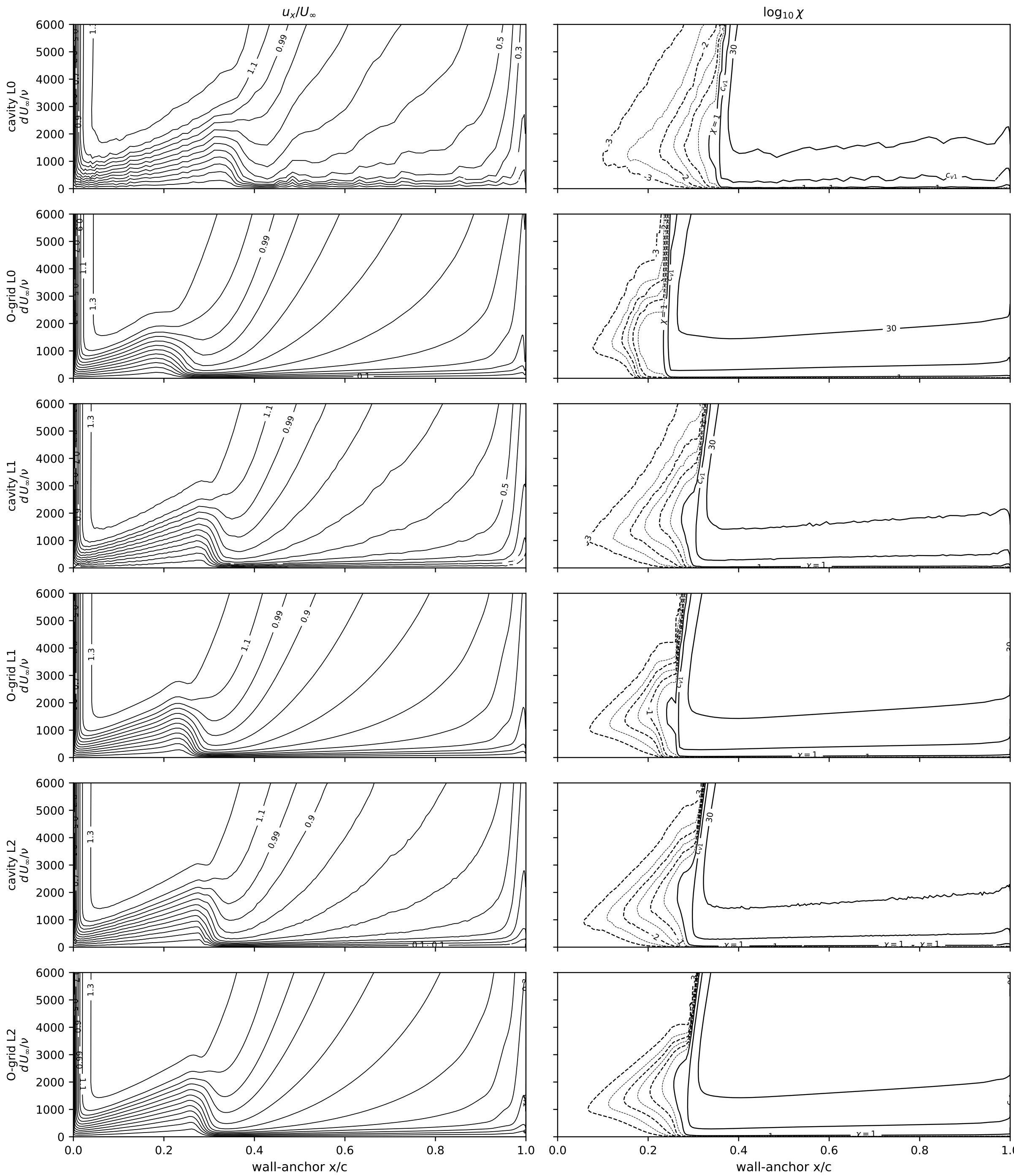

Fig. A9 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = 4°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

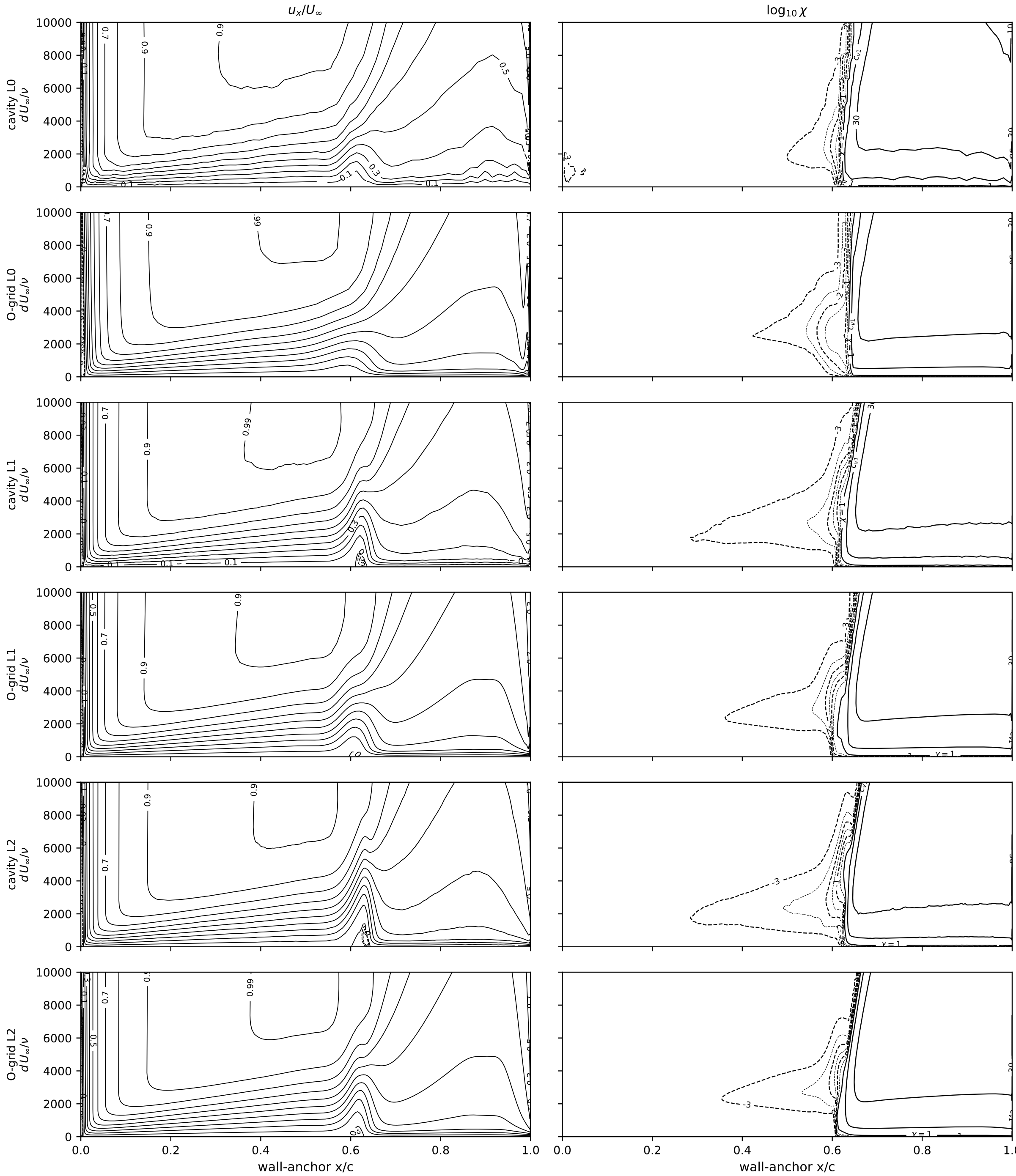

Fig. A10 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = 4°$, lower surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

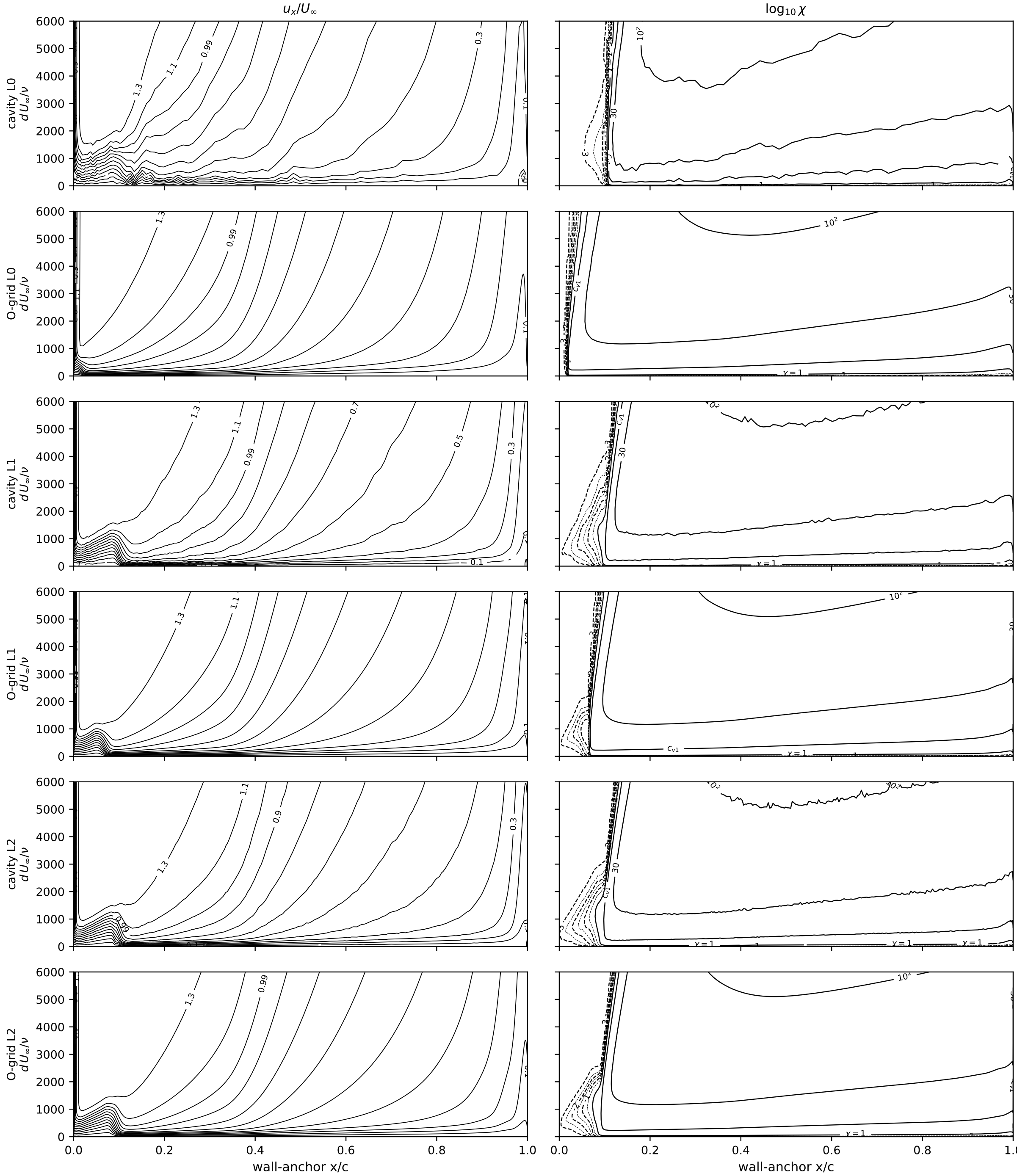

Fig. A11 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = 9°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

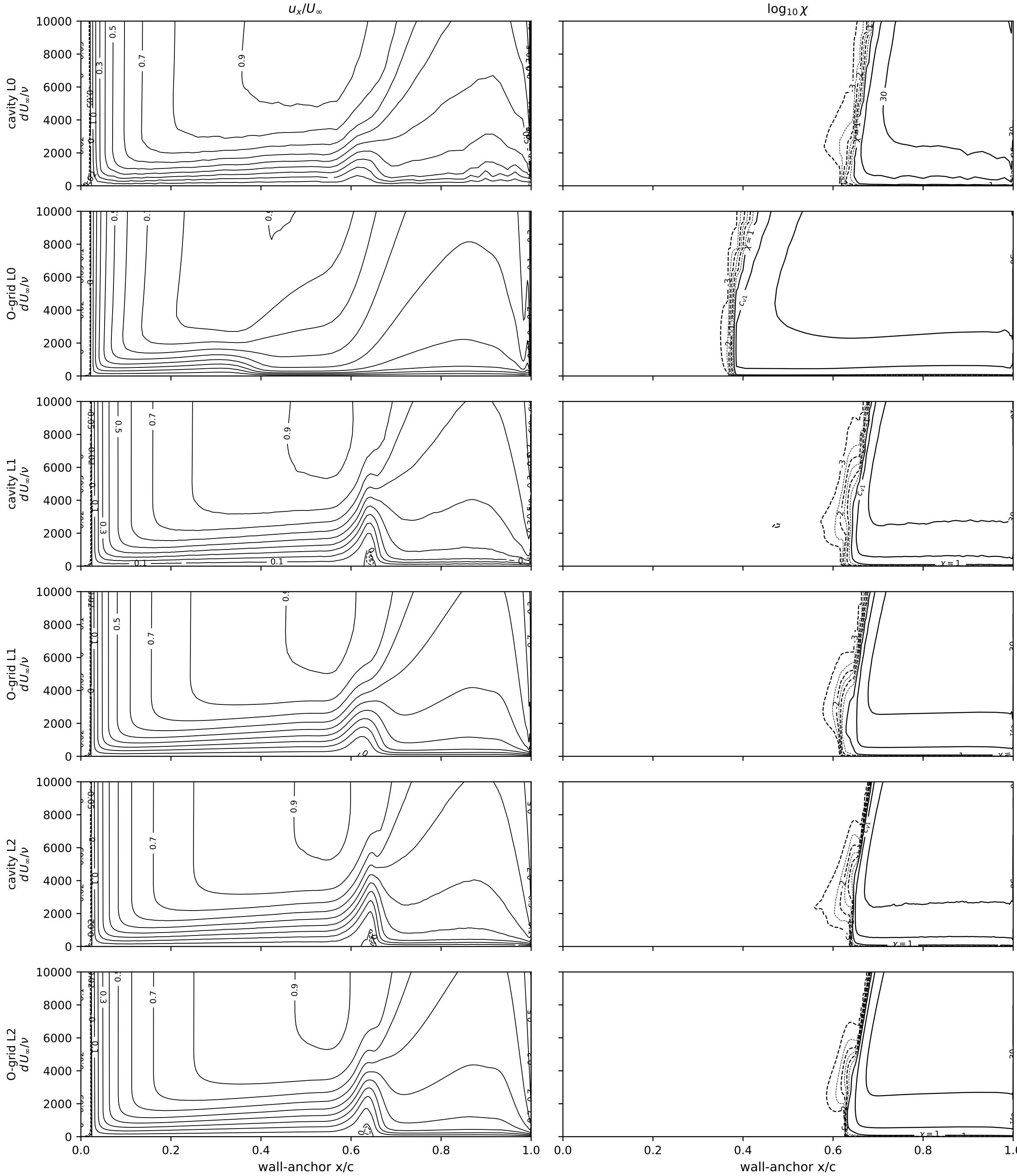

Fig. A12 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = 9°$, lower surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

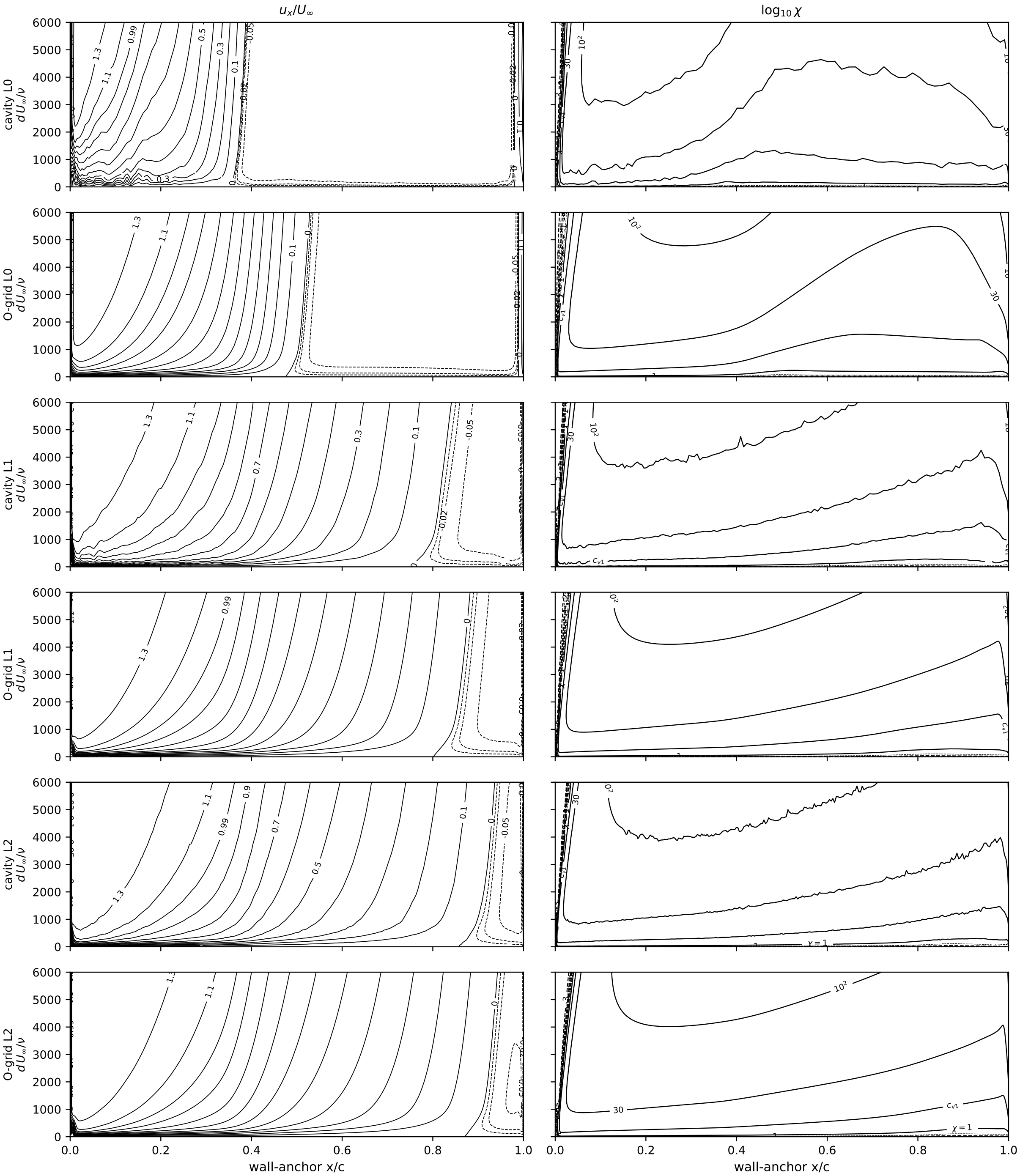

Fig. A13 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = 15°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

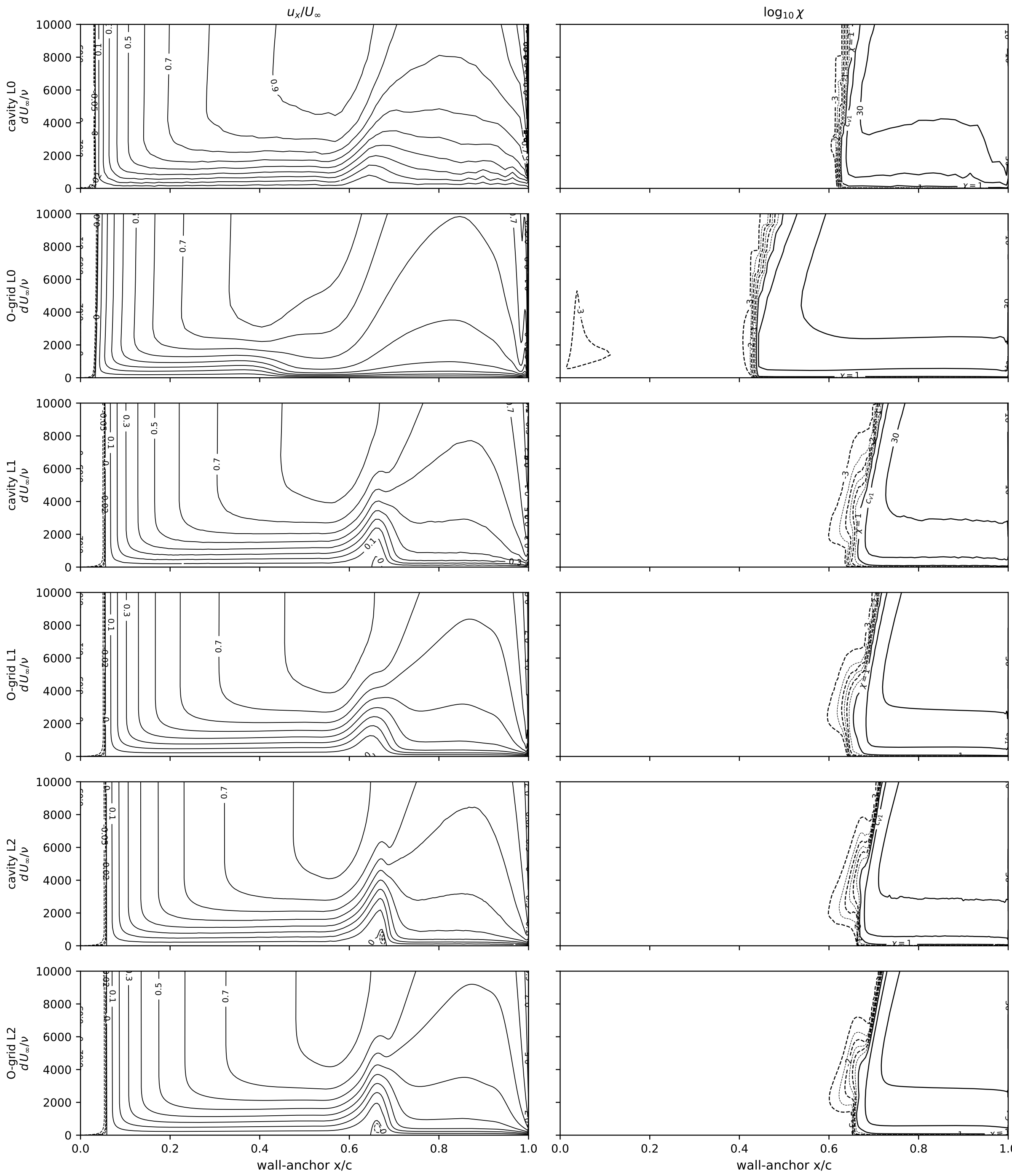

Fig. A14 NLF(1)-0416, $Re = 4 \times 10^6$, $\alpha = 15°$, lower surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

Appendix B: the Eppler 387 benchmark—meshes and contour sheets

The sheets of this appendix cover the Eppler benchmark of Sec. V (mesh close-ups: the grids follow the NLF recipe of Appendix A, Figs. A1 and A2; Table 5 gives the Eppler resolution): the α sweep at $Re = 2 \times 10^5$ in ascending incidence, including the $-2°$ and $8.5°$ extension pair (finest grids only), upper surface only—the lower surface stays attached and featureless throughout the sweep (wall-normal frame $0.0335c$, the 5×-raised handover frame of the Appendix A conventions).

The $\alpha = 5°$ sheet (Fig. B4) also resolves the anatomy of the double peak that the $\max_y \hat{\Omega}\hat{I}$ trace of Fig. 12 (row 2, a wall-normal probe of fixed depth $0.01\,c$, neighbor-smoothed as stated in Sec. IV) shows inside the bubble. The first peak is the lifted laminar shear layer over the recirculation core: at its maximum (0.92 at $x \approx 0.50$) the rate-carrying layer has climbed to the top of the probe window ($0.01\,c$, roughly $y^+ \approx 40$ in the viscous units of the reattached wall)—visible in the sheet as the elevated high-shear band arcing over the bubble. The notch that follows (minimum ≈0.28 at $x = 0.54$; deeper in the raw pointwise trace) is the layer leaving that fixed window, not a collapse of the physical field: probed at full depth, the lifted layer stays saturated ($\hat{\Omega}\hat{I} \approx 1$) just above the window until $x \approx 0.6$, and the full-depth envelope never drops below 0.9 between the peaks. With the strong outer branch out of frame, the trace shows the still-weak near-wall region, then re-rises as the layer descends through reattachment: the second peak (0.82 at $x = 0.615$) is the new near-wall reattachment layer (at a few thousandths of a chord, $y^+ \sim 6$, the sublayer of the forming turbulent wall layer), with concentrated shear at the low advection speed of the reattachment region saturating $\hat{\Omega}\hat{I}$ once more before the attached-boundary-layer level (≈0.08) takes over downstream. The handover crossings ($\chi = 1$ at 0.48, c_{v1} at 0.54, 30 at 0.59) bracket the notch because both track the bubble's growth toward reattachment; the alignment is correlational, not causal.

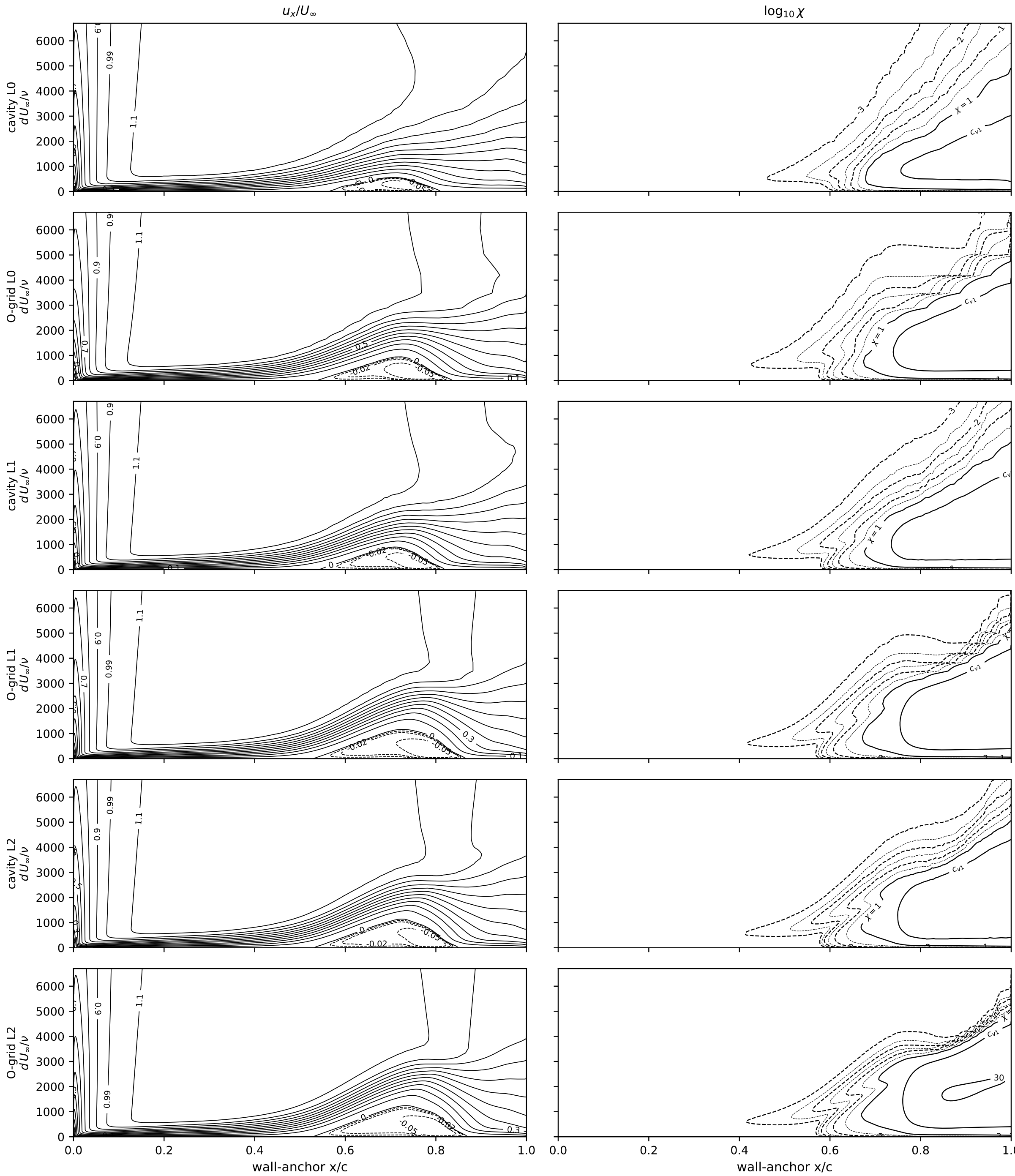

Fig. B1 Eppler 387, $Re = 2 \times 10^5$, $\alpha = -2°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids. The bubble sits at its aft-most station of the sweep ($x/c \approx 0.55$–0.80), and the location is grid-independent across the ladder.

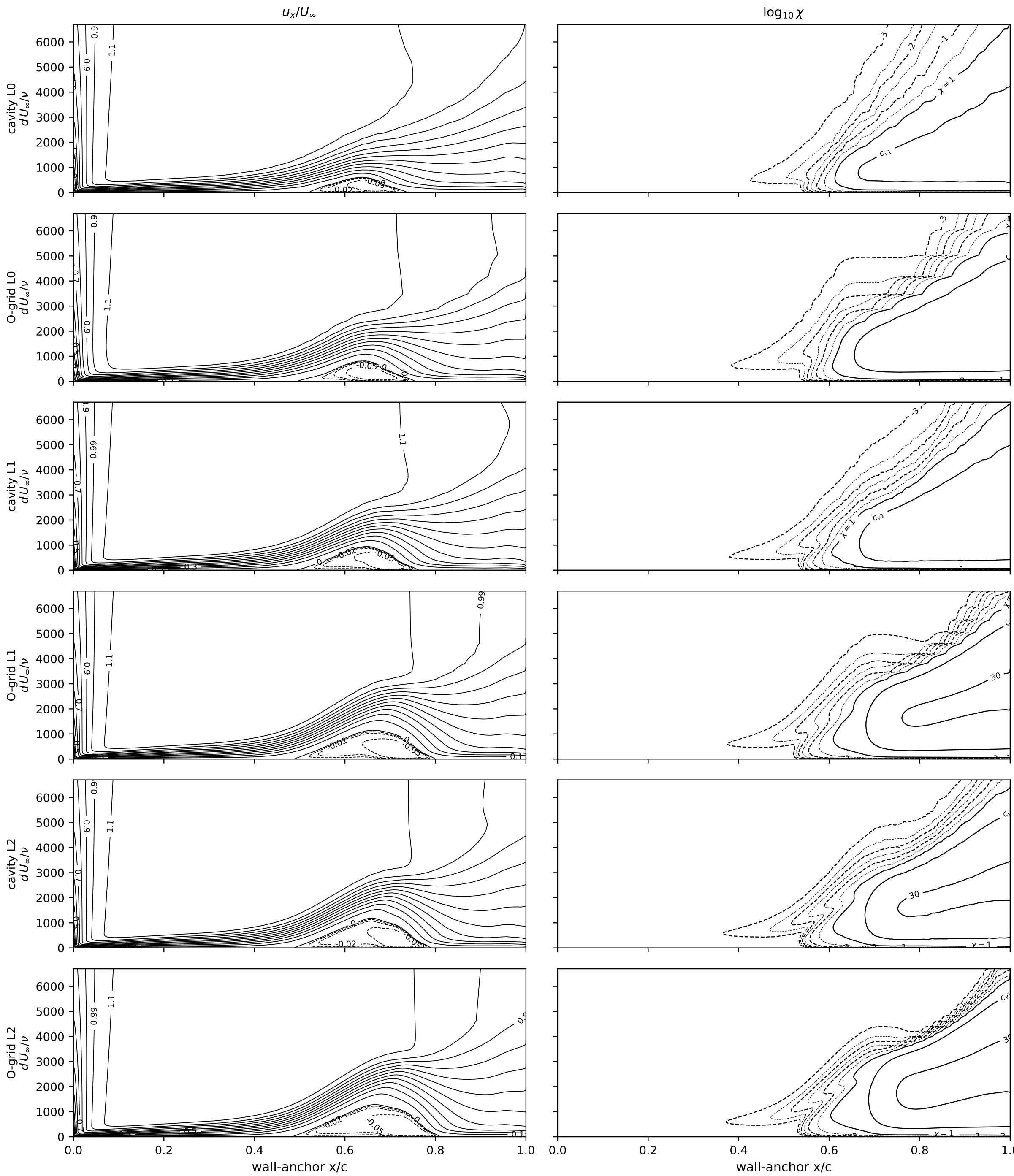

Fig. B2 Eppler 387, $Re = 2 \times 10^5$, $\alpha = 0°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

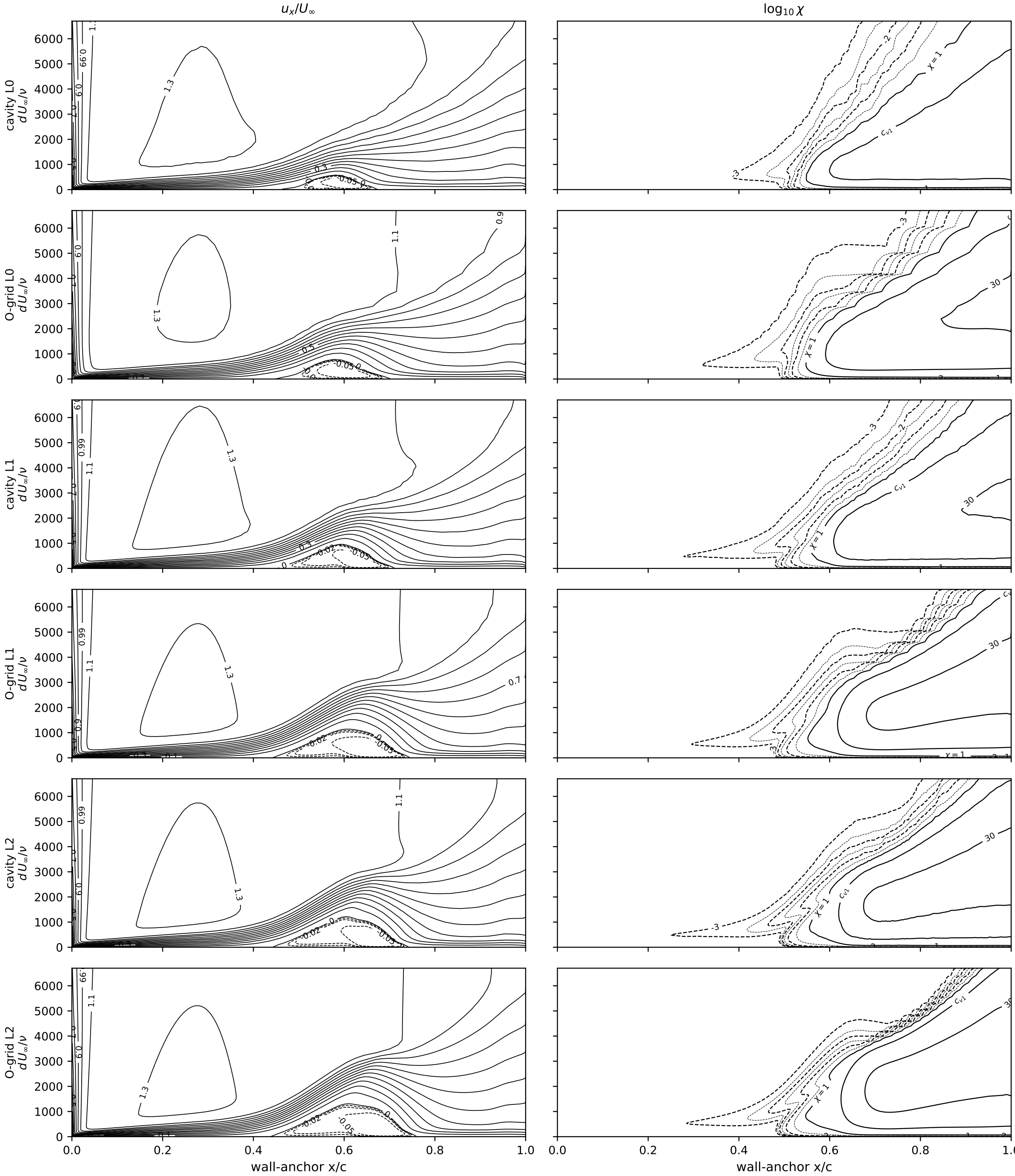

Fig. B3 Eppler 387, $Re = 2 \times 10^5$, $\alpha = 2°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

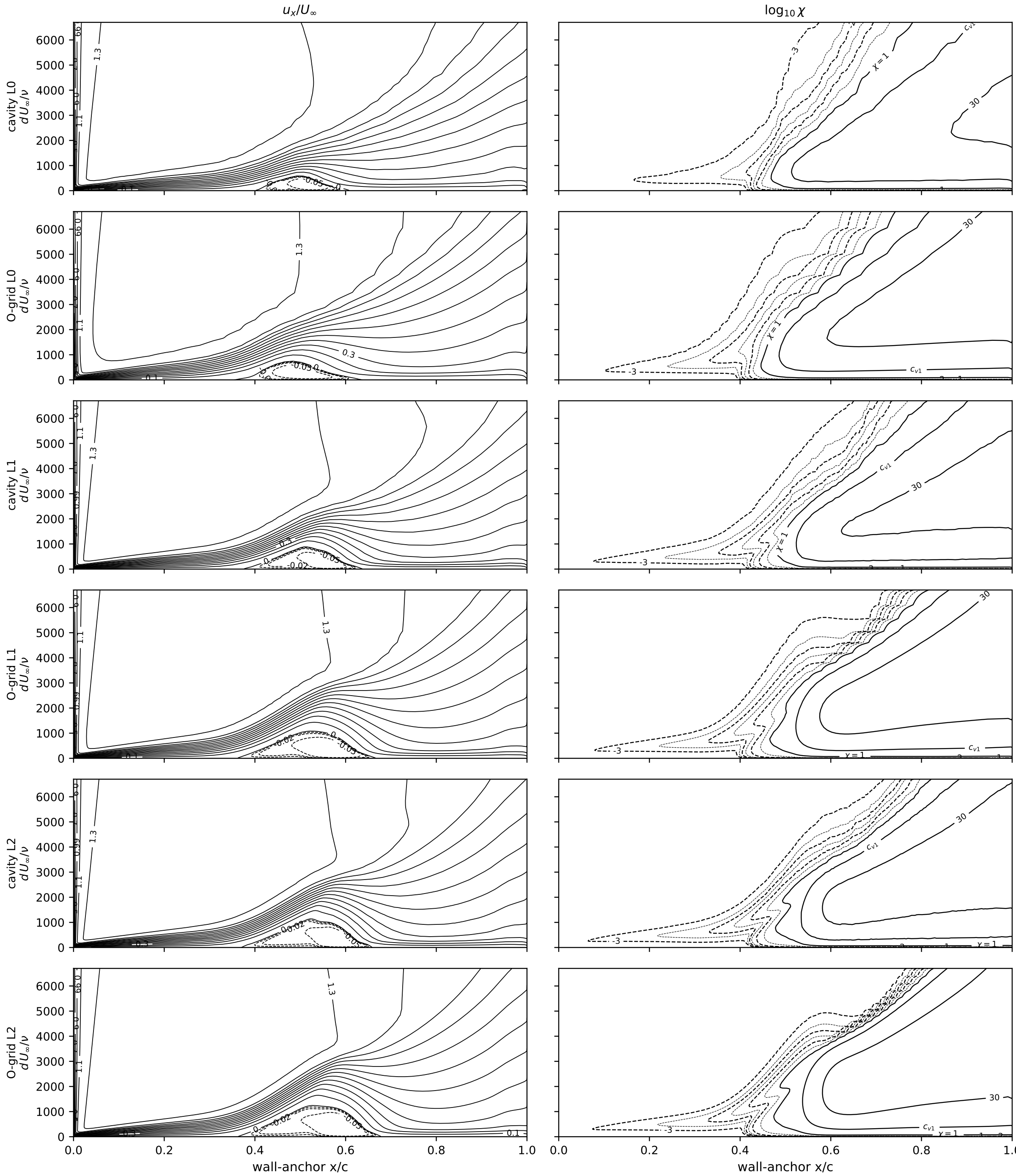

Fig. B4 Eppler 387, $Re = 2 \times 10^5$, $\alpha = 5°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

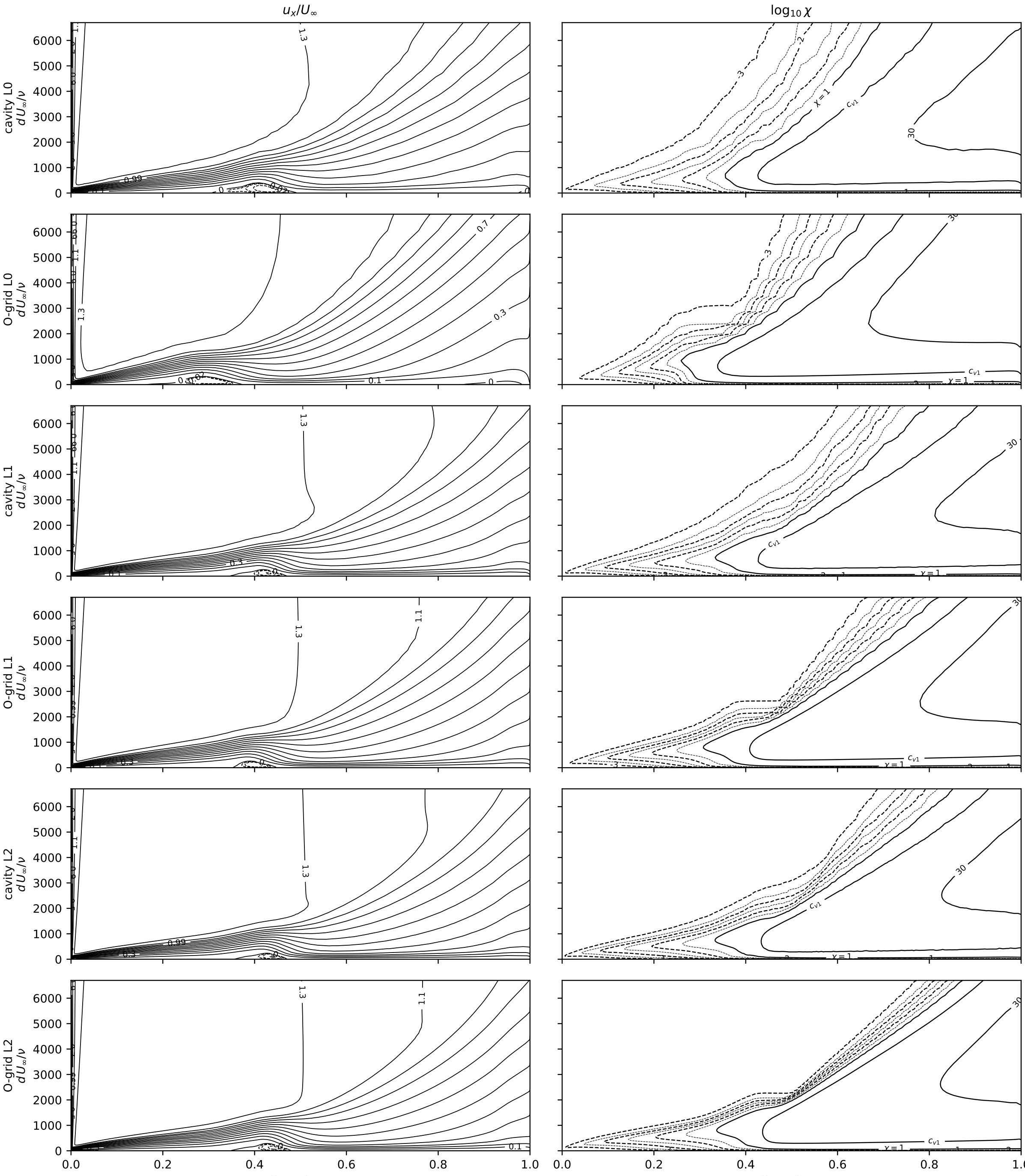

Fig. B5 Eppler 387, $Re = 2 \times 10^5$, $\alpha = 7°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

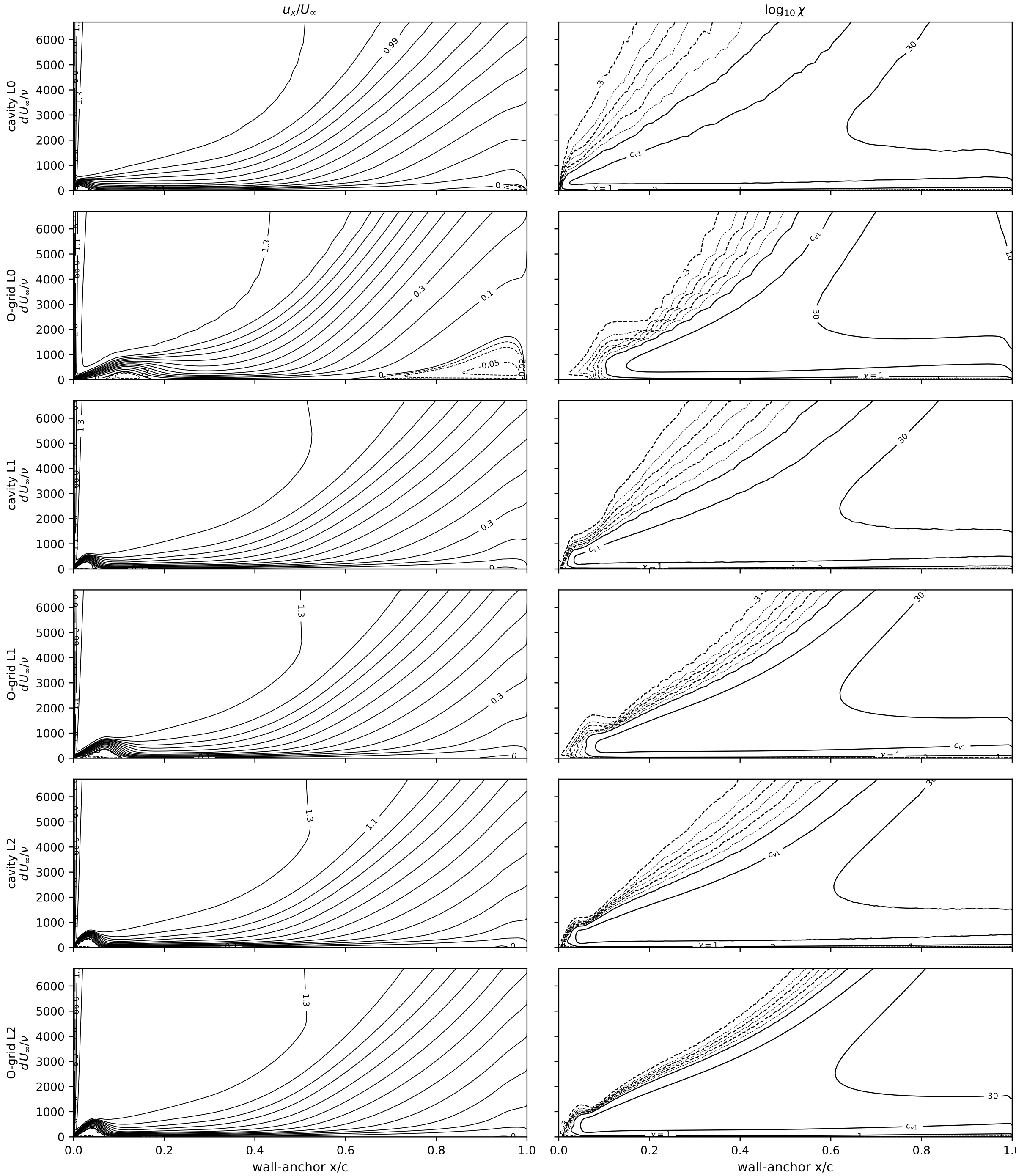

Fig. B6 Eppler 387, $Re = 2 \times 10^5$, $\alpha = 8.5°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids. Past the experimental stall: the layer trips in the leading-edge bubble within the first few percent of chord—closing roughly 4× shorter than the measured 0.03–0.18 c recovery (Fig. 14)—and runs turbulent and attached over the rest of the surface (Fig. 13).

Appendix C: the Eppler 387 Reynolds sweep—contour sheets

One upper-surface sheet per swept Reynolds number of Sec. VI (the $Re = 2 \times 10^5$ anchor is the $\alpha = 5°$ sheet of Appendix B), each covering the full L0–L2 suite on both mesh families; the wall-normal frame scales as $\sqrt{2 \times 10^5/Re}$. At $Re = 10^5$ two additional rows show each family's warm-started L2 solution after its 40 000-iteration extension (Sec. VI): the fields agree with the cold-start L2 rows above them to within the cavity family's residual unsteadiness—the initialization independence of the converged state, seen in the field variables. Table C1 tabulates the sweep's L2 section coefficients against the e^9 reference and the tunnel data (discussed in Sec. VI).

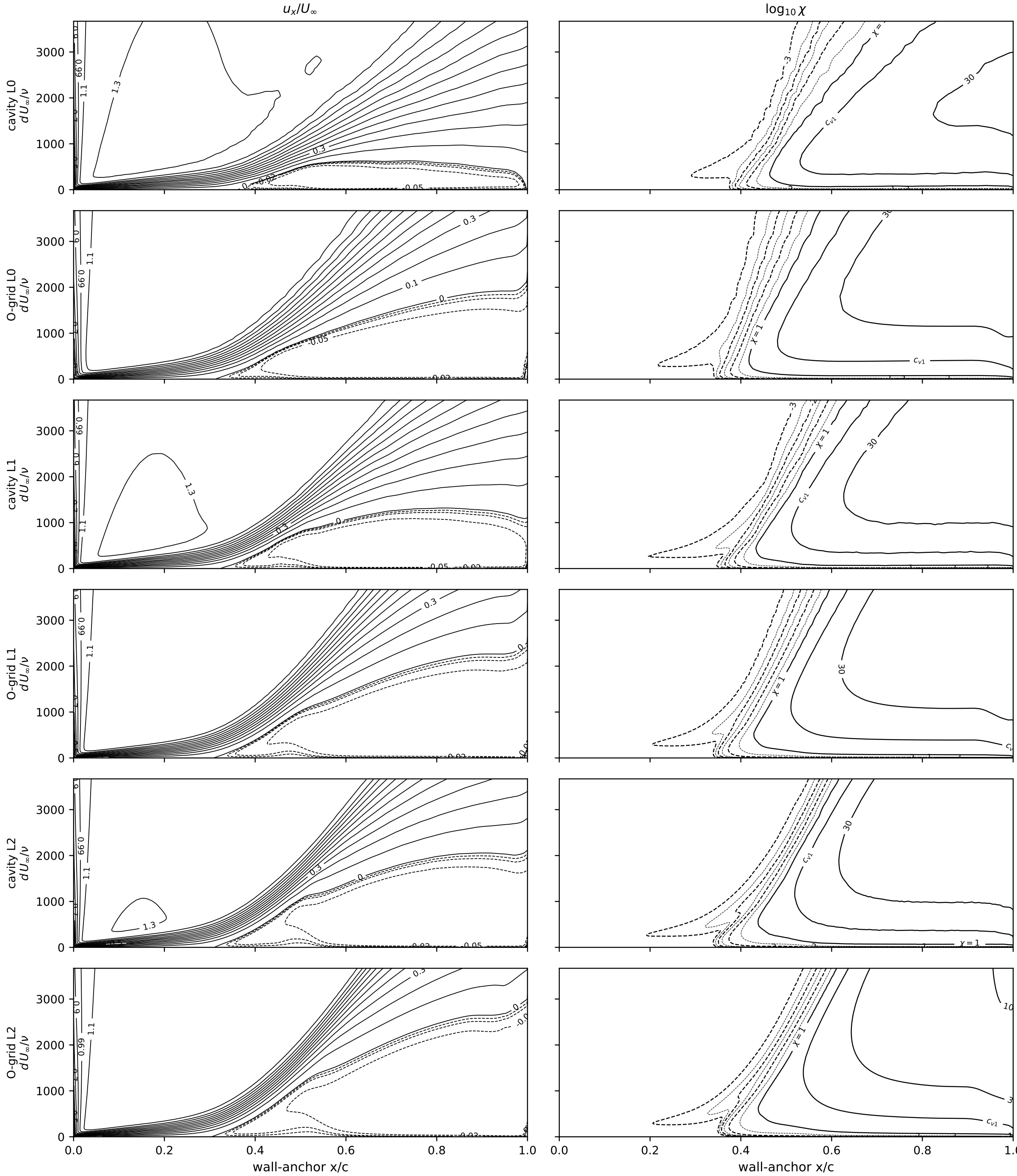

Fig. C1 Eppler 387, $Re = 6 \times 10^4$, $\alpha = 5°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

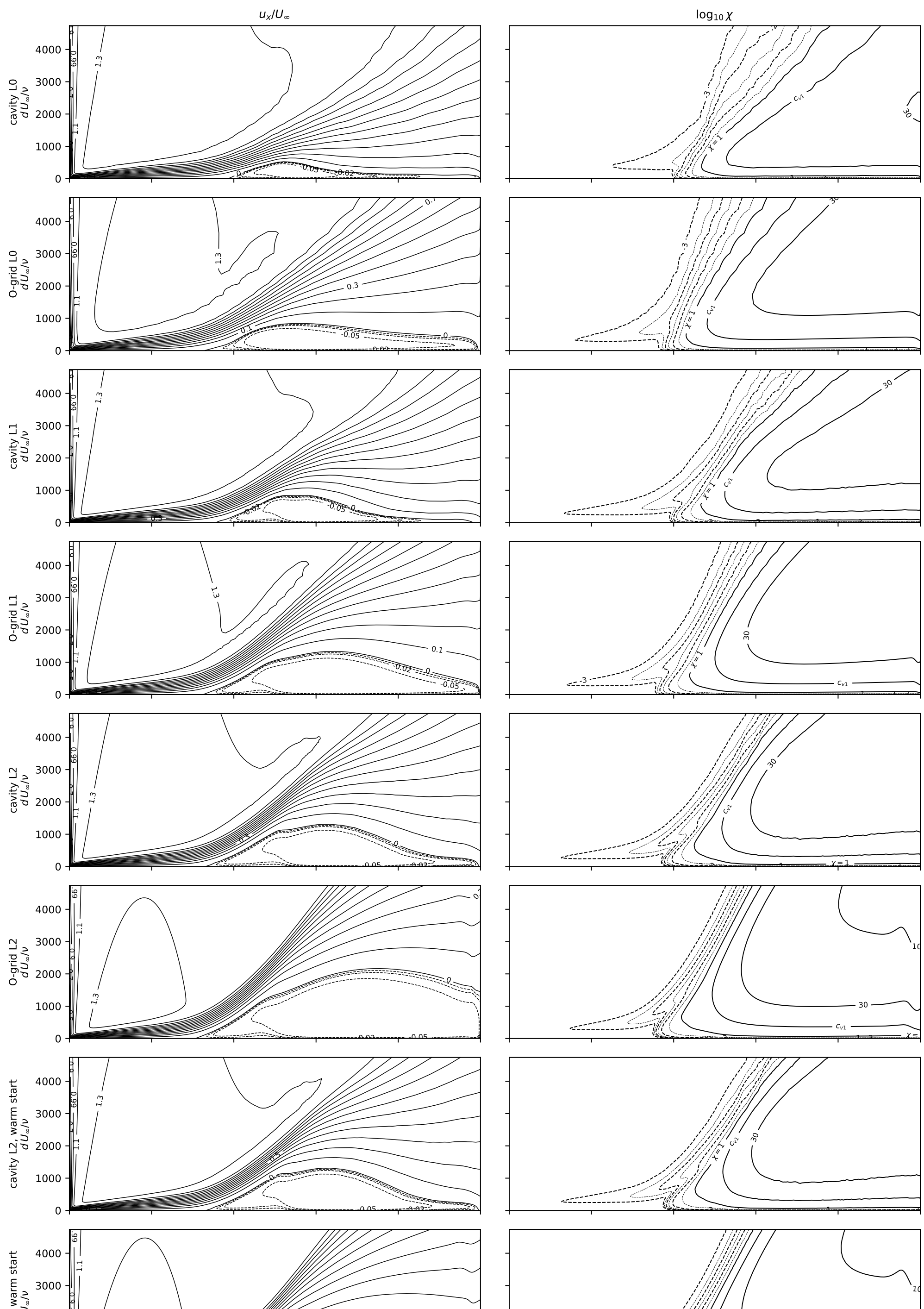

Fig. C2 Eppler 387, $Re = 10^5$, $\alpha = 5°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids plus the two extended warm-started L2 solutions (bottom rows), agreeing with the cold-start L2 rows to within the residual unsteadiness (Sec. VI).

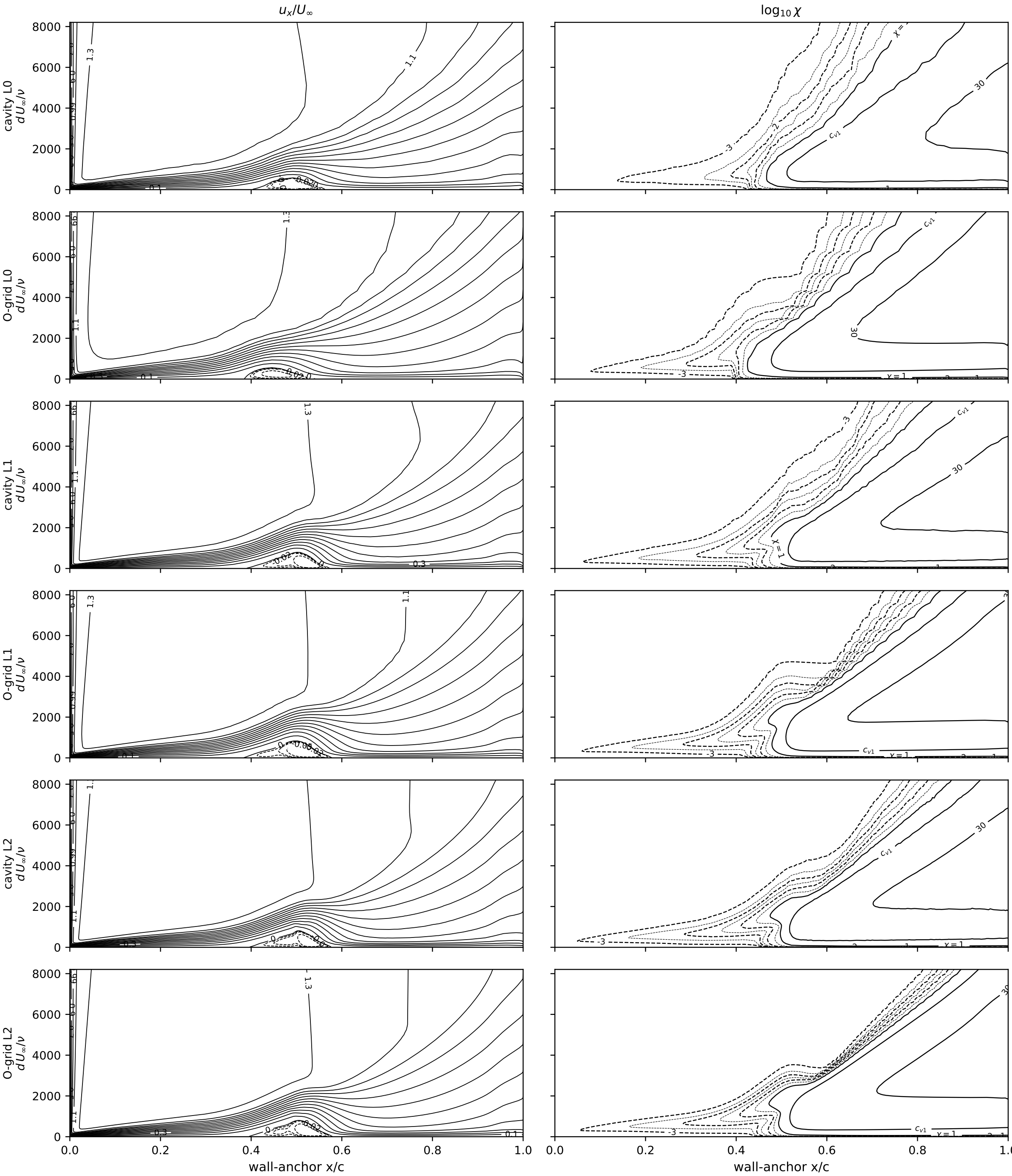

Fig. C3 Eppler 387, $Re = 3 \times 10^5$, $\alpha = 5°$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

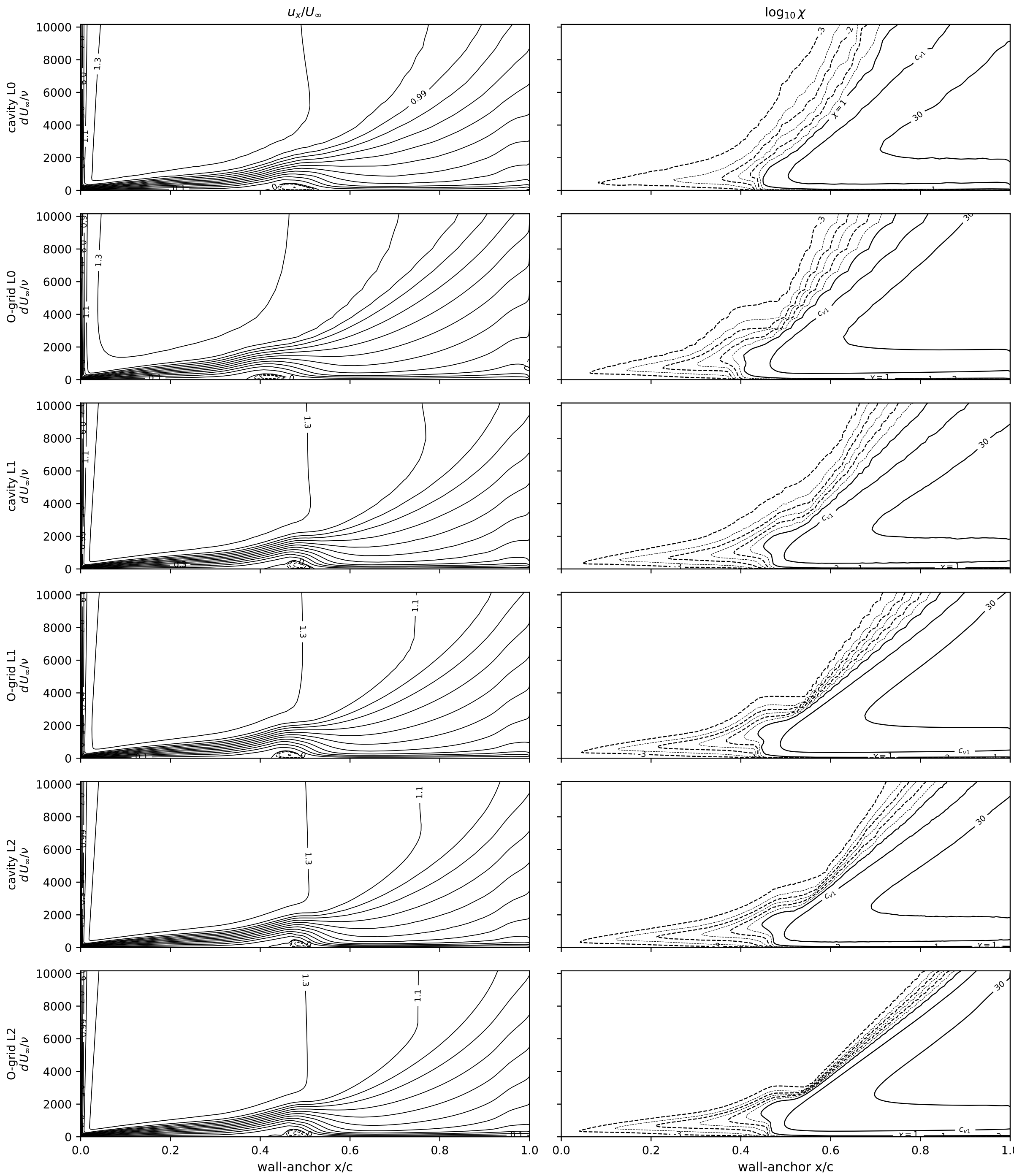

Fig. C4 Eppler 387, $Re = 4.6 \times 10^5$, $\alpha = 5^\circ$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on all six grids.

Table C1 Section coefficients for the Eppler 387 at $\alpha = 5°$ across the Reynolds sweep: lift, drag, and quarter-chord pitching moment from the two mesh families' finest (L2) grids (the same grids as the $Re = 2\times10^5$ study; the L0/L1 companions appear in Figs. 16–17 and 15). The families' L2 drags agree to 0.1–0.2×10^{-3} at 2–4.6×10^5 and diverge at 10^5 as described in the text; the 10^5 states are initialization-independent (Sec. VI). Also listed: the e^9 panel reference (mfoil; XFOIL at 6×10^4, where the two implementations visibly differ, and at $Re = 4.6\times10^5$, where mfoil's stagnation-point iteration diverges on the thin high-Re bubble), and the LTPT measurement (McGhee et al., NASA TM-4062, Table B1, read at the tabulated angle nearest 5°: $\alpha = 4.99°$–$5.06°$). The computed moment is transferred from the solver's leading-edge reference to the quarter chord, $c_m = c_{m,\mathrm{LE}} + \frac{1}{4}(c_l \cos\alpha + c_d \sin\alpha)$. Repeat experimental points at $Re \geq 10^5$ agree to within 0.006 in c_l and 0.0003 in c_d; at $Re = 6\times10^4$ the flow is bistable and the repeat scatter is an order of magnitude larger (see text).

	SA-AI, structured L2			SA-AI, unstructured L2			e^9 panel			Exp (LTPT)		
Re	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m
6×10^4	0.636	0.0533	−0.098	0.712	0.0506	−0.103	0.869‡	0.0411‡	−0.102‡	0.838	0.0439	−0.114
1×10^5	0.795	0.0410	−0.102	0.877	0.0307	−0.096	0.952	0.0213	−0.087	0.873	0.0237	−0.089
2×10^5	0.927	0.0145	−0.081	0.938	0.0143	−0.083	0.960	0.0128	−0.082	0.891	0.0138	−0.081
3×10^5	0.941	0.0107	−0.080	0.950	0.0108	−0.082	0.962	0.0103	−0.082	0.901	0.0114	−0.080
4.6×10^5	0.950	0.0087	−0.081	0.959	0.0089	−0.083	0.966	0.0086	−0.082	0.914	0.0093	−0.081

‡The only condition in this paper where the e^9 implementations visibly differ; XFOIL, whose forces sit closer to the measurement, is quoted as the reference. mfoil gives $c_l = 0.889$, $c_d = 0.0368$, $c_m = -0.097$—11% less drag—commensurate with the bistability of the flow at this Reynolds number, while FlexFoil sides with XFOIL: $c_l = 0.868$, $c_d = 0.0406$, $c_m = -0.102$ (the cross-check statement is in Sec. IV).

Appendix D: the Daedalus wing—definition, temporal-units conversion, and force tables

Geometry, meshes, and solutions. The wing blends the DAE-11, DAE-21, and DAE-31 sections [75] along the span (transition stations $\eta = 0.30$ and 0.60), holds the 0.917 m chord to $\eta = 0.88$, and tapers linearly to 0.35 m at the tip with a straight quarter-chord line. The structured family stacks two-dimensional O-grids spanwise, pinching the section to a line beyond the tip. The unstructured family is a classical three-dimensional unstructured mesh, distinct from the two-dimensional cavity construction of Appendices A and B: anisotropic boundary-layer prisms are grown from the triangulated wing surface, the far field is an octree of hexahedra, and tetrahedra join the two, with the tetrahedron size at the interface seeded from the area-equivalent edge length of the surface triangles and the prism stack terminated where its height reaches the local isotropic scale. Sizes, spanwise-station and around-the-section resolution are tabulated in the main text (Table 6). All solutions use the airfoil-study solver settings with the final kernel (Sec. IX) and are converged to steady or small-limit-cycle force histories (largest cycle: the structured-L1 4° case at 1.4×10^{-2} peak-to-peak in C_L, an order above the other seventeen); tabulated coefficients are means over the final 500 pseudo-steps (51 logged samples).

Temporal-units conversion. The comparisons of Sec. IX and of this appendix's ledger between the kernel ceiling and the e^N reference convert Drela's envelope slope into a growth rate per unit time in multiples of the profile's peak vorticity,

$$a_{\text{eff}} \;=\; \underbrace{\frac{dN}{dRe_\theta}\,\frac{m(H)+1}{2}\,\ell(H)}_{dN/dx\,\cdot\,\theta} \;\cdot\; \frac{c/U_e}{\omega_{\text{peak}}\theta/U_e},$$

with $\ell(H)$ and $m(H)$ the Falkner–Skan similarity factors of Drela's spatial conversion and c the packet speed, taken as the flow speed at the profile's inflection point. On the separation-limit profile ($H = 3.98$: $\theta/\delta_s = 0.585$, $f''_{\max} = 0.361$, hence $\omega_{\text{peak}}\theta/U_e = 0.212$, $c = 0.503\,U_e$) this gives $a_{\text{eff}} = 0.066 = 0.35\,a_{\max}$: the anchor condition of the predecessor kernel (a shaped, sigmoidal rate with its own three-anchor constants; cf. the linear clip of Eq. (8)) pinned its sigmoid's flank at a third of the ceiling. Profiles extracted from the wing's bubble (the predecessor-kernel campaign's structured L2, $\eta = 0.31$, $\alpha = 5°$, $x/c = 0.48$–0.53—that kernel's whole bubble; the final-kernel bubble at the same station spans 0.47–0.64 c) measure $\omega_{\text{peak}}\theta/U_e = 0.212$—unchanged from the attached value; a marginal bubble has not de-concentrated its vorticity—with the shear-layer center near $3.5\,\theta$ at $0.45\,U_e$. mfoil's envelope slope there (≈ 67 per chord) converts to $a_{\text{eff}} \approx 0.09$, half the ceiling, while the model's solved takeoff (≈ 200 per chord) rides its own $\tilde{\nu}$ peak at $2.4\,\theta$ and $0.22\,U_e$ and so corresponds to $a \approx 0.13$: the factor-of-three spatial excess decomposes into 1.45× in temporal rate (saturated sigmoid versus wall-damped mode) and 2.05× in ledger speed (the $\tilde{\nu}$ peak rides below the physical critical layer at half its convection speed). This ledger, measured on the predecessor-kernel campaign (the final-kernel recomputation of Sec. IX supersedes those solutions but not this diagnostic, which is a property of the ungated rate), is the in-place excess intrinsic to the advected temporal rate (§II.A); on frozen separated profiles the resolved band returns the spatial growth to 0.9–1.9× the correlation. The correlation itself is the weaker reference at

Table D1 Daedalus wing totals by grid level: SA-AI on both mesh families against the AVL+XFOIL reference trimmed to the RANS lift (finest structured level) at each incidence: Trefftz-plane induced drag at the trim plus XFOIL profile drag at the trimmed strip loading, matched local c_l and N_{crit} = 13.6, integrated over span (`daedalus/avl_fixed_cl_reference.py`; the reference drag at the marginally shifted trim lifts is unchanged at the printed digits).

		structured O-grid		unstructured	
α	level	C_L	C_D	C_L	C_D
4°	L0	1.0244	0.01958	1.0156	0.02134
	L1	1.0279	0.01946	1.0232	0.01941
	L2	1.0284	0.01969	1.0219	0.01976
	AVL+XFOIL	1.0284	0.02125		
5°	L0	1.1254	0.02219	1.1145	0.02427
	L1	1.1307	0.02197	1.1248	0.02201
	L2	1.1290	0.02212	1.1236	0.02223
	AVL+XFOIL	1.1290	0.02315		
6°	L0	1.2262	0.02505	1.2171	0.02713
	L1	1.2321	0.02465	1.2256	0.02481
	L2	1.2282	0.02474	1.2238	0.02491
	AVL+XFOIL	1.2282	0.02530		

these depths: Orr–Sommerfeld rates computed on the wall-bounded tanh family of Dini et al. [79] (their stand-in for Green's profiles [80], the family bubble measurements support) saturate with depth and side with the kernel—at $H \approx 13$ the kernel's inviscid limit matches the Orr–Sommerfeld rate while the linearly extrapolated correlation is ~2× above both (`repro/analytic/explore_green_tanh_family.py`).

Vortex-lattice + section-polar reference and diagnostics. The AVL model uses 12×40 cosine-spaced vortex-lattice panels per semispan with the blended sections at five stations, trimmed with AVL's fixed-lift constraint to the RANS C_L at each incidence; induced drag is taken from the Trefftz-plane total at that trim, distributed as $c_l \alpha_i$. (The near-field induced-drag total is a small difference of large terms at this aspect ratio and demands double precision: the single-precision AVL build returned $C_{D\text{ind}}$ 31% above $C_{D\text{ff}}$ on the 12×40 lattice and NaN on 24×80, while a double-precision rebuild returns $C_{D\text{ind}} = C_{D\text{ff}}$ to within a tenth of a count at every lattice; $C_{D\text{ff}}$ itself is insensitive to both precision and lattice, moving a quarter percent from 12×40 to 24×80.) Profile drag is XFOIL at each station's local c_l, chord Reynolds number, and $N_{\text{crit}} = 13.6$, integrated over span. RANS transition fronts are read per spanwise strip as the C_f-rise location, and bubbles as intervals of negative streamwise C_f. The χ columns of Figs. 23, D1, and D2 map max $\tilde{\nu}/\nu$ over a band extending 5% of local chord from the wall, each volume node scatter-reduced to its nearest wall node; the e^N strip rows of the same figures use FlexFoil at each station's AVL c_l and chord Reynolds number, its amplification drawn as $\chi_\infty e^N$ through the identical color map so the two transition contours ($\chi = 1$ and $N = N_{\text{crit}}$) are directly comparable. The $\alpha = 5°$ and 6° surface maps (Figs. D1 and D2; companions to Fig. 23 of the main text) follow.

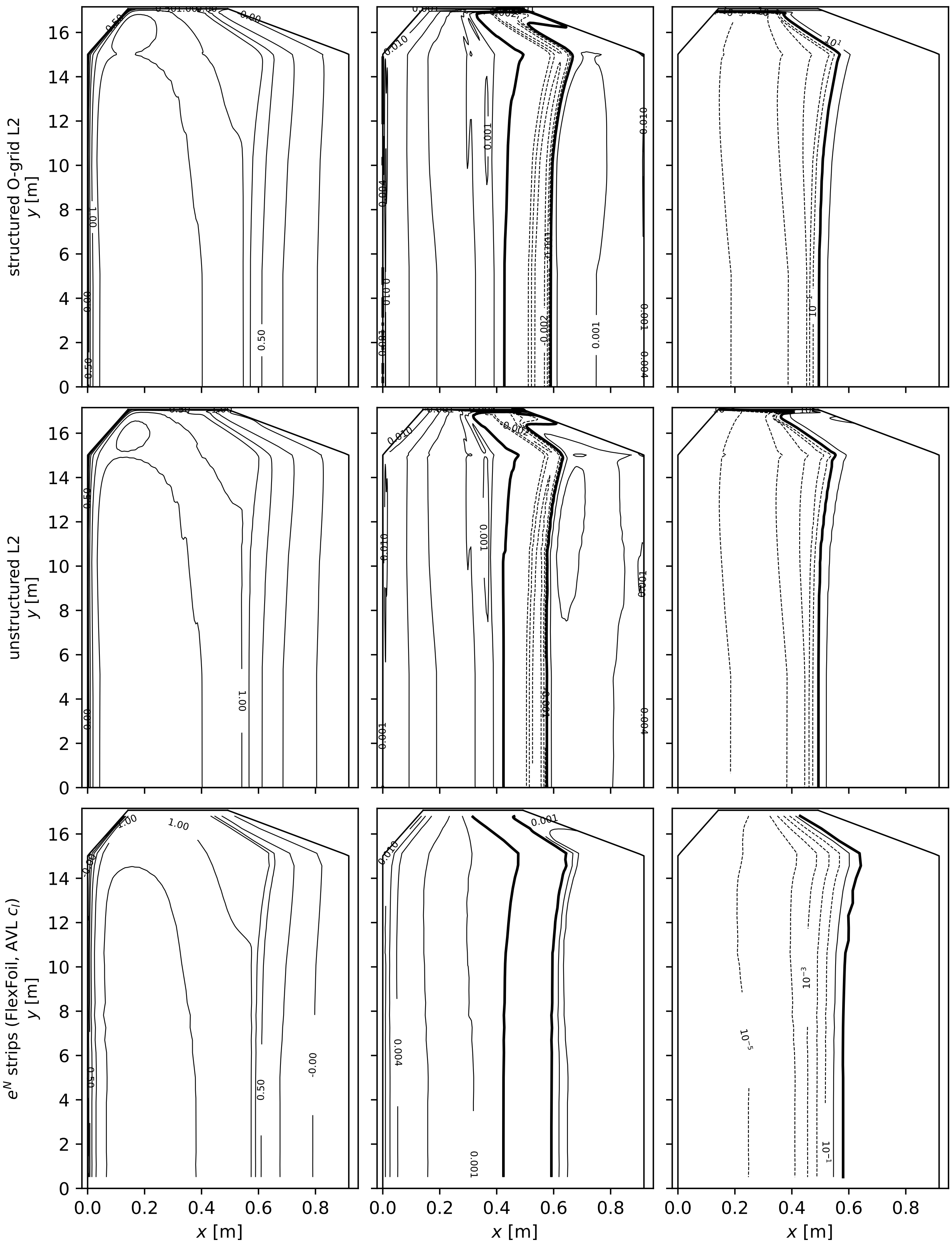

Fig. D1 As Fig. 23, at $\alpha = 5°$.

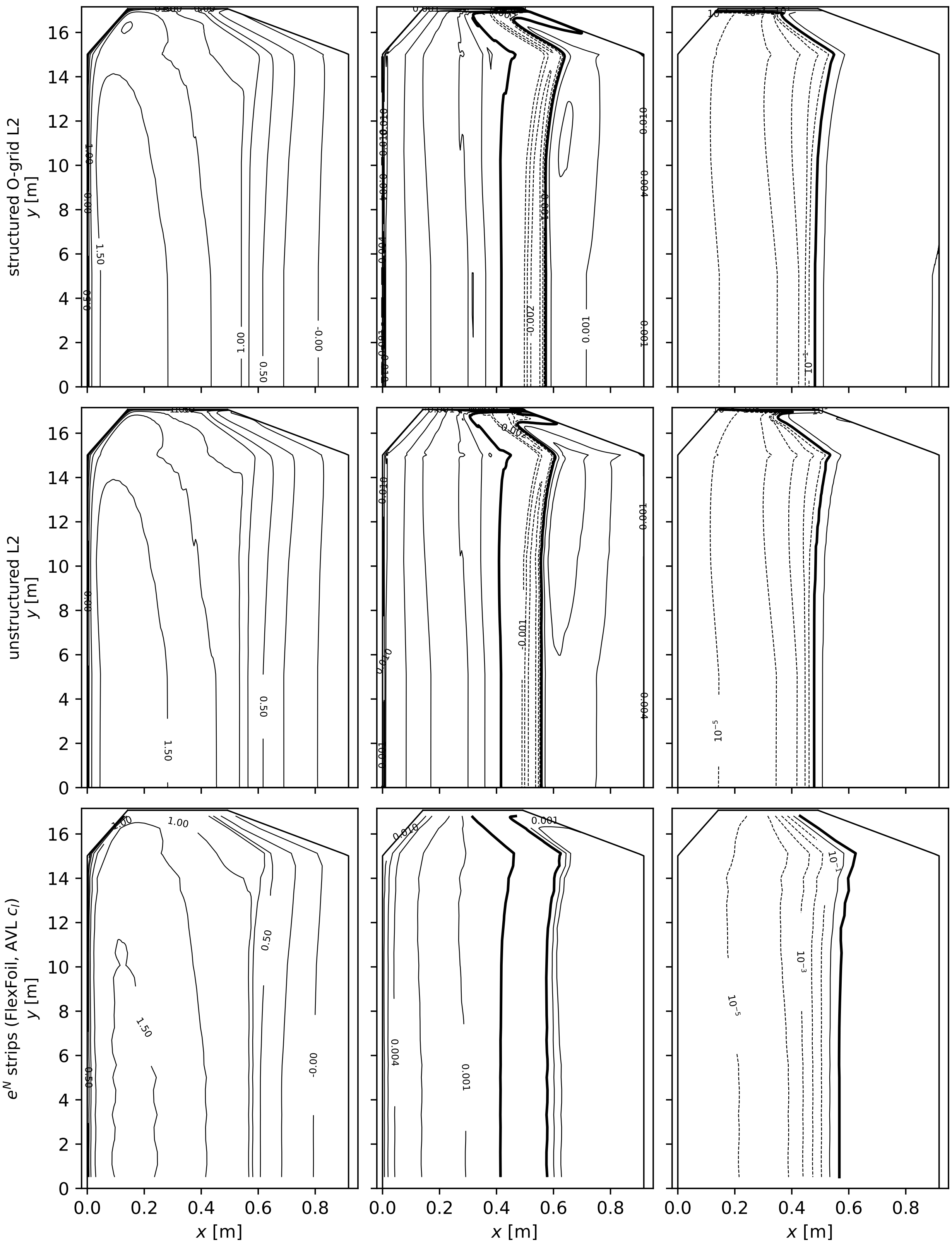

Fig. D2 As Fig. 23, at $\alpha = 6°$.

Sectional diagnostic sheets. Figures D3, D4, and D5 carry the section-cut diagnostics behind the drag-slope discussion of Sec. IX: at $\eta = 0.10$, 0.75, and 0.92 (the last in the taper, at its locally reduced chord Reynolds number), columns $\alpha = 4°$, 5°, 6°, five rows per station—the probe-maximum Re_Ω against the onset threshold $Re_\Omega^c(\hat{\Omega}\hat{I})$ evaluated at the local rate (dash-dotted; drawn from the finest-grid probes), the probe-maximum $\hat{\Omega}\hat{I}$ rate (neighbor-smoothed before the maximum, as the airfoil suites; log scale, as Fig. 2), the near-wall max χ with the strip's e^N amplification, $-C_p$, and streamwise C_f. Solid lines are the structured family, dashed the unstructured, all three grid levels with line weight increasing under refinement (the line-weight convention of the airfoil suites); dotted the AVL-trimmed FlexFoil/XFOIL strip at the same station.

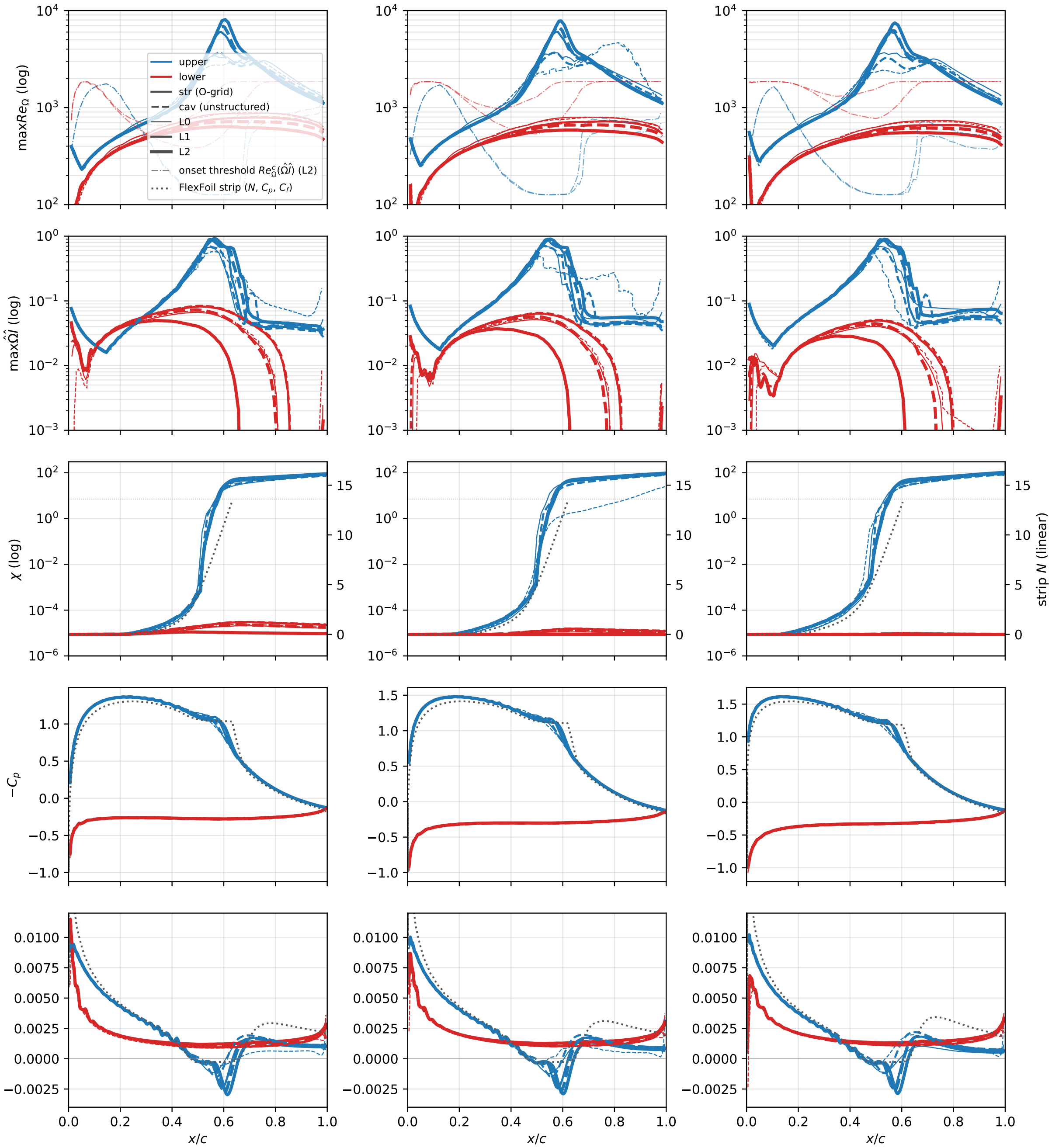

Fig. D3 Daedalus wing section cut at $\eta = 0.10$, $Re_c = 5.0 \times 10^5$: sectional diagnostics on all six grids (structured solid, unstructured dashed; line weight L0→L2 by refinement) against the AVL-trimmed FlexFoil/XFOIL strip (dotted) at $\alpha = 4°, 5°, 6°$ (columns).

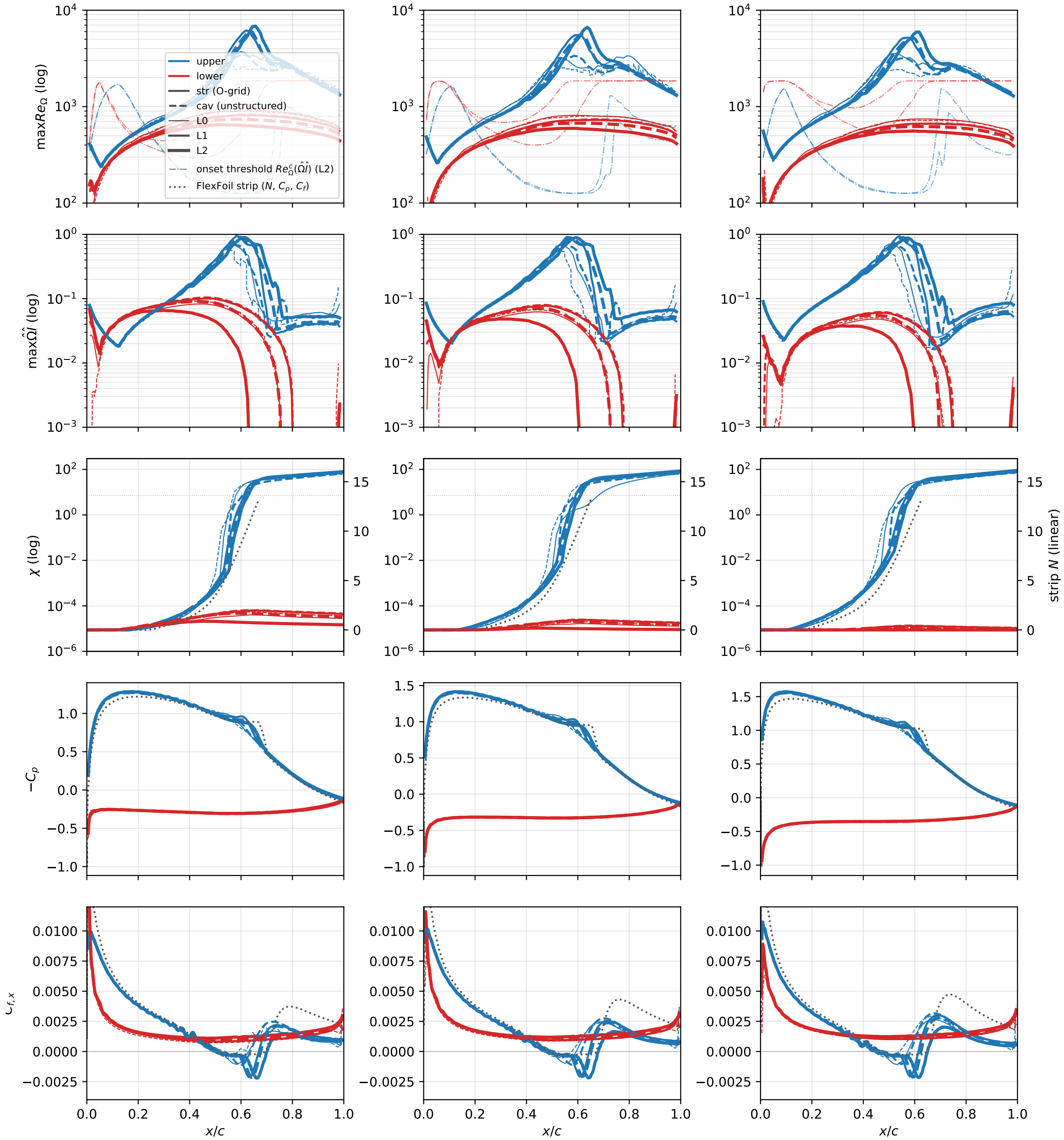

Fig. D4 As Fig. D3, at $\eta = 0.75$.

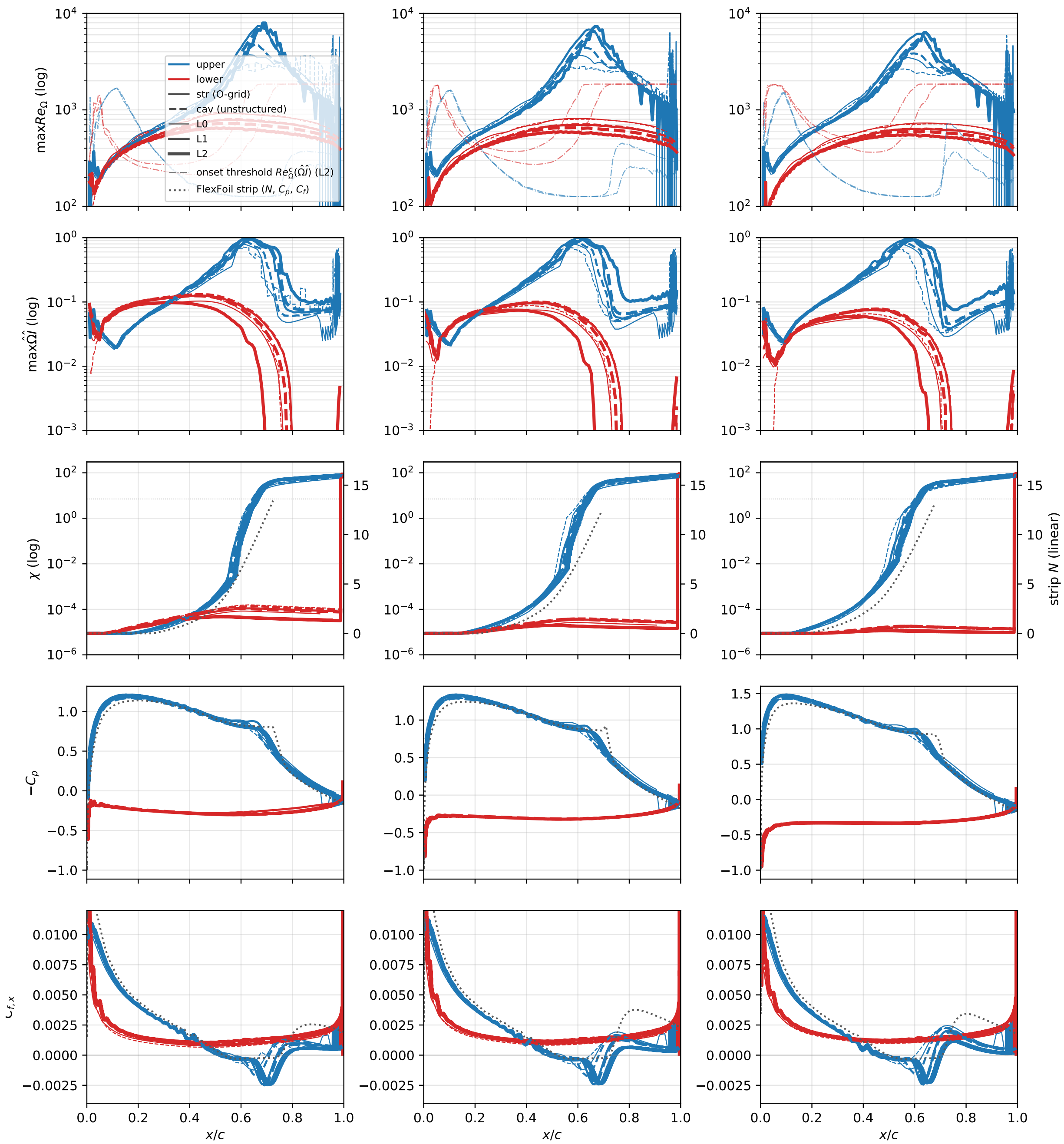

Fig. D5 As Fig. D3, at $\eta = 0.92$—inside the taper, chord 0.728 m and $Re_c = 4.0 \times 10^5$.

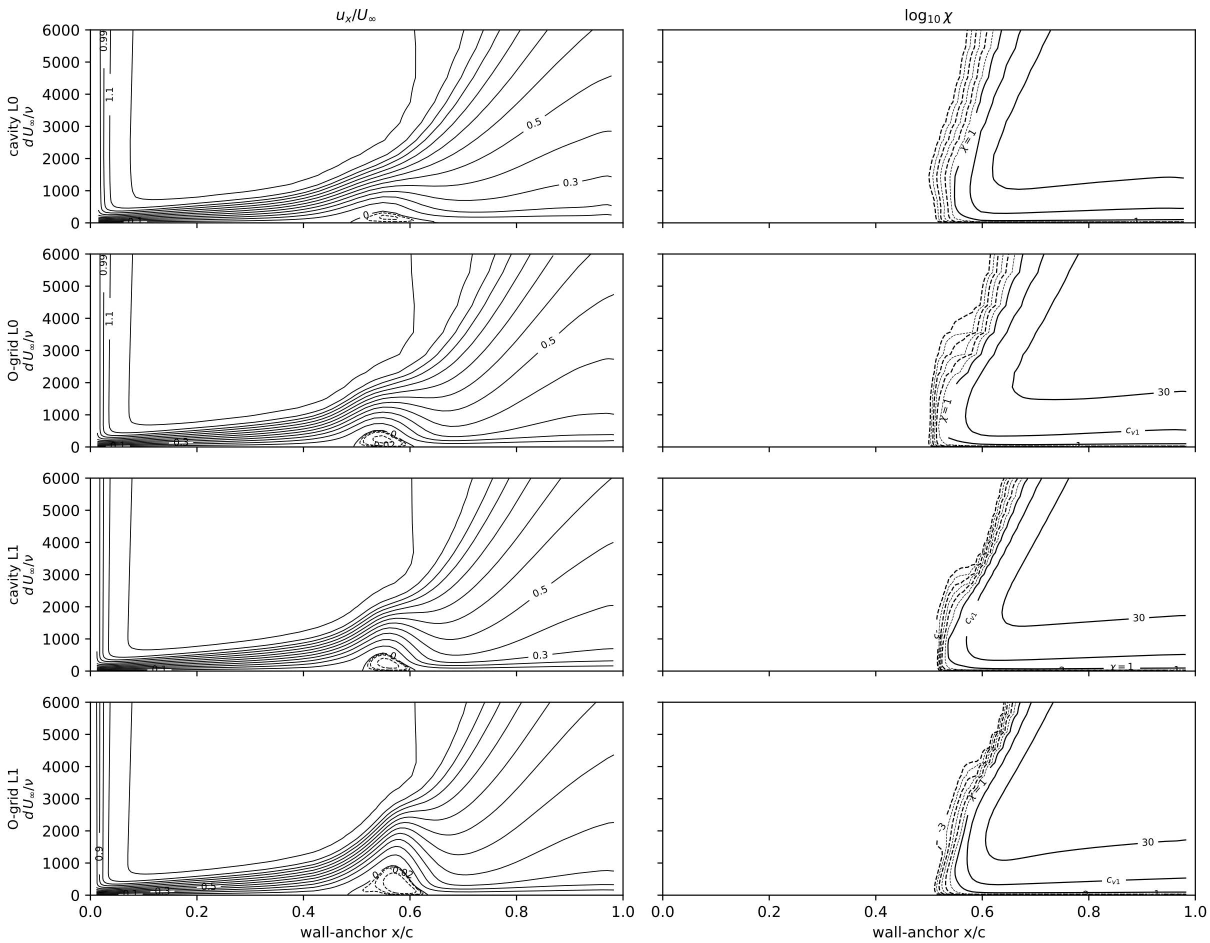

Fig. D6 Daedalus wing, $\alpha = 4°$, section $\eta = 0.10$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right) on the archived meshes (volume dumps present for L0/L1).

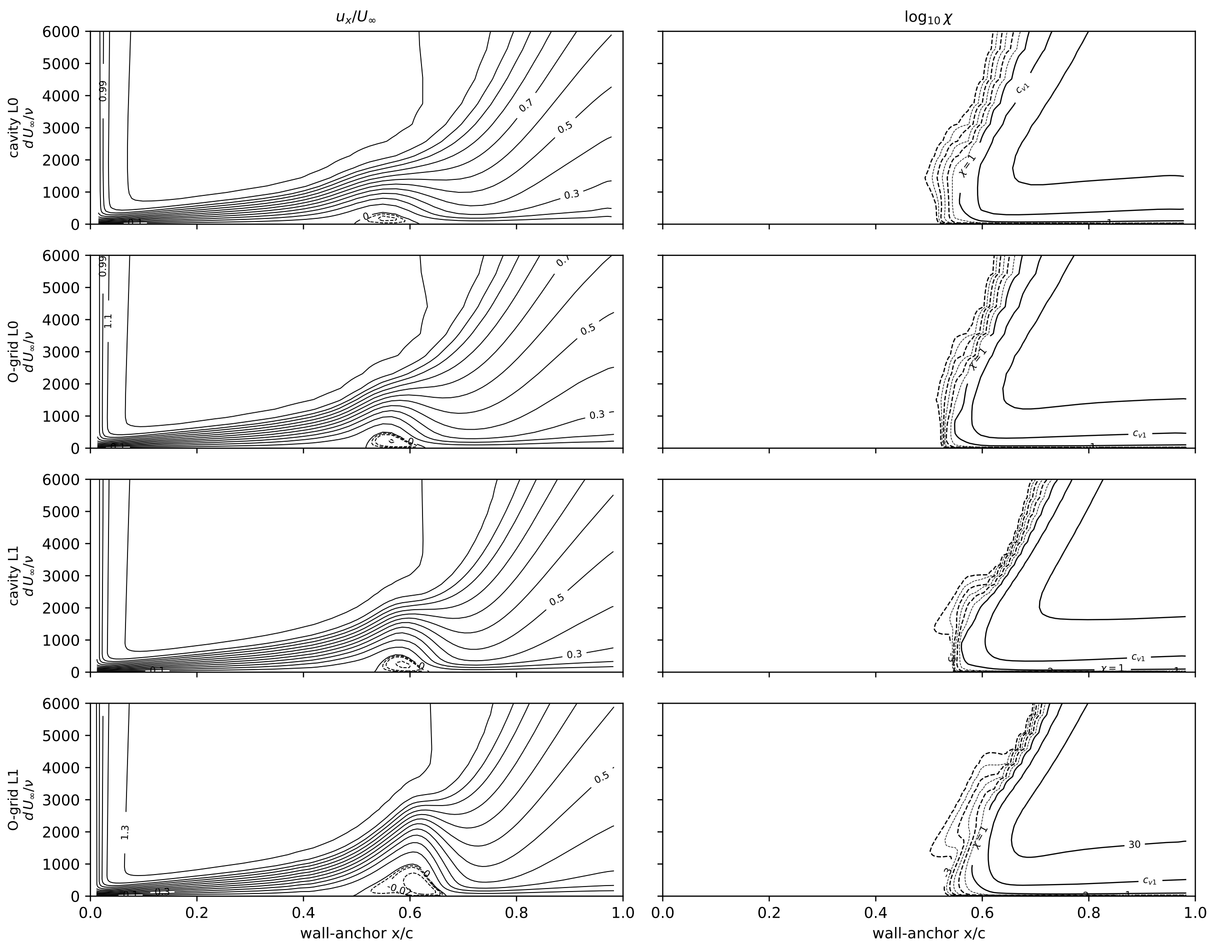

Fig. D7 As Fig. D6, at η = 0.75.

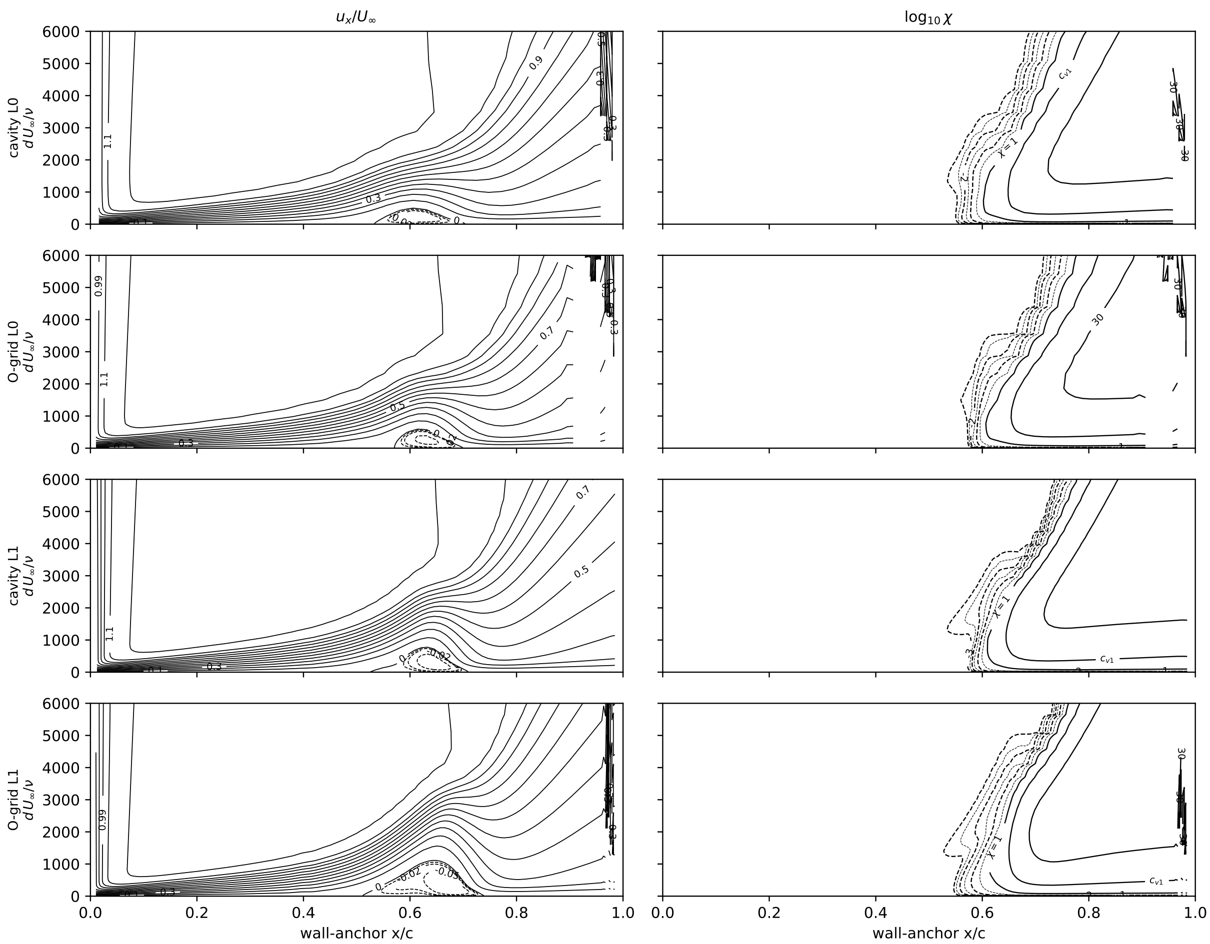

Fig. D8 As Fig. D6, at η = 0.92.

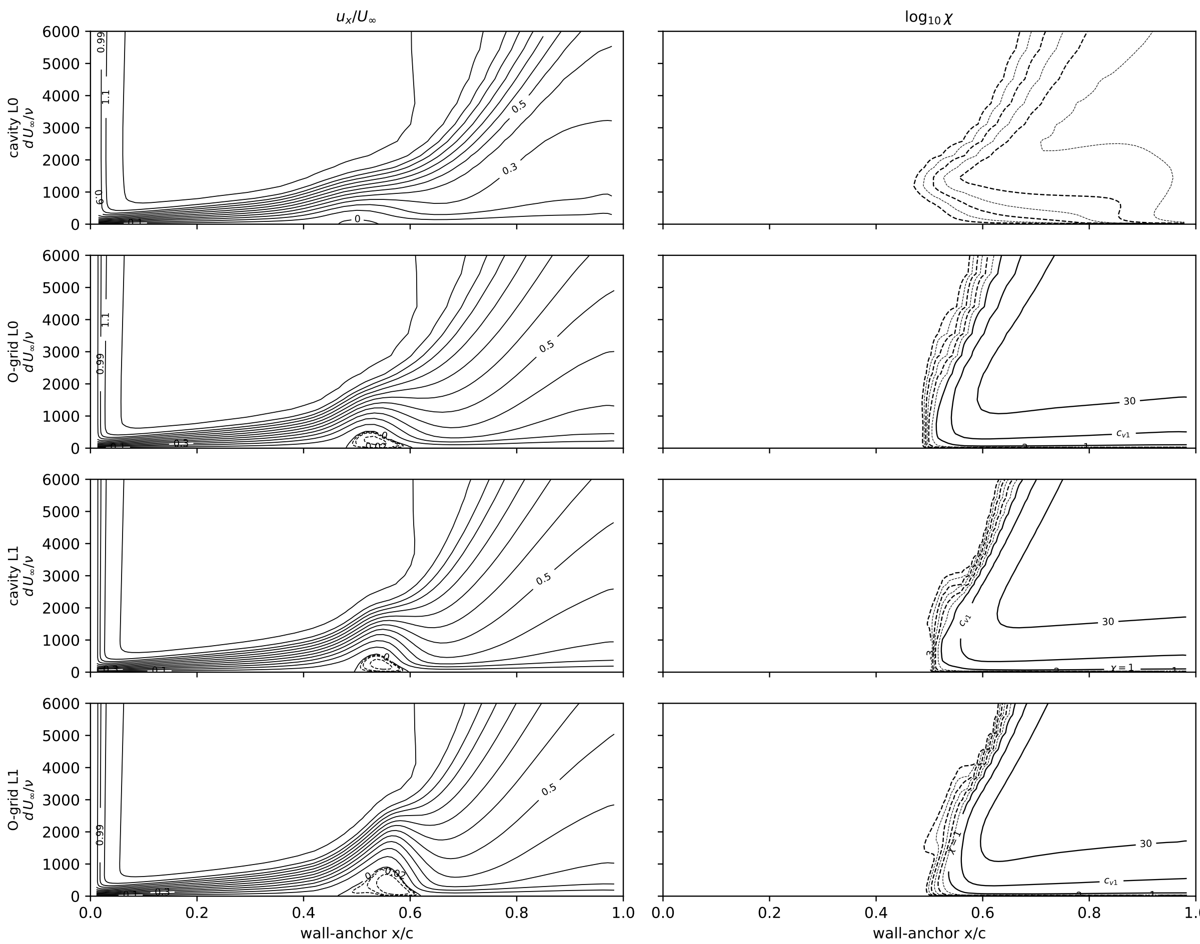

Fig. D9 Daedalus wing, $\alpha = 5°$, section $\eta = 0.10$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right).

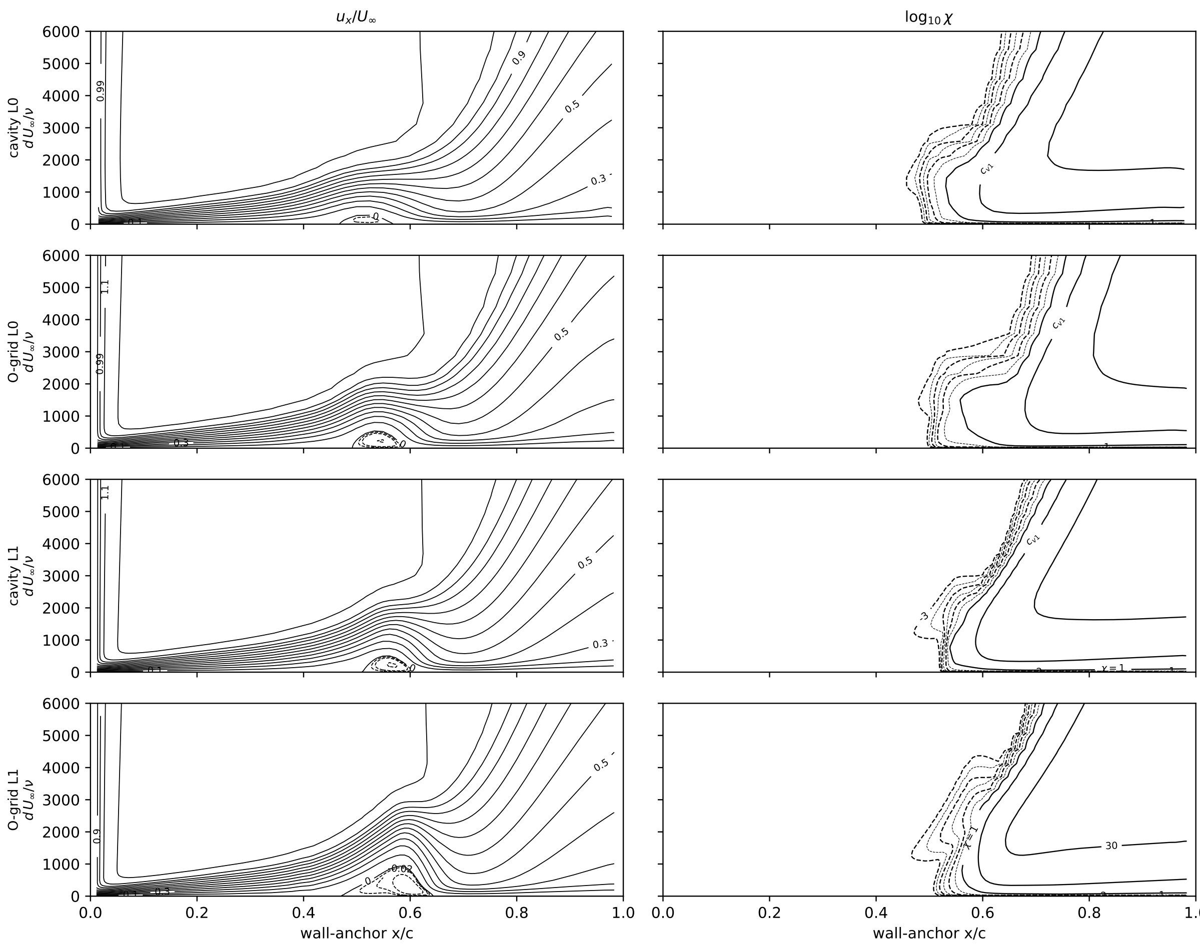

Fig. D10 As Fig. D9, at η = 0.75.

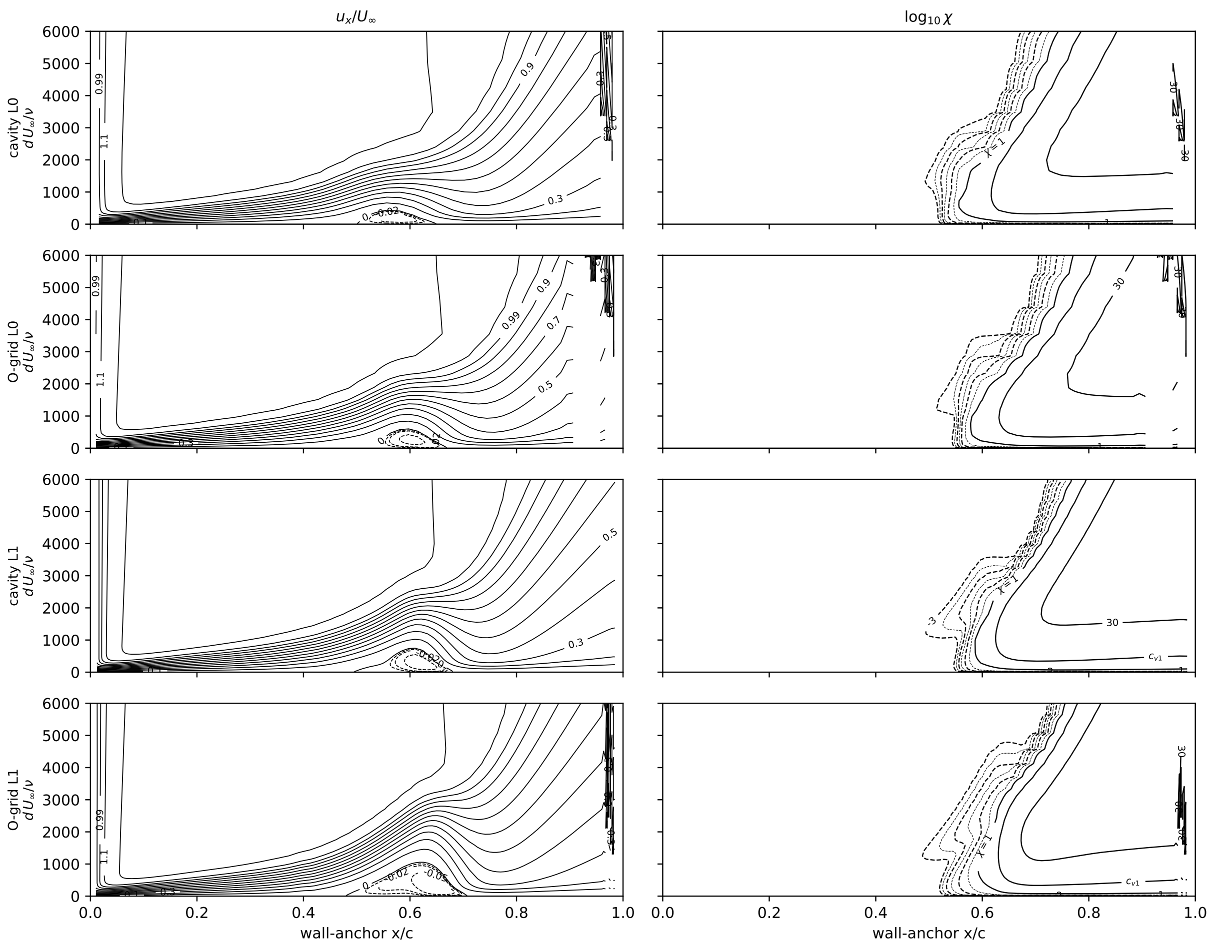

Fig. D11 As Fig. D9, at η = 0.92.

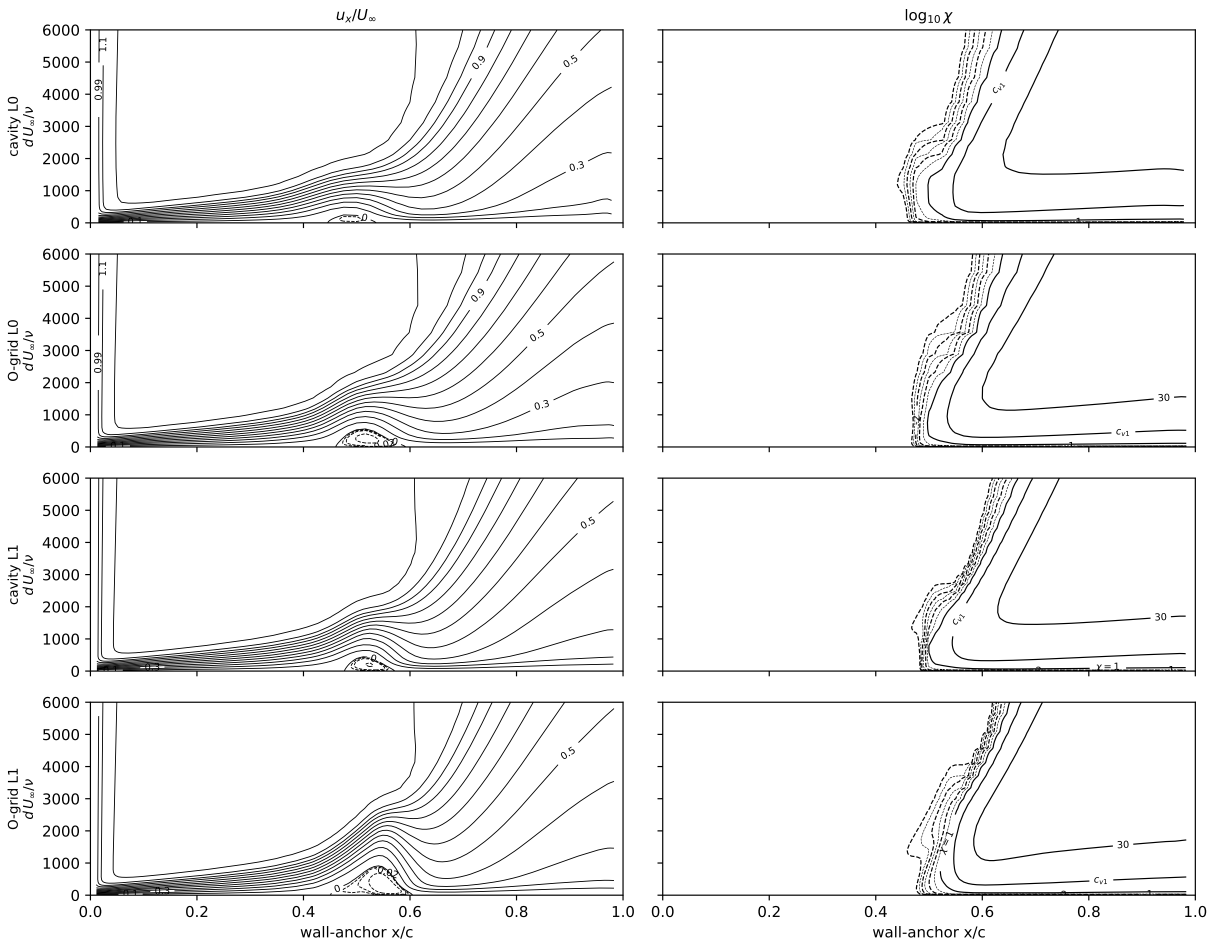

Fig. D12 Daedalus wing, $\alpha = 6°$, section $\eta = 0.10$, upper surface: streamwise velocity u_x/U_∞ (left) and $\log_{10}\chi$ (right).

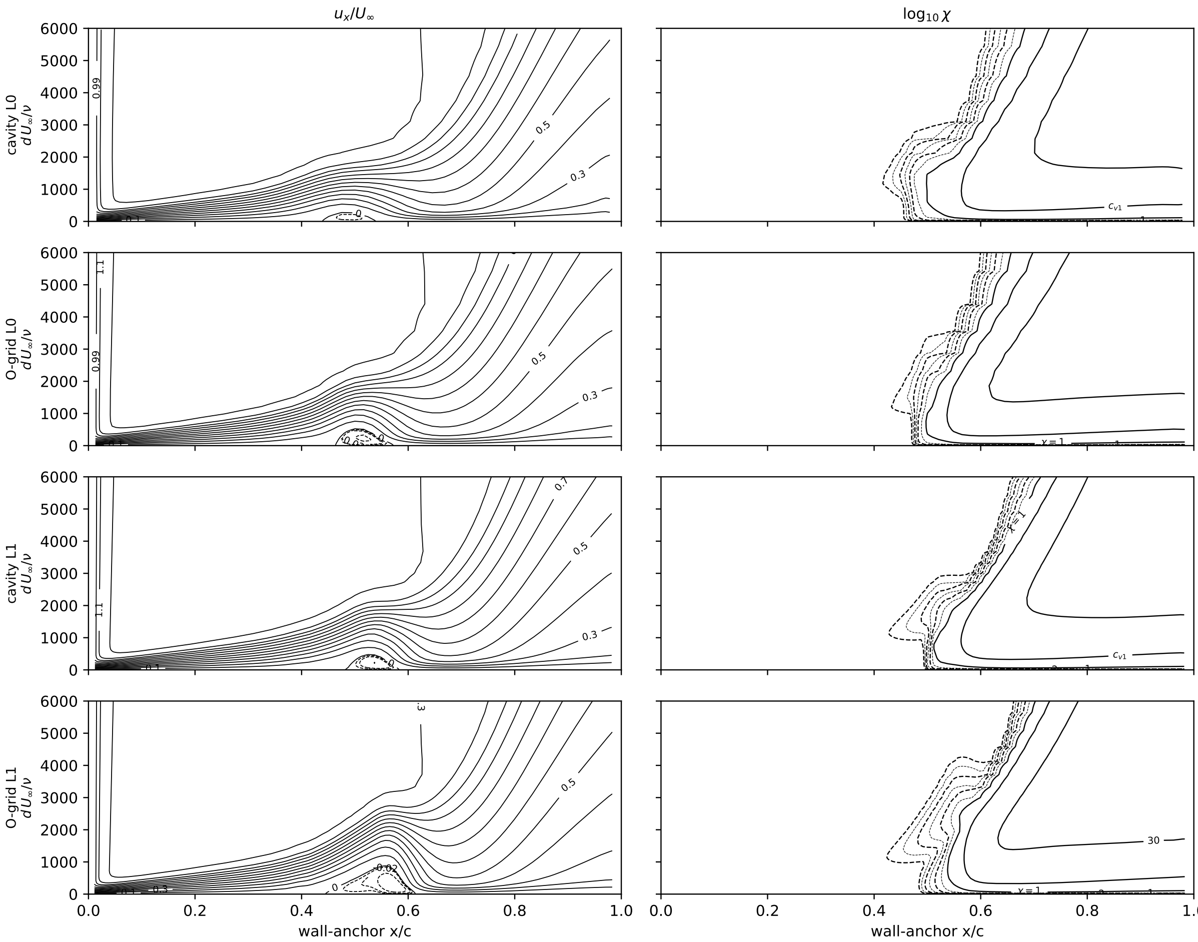

Fig. D13 As Fig. D12, at η = 0.75.

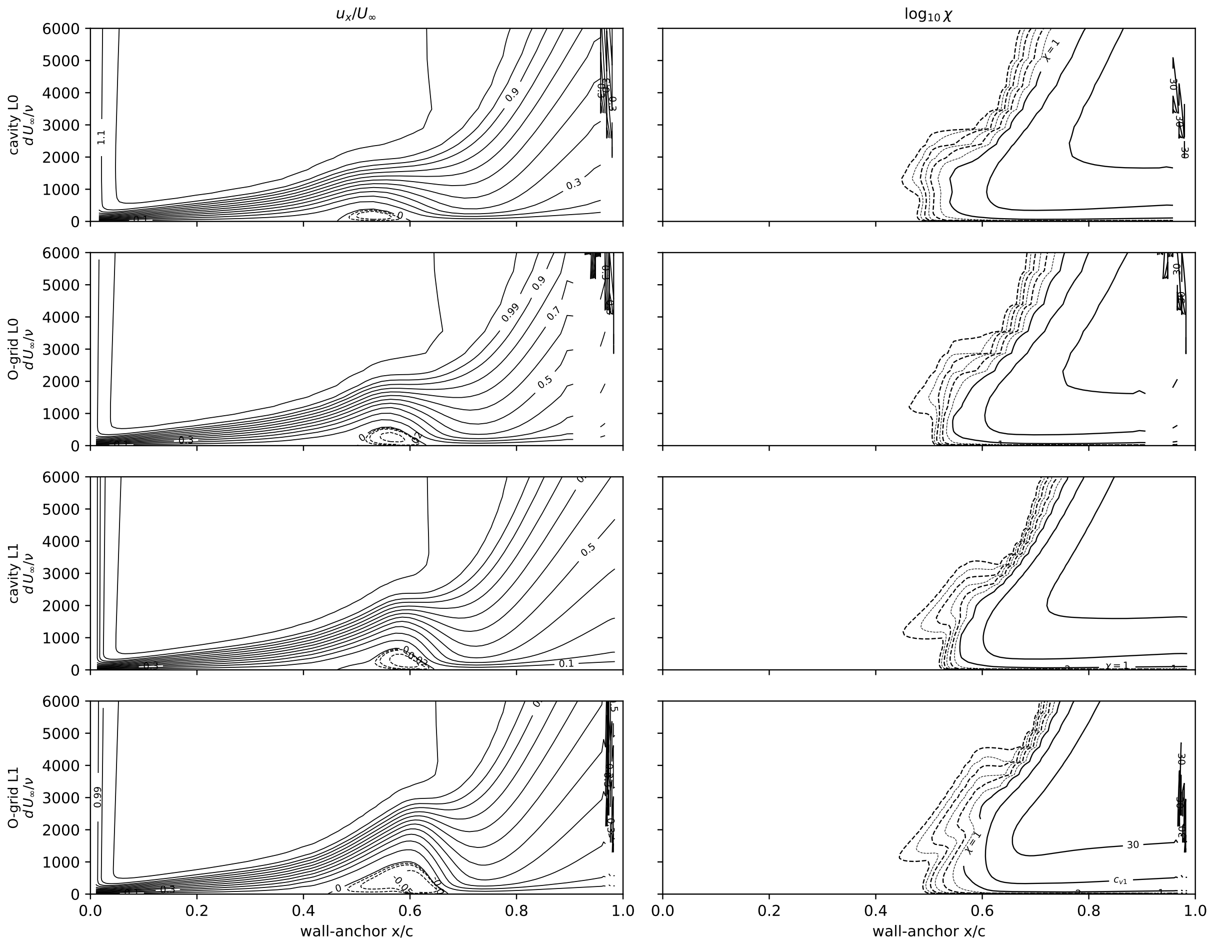

Fig. D14 As Fig. D12, at η = 0.92.

Appendix E: the assembled one-equation model

For reference, the complete model in one place. The working variable $\tilde\nu$ obeys the single transport equation

$$\frac{D\tilde\nu}{Dt} = P - D + \frac{1}{\sigma}\nabla\cdot\left[(c_{\nu,\mathrm{ai}}\,\nu + \tilde\nu)\,\nabla\tilde\nu\right] + \frac{c_{b2}}{\sigma}\,|\nabla\tilde\nu|^2, \tag{E1}$$

identical in form to Spalart–Allmaras [22] except in four places—the blended production P (its amplification branch built on $\hat\Omega\hat I$), the tied destruction D, the factor $c_{\nu,\mathrm{ai}}$ on the molecular diffusion, and the lifted-layer bypass weight b in the eddy-viscosity assembly; $\sigma_P \equiv 1$ (and with it $\sigma_D = 1$), $c_{\nu,\mathrm{ai}} = 1$, and $b \equiv 0$ recover baseline SA exactly.

Terms. The production and destruction are

$$P = \max\left[\,(1-\sigma_P)\,a\,S\,\omega\,\tilde\nu,\;\; \sigma_P\,P_{\mathrm{SA}}\,\right], \qquad D = \sigma_D\,D_{\mathrm{SA}}, \tag{E2}$$

with the handover weights and the amplification rate restated below (from Eqs. (4)–(16) and (19)). The baseline SA production, destruction, and closure functions, carried over unchanged, are

$$\begin{gathered} P_{\mathrm{SA}} = c_{b1}\tilde S\,\tilde\nu, \qquad D_{\mathrm{SA}} = c_{w1} f_w \left(\frac{\tilde\nu}{d}\right)^2, \qquad \tilde S = \omega + \frac{\tilde\nu}{\kappa^2 d^2}\,f_{v2}, \qquad f_{v2} = 1 - \frac{\chi}{1+\chi f_{v1}}, \\ f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}, \qquad f_w = g\left(\frac{1+c_{w3}^6}{g^6 + c_{w3}^6}\right)^{1/6}, \qquad g = r + c_{w2}(r^6 - r), \qquad r = \min\left(\frac{\tilde\nu}{\tilde S\,\kappa^2 d^2},\,10\right), \end{gathered} \tag{E3}$$

with ω the vorticity magnitude, $\chi = \tilde\nu/\nu$, d the wall distance, and eddy viscosity $\mu_t = \rho\tilde\nu\,[(1-b) f_{v1} + b]$ with the lifted-layer bypass weight $b = s(\chi)G(q)$ of Eqs. (17)–(18), where $s = \mathrm{clip}(\chi - 1, 0, 1)$, $G = \mathrm{clip}((q-2)/2, 0, 1)$, and $U^+(y^+)$ in Eq. (17) is Reichardt's law of the wall [81], $U^+ = \frac{1}{\kappa}\ln(1+\kappa y^+) + 7.8\,(1 - e^{-y^+/11} - \frac{y^+}{11}e^{-y^+/3})$ (the wall-function auxiliary g is Spalart's, with its literature symbol restored now that the coordinate rename freed it below). We retain the non-negative $\tilde S$ modification and the negative-$\tilde\nu$ safeguards of Ref. [23]. The handover weights are

$$\sigma_P = \sigma_t(\chi;\tau), \qquad \sigma_t(\chi;\tau) = \max\left[\,1 - e^{-(\chi-1)/\tau},\; 0\right], \qquad \sigma_D = 1 - \frac{c_{b1}}{\kappa^2 c_{w1}}\,(1-\sigma_P),$$

and the amplification rate is built from four local vector-calculus objects and their invariant pairings: the velocity $\mathbf{u}$, measured relative to the nearest wall point (which makes the indicators Galilean invariant; the absolute velocity for the static-wall cases of this paper), the vorticity $\boldsymbol\omega$, the wall-distance vector $\mathbf{d} = d\,\nabla d$ (so $|\nabla d| = 1$ and $\langle\mathbf{d},\mathbf{d}\rangle = d^2$), and the curvature vector $\boldsymbol\ell = \frac{1}{2}\langle\mathbf{d},\mathbf{d}\rangle\,\nabla^2\mathbf{u}$, with the divergence-free identity $\nabla^2\mathbf{u} = -\nabla\times\boldsymbol\omega$ supplying the second derivative.

The indicator triple is their Gram form,

$$(X,\,Y,\,Z) \;=\; \Big(\langle\mathbf{u},\mathbf{u}\rangle,\;\sqrt{\langle\mathbf{u},\mathbf{u}\rangle\,\langle\mathbf{d},\mathbf{d}\rangle\,\langle\omega,\omega\rangle},\;\langle\boldsymbol{\ell},\mathbf{u}\rangle\Big), \tag{E4}$$

in which every entry is an invariant pairing, no unit vector appears, no division by a vanishing magnitude can occur, and the only radical sits over a non-negative product. From the triple,

$$\hat{\Omega} = \frac{Y}{\sqrt{X^2+Y^2}}, \qquad \hat{I} = \frac{Y-X-Z}{\sqrt{X^2+Y^2+Z^2}}, \qquad Re_{\Omega} = \frac{d^2\,|\omega|}{\nu};$$

both $\hat{\Omega}$ and $\hat{I}$ are homogeneous of degree zero in the triple, so any common positive factor drops out, and the solver's equivalent ratio realization $\big(|\mathbf{u}|,\ d|\omega|,\ \tfrac{1}{2}d^2(\nabla^2\mathbf{u})\cdot\hat{\mathbf{u}}\big)$—the arithmetic that produced every number in this paper—coincides with Eq. (E4) wherever $\mathbf{u} \neq 0$.

On a parallel layer $\nabla^2\mathbf{u} = -\nabla\times\boldsymbol{\omega} = u''\,\hat{\mathbf{t}}$, so the curvature projection reads $\mathrm{sign}(u)\,u''$—the wall-normal derivative of the vorticity magnitude, $\hat{\mathbf{n}}\cdot\nabla\omega$, in that limit—and the ratio triple reduces to $(|u|,\ d|u'|,\ \tfrac{1}{2}d^2u''\,\mathrm{sign}(u))$. How these magnitude-based components realize the signed algebra of Sec. II (whose indicators, Eq. (4), are signed profile derivatives) is worth stating precisely: the implemented parallel-layer triple is $\big(\,\mathrm{sign}(u)\,u,\ \mathrm{sign}(u')\,d\,u',\ \mathrm{sign}(u)\,\tfrac{1}{2}d^2u''\big)$. Wherever the velocity and the shear share a sign—every attached profile, and all of a reversed profile except the layer between its backflow extremum and its velocity zero—the three components carry one common sign, so the triple is a legitimate projective representative of the signed state of Eq. (4), and every even product, $\hat{\Omega}\hat{I}$ included, coincides with the theory exactly: the magnitudes realize the antipodal identification rather than break it (a globally mirrored layer lands on the identified antipode), which is why the calibration instruments of Secs. II.C–II.D, marched on attached profiles, transfer to the solver unchanged. In the remaining mixed-sign layer the implemented direction deviates from the signed class in a single component's sign, and the deviation switches on and off continuously: at the backflow extremum the affected component Y itself vanishes, and at the velocity zero the state reaches the $X=0$ meridian, where $\hat{\Omega}$ remains smooth and the curvature projection changes sign with amplitude $\tfrac{1}{2}d^2|u''|$—small, because the velocity zero of a free-shear-like reversed profile lies close to its inflection point (for the antisymmetric layer the two coincide and the amplitude is zero). The same four vectors admit two other independent pairings, recorded here for completeness: the signed pairing $\langle\boldsymbol{\omega}\times\mathbf{d},\mathbf{u}\rangle$, whose Cauchy–Schwarz envelope is exactly Eq. (E4)'s middle entry in two-dimensional flow, and its lower bound in three (by the Lagrange identity $|\boldsymbol{\omega}\times\mathbf{d}|^2 = \langle\mathbf{d},\mathbf{d}\rangle\langle\omega,\omega\rangle - \langle\omega,\mathbf{d}\rangle^2$)—realizations built on the signed pairing itself were implemented and rejected, as they rectify vorticity-direction noise in weak shear—and the odd (pseudoscalar) pairing $\langle\nabla d,\mathbf{u}\times(\boldsymbol{\omega}\times\mathbf{d})\rangle$, which vanishes identically in two-dimensional flow, measures crossflow, and is taken up in the outlook.

The rate a (as in Eq. (8)) and its onset gate S (as in Eq. (11)) are

$$a = a_{\max}\ \mathrm{clip}\langle\hat{\Omega}\hat{I}\rangle_0^1, \qquad S = \mathcal{S}\big(Re_\Omega / Re_\Omega^{\mathrm{c}}(\hat{\Omega}\hat{I})\big), \qquad \mathcal{S}(\zeta) = \tfrac{1}{2}\big[1 + \tanh\big((\zeta - 1)/w\big)\big],$$

$$Re_\Omega^{\mathrm{c}}(\hat{\Omega}\hat{I}) = k\ \mathrm{softmin}_2\big(C,\ A + B\,\langle\hat{\Omega}\hat{I}\rangle_+^{-2}\big), \qquad \mathrm{softmin}_2(x_1, x_2) = \frac{x_1 x_2}{\sqrt{x_1^2 + x_2^2}},$$

with $\langle x\rangle_+ = \max(x, 0)$.

Constants. The baseline SA constants are unchanged:

$$c_{b1} = 0.1355, \quad \sigma = \tfrac{2}{3}, \quad c_{b2} = 0.622, \quad \kappa = 0.41, \quad c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma}, \quad c_{w2} = 0.3, \quad c_{w3} = 2, \quad c_{v1} = 7.1.$$

The SA-AI constants are

$$a_{\max} = 0.19, \qquad (C,\ A,\ B;\ k) = (2600,\ 175,\ 2;\ 0.712), \qquad w = 0.35, \qquad \tau = 4, \qquad c_{v,\mathrm{ai}} = \tfrac{1}{6},$$

with the freestream seed set by Mack's receptivity map, $N_{\mathrm{crit}}(Tu) = -8.43 - 2.4 \ln Tu_{\mathrm{frac}}$, $\chi_\infty = c_{v1} e^{-N_{\mathrm{crit}}}$ (Eq. (20)).

Appendix F: tabulated data for the one-dimensional comparison figures

Tables F1–F9 tabulate the computed values plotted in the paper's one-dimensional comparison figures—forces, transition locations, and bubble stations—so exact numbers need not be read off the plots. Conventions are those of the corresponding figures: forces are medians over the final 20% of the pseudo-step history, transition locations are the near-wall $\chi = 1$ front stations, and bubble stations are the signed-C_f zero crossings. The Daedalus and spheroid totals are already tabulated in Tables D1 and 7.

Digitized references. Every reference curve or symbol taken from a published figure enters the paper's figures through a committed data file; this list is the extraction provenance of record, and the captions carry only curve identity.

- `data/aft_nlf0416_digitized.json`: the AFT (OVERFLOW) NLF(1)-0416 incidence sweeps of Coder's dissertation [31]—transition fronts and polars at the nominal $N_{\text{crit}} = 10.07$ and the source's recalibrated 7.18, its XFOIL reference, and the LTPT microphone measurements—extracted vector-exact from the source figures' embedded paths (not pixels). The digitized AFT polars end at the source axis limit $c_d = 0.025$.
- The experimental section characteristics of Fig. 7 (Somers, NASA TP-1861, Fig. 11d [36]) are pixel-calibrated in `repro/cfd/regen_nlf_polar.py`; the transition-orifice onsets on the C_f rows (its Fig. 9d) are read at the computed section c_l, extrapolation marked open.
- `data/workshop_nlf_submittals.json`: all fourteen submittals to the First AIAA CFD Transition Modeling and Prediction Workshop's NLF(1)-0416 incidence sweep (α = −4°–8°), digitized per submittal from the workshop-summary overlay [32]—the only published collection of all participants—and placed on the lift axis through the computed structured-L2 lift curve. Fourteen submittals, nineteen curves: two participants supplied second model variants and three line-only series carry no letter code; segments break where the source's marker identity is ambiguous.
- `data/workshop_nlf_envelope.json`: the SA-LM2015 fine-grid sweep of Piotrowski and Zingg [33], vector-digitized at its native c_l ordinate.
- `data/lm_nlf0416_transition_digitized.json`: the Langtry–Menter γ–Re_θ (OVERFLOW, fine-grid) sweep of Denison [34], digitized from its Fig. 11. The workshop condition is $M = 0.2$, $Tu = 0.15\%$—immaterial on transition-vs-c_l axes.
- `data/lmbcm_eppler387_digitized.json`: the independent Selig et al. Eppler 387 measurements and the γ–Re_θ and SA-BC polar and bubble curves, digitized vector-exact from Ref. [39].
- `data/colemueller1990_e387_bubble_stations.json`: the Cole–Mueller pressure-station bubble stations [42] (±1% chord; their $\alpha \geq 7°$ entries are attached-transition values and plot no bubble).
- `data/ijsrp2019_e387_polars.json`: the k-k$_L$-ω and transition-SST polars of Ref. [45]; that source's $6{\times}10^4$ series and as-plotted experiment are corrupt in the source and excluded.

Table F1 Flat-plate transition-onset Re_θ (integrated from the computed profiles) at the $\chi = 1$ and $\chi = c_{v1}$ crossings, Flow360 and the OpenFOAM replication (OF), against the Abu-Ghannam & Shaw correlation (Fig. 5).

Tu [%]	AGS Re_θ	$\chi = 1$	$\chi = c_{v1}$	OF $\chi = 1$	OF $\chi = c_{v1}$
0.04	1126	1149	1203	1098	1154
0.08	1088	1002	1069	949	1004
0.16	1017	845	934	798	887
0.3	905	700	815	656	739
0.6	713	524	685	487	589

- `data/stock2006_fig15a_digitized.json`: the DFVLR measured transition points and Stock's computed free-vortex separation line (Stock Fig. 15a at $Re = 1.52\times10^6$ [78]), digitized to $\pm 0.005\,x/L$.
- `data/stock2006_fig2_digitized.json`, `data/stock2006_fig3_digitized.json`: the DFVLR measured C_p chains of Stock's Figs. 2–3 (α = 10° and 29.7°) [78], digitized from the raster scans by per-station exclusive continuity tracking (`repro/cfd/digitize_stock_waterfalls.py`, which records every truncation). Waterfall displacements $0.14 + 0.42i$ removed: the 0.42 step is the printed-axes value, and the 0.14 base (the displacement Stock's text states) is derived by anchoring the printed chains and Stock's own printed potential-theory curves to exact potential theory. Chains truncated where the printed symbol tangle makes chain identity unrecoverable ($\alpha = 29.7°$ leeward dives; the windward rises, suction peaks, and dive onsets are retained).
- `data/stock2006_fig4_digitized.json`, `data/stock2006_fig5_digitized.json`: the DFVLR measured C_{ft} and γ_w chains of Stock's Figs. 4–5 [78], same pipeline and truncation record; waterfall displacements (-1.5×10^{-3}/$-15°$ and -3×10^{-3}/$-20°$ per station) removed; the $\gamma_w = 0$ symmetry values at $\phi = 0$ confirm these two figures carry no base displacement.

Table F2 NLF(1)-0416, $Re = 4\times10^6$: computed forces and transition locations behind Figs. 6, 7, and 8.

α	grid	structured O-grid				unstructured cavity			
		C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}
−8	L0	-0.4426	0.01830	0.195	0.011	-0.4689	0.01869	0.140	0.012
−8	L1	-0.4656	0.01093	0.538	0.006	-0.4397	0.01150	0.571	0.011
−8	L2	-0.4646	0.00962	0.559	0.004	-0.4426	0.00964	0.561	0.010
−4	L0	0.0469	0.01078	0.435	0.100	0.0197	0.01025	0.476	0.083
−4	L1	0.0142	0.00790	0.435	0.102	0.0357	0.00832	0.452	0.126
−4	L2	0.0208	0.00718	0.447	0.109	0.0363	0.00739	0.453	0.122
0	L0	0.5458	0.00803	0.400	0.590	0.4896	0.00808	0.355	0.592
0	L1	0.5048	0.00607	0.395	0.553	0.5209	0.00660	0.392	0.579
0	L2	0.5139	0.00564	0.387	0.566	0.5219	0.00606	0.391	0.569
4	L0	1.0001	0.01090	0.250	0.645	0.9513	0.01062	0.330	0.621
4	L1	0.9678	0.00800	0.238	0.610	0.9839	0.00805	0.269	0.614
4	L2	0.9805	0.00730	0.255	0.610	0.9848	0.00772	0.260	0.621
9	L0	1.4483	0.02252	0.015	0.385	1.4251	0.02345	0.105	0.641
9	L1	1.4886	0.01436	0.065	0.625	1.5112	0.01353	0.085	0.635
9	L2	1.5146	0.01245	0.073	0.628	1.5139	0.01286	0.078	0.638
15	L0	1.5172	0.07109	0.010	0.440	1.3976	0.11602	0.005	0.623
15	L1	1.9317	0.03159	0.004	0.650	1.9232	0.03454	0.001	0.659
15	L2	2.0057	0.02628	0.005	0.659	1.9924	0.02772	0.000	0.662

Table F3 NLF(1)-0416, $Re = 4\times10^6$: the independent cell-centered incompressible (OpenFOAM) implementation on the paper's structured L1 and L2 grids—the steel-blue circles of Figs. 6 and 7. Conventions as in Table F2, with the transition locations read from the OpenFOAM fields at the same near-wall $\chi = 1$ crossing; the port omits the bypass of Eq. 18. At L2 every front lies within $0.016\,c$ and the lift within 0.9% (0.003 absolute at the zero-lift −4°) of the corresponding Flow360 values in Table F2. Drag agreement is not claimed: the codes differ by 14% in C_D at −8° at matched fronts, and by 1.6–3.5% elsewhere at L2.

α	L1				L2			
	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}
−8	-0.4785	0.00781	0.576	0.000	-0.4606	0.00826	0.568	0.000
−4	0.0005	0.00634	0.464	0.117	0.0239	0.00693	0.459	0.111
0	0.5034	0.00603	0.393	0.572	0.5153	0.00573	0.395	0.568
4	0.9670	0.00778	0.266	0.616	0.9793	0.00749	0.261	0.625
9	1.5066	0.01273	0.076	0.633	1.5151	0.01270	0.075	0.644
15	1.9780	0.02754	0.000	0.665	2.0221	0.02564	0.000	0.667

Table F4 Eppler 387, $Re = 2\times10^5$: computed forces behind the polar of Fig. 11. All three levels carry the four ladder incidences and the two that bound the sweep (−2°, 8.5°); the four interior extension incidences (1°, 3°, 4°, 6°) exist on the finest pair only.

α	grid	structured O-grid		unstructured cavity	
		C_L	C_D	C_L	C_D
−2	L0	0.1921	0.01687	0.1859	0.01519
−2	L1	0.1835	0.01333	0.1918	0.01153
−2	L2	0.1845	0.01178	0.1855	0.01147
0	L0	0.3869	0.01313	0.3841	0.01203
0	L1	0.3855	0.01101	0.3971	0.01032
0	L2	0.3879	0.01067	0.3908	0.01054
1	L2	0.4979	0.01136	0.5023	0.01122
2	L0	0.6083	0.01411	0.6055	0.01255
2	L1	0.6056	0.01228	0.6207	0.01157
2	L2	0.6068	0.01223	0.6126	0.01202
3	L2	0.7147	0.01322	0.7219	0.01291
4	L2	0.8207	0.01442	0.8298	0.01384
5	L0	0.9075	0.02051	0.9246	0.01610
5	L1	0.9214	0.01626	0.9462	0.01459
5	L2	0.9255	0.01552	0.9365	0.01481
6	L2	1.0318	0.01553	1.0438	0.01510
7	L0	1.0691	0.02675	1.1248	0.02027
7	L1	1.1268	0.01681	1.1596	0.01554
7	L2	1.1411	0.01442	1.1532	0.01425
8.5	L0	1.1129	0.03863	1.2309	0.02992
8.5	L1	1.2223	0.02510	1.2575	0.02465
8.5	L2	1.2353	0.02314	1.2503	0.02333

Table F5 Eppler 387, $Re = 2\times10^5$: computed upper-surface laminar-separation and turbulent-reattachment stations behind Fig. 14 (signed-C_f zero crossings; missing entries transition ahead of separation and have no bubble). The measured oil-flow reattachments (TM-4062 Table III) at $\alpha = 0°/2°/5°$ are 0.74/0.67/0.59 c.

α	grid	structured O-grid		unstructured cavity	
		x_{LS}/c	x_R/c	x_{LS}/c	x_R/c
−2	L2	0.573	0.822	0.567	0.815
0	L0	0.549	0.720	0.540	0.711
0	L1	0.526	0.771	0.526	0.737
0	L2	0.523	0.761	0.520	0.757
1	L2	0.500	0.740	0.499	0.733
2	L0	0.490	0.656	0.512	0.649
2	L1	0.479	0.713	0.483	0.682
2	L2	0.477	0.714	0.478	0.706
3	L2	0.454	0.691	0.454	0.679
4	L2	0.426	0.667	0.430	0.657
5	L0	0.393	0.568	0.429	0.558
5	L1	0.394	0.624	0.415	0.600
5	L2	0.402	0.634	0.404	0.623
6	L2	0.388	0.565	0.391	0.560
7	L0	0.250	0.343	0.351	0.458
7	L1	–	–	–	–
7	L2	0.388	0.407	–	–
8.5	L2	0.031	0.039	0.019	0.049

Table F6 Eppler 387, $Re = 2\times10^5$: the independent cell-centered incompressible (OpenFOAM) implementation on the paper's structured L1 and L2 grids—the steel-blue circles of Figs. 11 and 14. Transition locations are read from the OpenFOAM fields at the same near-wall $\chi = 1$ crossing as everywhere in this paper, and the bubble stations at the same signed-C_f zero crossings as Table F5; the lower surface stays laminar to the trailing edge in both codes at every incidence. At L2 the lift agrees within 0.5%, separation sits 0.030–0.037 c ahead of and reattachment 0.033–0.045 c behind the SA-AI stations (0.086 c at the near-stall 7°, both codes' hardest condition). The port omits the bypass of Eq. 18—worth −6 to −10 counts at $\alpha = 0$–5° here and −1 at 7° by the recomputation record of Sec. II.F—and the residuals are consistent with that omission in sign and magnitude: drag agreement is not claimed, with OpenFOAM above by 4.8/7.5/1.2 counts at $\alpha = 0°/2°/5°$ and below by 6.8 at 7°; the $\chi = 1$ fronts read 0.006–0.044 c ahead of the SA-AI values while the half-saturation $\chi = c_{v1}$ crossings of the two codes agree to 0.014 c below 7° (0.028 c there)—the stretched $\chi = 1 \rightarrow c_{v1}$ ramp is exactly the handover the bypass compresses.

α	L1					L2				
	C_L	C_D	x_{tr}^{up}	x_{LS}	x_R	C_L	C_D	x_{tr}^{up}	x_{LS}	x_R
0	0.3826	0.00978	0.605	0.495	0.775	0.3857	0.01058	0.621	0.486	0.806
2	0.6037	0.01174	0.543	0.448	0.727	0.6039	0.01229	0.564	0.443	0.752
5	0.9256	0.01394	0.468	0.371	0.651	0.9234	0.01463	0.487	0.372	0.667
7	1.1391	0.01323	0.329	0.353	0.453	1.1388	0.01363	0.379	0.351	0.493

Table F7 Eppler 387 $\alpha = 5°$ Reynolds sweep: computed forces and upper-surface stations behind Fig. 15 (x_R pinned near 1 means no closure ahead of the trailing edge; c_m and the e^9/experiment columns are in Table C1). The 2×10^5 rows are the figure script's own re-extraction of the benchmark histories and agree with Table F4's campaign-record values to within one unit in the last digit (tail-window rounding).

Re	grid	structured O-grid				unstructured cavity			
		c_l	c_d	x_{sep}	x_R	c_l	c_d	x_{sep}	x_R
60k	L0	0.6648	0.04879	0.317	0.994	0.8377	0.03560	0.374	0.996
60k	L1	0.6606	0.05064	0.329	0.996	0.7777	0.04373	0.344	0.998
60k	L2	0.6362	0.05330	0.319	0.997	0.7123	0.05065	0.325	0.999
100k	L0	0.8245	0.03320	0.360	0.994	0.8929	0.02349	0.401	0.977
100k	L1	0.8401	0.03336	0.347	0.997	0.9011	0.02475	0.382	0.951
100k	L2	0.7952	0.04103	0.328	0.999	0.8773	0.03071	0.349	0.992
200k	L0	0.9084	0.02007	0.393	0.568	0.9253	0.01596	0.429	0.558
200k	L1	0.9220	0.01577	0.394	0.624	0.9465	0.01444	0.415	0.600
200k	L2	0.9275	0.01451	0.402	0.634	0.9377	0.01427	0.404	0.623
300k	L0	0.9233	0.01718	0.391	0.494	0.9364	0.01351	0.442	0.529
300k	L1	0.9352	0.01213	0.423	0.545	0.9600	0.01155	0.435	0.548
300k	L2	0.9410	0.01073	0.435	0.551	0.9499	0.01083	0.437	0.549
460k	L0	0.9341	0.01535	–	–	0.9453	0.01154	–	–
460k	L1	0.9444	0.00998	–	–	0.9704	0.00959	0.462	0.490
460k	L2	0.9499	0.00870	0.467	0.482	0.9589	0.00886	0.471	0.482

Table F8 Circular cylinder, steady-branch drag-crisis traverse: the clean systematic $Tu = 0.2\%$ up-ladder of Fig. 18 (single seed, matched Re_D grid, warm-started up-continuation over four grid families). C_d is the median over the final-stage tail window; bracketed values are the tail-window C_d peak-to-peak for the cases the steady monitor did not accept (the open rings of the figure). θ_{tr} is the $\chi = 1$ radial-ray transition angle and θ_{sep} the first wall-separation angle; $|C_L| < 2 \times 10^{-6}$ on every case (symmetric protocol).

Re_D	grid	C_d	θ_{tr} (°)	θ_{sep} (°)
1	lowre	10.665	–	–
3	lowre	5.330	–	–
10	lowre	2.797	–	144.3
30	lowre	1.710	–	127.2
100	lowre	1.123	155.0	112.9
300	lowre	0.934	139.5	102.4
1000	lowre	0.874	123.5	93.4
3000	pilot	0.857	108.0	86.3
10^4	pilot	0.843	96.5	80.6
3×10^4	pilot	0.845	88.0	76.7
10^5	pilot	0.839	81.5	73.7
1.78×10^5	pilot	0.820	79.0	73.0
3×10^5	pilot	0.773	79.0	73.8
5.62×10^5	pilot	0.565 [0.075]	85.0	80.7
10^6	pilot	0.307 [0.044]	97.0	94.8
1.78×10^6	pilot	0.228	98.0	98.3
3×10^6	highre	0.217 [0.010]	95.5	118.6
5.62×10^6	highre	0.213	90.5	118.9
10^7	highre	0.212	86.5	119.2
3×10^7	ultra	0.256 [0.171]	100.5	117.6
10^8	ultra	0.188	79.5	120.4
3×10^8	ultra	0.198	42.5	121.4
10^9	ultra	0.200	17.0	121.4
3×10^9	ultra	0.195	7.5	121.3
10^{10}	ultra	0.185 [0.009]	2.5	121.3

Table F9 6:1 prolate spheroid, $Re_L = 1.5 \times 10^6$ (measured 1.52×10^6), $\alpha = 10°$: computed front stations behind Fig. 24 at the measured azimuths (DFVLR hot films, Stock Fig. 15a).

ϕ [deg]	measured	C_f-rise, $k = 1.5$			L2 band	$\chi = 1$	$\chi = c_{v1}$
	x/L	L0	L1	L2	$k = 1.25$–2	L2	L2
131.8	0.396	0.442	0.440	0.607	[0.596, 0.623]	0.959	0.972
119.3	0.481	0.511	0.764	0.613	[0.591, 0.705]	0.928	0.933
112.9	0.567	0.571	0.868	0.680	[0.660, 0.726]	0.915	0.918
108.7	0.653	0.628	0.652	0.717	[0.700, 0.749]	0.917	0.921
99.2	0.739	0.779	0.687	0.800	[0.784, 0.823]	0.944	0.951
93.2	0.823	0.848	0.756	0.830	[0.819, 0.846]	0.954	0.961
81.8	0.882	0.869	0.833	0.905	[0.895, 0.937]	0.934	0.940
63.3	0.934	0.892	0.886	0.913	[0.910, 0.918]	0.902	0.918

Appendix G: cross-solver replication detail

Figure G1 shows the flat-plate sweep of Fig. 5 computed end to end by the independent implementation of the model referenced in Sec. III—OpenFOAM, cell-centered, pressure-based, incompressible, with the plain compact Laplacian for the profile curvature, the 0.3Ω floor for $\tilde{S}$ in place of Spalart's c_2/c_3 continuation, and no freestream-decay compensation—on the same 320×80 grid specification, in the identical layout for panel-by-panel comparison. The summary crossings of both solvers are tabulated in Table F1.

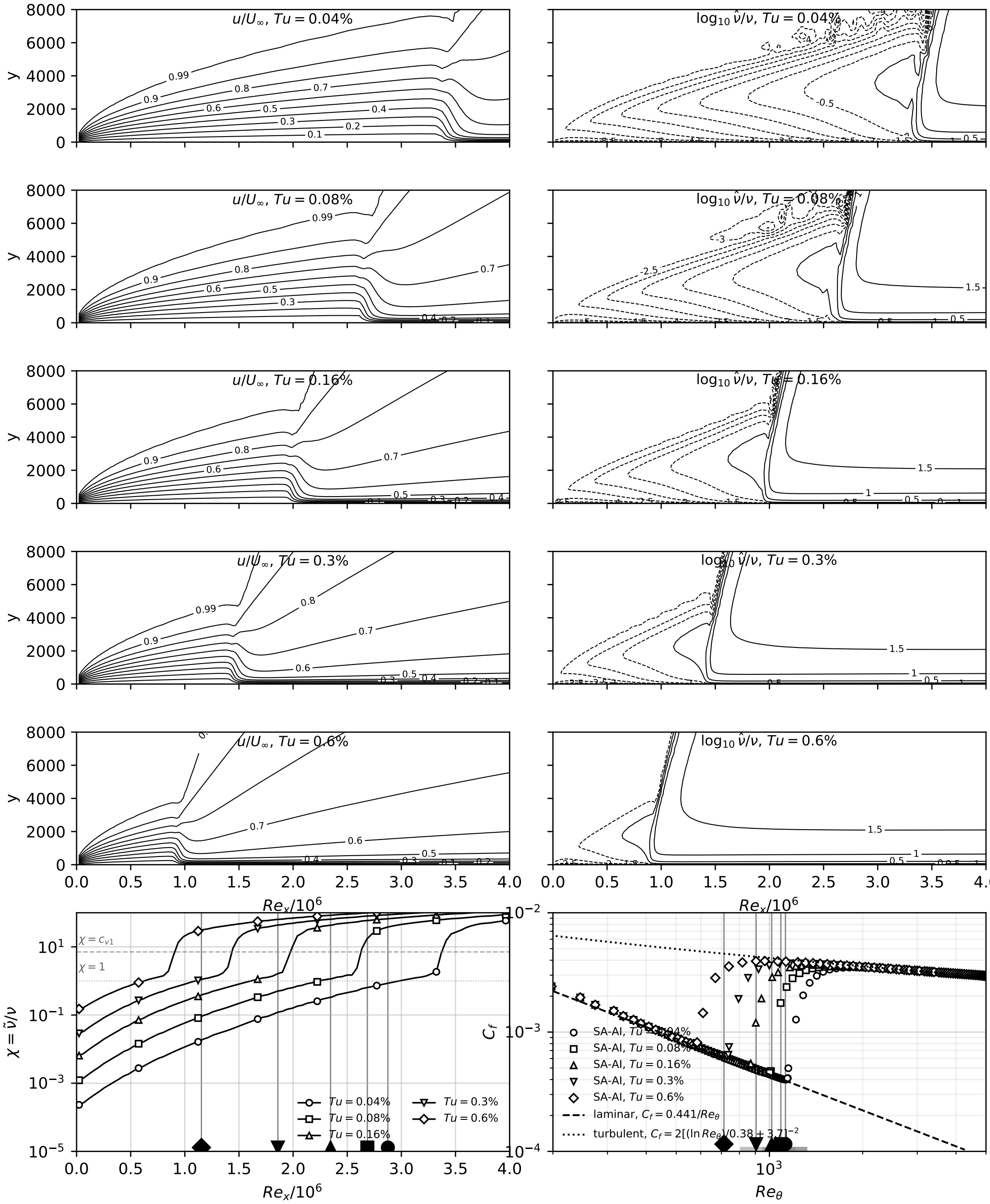

Fig. G1 The freestream-turbulence sweep of Fig. 5, recomputed by the independent cell-centered incompressible (OpenFOAM) implementation on the same grid specification; layout and conventions as in Fig. 5.

Appendix H: spheroid pressure stations and supplementary maps

Figures H1 and H2 present the computed spheroid solutions at the thirteen DFVLR pressure-tap stations, in the waterfall presentation of the measured campaign [77, 78] (Stock's Figs. 2–3): curves displaced +0.42 in $-C_p$ per station—the per-station step of the printed figures. The first pressure station ($X/a = -0.9988$) is read at the probe-grid edge ($x/L = 0.004$). Open red circles are the DFVLR measured chains, as in the hot-film waterfalls of Sec. X (Figs. 26 and 27). In the printed C_p waterfalls the chains sit a uniform 0.12–0.15 above exact potential theory at both incidences—as do Stock's own printed potential-theory curves—while exact potential theory matches the computed L2 station C_p to ~0.005 away from nose and tail; the printed figures therefore carry, in addition to the 0.42 step, a uniform base displacement, equal to the 0.14 Stock's text states, and the digitization removes $0.14 + 0.42i$ per station (which closes both comparisons to ~0.01).

Figures H3 and H4 complete the surface-map views of Sec. X: the $\alpha = 29.7°$ unrolled map, and the $\alpha = 10°$ skin-friction magnitude rendered on the body.

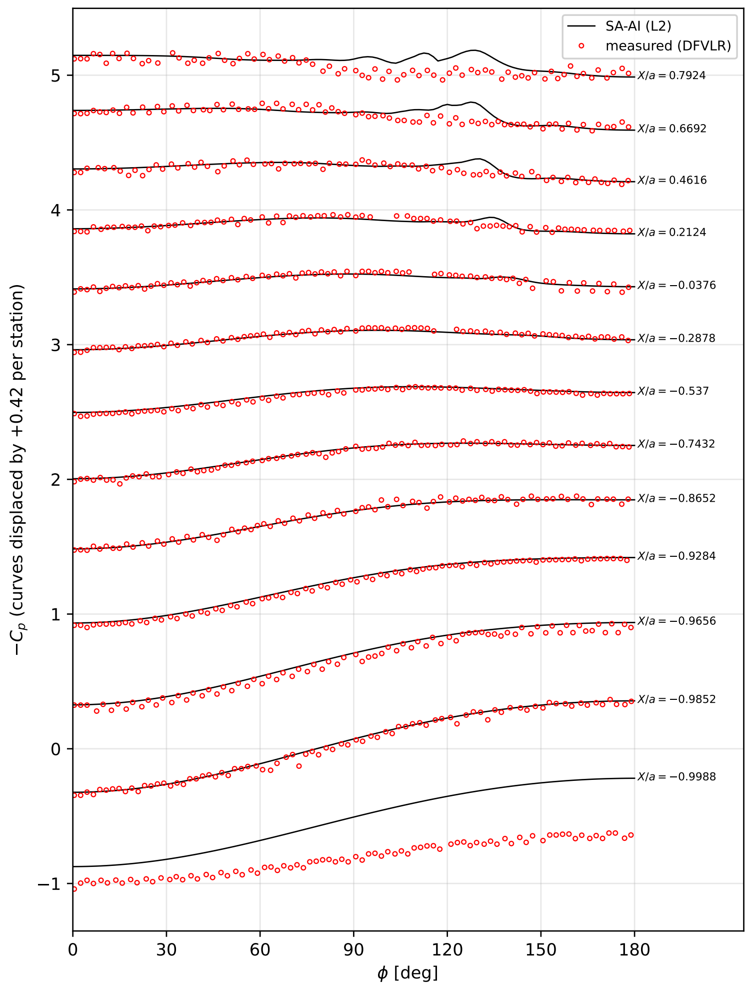

Fig. H1 6:1 spheroid, $Re_L = 1.5\times10^6$, $\alpha = 10°$ (L2): $C_p(\phi)$ at the DFVLR pressure stations, presentation of Stock's Fig. 2. Lines: computed; open red circles: DFVLR measurement, digitized from the printed figure (Appendix F).

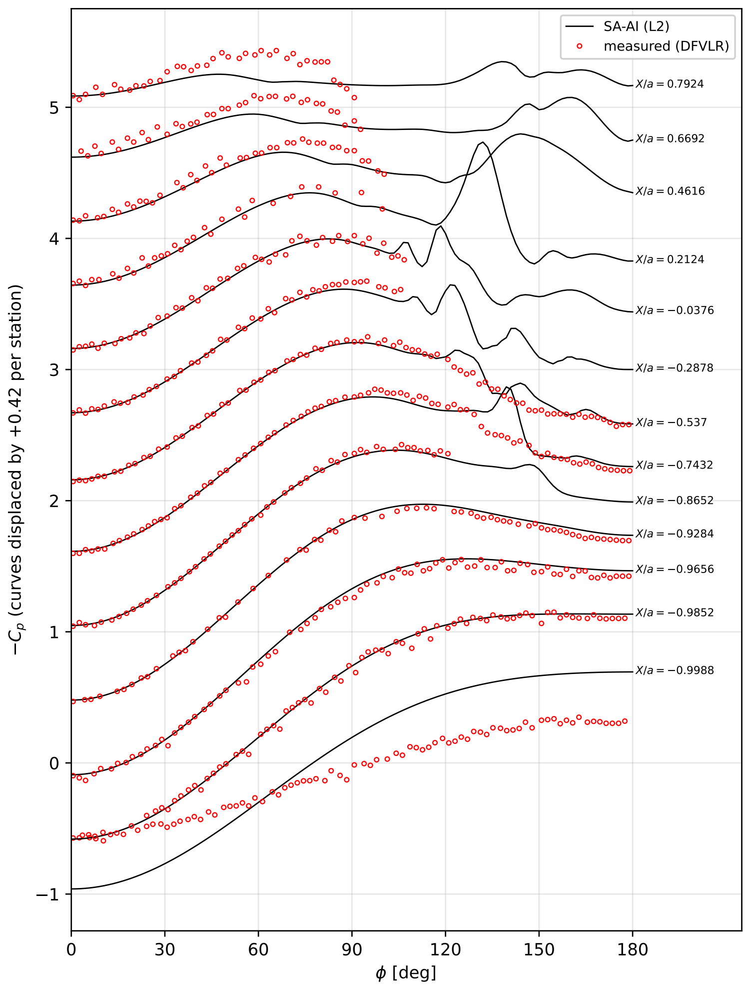

Fig. H2 As Fig. H1, at $\alpha = 29.7°$ (presentation of Stock's Fig. 3); the measured chains of the seven aft- and mid-body stations are truncated in the leeward dive ($\phi \approx 90$–$122°$) where printed chain identity is unrecoverable.

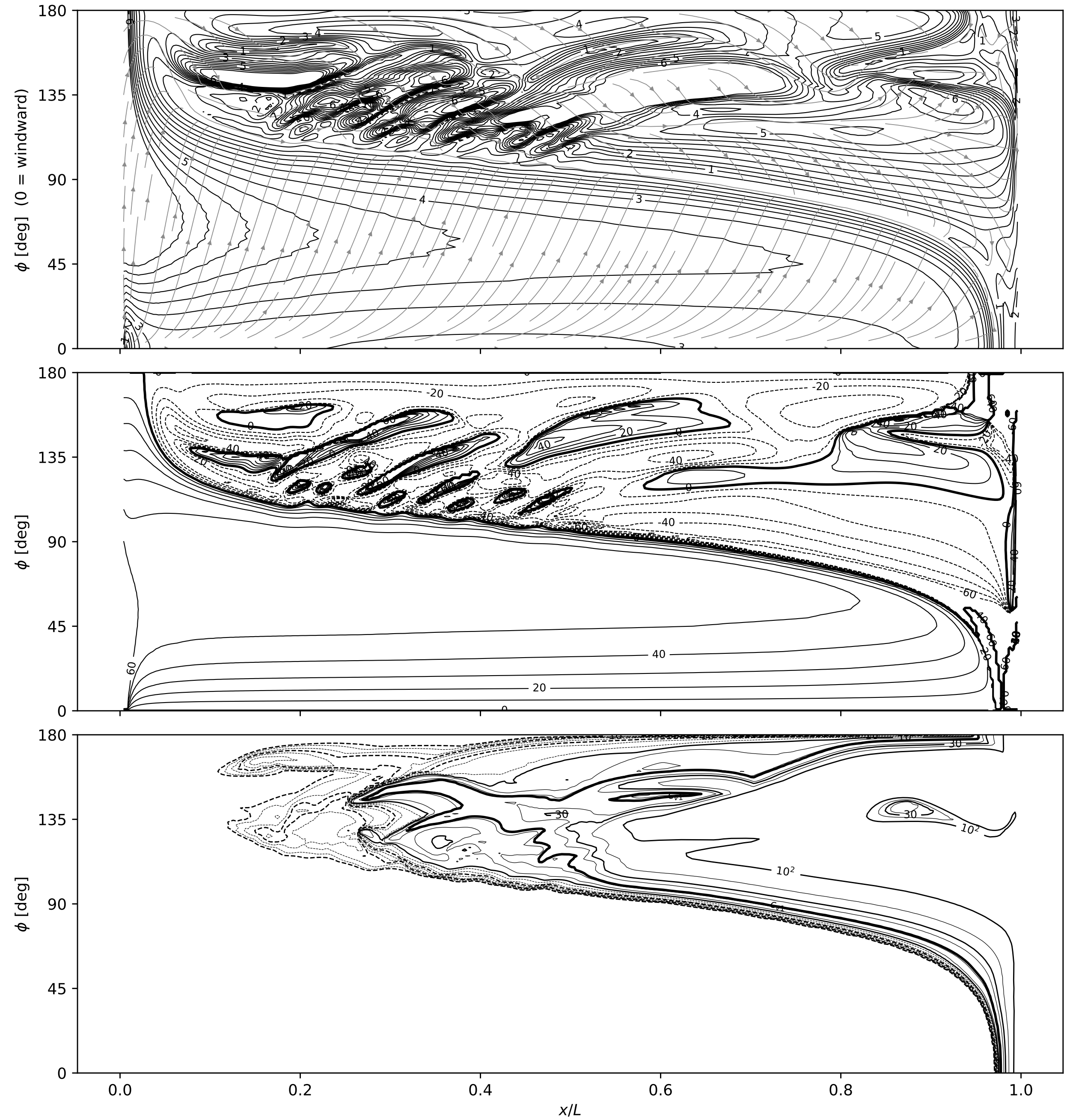

Fig. H3 As Fig. 25, at $\alpha = 29.7°$ (finest grid): the open free-vortex separation covers most of the leeward half, the windward laminar run shortens, and the γ_w field carries the full vortex signature.

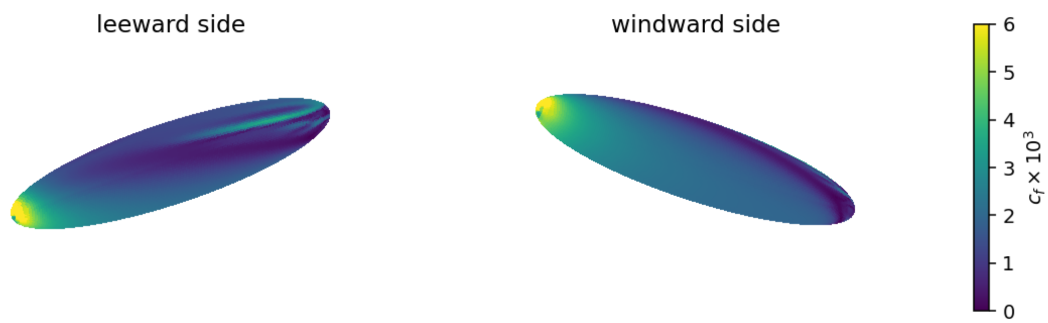

Fig. H4 The c_f magnitude of Fig. 25 rendered on the body (leeward and windward views), the presentation of the N-factor and skin-friction contour figures of the transition-prediction literature on this case.

Appendix I: the two-source rate — inviscid and viscous branches

The rate of the model is built on the single even coordinate $\hat{\Omega}\hat{I}$, the departure of the profile from its osculating parabola. That coordinate carries both the inviscid–inflectional and the viscous Tollmien–Schlichting content, because the two co-vary monotonically across the Falkner–Skan family it is calibrated on. They separate on exactly one profile—a parabola, where $\hat{I} \equiv 0$ identically. Two consequences bound the single-coordinate form. Plane Poiseuille flow is exactly parabolic in wall distance, so the rate returns identically zero at every Reynolds number, while plane Poiseuille is linearly unstable—by a purely viscous mechanism—at $Re_c = 5772$ (Orszag 1971). And on the strongly favorable Falkner–Skan branch the same collapse appears as the rate deficit of Fig. H5: by $\beta = 0.35$ the booked amplification is an order of magnitude below the Drela–Giles envelope, which retains a floor near 5×10^{-3} per unit Re_θ. Note that Blasius itself carries no inflection point, so the canonical Tollmien–Schlichting instability underlying natural-laminar-flow practice is already a viscous instability of a non-inflectional profile: the mechanism the missing branch represents is the main one, not an exotic case.

A two-branch rate separates the mechanisms. Alongside the inflectional coordinate $P_I = \hat{\Omega}\langle\hat{I}\rangle_+$ we form the curvature coordinate

$$P_c = \hat{\Omega}\left\langle -\hat{Z}\right\rangle_+, \qquad \hat{Z} = Z/R, \tag{E5}$$

positive wherever $u'' < 0$ (filled, non-inflected profiles), clipped to zero at an adverse wall, vanishing like d toward the wall and zero in the free stream. Each branch carries its own rate and its own onset, and the two complete sources—not the two rates—are combined:

$$P_{AI} = \omega\,\tilde{\nu}\ \operatorname{softmax}_2\Big[a_{max}\,P_I\,S\big(Re_\Omega/Re_\Omega^{c,i}\big),\ \ a_{visc}\,P_c\,S\big(Re_\Omega/Re_\Omega^{c,\nu}\big)\Big], \tag{E6}$$

$$Re_\Omega^{c,i} = k\,\operatorname{softmin}_2\Big(C,\ A + B\,P_I^{-2}\Big), \qquad Re_\Omega^{c,\nu} = A + B_c\,P_c^{-2}, \qquad \operatorname{softmax}_2(x,y) = \sqrt{x^2+y^2}.$$

The inviscid threshold is the model's existing gate unchanged, ceiling included; the viscous branch shares its floor A and adds one coefficient B_c. Constants: $a_{max} = 0.19$, $a_{visc} = 0.0276$, $A = 124.6$, $B = 1.424$, $B_c = 130$, $C = 1851.2$.

Decoupling the two gates is the substantive change from the form previously explored, which soft-maxed the two rates but soft-minimized the two thresholds into one shared gate. Kelvin–Helmholtz is inviscid and its threshold is essentially the floor A; Tollmien–Schlichting is viscous and its onset is far higher—on Blasius the two read 358 and 2598. A shared soft-minimum applies the lower of them to both rates, so the inviscid branch's low threshold switches on the viscous rate before the viscous mechanism is critical, worth +7.4% on the Blasius anchor. Under Eq. (E6) that cross-talk is absent, and the construction reproduces the single-source rate to within 0.7% over the whole calibrated family (Table F10), so a_{max}, k and the airfoil campaign are untouched: the branch adds amplification only where the single-source rate has none. Retaining the ceiling C is what buys that: dropping it, as the earlier form did, sends the

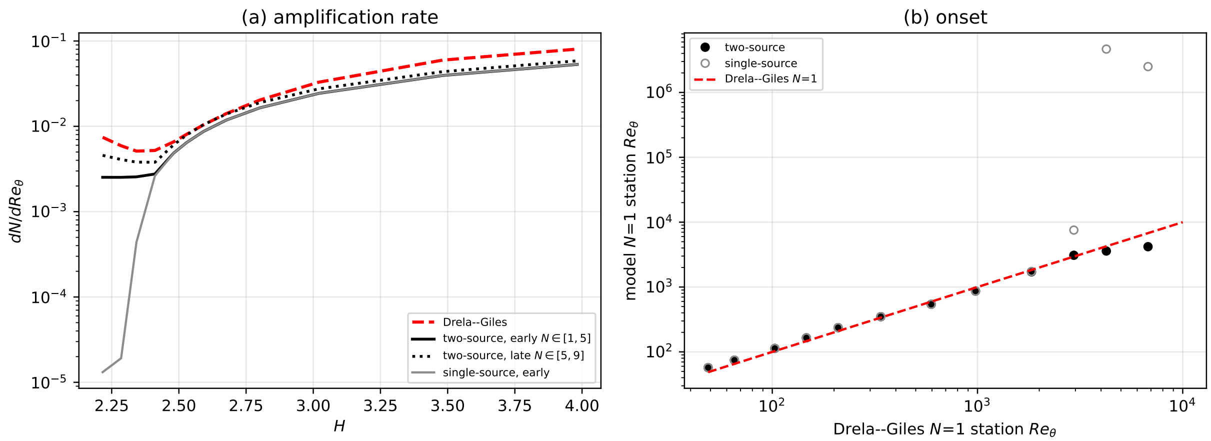

Fig. H5 Two-source rate against Drela–Giles across the Falkner–Skan family, $\beta = +1.0$ (stagnation, $H \approx 2.22$) to -0.1988 (incipient separation, $H = 3.98$); analytic march, no CFD. (a) amplification rate as early ($N \in [1,5]$) and late ($N \in [5,9]$) secants, with the single-source rate for comparison—the low-H collapse is the deficit the viscous branch removes. (b) the marched $N = 1$ station against the Drela–Giles $N = 1$ station $Re_{\theta 0} + 1/(dN/dRe_\theta)$; the diagonal is exact agreement.

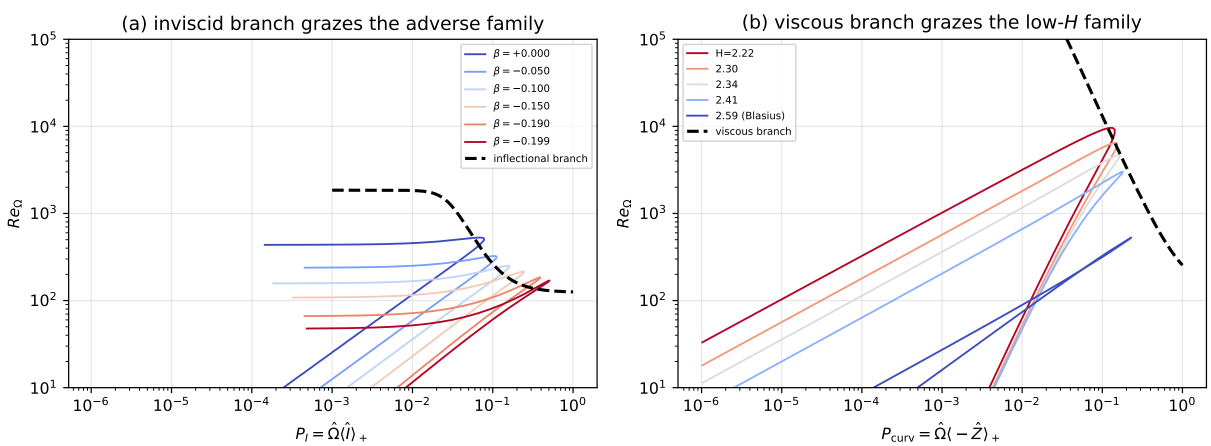

Fig. H6 Each onset branch grazes its own family in its own coordinate, each profile evaluated at its Drela–Giles critical Reynolds number $Re_{\theta 0}(H)$. (a) the inflectional branch $A + B/P_I^2$ against the adverse family; the strongly favorable members collapse to $P_I \to 0$ and leave the panel, which is the onset-resolution limit of a single-coordinate gate. (b) the curvature branch $A + B_c/P_c^2$ against exactly those low-H members, in the coordinate P_c of Eq. (E5).

$\beta = +0.20$ station to $0.63\times$ its single-source value.

The viscous branch's two constants are anchored as the inviscid branch's are: on canonical linear-stability results rather than on transition cases. $a_{\rm visc}$ follows the favorable Falkner–Skan family against the Drela–Giles envelope—itself constructed from Orr–Sommerfeld solutions, and the flow class in which viscous Tollmien–Schlichting amplification is both well characterized and the observed route to transition. Plane Poiseuille then supplies an independent check on B_c, exact and with no refitting: on a parabola $P_I \equiv 0$, so the channel constrains the viscous branch with no contamination from the inflectional one. Requiring the frozen-profile eigenvalue of Eq. (E6) to change sign at Orszag's $Re_c = 5772$

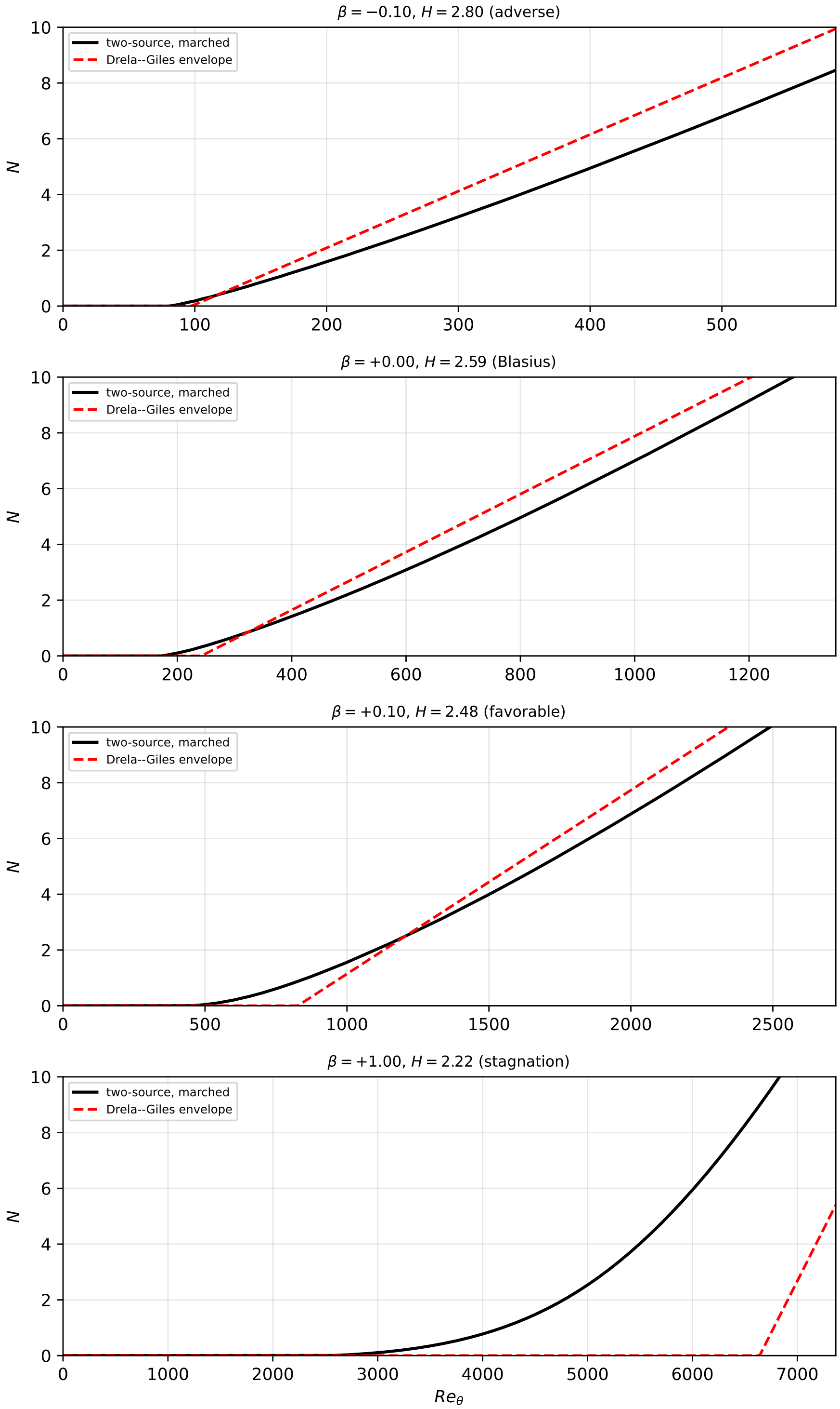

Fig. H7 Disturbance transport marched with Eq. (E6) on four wedges—adverse, Blasius, favorable, and the stagnation profile ($\beta = +1$) that the single-source kernel cannot ignite at all—against each profile's Drela–Giles envelope. On the stagnation wedge the branch ignites at $Re_\theta \approx 2.5 \times 10^3$ against the correlation's critical station $Re_{\theta 0} = 6.5 \times 10^3$: early, but finite where the single-source rate is identically zero. Same parabolized instrument and same $c_{\nu,\mathrm{ai}} = 1/6$ as the model's own calibration; the viscous branch touches the rate floor and the gate only.

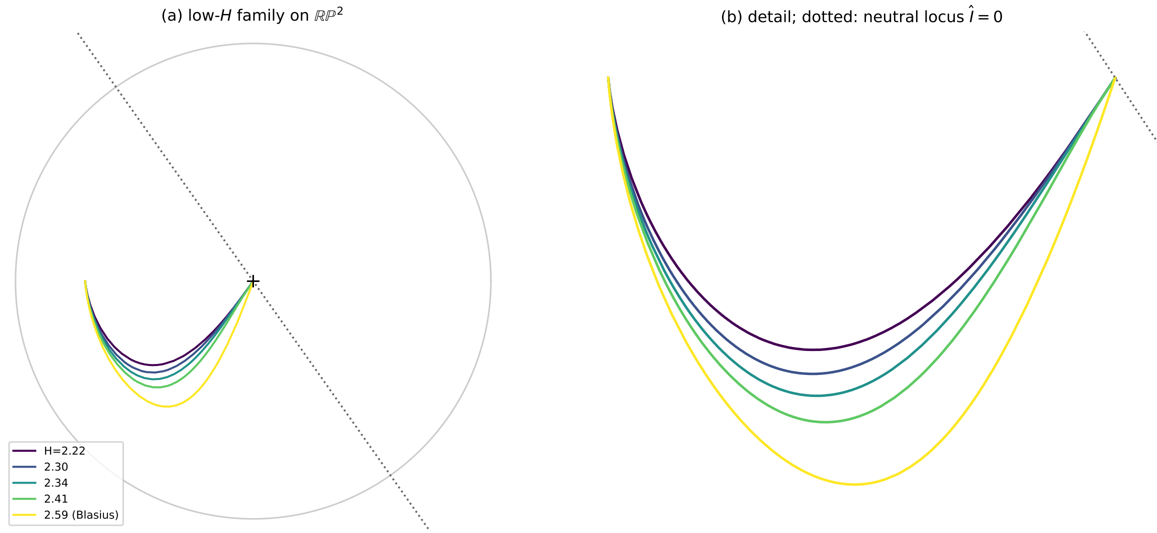

Fig. H8 The low-H Falkner–Skan family on the $\mathbb{RP}^2$ indicator sphere (orthographic, curvature vertical), from the stagnation profile $H \approx 2.22$ to Blasius $H = 2.59$, with (b) the near-wall detail. The family bunches against the neutral parabola locus, where $\hat{I}$—and with it P_I—is too small to distinguish members whose Drela–Giles critical Reynolds numbers differ by nearly a factor of two. The curvature coordinate P_c separates them.

Table F10 Peak dimensionless source $a\,S$ on Falkner–Skan wedges: the single-source rate, and the two-source form of Eq. (E6) at $(a_{\mathrm{visc}}, B_c) = (0.0276, 130)$.

β	Re_θ	single-source	two-source	ratio
0	500	0.01485	0.01485	1.000
0	1000	0.01485	0.01487	1.001
−0.10	400	0.03082	0.03083	1.000
−0.1988	300	0.09692	0.09764	1.007
+0.10	1000	0.00695	0.00696	1.000
+0.20	2000	0.00281	0.00282	1.004
+0.35	4000	0.00033	0.00405	12.4

gives Table F11: at the Falkner–Skan value $a_{\mathrm{visc}} = 0.0276$ the channel asks for $B_c = 91$ against the family's 130, and the family's constants place the channel's neutral point at 7409, 1.28× the true value—against a single-source kernel that never destabilizes it at all. Two unrelated flows agree on the same constant to 30%. The channel is a calibration and verification device only, never a validation case: its observed transition is subcritical and finite-amplitude, outside the envelope class this model represents.

Two limits of the construction are worth stating with it. The viscous rate $a_{\mathrm{visc}} P_c$ is Reynolds-independent, an assumption inherited from the inviscid branch where it is justified because the mechanism is inviscid; a viscous instability's rate need not be, and the Reynolds dependence of the plane-Poiseuille envelope is the natural test. And the branches are a decomposition of the rate, not a claim that each carries one physical mechanism exclusively: the viscous gate on Blasius opens only near $Re_\theta \approx 1190$, so Blasius remains carried by P_I in this parameterization.

Table F11 Plane-channel check: the B_c that would place the neutral point on Orszag's $Re_c = 5772$, and the neutral point actually obtained at the Falkner–Skan-anchored $B_c = 130$.

a_{visc}	B_c for $Re_c = 5772$	Re_c at $B_c = 130$
0.0200	81.0	8095
0.0276	91.0	7409
0.0350	99.9	6916
0.0600	127.9	5834